Meeting of the Board
17 – 20 February 2025
Songdo, Incheon, Republic of Korea
Provisional agenda item 10

**GCF/B.41/02/Add.08**

27 January 2025

# Consideration of funding proposals – Addendum VIII

## Funding proposal package for FP260


### Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration (FOREST Invest)";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat’s assessment;
e) Independent Technical Advisory Panel’s assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

**Table of Content**

# Funding Proposal

| | |
|---|---|
| Project/Programme title: | Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration (FOREST Invest) |
| Country(ies): | Republic of Serbia |
| Accredited Entity: | FAO |
| Date of first submission: | 2023/10/31 |
| Date of current submission | 2025/01/20 |
| Version number | V.5 |

GREEN CLIMATE FUND

## Contents



***Note to Accredited Entities on the use of the funding proposal template***

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

**Please submit the completed proposal to:**
fundingproposal@gcfund.org
**Please use the following name convention for the file name:**
"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

| LIST OF ACRONYMS | |
|---|---|
| AE | Accredited Entity |
| AFOLU | Agriculture, Forestry, and Land Use sector |
| APR | Annual Performance Report |
| ARA | Adaptation Results Area |
| BAU | Business-As-Usual |
| BH | Budget Holder |
| CAP | Common Agricultural Policy |
| CAS | Climate Adaptive Silviculture |
| CBAM | Carbon Border Adjustment Mechanism |
| CBD | Convention on Biological Diversity |
| CC | Climate Change |
| CE | Collect Earth |
| CPF | Country Programme Framework |
| DoF | Department of Forests |
| DPCE | Digital Platform on Circular Economy |
| EBRD | European Bank of Reconstruction and Development |
| EE | Executing Entities |
| EQ | Ellenberger Climate Quotient |
| ERP | Oracle-based Enterprise Resource Planning System |
| ESG | Environmental, Social, Governance |
| ESMP | Environmental and Social Management Plan |
| ETF | Enhanced Transparency Framework |
| ETS | Emissions Trading System |
| EU | European Union |
| FAI | Forest Aridity Index |
| FAO | Food and Agriculture Organization of the United Nations |
| FLO | Funding Liaison Officer |
| FLR | Forest and Landscape Restoration |
| GHG | Greenhouse Gas Emissions |
| GoS | Government of Serbia |
| GRMS | Global Resources Management System |
| HH | Household |
| IEA | International Energy Agency |
| IFAD | International Fund for Agricultural Development |
| ILFE | Institute of Lowland Forestry and Environment, Novi Sad |
| IPA | Pre-Accession Assistance |
| IPPU | Industrial Processes and Product Use |
| IRR | Internal Rate of Return |
| LCDS | Low Carbon Development Strategy |
| LTO | Lead Technical Officer |
| LULUCF | Land Use, Land Use Change and Forestry |
| MOAFWM | Ministry of Agriculture, Forestry and Water Management |

| | |
|---|---|
| ME | Ministry of Economy |
| MEP | Ministry of Environmental Protection |
| MESTD | Ministry of Education, Science and Technology Development |
| MME | Ministry of Mining and Energy |
| MoF | Ministry of Finance |
| MoI | Ministry of Interior |
| MoV | Means of Verification |
| MRA | Mitigation Results Area |
| MRV | Monitoring, Reporting and Verification |
| NAP | National Adaptation Plan |
| NDA | National Designated Authority |
| NDC | National Determined Contributions |
| NFMA | National Forest Monitoring and Assessment System |
| NGO | Non-governmental Organization |
| NPV | Net Present Value |
| NTFP | Non-timber Forest Product |
| NWFPs | Non-Wood Forest Products |
| ODA | Official Development Assistance |
| P | Monthly accumulated precipitation |
| PES | Public Enterprise Serbia Shume |
| PEV | Public Enterprise Vojvodina Shume |
| PFOA | Private Forest Owners Association |
| PKS | Chamber of Commerce |
| PMU | Project Management Unit |
| PSC | Project Steering Committee |
| PTF | Project Task Force |
| RCP | Representative Concentration Pathways |
| RES | Renewable Energy Sources |
| REU | FAO Regional Office for the Regional Office for Europe and Central Asia |
| SEEPEX | Serbian Power Exchange |
| SEP | Stakeholder Engagement Plan |
| SERBIO | Serbian biomass association |
| SFM | Sustainable Forest Management |
| SRP | Short Rotation Plantations |
| TA | Technical Assistance |
| TCP | Technical Cooperation Program |
| TNC | The Third Report of the Republic of Serbia under the United Nations Framework Convention on Climate Change |
| TJ | Tera Joule |
| UNCCD | United Nations Convention to Combat Desertification |
| UNFCC | United Nations Framework Convention on Climate Change |
| VCM | Voluntary Carbon Markets |
| WB | The World Bank |

## A. PROJECT/PROGRAMME SUMMARY

<table>
<tr><td>A.1. Project or programme</td><td>Project</td><td>A.2. Public or private sector</td><td colspan="2">Public</td></tr>
<tr><td>A.3. Request for Proposals (RFP)</td><td colspan="4">Not applicable</td></tr>
<tr><td rowspan="11">A.4. Result area(s)</td><td colspan="2"></td><td>GCF contribution</td><td>Co-financers' contribution</td></tr>
<tr><td colspan="2">Mitigation total</td><td>33 %</td><td>27 %</td></tr>
<tr><td colspan="2">☒ Energy generation and access</td><td>5.5 %</td><td>1 %</td></tr>
<tr><td colspan="2">☐ Low-emission transport</td><td>- %</td><td>- %</td></tr>
<tr><td colspan="2">☒ Buildings, cities, industries and appliances</td><td>5.5 %</td><td>1 %</td></tr>
<tr><td colspan="2">☒ Forestry and land use</td><td>22%</td><td>25 %</td></tr>
<tr><td colspan="2">Adaptation total</td><td>67 %</td><td>73 %</td></tr>
<tr><td colspan="2">☒ Most vulnerable people and communities</td><td>17 %</td><td>19 %</td></tr>
<tr><td colspan="2">☐ Health and well-being, and food and water security</td><td>- %</td><td>- %</td></tr>
<tr><td colspan="2">☒ Infrastructure and built environment</td><td>- %</td><td>52%</td></tr>
<tr><td colspan="2">☒ Ecosystems and ecosystem services</td><td>50 %</td><td>2 %</td></tr>
<tr><td rowspan="3">A.5. Expected mitigation outcome<br>(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)</td><td rowspan="3">-8.4 million $tCO_{2eq}$</td><td rowspan="3">A.6. Expected adaptation outcome<br>(Core indicator 2: direct and indirect beneficiaries reached)</td><td colspan="2">3,575,444<br>(1,823,476 women)<br>54 % of the total population</td></tr>
<tr><td>729,064 direct<br>(371,823 women)</td><td>2,846,380 indirect<br>(1,451,654 women)</td></tr>
<tr><td>11%</td><td>43%</td></tr>
<tr><td>A.7. Total financing (GCF + co-finance)</td><td>83,811,581 USD</td><td rowspan="2">A.9. Project size</td><td colspan="2" rowspan="2">Medium (Upto USD 250 million)</td></tr>
<tr><td>A.8. Total GCF funding requested</td><td>25,000,000 USD</td></tr>
<tr><td rowspan="2">A.10. Financial instrument(s) requested for the GCF funding</td><td colspan="4">Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</td></tr>
<tr><td colspan="2">☒ Grant Enter number<br>☐ Loan Enter number<br>☐ Guarantee Enter number</td><td colspan="2">☐ Equity Enter number<br>☐ Results-based payment Enter number</td></tr>
<tr><td>A.11. Implementation period</td><td>7 years (84 months)</td><td>A.12. Total lifespan</td><td colspan="2">27 years</td></tr>
<tr><td>A.13. Expected date of AE internal approval</td><td>25 October 2024</td><td>A.14. ESS category</td><td colspan="2">B</td></tr>
<tr><td>A.15. Has this FP been submitted as a CN before?</td><td>Yes ☒ No ☐</td><td>A.16. Has Readiness or PPF support been used to prepare this FP?</td><td colspan="2">Yes ☒ No ☐</td></tr>
<tr><td>A.17. Is this FP included in the entity work programme?</td><td>Yes ☒ No ☐</td><td>A.18. Is this FP included in the country programme?</td><td colspan="2">Yes ☒ No ☐</td></tr>
</table>

| | |
|---|---|
| **A.19. Complementarity and coherence** | Yes ☒ No ☐ |
| **A.20. Executing Entity information** | The Executing Entities will be FAO, the Ministry of Agriculture, Forestry and Water Management (MOAFWM), Serbia Shume (PES) and Vojvodina Shume (PEV). GCF funds will be managed solely by the Food and Agriculture Organization of the United Nations (FAO) while the other executing entities will manage interventions to be implemented through their co-financing. |

**A.21. Executive summary**

i. **Context and climate vulnerability:** The Republic of Serbia is recognized as a global hub of plant diversity, with forests playing a crucial role in its economy. Forests are vital for Serbia's energy security, providing for approximately 40% of its population, as well as supporting rural livelihoods and ecosystem services. Climate change and excessive fuelwood extraction are degrading forests, undermining the country's efforts to adapt to and mitigate climate change and exacerbating poverty. Rising energy prices, due to the current regional conflicts, further worsen the situation by increasing demand for fuelwood in both rural and urban areas.

ii. From 1980 to 2019, average temperatures have shown a significant increasing trend, with a rise of 0.6 °C per decade. Greater increases were observed in altitudes above 1000 m and during the summer months (June - August). Frost and ice days decreased, while summer days, tropical nights and heat waves increased. Projections highlight that between 2020-2060, average temperatures are expected to increase each decade by +0.29°C (RCP 4.5) and +0.48°C (RCP8.5) [Annex 24; TNC, 2020]. Precipitations are projected to remain stable with large yearly variations. Summer days will increase continuously between 2016-2035 through the end of the century [TNC, 2020;]. Extreme heat waves could occur at least 2-3 times a year compared to the 1986-2005 period, when considered rare events, while frost and ice days will continue decreasing, especially at higher altitudes (RCP4.5 and RCP8.5).

iii. **Key Challenges:** Damage caused by pests, diseases, fires, and climatic hazards has led to a 268% increase in the loss of wood volume between 1966 and 2019. Moreover, the largely uncontrolled extraction of fuelwood (>7 mcm yearly) and outdated forest management processes are worsening adverse climate change impacts on forests. Fuelwood is still the main source of heating energy in rural areas (77% of households) and for the poor (10.5 % of total population in rural areas) who cannot afford other sources of energy. Climate change, rising energy prices and current geopolitical tensions threaten Serbia's energy security, further increasing Serbia's dependency on imports of energy and fossil fuels, with a possible additional impact on demand for fuelwood, estimated at 10% compared to Business-As-Usual (BAU).

iv. The energy sector is the main contributor to greenhouse gas (GHG) emissions (78% of total in 2018). The Agriculture, Forestry, and Land Use sector (AFOLU) contributed the second highest share with 7%. In the BAU scenario, total GHG emissions, including Land Use, Land Use Change and Forestry (LULUCF), are expected to increase by 4.5% by 2030 and 14.4% by 2050 (compared to 2010), and the more forests are damaged, the more total emissions will increase, resulting in additional costs for low-carbon development strategies. The country has set climate and energy targets that include increasing carbon removals and sinks from forests by 17% by 2030 and 22% by 2050 compared to 2010 levels. Failing to achieve these targets would have severe consequences for Serbia's adaptation and mitigation commitments.

v. **The main barriers** to resilient forest-energy security and decarbonization are (**I**) High climate change adaptation deficit[1] of forest stakeholders to guarantee key ecosystem services (energy, protection, livelihood, among others); (**II**) Incomplete mechanisms for forest management and carbon governance; and (**III**) Limited capacities and incentives for private sector and community engagement in sustainable forest management (SFM) and decarbonization.

vi. **Proposed Interventions:** The project opts for a nexus approach and addresses bottlenecks to climate change adaptation and mitigation through three components: **Component 1** - National level upscaling of sustainable and climate adaptive silviculture (CAS) and carbon finance framework; **Component 2** - Improving energy security and livelihood from climate resilient forest ecosystem and GHG emissions reductions from increased carbon sinks and decarbonization opportunities; and **Component 3** - Engaging private sector in CAS and decarbonization investments. Taken together, **these actions aim to enable Serbia to reduce the exposure and vulnerability of its forestry sector and enhance resilience and energy security of households, while increasing total $CO_2$ removals from forestry and biodiversity.** Given the ownership and management structure of forests in Serbia, the project will operate at the national level while considering regional specificities. The project will work in synergy with projects of the GEF programme and other donors, and will introduce a number of innovations, including: (i) CAS practices, management and carbon finance, technologies and investments; (ii) a crosscutting approach to ensure that investments in adaptation also yield mitigation and poverty alleviation results; (iii) linkages of national banks and investors with private sector companies interested in decarbonization and forestry; and (iv) digital tools to aggregate and engage private owners of land and forests. The last point will allow the scaling up of technologies, practices and investments to the national scale and transform the private sector into an actor of change. The project shows solid parameters, with a 20-year Economic IRR of 15.0% (higher than the social discount rate), a Net Present Value (NPV) of US$ 78.4 million, and a 40-year Economic IRR of 17.6 % with an NPV of US$227.5 million. Section D and Annex 3 outline various scenarios considering the impact that mobilization of leveraged loans through the project's activities would have on the overall results. While the target is USD 50 million, even in a scenario where no loans are disbursed, the project's aggregated economic results remain positive over a 20-year horizon, with an EIRR of 11.5% and an NPV of US$39.7 million.

vii. **Climate results and co-benefits:** The project aims to directly enhance the resilience of 729,064 persons (371,823 women) while increasing carbon removals from the forestry sector (7.6 $MtCO_{2eq}$ (27Y) and reducing net emissions by the private sector (0.8 $MtCO_{2eq}$ in 27y). Indirectly the project will positively impact 2.8 million people (1.5 million women). Main co-benefits of the project will be biodiversity conservation, gender empowerment and poverty reduction**.** The project will directly contribute to GCF USP-2 portfolio-level targets on mitigation and adaptation results and to the USP-2 targeted results areas: 5 (Ecosystems), 6 (Infrastructure) 7 (Clean Energy), and 10 ( Local private sector early-stage ventures and MSME innovation).

## B. PROJECT/PROGRAMME INFORMATION

### B.1. Climate context

1. **Context, Climate Trends and Projections**: Serbia's topography features a wide range of elevations, ranging from 17 meters above sea level (a.s.l.) in the plains of Vojvodina to 2,660 meters a.s.l. in the southern mountain ranges. Serbia has a warm-summer, humid continental climate with regional differences due to its geographical location in southeastern Europe. The climate is characterized as continental, with four distinct seasons: spring, summer, autumn, and winter. Climate change has already impacted Serbia [FAO, 2021][2]: Annual average temperatures show significant increasing trends between 1980 and 2019 with + 0.6 °C per decade [FAO, 2021]. Greater increases were observed in altitudes above 1000 m and during the summer months (June - August) [SNC, 2017[3]; UNDP, 2018[4], TNC, 2020; FAO 2021]. Daily meteorological data from the National Meteorological Institute[5] highlight that frost days and ice days decreased by 8 days and 3 days, respectively, every decade between 1980 and 2019 [FAO, 2021]. Summer days and tropical nights on the other hand increased by 8 and 1 day(s) per decade. Heat waves[6] increased by 20 days and up to 30 days in lowlands and western parts [TNC, 2020]. No significant trends were observed in the total annual rainfall for the period 1980-2019, and the duration of dry spells remained stable, with an average of 27 days per year [FAO, 2021]. However, in the same period, there was a clear trend towards increased rainfall variability, with +24mm/dec variability of accumulated precipitation. The average standard deviation has risen from approximately 57 mm during the 1970-1980 period to 150 mm during the 2009-2019 period. Furthermore, an increasing number of meteorological, agricultural, and hydrological drought events can be detected. The average number of drought months per year has increased to +0.9 to +1.4 months per decade depending on the type of drought (SPEI indices) [Annex 24, p. 83-85].

2. Projections highlight that climate change will affect Serbia even more in the future. Between 2020-2060, average temperatures are expected to increase by +0.29°C (RCP 4.5) and +0.48°C (RCP8.5) each decade [FAO, 2021; TNC, 20207]. Minimum and maximum temperatures are expected to increase in a similar way over the next 40 years. July, August and September will be the most critical months regarding average temperature increase. Warming will be higher in July-August (up to +4.7°C) and during the fall (1986-2005 reference period) [UNDP, 20188]. Summer days will increase continuously, reaching an additional 15 days between 2016-2035 and +30 days by the end of the century [TNC, 2020]. These trends will be more pronounced in lower altitudes, especially in Vojvodina and Central Serbia. Extreme heat waves could occur at least 2-3 times a year compared to the 1986-2005 period, when considered as rare events [UNDP, 2018], while frost and ice days will continue decreasing, especially at higher altitudes (-15 days RCP4.5 and - 30 days RCP8.5). Reportedly [TNC 2020; FAO, 2021], annual precipitation trends are predicted to remain stable (RCP 4.5), and become negative, especially in summer (-13 mm per decades) between 2020-2060 (RCP 8.5). Analysis of the Ellenberger Climate Quotient (EQ)[9] revealed that at country level, the EQ is projected to increase by +0.75°C/mm per decade under the RCP4.5 scenario, and by +1.88°C/mm per decade under the RCP8.5 scenario [FAO, 2021]. Similarly, the analysis of the Forest Aridity Index (FAI)[10] confirms that at country level, FAI trends are projected to significantly increase between 2020 and 2060 under RCP8.5 scenarios by +0.42°C/mm per decade [ibid].

3. **Exposure and Susceptibility**: The three national communications to the United Nations Framework Convention on Climate Change (UNFCCC) concur in identifying agriculture and forestry as the most vulnerable subsectors in Serbia. These sectors are vital for the country's economy, and potential damage to forests is expected to have far-reaching consequences on energy security, livelihoods, and poverty levels in both rural and urban communities, as well as to environment and biodiversity. Forests, which cover approximately 30% of the country, are a crucial sector for the population due to the following: (i) provision of fuelwood, an essential energy source for about 47% of the population, particularly in rural areas, where it is used by 77% of households [ROS, 2021]; (ii) it acts as a primary defense against floods and landslides; and (iii) it represents, after agriculture, the most important sector for economic activities in rural areas. Fuelwood extraction is largely uncontrolled and poorly regulated by outdated forest management processes, partly explaining why official statistics generally represent a fraction of its consumption. Climate change, rising energy prices and current geopolitical tensions threaten Serbia's energy security, amidst a context in which prices have already been rising significantly over the last several years (see Annex 2, par. 110 - 112).

4. The price increases have particularly strong impacts on poor and vulnerable people. An electricity tariff increases of 16.3% will lead to an increase in the share of household income spent on electricity by 0.5 percentage points. At the same time, the overall poverty rate can increase by 1% under these conditions [World Bank, 2022a]. Despite government interventions, energy costs are already a high burden for the country, as evidenced by the fact that households currently spend more than 10% of their average expenditure on energy (the threshold to be considered "energy poor") [World Bank, 2017] and that Serbia ranks among the top 10 continental European countries with the highest share of households reporting that they could not keep their home adequately warm [World Bank, 2022a]. These trends reduce households' purchasing power and slow poverty reduction.

5. Given that fuelwood is still the cheapest and most easily available source for heating, rising energy prices suggest that its consumption will increase further. According to official statistics, the share of wood fuels in final energy consumption has increased from 11% in 2017 to 11.86% in 2020. Private and public forests are under pressure to increase logging to meet growing demand, and current forest use is already unsustainable [Glavonjić, B. 2017]. In the BAU scenario, fuelwood extraction will continue to be the main anthropogenic driver of forest degradation. As most of GHG emissions stem from the energy sector, the expansion of renewable energy from biomass plays an important role for the country's efforts to mitigate its climate change impact and achieve its development goals. As a European Union (EU) candidate country, Serbia aligns its actions to EU-directives and policies and generally to the EU *acquis* (see Annex 2, par. 112) and intends to increase RES from the current 26% to 34% in 2030, with biomass energy being an important contributor. There is therefore an urgent need to organize the sector in a sustainable and integrated manner to allow efficient resources use and to utilize the potential of other biomass sources - like short rotation plantations (SRP) and agricultural harvest residues for energy purposes. If properly and sustainably addressed, biomass, including fuelwood, can act as a central element in fulfilling national energy, climate and development objectives and can also serve as a tool to alleviate poverty in rural areas.

6. The health and diversity of forests are essential to secure livelihoods for rural communities in Serbia, given that the latter depend on job opportunities and production of non-wood forest products (NWFPs) such as mushrooms, herbs, honey, and berries. These products are estimated to generate an annual value of approximately USD 39.6 per ha per year (Annex 3b). Forests are the most important asset in reducing Serbia's total net emissions [TNC, 2020], but removals are decreasing over time due to both anthropic and climatic adverse impacts.

7. **Vulnerability:** The vulnerability of Serbian forests arises from a combination of climatic and anthropogenic factors, as well as from the lack of preparedness among public and private stakeholders, and inadequate integration of the forest-energy-climate nexus across sectors. Temperature increases and precipitation variabilities have already led to increased periods of droughts and a faster expansion of forest fires [SNC, 2017], as well as to increased pest outbreaks, decreased forest vitality and decreased soil water content [TNC, 2020]. Reportedly [Djurdjevic, 2020], the increasing number of climate related outbreaks of herbivorous insects like *Corythucha arcuate* and *Pytiogenes chalcographus* have led to extensive mortality of *P. abies,* causing severe ecological and economic losses [Stojanovich, 2021]. Research shows that changes in abiotic factors, such as temperatures and precipitation, change vegetation patterns and the provision of ecosystem services [Laco, 2021]. Recent studies highlight furthermore that globally climate change has significantly decreased the resilience of temperate forests globally, such as the one present in Serbia[11] [Forzieri, 2022]. This is confirmed by data from the Ministry of Agriculture, Forestry and Water Management (MOAFWM) showing that the adverse impacts from pests,[12] diseases, fires and climatic hazard on Serbian forests continue to increase (+268% of $m^3$ of losses between 1966 and 2019). More information on vulnerability is in Annex 2. Par. 89-90.

8. **GHG Profile of Serbia:** Total GHG emissions without removals amounted to 64,592 kt $CO_{2eq}$ in 2020, [2BUR, 2023]. The energy sector is the main source of emissions, representing 78.6% of the total (2020 data, 2BUR, 2023) with 50,739 kt$CO_{2eq}$. The second largest GHG emitting sector is the AFOLU sector with around 9%, followed by the Industrial Processes

and Product Use (IPPU) and the waste management sectors. The forestry sector contributed to $CO_2$ removals for 8.4% of Serbian's emissions equivalent [GFA, 2019]. At the same time, the forestry sector decreased by 16% between 2010 and 2020 in regard to $CO_2$ removals from the atmosphere. In the BAU scenario, total GHG emissions, including AFOLU, are expected to increase by 4.5% by 2030 and by 14.4% by 2050 (compared to 2010 levels) and the more forests will be damaged the more total net emissions will grow.

9. **Proposed Interventions:** Based on a recent analysis performed by the Forestry Institute of Serbia (Annex 16) as well as from the findings of an analysis performed under the FAO/GEF project "Contribution of Sustainable Forest Management to a Low Emission and Resilient Development", forest areas that are in need of urgent climate proofing are distributed across the regions of Central Serbia and Vojvodina. Therefore, the project will work in these regions with both national (Department of Forestry at the Ministry of Agriculture, Forestry and Water Management and academia) and local institutions (provinces, municipalities, civil society organizations (CSOs) and academia). Detailed descriptions of forests in both regions are available in Annex 2, Appendices 3-5). Maps of project locations are inserted in Annex 16.

10. Environmental and climate finance in Serbia has largely focused on water management, energy, and biodiversity. While these sectors are of great importance to the country and its adaptation path, there have been limited forestry projects in the Republic of Serbia, and currently, none of them specifically address climate change and its adverse impacts on the forestry-energy-decarbonization nexus. A complete list of ongoing projects and their complementarities and synergies with this project are available in Annex 2, Table 30. The project will seek and enhance collaboration for addressing climate-related needs by combining unique contributions from various partners and maximizing in this regard existing local insights, specialized skills, and financial resources.

11. The activities of the proposed intervention build on the initiative "Contribution of sustainable forest management to a low emission and resilient development in Serbia", financed by GEF6 and implemented by FAO. This initiative was successful in developing new approaches and improved methodologies for SFM with benefits on biodiversity and climate change mitigation, as well as in its implementation of, among others, the 2nd National Forestry Inventory. The terminal evaluation of the project recommended to increase efforts to institutionalize sectoral tools and guidelines, in particular the introduction of a permanent national forest monitoring system (NFM) and to fully engage private forest owners and other stakeholders in management and investments strategies. These recommendations have therefore been considered of primary importance and as guiding principles in the design of this GCF project.

## B.2 (a). Theory of change narrative and diagram

12. **Goal Statement**: **IF** knowledge and technology, coupled with climate-sensitive forestry investments, are made available in Serbia, and transparent, inclusive governance mechanisms for forests and carbon are established, **THEN** the country will have an enabling environment to foster cooperation between private and public actors to increase and sustainably and adaptively manage forests, reduce the exposure of the most vulnerable, and achieve its strategic adaptation and mitigation targets, **BECAUSE** these investments will strengthen forest ecosystems, enhance energy security, and support vulnerable communities—particularly women and girls—while increasing carbon storage and advancing decarbonization within the agrifood sector.

13. **Assumptions**: Forests in the Serbia have a vital role in ensuring energy security and other ecosystem services that are key to the economy and the survival of communities. They are the second most significant source of livelihood in rural areas, following agriculture, and play a pivotal role in enabling both low-carbon development of the economy and private sector engagement in decarbonization. Nonetheless, similar to other Balkan and European countries [Forzieri, 2022], forests in Serbia are increasingly vulnerable to climate change[13] (Annex 2 par 89-90). As outlined in Chapter B.1 and further explained in Annex 24, Serbia is already experiencing the impacts of climate change. Despite the existence of laws, strategies, and policies that have forest ecosystems services (e.g. energy, livelihoods, non-timber forest product NTFP and carbon removal) at their core[14], investments aimed at adapting forest ecosystems, reducing the vulnerability of communities reliant on these ecosystems, and maximizing the contribution of forests towards Serbia's climate change targets remain limited. In the BAU scenarios, variability in precipitation and increasing temperatures negatively affect forests' ecosystems in terms of tree growth and health, natural expansion, $CO_2$ removal (-10% in 2030 compared to 2005) and other ecosystem services. National net GHG emissions are predicted to increase by 4.5% by 2030 (compared to 2015) while forests are inefficiently used for fuel or encroached by agriculture and are on the brink of overexploitation, with adverse impacts on energy security of the poor and increasing the overall cost of decarbonization by 22%. Finally, there is a strong nexus between forestry energy security and decarbonization. The current limited capacities of public and private actors as well as the weak governance of energy, forestry and decarbonization are preventing the country's low carbon and climate resilient development.

14. Along with climate change impacts (section B1 and Annex 24), in the BAU scenario, fuelwood extraction is and will remain the main anthropogenic driver of forest degradation, affecting the energy security, livelihoods and decarbonization efforts of Serbia. Even so, as established in a number of national strategies and policies (table 11), climate adaptive and sustainably managed forests constitute a central and non-exchangeable element for the fulfilment of key adaptation, poverty alleviation, energy security and climate mitigation objectives. As the nexus between forest-energy-livelihoods and decarbonization is not exchangeable, there is an urgent need to transform the sector from the current unsustainable and maladaptive scenario to one that is sustainable, integrated and climate adaptive, to allow the country to fulfil its climate change adaptation and mitigation goals. This will also require the inclusion of all stakeholders and the creation of mechanisms and incentives to increase private sector participation in forestry. The adaptation of forests and their associated ecosystem services represents a tangible economic opportunity as it will guarantee access to low-cost sustainable energy sources, livelihoods, and carbon removals. The connection between adaptation, mitigation and sustainable development is also explicitly recognized by the country's Low Carbon Development Strategy (LCDS), which includes adaptation measures of the forestry sector as a priority for the achievement of national climate change mitigation goals, respecting social and environmental aspects[15]. Additionally, the proposed approach will secure a wider participation of stakeholders into forest governance with clear and measurable co-benefits in terms of adaptation and low emission development.

15. **Barriers:** The main root causes and bottlenecks to a sustainable, adaptive, efficient, effective and transparent management of the forest-energy security and decarbonization nexus are: (I) High climate change adaptation deficit[16] of forest stakeholders to guarantee key ecosystems services (energy, protection, livelihood, among others); (II) Incomplete mechanisms for forest management and carbon governance; and (III) Limited capacities and incentives for private sector and community engagement in SFM and decarbonization. Further information on how the project will address these barriers is included in Table 23 of Annex 2.

16. **Main risks:** The combined adverse effects of identified barriers (Table 2) impede the forestry sector from adapting to climate change (B1) and contribute efficiently to the main sustainability and climate related target. Furthermore, under a BAU scenario, the identified barriers do not allow the AFOLU sector to fully express its potential to (i) secure ecosystem services, such as provisions of energy and NWFP, (ii) increase carbon sequestration and (iii) engage the private sector in SFM and decarbonization. Consequently, in a BAU scenario, forests will continue to degrade, increasing the exposure of vulnerable communities and threatening the energy security and livelihoods of the poor and rural communities, further compromising Serbia's National Determined Contributions (NDCs) targets, biodiversity and the role of forests to protect and sustain communities. This hinders the country's efforts to become energy secure, reduce poverty and decarbonize its economy as planned through the Energy Security Strategy (2022), the LCDS (2023), the Law on Climate Change (2020), the updated NDC (2021) and the upcoming Integrated Energy and Climate Action Plan. Furthermore, as the country's climate and energy targets have at their core the increase of carbon removal and sink from forests by 17% by 2030 and by 22% by 2050 (compared to 2010) it will increase the overall cost of decarbonizing the economy of Serbia.

17. **Options analysis:** Given the specificity of the sector and its relevance for key productive sectors in Serbia, the existing policy framework, including NDC and LCDS, the available options to address identified barriers and mitigate risks are limited. These can be summarized as follows: (I) No action, (II) Sector Approach, and (III) Nexus approach.

18. **No action approach:** Failure to reach the forestry targets would have drastic impacts on Serbia's adaptation and mitigation commitments. It would also further aggravate the overall economic impact that climate change has had on Serbia (climate change has caused EUR 6.8 billion in damages over the 2000-2020 period [NDC, 2021]. From an adaptation perspective, a no action approach would also prevent rural communities and the poor from being energy secure and it would further increase internal migration dynamics (rural to urban). Significant increases in fuelwood consumption over the last decade has been detected in studies[17] and official statistics[18], and is expected to grow further by an average of at least 0.5% per year in the coming years. Furthermore, non-adaptation of forests, especially in the lowlands, will exacerbate the occurrence of natural disasters impacted by climate change, and the damage and loss caused by them. This is expected to decrease GDP by -0.15% to -0.32% per year in RCP4.5 and RCP8.5 scenarios respectively. Direct contribution of the production of the forestry sector to the GDP, on the other hand, is expected to decrease annually on average by 0.3% (2037) to 1.6% (after 2067) with a temperature increment of 3C [UNDP, 2019; UNEP 2022]. In a BAU scenario the LULUCF sink is expected to miss -1,526 kt$CO_{2eq}$ compared to the foreseen Low Carbon Development Scenario. To balance this missing sink in such a scenario, the other economic sectors would have to further increase by 22% the overall GHG reduction efforts, resulting in a significant increase in the additional investment costs required for the implementation of the strategies' low carbon development pathways, which already correspond to 6.5 billion EUR (approx. USD 6.5 billion).

19. **Sector approach:** This more traditional option consists of focusing solely on the forestry sector, addressing the adaptation deficit of public stakeholders with limited actions focused on capacity development and infrastructure investments, and will not allow the application of CAS to forestry investments. Applying a traditional sectorial approach will have a positive

short-term impact due to the improved forest governance, but in the medium-long term will *de facto* have similar consequences of the no action option. A BAU sectorial approach will have limited sustainability and replicability due to the lack of inclusiveness and diversity of stakeholders. Finally, a traditional sectoral approach will not align with the 2021 NDC update, which explicitly highlights the connection between the need to adapt forest management and the co-benefits of mitigation.

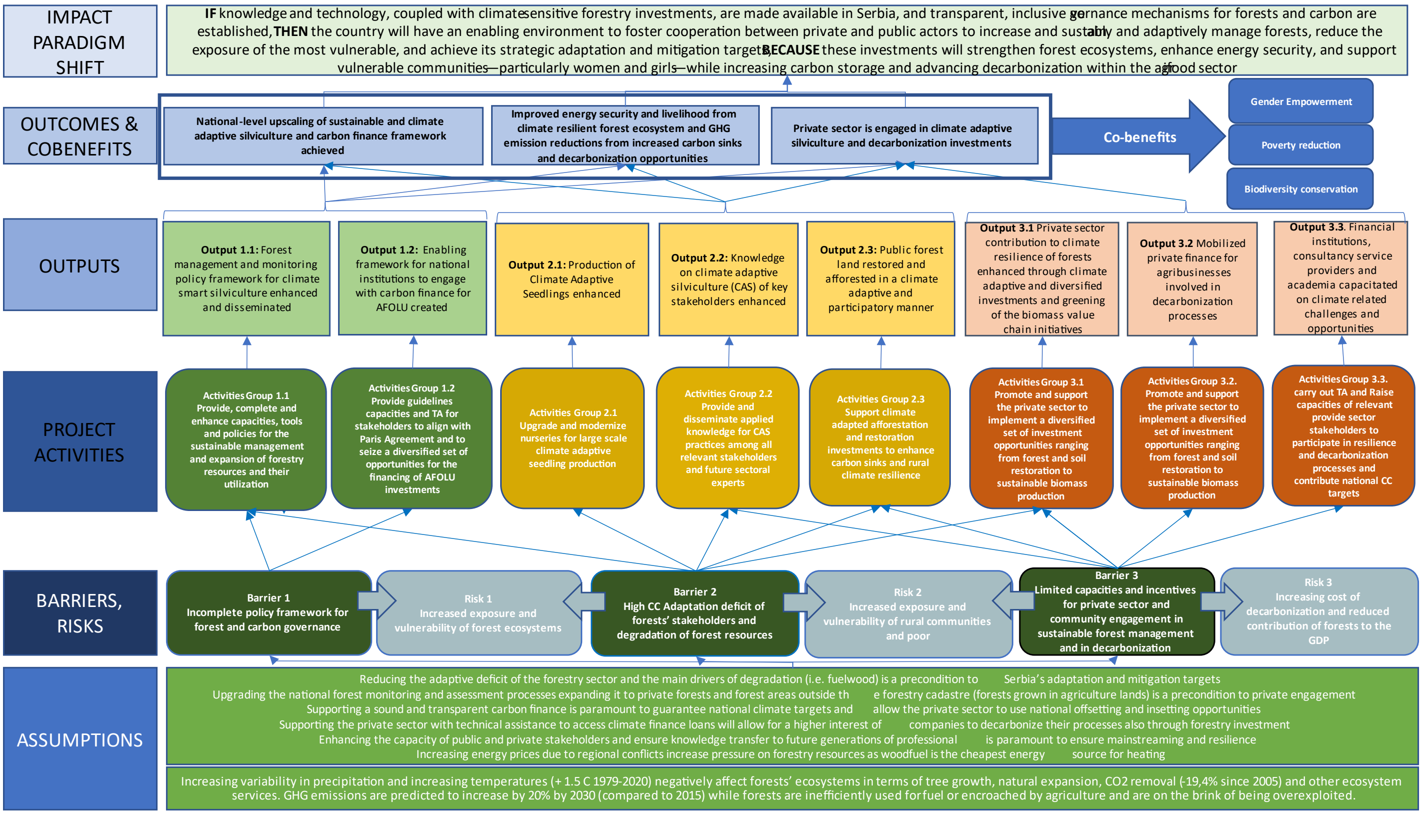
20. **Nexus approach**: This approach will integrate all features of the sector approach, expanding its range of activities to focus on ecosystem services, including energy security, livelihood, protection and decarbonization, applying an inclusive livelihoods approach targeting specifically private sector operators and communities that are not traditionally involved in forest management. Particularly, the participation of agrifood sector operators, local communities and private forest owners' participation will help ensure the envisaged paradigm shift, the sustainability of the activities and the integration of the forest-energy security-decarbonization nexus. The beneficiaries are described in more detail in section D.2. Under this option activities will involve: (I) forest owners (natural forests and coppicing stands) and their associations; (II) biomass fuels producers (i.e. fuelwood and pellets) and their associations; (III) land owners that have halted farming their lands due to soil degradation; (IV) land owners, companies and producer associations who want to establish windbreaks / shelterbelts to protect their crops; (V) enterprises from the agri-food and other sectors in investing in the adaptation and decarbonization process; (VI) public and private financial institutions; and (VII) public institutions not directly involved in forestry (e.g. Water Authorities) and municipalities. An inclusive approach involving all stakeholders directly and indirectly related to forests will allow the country to expand forests, address drivers of forest degradation while enhancing the carbon removal, socio-economic, biodiversity and disaster risk prevention potential of forests.

21. **Component 1 and Component 2** are preconditions for the country's adaptation and mitigation to climate change through the forestry sector and increase rural communities' resilience. The sustainable utilization of Serbia's forest resources is an important contribution to secure the country's climate and energy targets. Further, one of Serbia's key climate objectives is to increase the carbon sink from forests by 17% by 2030 and 22% by 2050, compared to 2010. The main national policy and strategy documents, including Energy Security of the Republic of Serbia, 2022; 2BUR, 2023; NDC, 2021; LCDS, 2023 acknowledge the forestry sector's centrality in achieving its goals (further information of synergies with national policies can be found in Table 24 in Annex 2), due to the significant potential of forests to enhance socio-economic outcomes, reduce the risk of climate-related disasters and mitigate climate change. The country's high vulnerability to climate change and increasing forest degradation are therefore jeopardizing Serbia's efforts to reduce the risk of extreme poverty for about 25% of the population by ensuring energy security via fuelwood and other livelihoods options and decrease its net emissions and initiate decarbonization. The first two components include the: (a) actions needed to remove bottlenecks in the regulatory framework and transfer knowledge and technologies that are paramount to adapt the forestry sector; (b) the enhancement of capacities of forest operators to provide energy and NWFP to communities, while reducing the main drivers of forest degradation (i.e. inefficient and uncontrolled fuelwood consumption); (c) creation of tools and processes to increase private forest owners' and communities' (i.e. municipalities) participation in the climate adaptive management of forests; and (d) upgrading the key policies and monitoring tools that will ensure the reliability, transparency and accountability of carbon offsets governance; and accounting of ecosystems services (including the national carbon accounting mechanisms) and the economy. A detailed description of the private sector involved is available in Annex 8 (Table 2). The use of CAS approaches along with measures to improve forest management and governance will increase climate resilience and biodiversity, avoid competition for resources, and ensure the growth of hardened and more resilient trees [Allen, 2010; Abatzoglu and Williams, 2016; Ager et al., 2016; Gustafson, 2020; Prichard et al., 2021; Muller 2021].

22. **Component 3** will support the private sector and communities to manage forests more sustainably and efficiently by allowing smallholder forest owners and municipalities to aggregate and engage in climate adaptive forest management activities, greening the fuelwood value chain (the main driver of forest degradation) and introducing agroforestry in agriculture areas (i.e. shelterbelts) as a crosscutting investment to cope with climate change impacts while contributing to decarbonization. Further, the project will work with national and international financial institutions[19] to provide technical assistance to private companies (e.g., agrifood sector) in defining their adaptation and decarbonization strategies and ensured-constraining access to finance to invest in carbon-reduction [Output 3.2] also via forest-related projects. This will ensure the engagement of the private sector in the forestry sector and reduce the dependence of the forestry sector on the national budget or international financial support to implement nature-based-climate action projects such as the ones presented in Component 2. Therefore, Component 3 is important to kick-start and sustain the process – even beyond the project duration — and reduce technical barriers for the private sector to engage with forestry, adaptation, and decarbonization. Policy support to prepare the carbon finance and complement financing opportunities for the activities of component 3 are integrated into component 1.

23. **Co-Benefits:** This proposal will also contribute to generating several co-benefits. The main ones are: (a) Biodiversity conservation by promoting forestry activities following clear protocols aiming to restore local forests in a climate-adaptive way); (b) poverty reduction through increased livelihood and rural job opportunities and enhancing rural energy security); and (c) gender empowerment, as all activities will prioritize the participation of women. A detailed description of co-benefits is provided in Section D3.

**Figure 1 Theory of Change (Annex 23)**




## B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

| Outcome number | GCF Mitigation Results Area (MRA 1-4) | | | | GCF Adaptation Results Area (ARA 1-4) | | | |
|---|---|---|---|---|---|---|---|---|
| | **MRA 1** Energy generation and access | **MRA 2** Low-emission transport | **MRA 3** Building, cities, industries, appliances | **MRA 4** Forestry and land use | **ARA 1** Most vulnerable people and communities | **ARA 2** Health, well-being, food and water security | **ARA 3** Infrastructure and built environment | **ARA 4** Ecosystems and ecosystem services |
| Outcome 1 National-Level Upscaling of sustainable and climate adaptive silviculture and carbon finance framework | ☐ | ☐ | ☐ | ☒ | ☒ | ☐ | ☐ | ☒ |
| Outcome 2 Forest structure, composition and cover enhanced with national and local stakeholders to enhance contribution to national climate change resilience and mitigation goals | ☐ | ☐ | ☐ | ☒ | ☒ | ☐ | ☒ | ☒ |
| Outcome 3 Private sector is engaged in climate adaptive silviculture | ☒ | ☐ | ☒ | ☒ | ☒ | ☐ | ☒ | ☒ |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| and decarbonization investments | | | | | | | | |

| Co-benefit number | Co-benefit | | | | | |
|---|---|---|---|---|---|---|
| | **Environmental** | **Social** | **Economic** | **Gender** | **Adaptation** | **Mitigation** |
| Co-benefit 1 Biodiversity conservation | ☒ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Co-benefit 2 Poverty reduction | ☐ | ☐ | ☒ | ☐ | ☐ | ☐ |
| Co-benefit 3 Gender empowerment | ☐ | ☒ | ☐ | ☒ | ☐ | ☐ |

## B.3. Project/programme description

**OUTCOME 1 - National-level upscaling of sustainable and climate adaptive silviculture and carbon finance framework achieved**

24. Serbia recognizes that forest ecosystems are paramount in its economy. Forest health, richness and cover are preconditions to fulfilling its NDC and other strategic development targets. Although policies and long-term strategies for the forestry sector are in place, the lack of reliable and up-to-date information on the status and dynamics of forest ecosystems seriously hampers evidence-based decision-making in relation to forest governance and management at local, regional, and national levels. (More information on laws, policies, and governance can be found in Annex 2, Appendix 11). Benefits from existing forest monitoring and governance remain rudimentary and are insufficient to ensure the transparency and accuracy that the country needs to assess its targets, and to generate up-to-date data to inform its adaptation and mitigation measures. The project will enable Serbia to ensure that forests' benefits – including biodiversity, protection (i.e. DRR), livelihood and carbon – are monitored, assessed and reported according to the latest guidelines produced by the UNFCCC, the Convention on Biological Diversity (CBD) and other international references to allow an evidence-based adaptation and mitigation process. The continuous provision of up-to-date information on forest dynamics in the country during the lifetime of this project and beyond will also permit to bring the existing carbon monitoring, reporting and verification (MRV) system up to international standards. Within component 1, the project will also assist Serbia in reducing information and capacity barriers and securing carbon finance for forestry/LULUCF to raise mitigation ambition and to provide the tools for additional finance during and beyond the project lifecycle.

**Output 1.1: Forest management and monitoring policy framework for climate adaptive silviculture enhanced and disseminated**

25. The project evaluated different options for the improvement of the monitoring and management of forestry resources and concluded that the implementation of a statistically sound and continuous monitoring and assessment system in combination with policy actions engaging the private sector in reducing degradation drivers and promoting biomass alternatives is the best approach for a sustainable planning and use of forest resources (Annex 2, table 18). Refer to Annex 4 and table 6 for co-financing details.

26. **Activity 1.1.1: Establish the National Forest Monitoring System (NFM).** Building on the methodology sampling design and preliminary results of the ongoing 2nd National Forest Inventory, the project will establish a continuous national forest monitoring (NFM) system using permanent and continuously assessed sample plots. Such a system has been proven to be an accurate approach to estimating dynamic variables such as change of forest area and growing stock and drain (mortality and number of cuts), carbon storage and other non-wood/carbon related functions such as biodiversity, NWFPs and disaster risk protection. The project will also deliver regular trainings and support to field teams in data collection to maintain common quality standards and ensure the credibility of results of the forest assessment and monitoring. The government will provide field work as contribution after the initial phase of the NFM, to ensure full ownership by the counterpart, in particular the Directorate of Forests of the MOAFWM, and the NFM's continuation after the project. A continuous forest monitoring system also allows for the evaluation of results on an annual basis to ensure that decision-making processes can rely on a permanent inflow of unbiased and relevant information on the actual condition of forests and latest trends in timber harvesting, carbon storage, etc. The approach used by the project will guarantee the technical sustainability of the visual interpretation of the NFM by 1) using a free and open-source software for land use monitoring, namely Collect Earth developed by FAO, 2) establishing baseline data on land categories and assessing land use changes on a rolling basis, and 3) continuously training and supporting staff of the Institute of Forestry in Belgrade in forest monitoring. (GCF financing: USD 767,901; co-financing: MOAFWM: USD 7,307,141).

27. **Activity 1.1.2: Develop guidelines for decision-makers on AFOLU to prevent soil degradation.** The project will downscale international policies to the Serbian context and develop guidelines for decision makers regarding AFOLU and prevention of soil degradation in accordance with IPCC 6AR. For this purpose, the activity will elaborate and publish guidelines on the related practices that can have significant benefits for climate change mitigation, food security and local economic development and disseminate the knowledge through a series of national and local workshops (GCF financing 20,750 USD; Co-financing: FAO 105,100 USD).

28. **Activity 1.1.3: Create national standard for biomass production / handling for energy purposes.** The utilization of wood fuel in Serbia often results in wasted energy due to its high moisture content, which lowers its calorific value. This issue is mainly attributed to the lack of awareness among the population on how to select the best available fuelwood, and the absence of incentives for suppliers to promote high-quality fuels. As a result, suppliers offer mostly raw wood or air-dried wood for a short period, which further reduces its energy output. To address this challenge, the project aims to introduce technical regulations that will establish appropriate rules, inspection mechanisms, and complaints procedures that sellers and distributors of firewood must follow. The implementation will also evaluate the possibility on how to impact fuelwood quality provided from the informal sector. The primary objective of the regulation is to mandate sellers to provide the market with firewood with a maximum moisture content of 30%, thus increasing the energy output of purchased wood. This intervention will benefit urban households, which often purchase and utilize raw wood immediately due to the lack of storage facilities. Furthermore, the regulations will enable market inspections, which will hold suppliers accountable for unethical practices and provide consumers with the necessary information to make informed decisions. Together with the other activities of the project this will contribute to an increased monitoring of the utilization of forest resources and hence to better control. The aforementioned will be complemented by an evaluation of the impact of the biomass standards on a pilot area and by Renewable Energy Days organized in 10 different municipalities throughout the country raising awareness on the correct user information and utilization of biomass for heating in combination with other renewable energy/energy efficiency measures. (GCF financing: 155,500 USD; co financing: 0 USD).

29. **Activity 1.1.4: Develop the national strategy, action plan and execution guidelines for Short Rotation Plantations.** Most bioenergy currently comes from natural or semi-natural forests or woodlands rather than from forest plantations. However, there is growing recognition that planted woody species are an important means for providing energy. Establishing plantations may also serve other purposes, such as land rehabilitation and erosion control, watershed protection and production of non-woody forest products. In addition, there is a wide variety of agroforestry, farm forestry or urban systems where trees are planted in non-forest conditions. The current level of production of wood fuel from plantations makes only a small contribution to the energy needs of Serbia, but plantations could be a very important source of wood energy in certain parts of the country. The project will, therefore, develop a national strategy, action plan and execution guidelines for sustainable wood energy plantations that can contribute to a diversified forestry sector. The guidelines will also analyze the potential of Short Rotation Plantations (SRP) for purposes competing with energy use, such as paper production. The activity will raise the capacities of experts in the field through workshops and training activities. (GCF Financing: 145,414 USD; Co-financing: 0 USD)

**Output 1.2: Enabling framework for national institutions to engage with carbon finance for AFOLU created**

30. Serbia must increase its efforts to reduce GHG emissions, as a signatory to the Paris Agreement, including as part of its accession process to the EU. Forestry and agriculture are critical sectors to achieve these mitigation efforts with carbon finance offering an important mechanism to tackle climate change in these sectors. The country has already signaled its commitment to these sectors and more broadly to its green agenda through national legislation and international treaties. The country recently adopted the LCDS (2023) including objective 3 to increase the carbon sink in Serbian Forests by 17% by 2030 relative to 2010 through SFM and restoration approaches. Further, in the agriculture sector, the LCDS includes promoting sustainable land use through several measures and incentives. These include Ecological Focus Areas through green direct payments. The EU's recently adopted plans for the Carbon Border Adjustment Mechanism (CBAM) will impose a cost for the carbon content of certain products exported to the EU from 2026, and while agriculture and forestry, apart from fertilizers, is not covered by CBAM, will increase the urgency to decarbonize and introduce domestic carbon pricing. As emerged from the option analysis (Annex 2- table 19), developing a domestic offset mechanism for forestry, and opening international carbon finance opportunities through the voluntary carbon market and evolving Article 6 strategies can unlock needed financing to upscale forest investments for crosscutting benefits, over and above those that can be delivered with public financing alone. This requires concerted investments in national capacities and the development of respective regulatory frameworks which will be supported by the project.

31. **Activity 1.2.1: Support the design of an offset mechanism [for AFOLU] as part of the domestic carbon pricing framework and institutionalize knowledge in the national curricula.** A robust carbon pricing framework will enable Serbia to meet its goals in the NDC – and beyond – and Low Carbon Development Strategy. It will help the country

eventually integrate with the EU ETS as well as comply better with the impending CBAM. The country currently does not have a well-defined carbon pricing framework overall. Stakeholder interviews reveal a knowledge gap around the LULUCF sector. Within this activity, the project will provide a policy framework and options for forest and soil carbon linked to Serbia's ETS plan, as it evolves. This will include developing specific policy options for domestic offset mechanism for forest and soil carbon linked to Serbia's evolving ETS plans. It will also involve strengthening the capacity of private services providers and academia to support national and international carbon finance for LULUCF. As demand for national and international carbon finance increases, private firms will require technical support in developing their strategies and systems around carbon finance. The project's technical assistance to service providers will help meet this demand. Further, as carbon finance evolves in the country, incorporating it in national curricula at the university level will help institutionalize some of the technical knowledge and will help feed into the demand in the longer term, supporting a cadre of future professionals who are equipped to engage with carbon finance in the LULUCF sector. (GCF financing: 296,850 USD; co-financing: 0 USD).

32. **Activity 1.2.2: Support the development of Serbia's Article 6 strategy related to "AFOLU" opportunities.** The project will provide technical inputs and augment national capacity on activities related to forestry and land restoration falling under Serbia's evolving Article 6 strategy (under the Paris goals). Serbia's Law on Climate Change (2021) clarified the mandate of its National Council on Climate Change. The Council has an advisory role on Article 6, falling under two related areas of the Law on Climate Change related to Article 6 on implementation of Paris alignment and guiding ministries on policies and regulations to meet EU and international standards. The project will support national capacity in these areas. First, it will formulate a set of guidelines and action plan for the forestry sector's access to carbon markets under Article 6. Second, it will provide capacity development for national negotiators on engaging with the LULUCF sector in Article 6 discussions. Third, it will provide technical inputs for Serbia's Article 6 strategy in this sector that can be considered and reflected by policymakers for the country's next NDC update and strategy documents going forward (GCF financing: 165,097 USD; co-financing: 0 USD).

33. **Activity 1.2.3: Promote and support knowledge-sharing at the regional level.** Serbia's regional commitments in the Western Balkans under the Sofia Declaration go beyond decarbonization and extend to depollution, the circular economy, biodiversity, and sustainable food systems.[lix] The project will facilitate knowledge exchanges between the different countries in the region on carbon finance governance and mechanisms. Western Balkans countries are at different stages of defining their carbon governance frameworks and have accessed carbon finance to different extents. For example, Montenegro already introduced its carbon pricing systems in line with EU's Emissions Trading System (ETS) in 2021.[lx] The Balkan countries – Serbia notwithstanding – need to accelerate their efforts to decarbonize through defining their offsetting and insetting mechanisms and carbon prices. Knowledge exchanges between peers from different countries can help policymakers tackle similar issues in Serbia (and vice versa), drawing lessons from existing policies and experiences. The project will reduce information barriers between countries in two ways. First, it will advance and operationalize one regional knowledge-sharing virtual platform to allow regional partners to share technical expertise, policies and lessons from carbon markets governance[4] as well as AFOLU investment in Serbia and CAS capacities of Serbian stakeholders (e.g. institutions, municipalities and private sector operators). This will allow Serbia to learn from neighboring countries and to further expand its capacity to apply innovative approaches to CAS investments and involvement of different stakeholders. The platforms' audience not only includes policymakers from Ministries of Environments and Agriculture but also potential buyers of carbon credits such as energy utilities. Second, it will promote the platform through regional communication campaigns and use it to disseminate information around policy developments and technical guidelines. (GCF financing 0; co-financing: FAO: USD 48,500 USD).

34. **Activity 1.2.4: Support access to voluntary carbon finance for forestry to enable sequestration beyond NDC targets and to ensure long-term sustainability**. Under its current NDC targets, Serbia commits to reducing its emissions by 33.3% by 2030, compared to its 1990 levels. Agriculture and forestry are included in the mitigation co-benefits of the NDC, and forestry has been designated as a priority area in the LCDS. In this regard, voluntary carbon markets (VCM) can support sequestration beyond existing targets. The project will support Serbia in defining more ambitious / enhanced climate targets in the forthcoming update of the NDCs in 2025 and will facilitate access to VCMs, through identifying gaps and guidelines on entry points for access linked to evolving, high-integrity standards in VCMs, preparing a capacity development plan and the cost implications of addressing these gaps, and providing technical assistance to strengthen some of the enablers of participation such as data systems and capacities to maintain them. The capacity development will take place through training and incorporating guidelines on upgrading and maintaining data systems in university curricula in Serbia. Multilateral organizations have and are currently supporting Serbia in the development of its Monitoring, Reporting and Verification (MRV) system. For example, UNDP helped to develop an MRV system and the World Bank's Green Transition Development Policy Operation includes support to the country's MRV system as well. Based on progress achieved under ongoing activities, the project will address remaining national capacity gaps. It will develop a capacity development plan to address these identified gaps, building on the work done or underway by other multilaterals in the MRV space in Serbia. A

key distinction between the project's activities and the existing ones is that while the project focuses on a specific aspect of carbon markets – the AFOLU sector – other projects including those on MRV focus on carbon markets broadly. (GCF financing: 77,250 USD; co-financing: 0 USD)

35. **Activity 1.2.5: Enable insetting as part of company decarbonization strategy support.** The project will reduce barriers such as lack of information and knowledge to facilitate insetting at the company level. First, it will document policy options to provide incentives for insetting at national level and develop appropriate mechanisms if conducive to enhancing climate results beyond Serbia's current NDC targets. Second, it will produce guidelines for companies on good practices for carbon accounting and MRV, focused on insetting. Finally, it will also strengthen the capacity of local service providers and consulting companies in the field of decarbonization on their technical expertise around insetting. The support will target areas such as carbon accounting and measurement, issues with quantification, using tools and systems for reporting and their cost implications. (GCF financing: 114,750 USD; co-financing: 0 USD)

**OUTCOME 2 - Improved energy security and livelihood from climate resilient forest ecosystem and GHG emission reductions from increased carbon sinks and decarbonization opportunities**

36. Transferring technology and knowledge with on-the-job training to forests' stakeholders is a precondition to secure the highest possible survival rate of forestry investments and to ensure their long-term climate proofing of forestry investments. As discussed further in the option analysis (table 20 of Annex 2), addressing only a few steps of the process (e.g. planting of seedlings or maintenance of forests) is not functional and does not respond to the existing challenges linked to climate change, the needs of people needs and the health of ecosystems. Therefore, the project will work with Serbian stakeholders to ensure a 360-degree approach to forestry where climate change and its adverse impacts are addressed to the greatest extent possible. The project will support forest stakeholders with: (I) technology and knowledge transfer to ensure climate adaptive forestry investments; and (II) tailored forest investments through afforestation and forest restoration.

**Output 2.1: Production of climate-adaptive seedlings enhanced**

37. The current production of seedlings in the two national public nurseries and seed centers is fully dedicated to routine afforestation and reforestation activities, but it does not contribute to increasing the biodiversity of forests or upscaling efforts and is therefore unable to help contribute to the achievement of targets identified by Serbia in relation to forest expansion, resilience and contribution to energy security. While the country possesses a basic technical know-how, skilled staff and technical infrastructure, under a BAU scenario, the state and private nurseries are only capable of producing a limited number of species with often limited survival rate after planting in the field, favoring single-species forests with lower relevance in terms of providing ecosystem services and diverse wildlife habitats.

38. **Activity 2.1.1: Upgrade public nurseries (Vojvodina/C. Serbia).** As the project aims to select species for forestry investments from more than 20 autochthonous trees and shrubs (Annex 7b), it will be necessary to upgrade the existing seed center and the two state nurseries, as well as further develop capacity of the operators. In order to afforest at least 7,000 ha of land and restore at least 51,000 ha degraded forest land and establish planted forest on at least 500 ha of unfarmed private lands for energy use, agro-forestry or soil rehabilitation purposes, and at least 500 ha of shelterbelts on agricultural lands, the nurseries in Pozega and Ratno ostrvo – in the absence of other larger nurseries in the country - will have to produce at least 33,000,000 container seedlings, at least 1,500,000 bare-root seedlings and at least 500,000 vegetative planting materials (cuttings), mainly poplar and willow cuttings. (GCF financing: 4,078,779 USD; co-financing: MOAFWM: 455,710 USD).

39. **Activity 2.1.2: Train and support operators of public and private nurseries in the production of diverse and climate adaptive forestry seedlings.** In the second half of year one, the project will organize four training courses of two days each, (two in Kać and two in Pozega nursery) on the production of high-quality plant material (seeds, seedlings and cuttings) to be delivered by the Chamber of Forestry Engineers with support by an international expert, to refresh: the knowledge of all staff of PES and the Institute of Lowland Forestry and Environment (ILFE) in charge of the nursery works and to train staff and workforce from private nurseries, other forest enterprises (both public and private) and related non-governmental organizations (NGOs) for spreading knowledge on up-to-date seedling production methods and technologies and their use beyond the project. The training program will include the following modules with specific information about the selected native species: a) Module 1 – High quality plant material, and b) Module 2 – High quality plant production. **Selection criteria:** Beneficiaries are involved in the seedlings production activities and have been freed from their employers to attend the training. (GCF financing: 86,144 USD; co-financing: 0 USD).

**Output 2.2: Knowledge on climate adaptive silviculture (CAS) of key stakeholders enhanced**

40. Although Serbia has a long-standing tradition of forestry, the available knowledge, technical capacity and infrastructure remains outdated in the context of climate change, impacting how well identification, prevention and management measures are implemented. Forest management plans (FMP) and procedures partly still in place in Serbia

are rigid, not allowing to factor in climate change impacts and incentivizing forestry investments in an almost monoculture approach. As a result, the planned forest interventions remain modest and similar across all management classes with unnecessary health risk for forest ecosystems. Forests are viewed mostly as sources of wood and timber and not to adapt to and mitigate climate change. With CAS, the project will help existing forest communities to diversify their practices through appropriate restoration practices emphasizing spatial heterogeneity and diversity. It will help increase the share of species that are better suited to current and future climatic conditions. Furthermore, seeds and seedlings are currently not selected, prepared and hardened in a climate adaptive way, resulting in a low survival rate after planting and a higher exposure to the adverse impacts of climate change on new forests. Climate adaptation considerations will not only guide the final decisions concerning the individual seed stands, but also ensure that, from seed production to the maintenance of forestry operations, all measures are executed according to specific protocols. The guidelines to be developed under activity 2.2.1 will cover: 1) climate adaptive nursery production including seed selection; 2) soil preparation for planting on extreme sites; 3) effective and efficient planting methods; and 4) maintenance after planting and first thinning operations.

41. **Activity 2.2.1: Produce guidelines on climate adaptive silviculture.** The activity envisages developing four guidelines for stakeholders in the forestry sector that will be published online and utilized within training activities, to support the smooth and swift transition to climate-adaptive silvicultural approaches. The four guidelines will cover the following: 1) climate smart nursery production including seed selection; 2) soil preparation for planting on extreme sites; 3) effective and efficient planting methods; and 4) maintenance after planting and first thinning operations. The project aims to ensure the application of guidelines developed by relevant institutions in regular forest management planning, transferring knowledge through training and capacity development for over 600[20] individuals in rural communities. Priority will be given to women involved in the sector, poor individuals and those at risk of poverty involved in the sector, and workers of forestry companies. Additional criteria will be identified at project start-up with local stakeholders. The project also plans to establish partnerships learning tours to countries with successful experiences in Forest and Landscape Restoration (FLR). Countries for the tours will be selected at project start-up together with stakeholders. Priority will be given to women involved, actors with recognized leadership (e.g. from NGO/CBO/Academia), and university and vocational schools professors involved in the sector. With the developed capacities and guidelines, there is potential for scaling up and replicating the interventions beyond the project areas. These investments will ensure long-term sustainability in achieving national climate change adaptation and mitigation goals. (GCF financing: 21,600 USD; co-financing: 0 USD).

42. **Activity 2.2.2: Train public and private stakeholders in climate adaptive silviculture (CAS).** Through the collaboration with the Ministry of Education, Science and Technological Development of Serbia (MESTD), the Forestry High Schools in Kraljevo and Sremska Mitrovica, as well as with the Faculty of Forestry and the Forestry Chamber, this activity will contribute to capacity strengthening by establishing a consistent process for professional training and education on climate change-related issues. The project will conduct various training activities to enhance the capacity of professionals, forest workforce, and stakeholders in integrating sustainable and climate-adaptive approaches in their work, with a focus on seed/plant material collection, climate adaptive forest investments and ecosystem management, forest management approaches, and silvicultural practices, and will include study tours for private forest/land owners, prioritizing women and single women-headed households, to demonstration plots established by the responsible agencies, supported by an international expert and collaborating with relevant institutions. (GCF financing: 0 USD; co-financing: FAO 321,350 USD)

43. **Activity 2.2.3: Upgrade the National curricula of the faculty of forestry and vocational schools working on forestry, agriculture and accounting with introduced practices and technologies.** Up-to-date knowledge and skills will also be transferred to future generations of professional staff working in relevant sectors by revising national curricula related to forestry and forest plant production of the Faculty of Forestry, as well as vocational schools on forestry (Kraljevo and Sremska Mitrovica), agriculture (26 agricultural high schools) and accounting. The newly introduced practices and technologies will be incorporated in advanced trainings for the workforce offered by the aforementioned institutions. Special attention will be paid during project implementation to involve and empower younger professionals for sustainability. As part of this, FAO as EE will sign a collaboration agreement with the MESTD. This formal collaboration will further contribute to the sustainability of the project's results, by investing into the future generation to apply sustainable and climate adaptive forest management practices all over Serbia, both in public and private forests. The project expects that about 3000 students per year will learn about practices and technologies introduced by the project. The activities will also include training and capacity development actions for teachers and professors at the above-mentioned institutions to ensure full mainstreaming of the introduced technologies and practices in their teaching modules. (GCF financing: 39,000 USD; co-financing: 0 USD)

44. **Activity 2.2.4: Facilitate regional knowledge-sharing dialogue on CAS.** Through its institutional, academic and civil society partners, the project will favor a regional knowledge-sharing environment to exchange experiences on CAS approaches used both within this and other relevant projects in Serbia and neighboring countries. The activity will support regional practitioners in sharing relevant data and information by bridging experts, institutions, and universities. Further, the

main aim of the regional dialogue is to improve and enhance the framework for AFOLU investment in Serbia and the CAS capacities of Serbian stakeholders. The modalities for knowledge sharing will be defined by a communication specialist and will make use of existing platforms, meetings and others to develop as many synergies as possible to existing coordination mechanisms (GCF financing: 9,500 USD; co-financing: 0 USD)

**Output 2.3: Public forest land restored and afforested in a climate adaptive and participatory manner**

45. To afforest 7,000 ha, FAO and the GoS pre-identified approximately 20,000 ha of geo-referenced potential areas for planting of seedlings on both state-owned forest land (about 17,400 ha) and underused or not managed privately owned land (2,600 ha) based on criteria agreed upon within the framework of an LoA between FAO and the Institute of Forestry in Belgrade during formulation (Annex 2, Appendix 7). The selection criteria for the preliminary identification of potential afforestation areas were: (i) Clear land ownership; (ii) Absence of land tenure conflict and of environmental protection restrictions (i.e. protect areas, parks and others), and (iii) Absence of land cultivation or for pasture use (Annex 16). In these areas, the project will plant the new climate adaptive seedlings working with beneficiaries trained in climate adaptive practices and technologies. Therefore, the project will: (i) consult with the Ministry of Environmental Protection (MEP), local administrations and communities to ensure local engagement and finalize an inclusive afforestation/forest restoration plan; (ii) prepare the area for planting and ensure all logistics are in-place; (iii) plant the new seedlings according to CAS approaches; (iv) ensure the logistics and execution of the needed maintenance and replanting operations to be carried out after the project life-cycle are in place and will be carried out independently by the governmental authorities according to the climate adaptive practices introduced in component 1.

46. Another restoration activity planned under this output will address the issue of degraded coppice forests and how to manage those forests in an efficient manner. Originally these degraded coppice forests were high forests, and it is the declared objective of the Government to bring them back to this stage (Forest Development Strategy of Serbia, 2006). The project will apply a step-by-step process starting from the current forests which will transform them into “coppice forests with standards” as a transitional stage and ultimately in the long run into high forests (after 80 – 100 years). Demonstration sites, where adaptive management measures will be applied and monitored, will be established at the beginning of the project to provide first-hand experience to staff of PES and PEV, but also to private forest owners, on alternatives to coppicing with the aim to bring back degraded forests into a healthy state, regaining their ecological functions and climate resilience, and thus enhancing human well-being. Inefficient coppicing practices in the public and private sectors are currently carried out in first place due to the lack of knowledge on alternatives. The project will hence carry out awareness raising on modern forestry processes disseminating and emphasizing the economic advantages of healthy forests for communities and private sector operators involved in NWFP among others. Areas interested in forestry investment under these activities will not be subject to cutting or any other exploitation that could jeopardize its integrity and health. Maintenance will be secured by the Forestry Department according to the laws and regulations of the Republic of Serbia.

47. **Activity 2.3.1: Carry out afforestation activities on public land.** The project aims to carry out afforestation activities on 7,000 hectares of public land in Serbia (based on publicly available land cadaster). Afforestation plans will be developed, including site descriptions, historical information, proposed interventions, species selection criteria, detailed schedules, and monitoring plans within each annual working plan and budget. Local governance mechanisms will be established, and training will be provided to workers and foremen involved in the planting activities. The project aims to plant on average 1,400 hectares per year from year 2 to year 6, with planting densities designed to conserve water resources and promote resilience to climate change (Annex 2, appendix 7). The project monitoring and maintenance will be fully under the responsibility of the MOAFWM. The selection of species will be based on reference ecosystems and selected among a long list of species available in Serbia (Annex 2, appendices 3-5). Monitoring will be conducted through permanent plots, expert supervision, and satellite imagery analysis. (GCF financing: 6,882,659 USD, co-financing: MOAFWM: 7,530,255 USD; PES: 2,423,741 USD; PEV: 659,147). **Selection criteria of each plot are as follows**:

- Public (State or Municipal) ownership of the land is clearly defined from a legal point of view and verified by third parties (e.g. NGOs, / Municipalities, Companies). Potentially disputed plots will be excluded.
- High vulnerability to climate change of target regions.
- Higher potential to enhance ecosystem services (disaster risk reduction/protection, water and erosion control, etc.)
- Minimum size of the plot is at least 10 ha (preference assigned to largest ones facilitating planting and monitoring tasks).
- Accessibility of the plot (ensuring access for standard 4x4 vehicles used for transport of seedlings and workers).
- Steepness of the slope (sites with slope average over 30 degrees may be excluded due to the higher cost of operation).
- Distance from human settlements (avoiding forestation works in the perimeter surrounding of the most peripheral houses; a buffer zone without forest cover should be maintained to prevent potential forest fire expansion).
- Availability of workforce within 25 km radius to ensure that local communities could benefit from employment opportunities.

- Priority will be assigned to the possibility of establishing wildlife corridor(s) in newly established forests in Vojvodina region.

48. **Activity 2.3.2. Convert public degraded coppice stands into high forest.** The project will work with the MOAFWM and the respective public enterprises (PES) to pre-identify at least 33,000 ha of forest stands for project interventions, the majority of which will be degraded coppice stands for conversion into high forest, but also forest stands damaged by abiotic or biotic factors with urgent need of restoring their ecological functioning. Most coppice forests in Serbia are around 50-70 years old and lack proper management, resulting in underutilization of their potential for wood production and ecosystem services. Although there have been pilot studies on converting coppice forests into high forests, the actual conversion rates have fallen behind the government's target of 7,000 hectares per year. To address this, the project aims to upscale conversion efforts and promote sustainable conversion by employing a single tree-oriented management approach and enrichment planting with native species. State-owned forests will gradually transition from fuelwood cutting to adaptive forest management, incorporating climate change adaptation and mitigation measures. During the short- to medium-term, the quantity of fuelwood production for the rural population is expected to remain stable and will continue to be the primary source of revenue for forest owners. However, as the forest matures, there will be a shift from solely fuelwood to industrial timber production. This transition will afford forest owners the opportunity to increase their income by selling higher-value timber, rather than relying on fuelwood production only. To mitigate the anticipated reduction in fuelwood availability 30–40 years from now, future government initiatives[21] will promote alternative energy sources and energy-efficiency measures among the rural communities. These programs will encompass solar energy solutions and building insulation, among other options. Simultaneously, this GCF project (e.g. Activities 1.1.3, 1.1.4, 3.1.2, 3.1.4, 3.1.5) will address the widespread use of low-quality fuelwood: Introduction of biomass standards will compel suppliers to offer fuelwood with lower moisture content and greater energy efficiency. The project will provide training on CAS management and involve local communities in restoration actions. Climate proofing and maintaining of existing forest roads[22], through MOAFWM cofinancing, is critical to ensure access to restoration sites and allow sustainable forestry investments. Monitoring will be conducted through permanent plots, supervision missions, and satellite image analysis. In a first step, PES and PEV will identify and map potential intervention sites within those already identified at design, which will be reviewed and fine-tuned by the project before making a final decision on each individual site. (GCF financing: 3,228,576 USD; co-financing: MOAFWM: 20,707,500 USD; PES: 1,558,967 USD). **Selection criteria:**

- Areas of high biodiversity value remain untouched aside from regular maintenance operations - confirmed by MEP.
- High vulnerability to climate change of target region.
- Proven land ownership (disputed plots to be excluded).
- Minimum size of the plot at least 20 hectares (with preference given to large plots).
- Accessibility of the plot (ensuring access for standard 4x4 vehicles used for transport of seedlings, workers and water).
- Steepness of the slope (sites with slope average over 30 degrees may be excluded due to higher cost of operation).
- Availability of workforce within 25 km radius to ensure that local communities could benefit from employment opportunities.

**OUTCOME 3 - Private sector is engaged in climate adaptive silviculture and decarbonization investments**

49. The project will support the Serbia's adaptation and decarbonization efforts by enhancing the operationalization of the existing legal framework (e.g. Climate Change Law, 2021) and supporting the private sector engagement in forest management, adaptation and decarbonization efforts. Though at least two commercial banks in Serbia offer products tailored to climate action, take-up of these products remains low at the firm-level due to low awareness and know-how around navigating the decarbonization process. The project will work with both national financial institutions and co-financiers to provide Serbian companies with technical assistance to invest in best available technologies and approaches climate action (i.e. through forestry projects to adapt and reduce net emissions).

**Output 3.1: Private sector contribution to climate resilience of forests enhanced through climate adaptive and diversified investments and greening of the biomass value chain initiatives**

50. Forest restoration under an ecosystem-based adaptation approach is quite a new concept characterized by the complexity of holistically addressing the environmental, social and economic challenges of forest restoration in a climate change scenario. Considering that 47% of forests are under private ownership and that state-owned areas for afforestation are limited in scale, it is necessary to call for an active involvement of non-state actors to reach the national forestry targets stated in the LCDS (2023). Within the framework of this project, the private sector will be involved in forestry investment on a voluntary basis and all applicable rights are protected. The project will conduct awareness raising campaigns and establish a digital platform to collect and structure cadaster data and information on private forest owners interested in forestry investment, including conversion of coppice, establishing new forests or shelterbelts, or the management of their degraded coppice forests. This will allow to cluster private forest land parcels to reach a minimum scope of five (5) hectares for forest

interventions or to plan jointly with interventions in adjacent state-owned coppice forests. Thanks to the utilization of the digital cadaster, the PEs will ensure direct contact with private forest owners and ensure that women and single women headed households (representing less than 30% of forest owners) can be prioritized. Of particular importance will be the support to establishment of a technical partnership between the MOAFWM, the University of Belgrade and the University of Florence for transfer of technology and knowledge (refer to Annex 1.b for letters of support and Annex 2, par 167 for Forest Sharing). The option analysis presented in table 21 of Annex 2 concluded that the approach followed through this output is the most efficient one in engaging the private sector in sustainable forestry activities and greening the value chain. The choices offer a wide range of possibilities to carry out sustainable investments. This can lead to a decrease in degraded land areas (approx.6% of agricultural land) and to the rehabilitation of private coppice stands, which have unsatisfactorily low standing volume.

51. **Activity 3.1.1: Convert degraded private coppice stands into high forest.** The project will work with the MOAFWM, the respective PEs, and private landowners to pre-identify at least 18,000 ha of forest stands for project interventions, namely, to convert degraded coppice stands into high forest. Most private-owned coppice forests are primarily managed for repeated fuelwood harvesting, which overlooks the site's potential for higher wood production, both in quantity and quality. This limited approach also reduces the forest's ability to provide important ecosystem services, such as carbon storage. The activity will initiate, every year from year 2 to year 6, the gradual conversion of an average of 3,600 ha of private-owned coppice forests into high forests with planting densities of 300 seedlings/ha (avg.) for enrichment planting. As a first step, PES and PEV will confirm and map potential private-owned intervention sites, which will be reviewed and fine-tuned by the project before making a final decision on each individual site. Climate proofing and maintaining of existing forest roads [23,] through MOAFWM cofinancing is critical to ensure access to restoration sites and allow sustainable forestry investments. (GCF financing: 2,849,190 USD; co-financing: MOAFWM: 12,024,000 USD; PES: 239,400 USD) **Selection criteria:**

- Proven land ownership (disputed plots to excluded).
- High vulnerability to climate change of target regions.
- Plots situated within a cluster of at least 5 hectares to ensure efficiency of the investment and guarantee the best conditions for high forest growth.
- Total forest area under ownership limited to max 10ha.
- Potential accessibility of the plot (access for standard 4x4 vehicles used for transport of seedlings, workers and water).
- Steepness of the slope (sites with slope average over 30 degrees may be excluded due to higher cost of operation).
- Availability of workforce within 25 km radius to ensure that local communities could benefit from employment opportunities.
- Land plots that had forest in the near past with a clear need for active restoration interventions (well-preserved deforested areas where there is no natural regeneration, or degraded areas, e.g. after forest fires, with no signs of natural regeneration) may also be considered for re-establishing high forest.
- Ensuring that areas of high biodiversity value remain untouched – confirmed by MEP.

52. **Activity 3.1.2: Rehabilitate unfarmed private lands through forestry investments such as short rotation plantations, agro-forestry or soil rehabilitation purposes (Vojvodina).** Related to the rehabilitation of unfarmed private lands through forestry investments the project will work with the regional government, responsible for Agricultural Land, the Grain Producers Association, Serbian Chamber of Commerce (PKS), private land owners, local communities and other interested partners to identify at least 500 ha of unfarmed private lands to be cultivated with wooden species for energy use, agro-forestry, honey production, soil protection or rehabilitation, or a combination of the mentioned purposes. This project will cluster private land parcels to reach a minimum scope in terms of area for cultivation with wooden species. The project aims at a share of 10% energy, 60% agro-forestry including honey production and 30% soil rehabilitation sites for cultivation with wooden species (including shrubs) (Annex 2, Appendix 7). The concerned Municipalities will confirm and map potential intervention sites. (GCF financing: 869,053 USD; co-financing: MOAFWM: 2,031,400 USD). **Selection criteria**:

- Proven land ownership (disputed plots excluded).
- High vulnerability to climate change of target regions.
- Plots situation within a cluster of at least 5 hectares.
- Land plots that had forest in the near past with a clear need for active restoration interventions (well-preserved deforested areas with no natural regeneration or degraded areas, e.g., after forest fires, with no signs of natural regeneration).
- Availability of workforce within a 25 km radius, ensuring local communities could benefit from employment opportunities
- Possibility to contribute to wildlife corridor(s) with newly established forests in Vojvodina region.

53. **Activity 3.1.3: Establish shelterbelts in agricultural landscapes (Vojvodina).** The output also foresees investments in shelterbelts. The project will work with the regional government's secretariat responsible for agricultural land, the Grain Producers Association, PKS, private land owners, local communities and other interested partners to identify at least 835 km (equivalent to 500 ha) of agricultural lands for the establishment of shelterbelts, predominantly in almost treeless agricultural landscapes of the Vojvodina region with the main aim of soil protection (aeolian erosion control), increased biodiversity, provision of pollination services and (re)establishment of suitable wildlife habitats and migration routes. (GCF financing: 773,138 USD; co-financing: MOAFWM: 355,570 USD). **Selection criteria:**

- Proven land ownership (disputed plots to be excluded).
- High vulnerability to climate change of target regions
- Minimum size of the individual area for establishing shelterbelts at least 5 hectares.
- Area potentially protected from wind erosion
- Distance from human settlements (avoiding works in the perimeter surrounding most peripheral houses) of at least 500m
- Possibility to connect or establish a wildlife corridor in Vojvodina region

54. **Activity 3.1.4: Engage private actors in sustainable biomass value chains.** Many agrifood businesses generate biomass residues from their production, which can serve as a valuable source for solid biofuels or biogas, especially when available during off-season periods. In many cases, equipment such as dryers, cranes, storage bays, belt conveyors, hoppers, or silos is already in place and can be repurposed to support biomass production processes. The activity will analyze the current potential for the valorization in Serbia in the frame of a technical report and use it as a baseline to train private sector actors on the possibilities for the valorization of biomass residues for energy purposes. (GCF financing: 76,466 USD; cofinancing: 0 USD).

55. **Activity 3.1.5: Support platforms involving stakeholders of the forestry sector for a modern and transparent biomass energy value chain.** The activity will provide technical assistance to the Serbian Biomass Association (SERBIO), currently supporting the establishment of platforms aggregating and involving relevant stakeholders to produce and market high quality solid biofuels. These platforms have a production line for biomass as well as storage and logistic facilities. They represent the intermediator to organize local bioenergy value chains between suppliers providing raw biomass resources and customers of different scales from private households (HH) to large heat and power plants. This platform model has been shown to be very successful in countries like Slovenia, Austria and Germany. The most important factors for success of the platforms are (i) sufficient biomass supply covering the demands of current and future customers in the region; (ii) sufficient storage capacity for solid biofuels (e.g., fuelwood, wood chips and pellets), with the option to install drying facilities; and (iii) good access to transport infrastructure. The implementation of the activity, in line with successful experiences in other neighboring countries, will aim at involving the local administrations in the platform that can provide/lease public land to the private sector. Technical support will be provided to the SERBIO in all business activities, from feasibility studies, development of business and investments plans to marketing (GCF financing: 112,000 USD; cofinancing: 0 USD).

**Output 3.2. Mobilized private finance for agribusinesses involved in decarbonization processes**

56. The activities aim to increase agribusiness firms' capacities to assess the potential for decarbonization, to incorporate climate-related risks in their production decisions, assist them in defining their decarbonization strategies and facilitate linkages with financial institutions to meet their eventual funding requirements. The project will prioritize agribusinesses (e.g., farms, dairy and meat producers, processors) and forestry companies (e.g., forest owners, wood processing, and saw mills) interested in decarbonizing their processes, including through using forests to offset the emissions that cannot be avoided and/or engaging in insetting projects to climate proof their operations and reduce net emissions. Together, the activities aim to create a longer-term change in doing business, such that firms incorporate decarbonization in their activities and processes beyond the project life cycle. As can be seen from the option analysis (more details included in table 22 of Annex 2) the chosen approach is necessary to involve the private sector proactively in the decarbonization processes. In fact, as surveys have shown, most businesses, while interested in sustainability topics like circular economy, do not have the capacity and knowledge to engage in the process. The activities of this output aim at overcoming this barrier to facilitate access to finance for the private sector on low carbon pathways.

57. **Activity 3.2.1: Involve agribusiness and other companies in the decarbonization process of the private sector.** The activity aims to strengthen the network of stakeholders involved in decarbonization (e.g., ministries, regulatory bodies, financial institutions, and companies), disseminate knowledge to raise awareness and ultimately support the identification of decarbonization investment opportunities by private companies and financial institutions. The entry point for the private sector's involvement will be through existing outreach mechanisms, such as the already-operational Digital Platform on Circular Economy (DPCE), operated and managed by the PKS.[24] The project will support it by providing the information about decarbonization practices and technologies, different banks' financial product offerings around

decarbonization and greening, information about compliance and regulatory issues from different ministries and other relevant technical and practical resources on decarbonization. Using an existing platform rather than building a new one will facilitate implementation at lower costs to the project and strengthen national ownership. The platform will also bring together diverse actors, and help banks expand their potential pool of borrowers, helping to address what has been cited by the banks as a costly and challenging activity as these would be new clients. The project's collaboration with the DPCE will enrich its technical content and help expand its scope for decarbonization. This activity will augment and build upon the platform's outreach and communication activities and produce and disseminate technical materials. Through technical discussions, it will contribute to the identification of policy/ regulatory bottlenecks and facilitate policy dialogue between stakeholders to identify solutions to challenges preventing investments in decarbonization. (GCF financing: 1,376,000 USD; co-financing: 0 USD).

58. **Activity 3.2.2: De-constrain access to credit for agribusiness and other companies**. The project will provide technical assistance (TA) to agribusiness firms to develop decarbonization strategies and financial plans to achieve them. FAO, as EE, will sign Letters of Agreement with the identified firms. The value of TA per firm will not exceed a set ceiling, depending on the firm size, (15,000-50,000 USD per firm), in line with the average cost of decarbonization strategy preparation, and will require a minimum contribution of the firm ranging between 25 and 50 percent of the costs (self-financed TA). Firms requiring support beyond this ceiling will be either excluded or higher co-financing from the firms will be expected. The project TA could include specialists in agri-food sector, energy, GHG accounting and certification, adaptation and resilience considerations. The expertise mobilized through the project TA will depend on the specific needs of the individual enterprise's decarbonization strategy.

59. Commercial banks operating in Serbia interested in building a green portfolio cite the lack of firm-level capacity to develop business plans for decarbonization as a key constraint. The project will provide TA to firms to assess climate risk and monitor emissions or potential carbon sequestration from their activities, and potential costs of Environmental, Social, Governance (ESG) compliance. It will also help firms identify viable decarbonization investments and suggest technologies that maximize benefits while reducing GHG emissions associated with the firm's activities. Given the participation of financial institutions in the DPCE, firms will be able to assess suitable financial products. Firms will directly approach financial institutions as the connection facilitated through engagement in the platform, to apply for loans for decarbonization. The project will not assist the financial institutions in client selection (GCF financing: 1,500,000 USD; co-financing: 0 USD).

60. **Selection criteria:** All registered companies can participate in the activity, provided they are registered and cleared to operate in Serbia and do not have pending issues with the Serbian justice sector. The financial partner institutions will provide a shortlist of companies. The participation of the selected firms will be contingent on their confirmation of interest in carrying out decarbonization investments.

**Output 3.3: Financial institutions, consultancy service providers, and academia capacitated on climate-related challenges and opportunities**

61. This output aims to enhance national financial institutions' capacity on climate change and service providers to support agribusiness investment adhering to international standards on climate risks and decarbonization. Banks express willingness to increase decarbonization investments but they need capacity development and client outreach support. Banks recognize the potential benefits of decarbonization investments and potential increase in demands for green investments. Despite their willingness to enhance decarbonization investment, they face several challenges such as lack of information and technical know-how on assessing climate risks and decarbonization strategies and low awareness around compliance with international climate standards. The approach emerged as the best option (further details on the analysis are available in table 22 of Annex 2), as it will provide the basis for all further decarbonization processes of the private sector in the country. In fact, only with sufficient critical mass and knowledge will it be possible to support the private sector in contributing to national climate change commitments, while maintaining competitiveness in international markets.

62. **Activity 3.3.1: Support to the chamber of commerce and industry and the association of financial institutions to assess climate-related risk of banks and ensure client engagement.** This activity will aim at enhancing the capacity of the banks to assess climate-related and other risks. All banks can potentially receive TA or capacity development support on assessment of climate-related risks. TA will be available to all banks registered with the Central Bank that express interest in the support. With knowledge and skills gained, the banks will conduct their own due diligence processes including through their assessment of risks (environment, climate-related, reputational etc.). Financial institutions will retain their own credit disbursement decisions. (GCF financing: 100,000 USD; co-financing: 0 USD).

63. **Activity 3.3.2: Support to the chamber of commerce and industry and the association of financial institutions and academia to train decarbonization service providers.** This activity will include a capacity needs assessment and the development of training curriculum, and subsequent training/coaching, for decarbonization strategic

advisory service providers. Specifically, the project will provide training to at least five service providers in the decarbonization space providing engineering, GHG accounting and ESG compliance support. Building the capacity of local service providers will ensure sustainability beyond the project. Participation in the project will be voluntary and based on a call that the project will raise through the PKS. All applicable rights are protected. The activity will prioritize companies from the accounting/engineering sectors proposing qualified young women (< 40 years) as trainees or companies led by women. (GCF financing: 120,000 USD; co-financing: 0 USD).

## B.4. Implementation arrangements

64. **Accredited Entity (AE):** FAO has extensive experience in supporting countries in the field of climate change adaptation and mitigation, as well as CAS. Furthermore, FAO - via its Investment Center – has been collaborating for over four decades with International Finance Institutions (IFIs) such as the World Bank (WB), the International Fund for Agricultural Development (IFAD) and the European Bank of Reconstruction and Development (EBRD) to support countries with development of investment plans and strategies; technical, financial, and policy assessments; and studies that are essential for CC and green investments. Having under implementation more than 138 climate and resilience projects worth over USD 600 million, FAO has the experience and technical capacity to manage the requested grant and support Serbia in the forest sector and decarbonization processes. In Serbia, FAO has been operational since 2001 in various thematic areas, including in SFM, biodiversity and wildlife protection, sustainable management of natural resources, land degradation, disaster risk reduction and climate change adaptation. Activities include the capacity development of national institutions, NGOs, associations, companies and farmers as well as the formulation of investments in forestry (forest restoration, forest assessment and afforestation) and policy assessment. Annex 2, Par. 250 contains examples of recently implemented or ongoing projects in Serbia.

65. Following consultations with the MOAFWM, other national institutions, UN agencies, civil society and private sector stakeholders, GoS and FAO agreed to implement the UN Country Programme Framework (CPF) focusing on three priority areas during the period 2019-2022. Building on FAO's experience in the country and its comparative advantages - and in line with key national strategies. Additionally, the project will strengthen the EU accession process focusing on alignment to the EU Common Agricultural Policy (CAP), environmental protection and standards.

66. FAO will serve as the Accredited Entity (AE) for the Project. FAO will be responsible for overall oversight of the Project, including: i) All project evaluation aspects; ii) Administrative, financial and technical supervision throughout implementation of the Project; iii) Supervision of effective management of funds to achieve the results and objectives; iv) Quality control of Project monitoring and reporting to the GCF; v) Project closure and evaluation. FAO will ensure that the project is executed in compliance with GCF and FAO rules and regulations, policies, and procedures, including relevant requirements on fiduciary, procurement, monitoring and evaluation, environment and social safeguards, and other project performance standards.

**Table 4 Contribution of the project to the FAO Country Programme Framework 2022-2025**

| Outcome | Component | Main Contributing Activities |
|---|---|---|
| *Serbia adopts and implements climate change and environmentally friendly strategies that increase community resilience, decrease carbon footprint, and amplify equitable benefits of investments.* | Component 1<br>Component 2<br>Component 3 | SFM<br>Creation of shelterbelts.<br>Restoration of degraded and unfarmed agricultural lands via forestry investments.<br>Support the decarbonization of the agriculture sector.<br>Training and capacity development.<br>Support to national forestry regulatory and policy framework. |
| *Natural and cultural resources are managed in a sustainable way* | Component 1<br>Component 2<br>Component 3 | CAS practices in the restoration/natural regeneration/reforestation in public lands<br>Conversion of coppice stands into high stand forests.<br>Enhancing the fund-raising opportunities in the forestry sector.<br>Greening of the wood biomass value chain.<br>Training and capacity development. |
| *Equitable economic and employment opportunities are promoted through innovation* | Component 3 | Support to the decarbonization of the agrifood sector.<br>Greening of the wood biomass value chain.<br>Training and capacity development. |

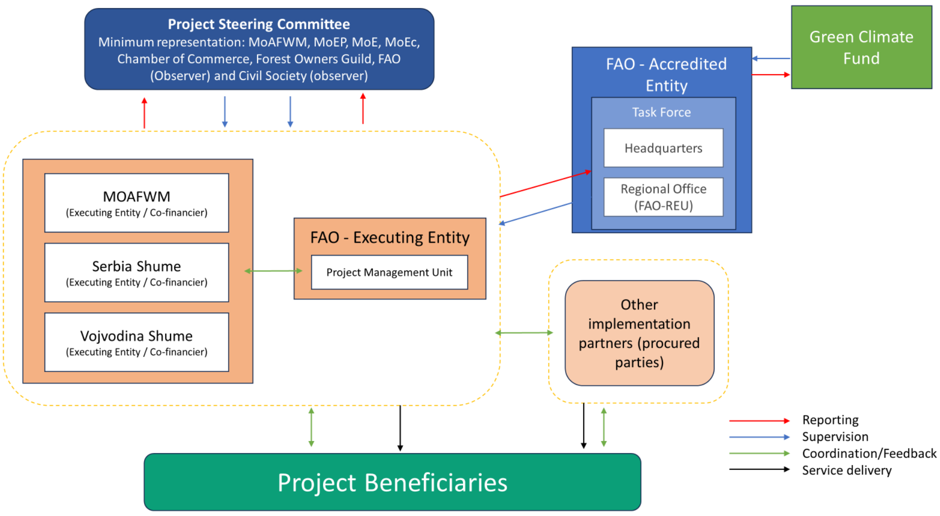
67. FAO will assume these responsibilities in line with the detailed provisions listed in the Accreditation Master Agreement (AMA) between FAO and the GCF. As Accredited Entity (AE) of the Project, FAO's supervision role will be attributed to the FAO Regional Office for the Regional Office for Europe and Central Asia (REU) in Budapest and relevant Offices and divisions at FAO headquarter (HQ), in Rome Italy, such as the FAO Office of Climate, Biodiversity, and Environment (OCB) and other technical divisions as required. To perform the AE functions, FAO will set up a dedicated FAO-GCF Project Task Force (PTF) comprising relevant staff from REU, FAO Office in Serbia and HQ. The segregation of responsibilities within FAO will ensure that the Organization can independently and effectively perform the AE functions.

The project implementation and supervision function will remain independent of the Executing Entity functions. The Project Task Force (PTF) will be established as a management and consultative body with a Formulator/Budget Holder (BH), a Lead Technical Officer (LTO) and a Funding Liaison Officer (FLO) / Technical Cooperation Program (TCP) Officer.

68. **Government Line Agencies.** The project will include several GoS line Ministries. Thus, the MOAFWM, the MEP, the Ministry of Interior (MoI), the Ministry of Finance (MoF), the MESTD and its affiliated departments at the Government level will be involved as beneficiaries. The MOAFWM and its affiliated Directorates (e.g. Forests) in the field will assist with the facilitation of activities in the provinces. The MEP, together with the MoF will participate in the activities connected with the development of carbon governance and related regulatory framework. MOAFWM will also play a supportive and facilitative role during project implementation and identify opportunities for building synergies with other projects dealing with climate change finance. Other essential entities to involve are the Ministry of Economy (ME) and the Ministry of Mining and Energy (MME), in particular on all activities in support of the biomass energy value chain. The participation and contribution of the Ministries will help to build their ownership and sustainability of project investments.

69. **The Project Steering Committee** (PSC) will be established for the overall strategic guidance of the project and housed at the MOAFWM. The MOAFWM as the NDA[1] for GCF in Serbia will notify the formation of the PSC and chair and convene regular six-monthly meetings to assess performance and issue appropriate guidance. The PSC will be composed of primary stakeholders such as the MEP, MoF, the MoI, the Vojvodina Undersecretary for Agriculture, forests and Water and the Under-secretariat for Environment and representatives of municipalities in target areas, the PKS/Finance Institutions, Forest Owners Guild, FAO (Observer) and Civil Society (Observer). The role of the PSC will be to: (i) Provide overall guidance and direction to the project; (ii) Ensure that co-financing support is provided in a timely and effective manner and report against its availability and use; (iii) Address project issues as raised by the Project Management Unit (PMU) and/or PSC members or EEs; (iii) Review the project progress, and provide direction and recommendations to ensure that the agreed deliverables are produced satisfactorily and within the approved project framework; (iv) Review and approve annual work plan and provide necessary strategic guidance for its implementation;(v) Appraise the annual project reports; (vi) make recommendations for subsequent work plans to build on achievements and address any shortcomings, etc. One representative from the PMU will act as Rapporteur to the PSC that should ensure through its overall leading and central role a strong country ownership.

**Figure 2 Implementation Arrangements**




70. **Nationally Designated Authority.** The MOAFWM in its capacity as the NDA and as the lead agency will use its convening power to facilitate consultations at the national level as well as assist in conducting knowledge exchange processes for scaling up the innovations and investments, enhancing impact and introducing a paradigm shift in the management, allocation and use of scarce water resources and improved adaptation to climate risks and efficient use of renewable sources of energy.

[1] The Ministry of Environmental Protection was appointed as the new National Designated Authority on 21 November 2024.

**Table 1 Role of the main institutional stakeholders**

| Institution | Description |
|---|---|
| MOAFWM | As NDA and responsible for developing and implementing policies in the field of forestry, agriculture and water management. The project will work mostly with the Forestry Directorate, with support and involvement of Agricultural Land Directorate, Rural Development Directorate, and others as necessary |
| MEP | Responsible for development and maintenance of the national mechanism for protection and conservation of the environment and its resources. MEP will be the main partner for the establishment of the offsetting / insetting mechanism and in the activities related to the decarbonization process. |
| MME | Responsible for increasing energy efficiency and energy security. The project will work with the MoE in greening the fuel biomass value chains and in the activities related to the decarbonization process. |
| MESTD | Responsible for the national education system. The MESTD will be main partner for the upgrade of the national universities and vocational schools' curricula that are relevant for the practices, technologies and methodologies introduced. |
| MoE | Responsible for elaborating the national economic, trade and industrial development policies and the strategies of economic security and sustainable development. The MoE will be engaged in forestry and decarbonization activities involving private sector. |
| MoF | Responsible for national finance, tax, and coordination of the management system and implementation of programs financed from European Union funds. The MoF will be involved in the carbon finance options and development of related guidelines, methodologies, standards and approaches. |
| Municipalities | Municipalities are in charge of municipal lands (including insignificant forest area) within the borders of their territory. The forested municipalities will ensure adherence to the CAS and SFM. |
| PKS / Industry | Both chambers and producers' owners' association will be engaged in identification of private sector actors and relevant activities. |
| SerBio | SerBio is an association of NGO's, companies and individuals in the field of biomass utilization and can facilitate interactions with various stakeholders and in the biomass mobilization and utilization. |

71. **Executing Entities.** FAO will act as the Executing Entity (EE) for all GCF-funded project activities and will be responsible for the GCF proceeds. FAO-REU will establish a dedicated PMU to oversee the execution of the project as a whole. The GoS, acting through the MOAFWM, PES and PEV, will be the EE for activities funded by their co-financing resources and will not execute any GCF Proceeds. The two public enterprises, PES and PEV are responsible for forest management in Serbia. PES oversees forest management of state-owned forests in Central Serbia, while PEV manages state-owned forests in the territory of AP Vojvodina. Both enterprises provide technical support to private forest owners, including drawing up of management plans, tree marking and other advisory functions (Annex 2, Appendix 7, paragraphs (para) 74-90). FAO will sign with each co-financier and EE a co-financing agreement which will detail their roles and responsibilities as co-financiers and EE. FAO will also sign contractual agreements with service providers/ procured parties, which may also include CSOs, identified for the project in accordance with FAO internal policy.

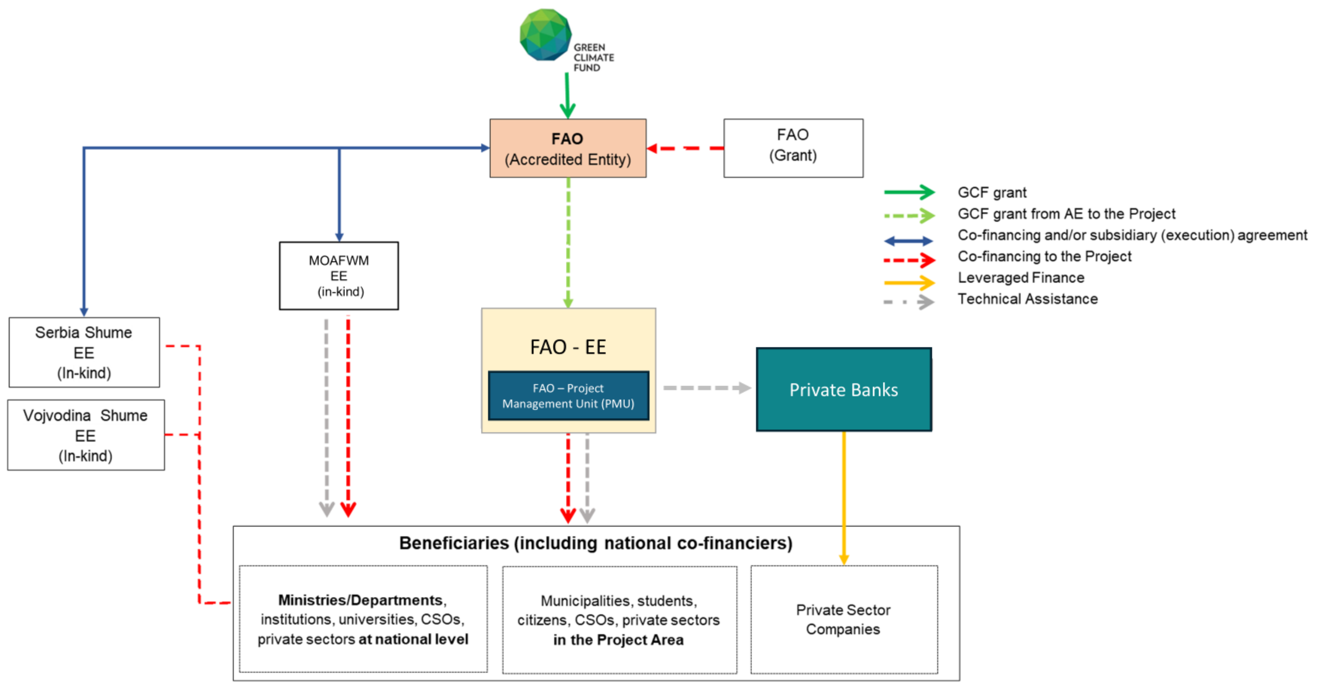
72. A dedicated **PMU** will be established and hosted by MOAFWM or FAO in (Belgrade) as determined during the project's inception phase. The PMU will coordinate directly with Directorate of Forests of MOAFWM, and the equivalent offices in the Province of Vojvodina and will be responsible for providing support to the execution of day-to-day activities with participating regional and local governments and other stakeholders. The PMU will be led by a project-recruited National Project Coordinator (NPC), responsible for overall project management and coordination with project stakeholders. FAO will either undertake direct responsibility for execution of the GCF financed activities through technical experts in the PMU and/or competitively outsource the services. Table 6 gives the responsibility for the achievement of the main project outputs. In each case, FAO will work in close coordination with the relevant ministries. FAO will competitively recruit service providers to carry out specific activities as detailed in the procurement plan and in the budget (Annex 11 and Annex 4).

**Table 2 Key Responsibilities and executing responsibilities**

| **Agency** | **Role** | **Specific Responsibility** |
|---|---|---|
| FAO | AE and Co-financier | Supervision & oversight<br>Grant contribution |
| FAO | EE | Execution of project activities |
| MOAFWM - Directorate of Forests | EE and Co-Financier | In-kind contribution and Policy Review<br>Forestry investments<br>Provision of seedings<br>Coordination and support in the field |
| Serbia Shume<br>Vojvodina Shume | EEs and Co-Financier | Forestry investments<br>Seedlings production |

73. GCF funds will be transferred to FAO in the form of a grant. GCF Proceeds will not flow through the proposed EEs. These institutions will only be responsible for the management of their respective co-financing. Table 32 in Annex 2 provides an overview of the EEs executing each sub-activity and the respective source of financing.

**Figure 3 Funds flow and contractual arrangements**




## B.5. Justification for GCF funding request

74. In 2022, Serbia's public debt accounted for more than 60% of its GDP [IMF, 2022; latest available data]. The government launched several programs to address structural weaknesses, increase public sector efficiency, and eliminate bottlenecks to private sector growth, while maintaining macroeconomic stability. These include a green growth program to support post-COVID-19 economic recovery efforts while addressing a shrinking population, labor shortages, and climate change[25]. Furthermore, in 2021 the country issued a green bond worth EUR 1 billion[26] to support green investments including for renewable energy, energy efficiency, sustainable water and wastewater management, pollution prevention and control, circular economy, protection of the environment and biodiversity and sustainable agriculture. The bond has been oversubscribed three times as of December 2022, and it has helped to fund 34 projects on environmental protection, biodiversity and sustainable agriculture. Therefore, as reported in the second NDC (2020), Serbia will be able to speed up its decarbonization process if financial support is provided for key investments. Furthermore, the country estimates the need for further climate investments at about EUR 19 billion (USD 23 billion) for the 2021-2030 period indicating additional support for investing in forestry, agriculture, and agrifood sectors. The performance of the EUR 1 billion green bond shows an overall interest of the economy (all sectors) in sustainability aspects. As of 2022, however, the allocation to energy efficiency and to environmental protection (including agriculture) did not pass 15% of the total allocation. The project is proposing support to investment in the forestry sub-sector, therefore, involving both public and private stakeholders. Due to their remoteness and limited size of the privately owned forest holdings, the transaction and investment costs are particularly high and require higher concessionallity (ref.: Annex 3 narrative). Furthermore, energy efficient projects (2% of the total share of proceeds of green bond) are largely focused on domestic appliances/ buildings. The project is instead promoting energy efficiency and decarbonization investment in the agriculture and agrifood sector, representing a large untapped potential for overall reduction in Serbia's emissions.

75. While the country is currently engaged in forest investments and substantially cofinancing both the adaptation and mitigation aspects of this proposal by focusing most of the Forestry Department's resources in project activities, the state does not dispose of the necessary resources to cover for additional costs of CC adaptation and support development of its decarbonization governance and path. Therefore, the GCF funding will support the upgrading of existing policies and procedures and cover the additional cost needed to guarantee the paradigm shift that will transform Serbian forests in the main adaptation and mitigation engine of the country (the project investment is equivalent to about USD 1.8m, from output 1.1 and 1.2). The GCF will also contribute to the provision of incentives (technical assistance) for the private sector to engage in forestry and decarbonization process. As reported in Annex 2, appendix 12 and supporting letters (Annex 1.b) the private sector and its organizations are ready to engage, dispose of credit as well as of the means to access it, but misses the capacity to define the investment needed. Therefore, as reported in both its NDCs, to ensure that expected

results are achieved and to contribute to identified goals, the country requires international technical and financial assistance. As reported in the updated NDC (2021), Serbia will reduce its GHG emissions by 33% by 2030 (compared to 1990) only with the appropriate international financial, technical, and capacity building support. Forestry has been identified as a priority area in the updated NDC (Section 3 (d)), and the country plans to invest over USD 9 million annually into the sector, including a majority of its forestry budget over the next 7 years into the target areas of the FOREST-Invest proposal, thereby contributing to its results. Nonetheless, the country has not included in its investment plans the costs of adaptation and decarbonization related to forests (decarbonization governance), which puts the overall government investment in forestry at risk. External funding for Serbia's forests is critical to help the country achieve its current NDC targets and decarbonize its economy. The public sector is the only major financier of Serbia's forests, and while the government has already demonstrated a strong financial commitment, the magnitude of additional funding required is too large for existing and committed funding to fill this gap on its own. By injecting initial concessional financing into the forestry sector through the GCF, the project can act as a catalyst to leverage additional funds from the government and external financiers (including the private sector) to improve carbon storage in forests and national capacities to manage them, to contribute reducing inefficiencies in the wood value chain, and to trigger emissions reduction by unlocking and de-constraining strategic private investment.

76. The available financing is not adequate given the magnitude of the challenge. For instance, the European Union and International Financial Institutions already support Serbia with some financial instruments for Pre-Accession Assistance to support energy efficiency and renewable energy. However, this does not adequately help Serbia achieve its targets.[27] Concessional financing through this project, in addition to Serbia's positive and stable economic outlook[28] will help attract much-needed additional funds and provide an impetus to Serbia's decarbonization efforts.

77. The project also requires external funding, especially given the current rise in global energy prices. The current geopolitical crisis in Ukraine and Russia could adversely affect Serbia's energy security. Specifically, in 2019, Serbia imported USD 2.5 billion worth of goods from Russia, or about 5% of its GDP.[29] Of this, it imported USD 1.4 billion in fuels alone, USD 359.9 million of which from Ukraine. The World Bank's Commodity Markets Outlook (April 2022) anticipated that energy prices will rise more than 50% in 2022 alone (World Bank Group, 2022c). While the World Bank's Commodity Markets Outlook for 2024 anticipates a slight decrease in global energy prices in 2024 by 3%, with further declines in 2025, it also anticipates that they will remain significantly higher than pre-pandemic levels, suggesting that while there may be some relief, energy prices in Serbia will likely stay elevated in the near term due to persistent global pressures from geopolitical conflicts and tight supply conditions. Further, rising fuel or gas prices are likely to affect prices of all other energy sources, many of which have already been experiencing rising prices since 2021.[30] For instance, the Serbian Power Exchange (SEEPEX) day-ahead electricity prices increased by more than five times between March 2021 and March 2022, from about EUR 50 to about EUR 270 per MWh. Investing in forests and in the efficiency of biomass value chain, and at the same time supporting decarbonization of private sector – and thus ensuring that fuelwood is more cost-effective and efficient – can serve as a potential hedge against these rising prices. Thus, the project's investments in forests can help ensure that rural populations are not priced out of using fuelwood in the short term and that alternative sources of energy such as biomass are developed in the medium term.

78. The forest sector in Serbia is highly fragmented with limited investment capacity. Out of the over 2,000 companies involved in forestry (mostly in wood processing), about 97% are micro/small enterprises with limited human capital and access to credit due to lack of collateral. Further, forest owners do not see investments in forestry as viable due to the limited scale emanating from their small plots. Beyond the public budget, there are limited resources to address these bottlenecks of the Serbian forestry sector and sustain the national decarbonization process. The technical assistance provided by the project is a necessary condition to ensure the financial attractiveness and viability of decarbonization investments at the firm level. These measures will contribute to increasing financial returns, augmenting scale and the affordability of financial instruments for decarbonization (provided by national and international financial institutions). The project will only cover the additional costs for adaptation and mitigation, beyond already-committed public investments. The proposed budget reflects the needs of the state to increase its forest cover while adopting sustainable and climate adaptive management practices, upgrade the technical capacity of public and private actors in forestry, and enhance private sector participation in decarbonizing the economy. Although Serbia is a middle-income country, the magnitude of funding required to achieve its NDC targets and address decarbonization challenge cannot achieve them with public funding alone.

79. **In the immediate term**, concessional financing into the project will help enhance the energy security of the rural poor and set Serbia on a more sustainable forestry-energy-decarbonization path. In the longer term – because Serbia is a middle-income country – concessional finance can help unlock additional necessary funding.

80. **In the short and medium term**, concessional financing in forests is critical to ensure continued energy security for Serbia. Forests account for a significant source of energy (11.86%) for the country. Populations in rural areas as well as the

most vulnerable segment are the primary users of firewood as a source of energy, and due to the more limited economic opportunities, are the most affected by fluctuation or spikes in energy cost. Nearly half of the lowest income decile households in Serbia have arrears in utility services, demonstrating that the poorest households are already stretched thin and not able to absorb rising energy prices [31] The present geopolitical crisis makes a strong case for immediate investment in forests and decarbonization efforts more broadly to ensure that rural populations do not get priced out of using firewood in the short term and that alternative sources of energy such as biomass are developed in the medium term without compromising the provision of other paramount ecosystem services (e.g. carbon removals, protection, NWFP and others). Similarly, the current forecast of sustained high global energy prices [World Bank 2024], will drive the market toward larger investments in decarbonization through energy savings and optimization.

81. **In the long term**, given the large amount of funding required to decarbonize, concessional financing will be critical for decarbonization. The magnitude of financing required is larger than the size of existing and committed financing, notwithstanding the increase over time in resource allocation to the sector. The project can catalyze funding from the government, providing initial concessional financing to unlock broader investments, and external financiers, including the private sector. For instance, the EU and IFIs already support Serbia with financial instruments for Pre-Accession Assistance to support energy efficiency and renewable energy. However, these are not enough to help Serbia achieve its targets.[32] Concessional financing through this project, in addition to Serbia's positive and stable economic outlook,[33] will help attain much-needed additional funds and spur decarbonization efforts. Stakeholder support for the project also indicates the value proposition of concessional financing. This is confirmed by the recently agreed increase in co-financing from the state to a total co-financing amount to over USD 51 million, bringing co-financing to 50% of the total investment. According to the OECD Creditor Reporting System, Serbia's forestry sector only received USD 441,000 in official development assistance (ODA) in 2019, only 2.4% of total ODA to the Agriculture, Forestry and Fishery sector. The combination of diverse sources of financing (concessional financing, public funding and ODA) will help de-constrain private investment envisaged in component 3, critical to trigger the transformative change of the agrifood and forestry value chains towards its NDC targets. The Serbian government and the main stakeholders involved in the project not only confirm their support but also their willingness to increase in-kind co-financing. In addition to the government, financial institutions such as large commercial banks in Serbia have also expressed their interest in contributing to the project's activities (see Annex 1.b). Banks indicate increasing demand for products aimed at decarbonization and energy efficiency and the value added of concessional financing and technical assistance in reducing information barriers and regulatory uncertainty. This will bring to the project some very important leveraged private finance that is expected to be around USD 50 million.

## B.6. Exit strategy

82. An increased adaptation and decarbonization agenda is necessary for Serbia to continue doing business with the EU and other markets where decarbonization policies are increasingly more stringent toward products that are high in terms of carbon emissions (among others). This has been recognized by the government as a priority in its national strategies where it targets a climate neutral and climate resilient economy. Each project activity is tailored to support national strategies and climate targets. Actions with both public and private actors have been designed with stakeholders based on needs, potential and execution capacity. The project will not add to the already existing or planned procedures and infrastructures. It will provide tools and capacities to execute strategies efficiently, transparently, and effectively.

83. **Sustainability and replicability** of the project has been considered in the design through the following: (i) processes and mechanisms introduced will be self-funded and will optimize actions and national expenditures in the forestry sector (ii) knowledge management processes will generate evidence for stakeholders and they will have the skills to progressively intensify and optimize its investments in forests, and decarbonization; (iii) the introduction of new forest management practices will increase the effectiveness of forest investments (e.g. high survival rate of planted material) and will, at the same time, decrease the overall cost of operations, supporting the country in ensuring the adaptation of the remaining forests and the additional forests planned by the state (48,000 ha of afforestation); (iv) the introduction of the new forest governance strategies will enhance the role and contribution of the private sector and its associations[34] and will allow for climate adaptive management of forests, reducing the adverse impacts of drivers of forest degradation and ensuring faster response of concerned stakeholders at no additional cost. Further, proposed actions will become part of the national policy and management framework of forests, ensuring immediate national scaling up of introduced practices and technologies and supporting the development of an offset mechanism will provide further opportunities for a continuous flow of financial resources for sustainable forest practices. Finally, proposed actions will enhance the role and contribution of the private sector and local administrations. Administration, municipalities, private landowners and companies/agribusiness require time and incentives to shift from the BAU scenario (e.g. forests as mining grounds) to a more engaged and transparent sector where stakeholders collaborate for the maintenance and expansion of forest ecosystems. While in

execution as well as after completion, actions will be monitored with the involvement of all national and local stakeholders (Annex 7 contains more information on monitoring activities).

84. **Private sector engagement:** The project will collaborate with the main Serbian sectorial associations (e.g. PKS, Serbia Livestock Association, Serbia Grain Association, SERBIO and others) and with national and international academia (e.g. University of Florence, the University of Belgrade) to establish, revamp and replicate in Serbia the use of successful approaches developed to secure an agglomeration of smallholders and to bring economy of scale in contexts where land fragmentation prevents sustainable and profitable management of forests. The AE, in collaboration with Serbian partners, analyzed a number of platforms in Serbia (e.g. SERBIO Biomass logistic and trade center[35] and the Digital Platform on the Circular Economy or CE-HUB[36]) and Europe to decide on the best examples to tailor to or expand in Serbia. Among the European examples, a strong candidate that has received a strong vote of confidence from the Serbia academia and of the Forest Directorate is ForestsSharing[37]. Practices and technologies introduced will become part of the national educational process integrating the knowledge derived from the project to reach relevant stakeholders even after the project life-cycle. During implementation and after completion, the intervention and achieved results will be monitored and scaled up by the MOAFWM, the MEP and by the MESTD (Table 7).

**Table 3 Sustainability roles and responsibility of the main stakeholders**

| Institution | Description |
|---|---|
| MOAFMW | Responsibility over the supervision and monitoring of all the forests in Serbia. Thanks to the knowledge acquired with the project, the ministry will dispose of more upgraded technologies and practices to monitor forests and report about their state and evolution. |
| MEP | Responsibility over the monitoring, reporting and verification of GHG emissions as well as over sustainable and low carbon development in Serbia. The ministry will be in charge of ensuring transparency and accountability of the newly created carbon mechanisms and to certify decarbonization efforts. |
| MESTD | Guarantee that the knowledge transferred with the project is passed to the new generations of professionals, and that education on forest is integrated under the environment and climate change education. |
| Private Forests' Owners Associations | Guarantee the involvement and participation of private landowners and liaison with the institutions and academia to guarantee knowledge transfer to and from the private sector. Participation of private forests' owners will also be guaranteed by replicating in Serbia the success of a platform created in Italy by the University of Florence (www.forestsharing.it) that will allow to support the Serbian private owners' associations (e.g. SERBIO, PKS) to aggregate small forest players and stimulate economy of scale in a very fragmented scenario. |

85. The innovative mechanism promoted by the project is also expected to create spill-over effects in other sectors willing to engage in decarbonization. For instance, environment certifiers, carbon accountants / MRV specialists and other service providers will also operate beyond the forestry sector. The incomes generated through the offsetting mechanism (purchase of carbon credits from the State) will ensure the market-driven nature and sustainability of the decarbonization facility and restoration/forestry expansion activities beyond the project's lifetime[38].

## C. Financing Information

### C.1. Total financing

| (a) Requested GCF funding (i + ii + iii + iv + v + vi + vii) | Total amount | Currency |
|---|---|---|
| | 25,000,000 | USD ($) |

| GCF financial instrument | Amount | Tenor | Grace period | Pricing |
|---|---|---|---|---|
| (i) Senior loans | Enter amount | Enter years | Enter years | Enter % |
| (ii) Subordinated loans | Enter amount | Enter years | Enter years | Enter % |
| (iii) Equity | Enter amount | | | Enter % equity return |
| (iv) Guarantees | Enter amount | Enter years | | |
| (v) Reimbursable grants | Enter amount | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| (vi) Grants | 25,000,000 | | | | | |
| (vii) Results-based payments | Enter amount | | | | | |
| **(b) Co-financing information** | **Total amount** | | | **Currency** | | |
| | 58,811,581 | | | USD ($) | | |
| **Name of institution** | **Financial instrument** | **Amount** | **Currency** | **Tenor & grace** | **Pricing** | **Seniority** |
| MOAFWM | In kind | 53,455,376 | USD ($) | Enter years<br>Enter years | Enter% | Options |
| Serbia Shume | In kind | 4,222,108 | USD ($) | Enter years<br>Enter years | Enter% | Options |
| Vojvodina Shume | In kind | 659,147 | USD ($) | Enter years<br>Enter years | Enter% | Options |
| FAO | Grant | 474,950 | USD ($) | Enter years<br>Enter years | Enter% | Options |
| **Total financing (c) = (a)+(b)** | **Amount** | | | **Currency** | | |
| | 83,811,581 | | | USD ($) | | |
| **(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)** | 86. The project will also attract USD 50 million in leveraged private finance from financial institutions operating in Serbia (ref. Annex 1.b for letters of interest). The institutions will disburse this amount in the form of loans for the purposes of investing in decarbonization across sectors. These funds are additional and not included in the Table above. Moreover, the project will leverage about USD 0.8 million of private forest landowners to support forest investment. | | | | | |

## C.2. Financing by component

| Component | Output | Indicative cost USD | GCF financing Amount USD | GCF financing Financial Instrument | Co-financing Amount USD | Co-financing Financial Instrument | Co-financing Name of Institutions |
|---|---|---|---|---|---|---|---|
| Component 1 | 1.1 | 8,501,806 | 1,089,565 | Grants | 7,307,141 | In-kind | MOAFWM |
| | | | | | 105,100 | Grants | FAO |
| | 1.2 | 694,047 | 646,147 | Grants | 48,500 | Grants | FAO |
| Component 2 | 2.1 | 4,620,633 | 4,164,923 | Grants | 455,710 | In-kind | MOAFWM |
| | 2.2 | 391,450 | 70,100 | Grants | 321,350 | Grants | FAO |
| | 2.3 | 42,990,845 | 10,111,235 | Grants | 28,237,755 | In-kind | MOAFWM |
| | | | | | 3,982,708 | in-kind | PES |
| | | | | | 659,147 | in-kind | PEV |
| Component 3 | 3.1 | 19,330,217 | 4,679,847 | Grants | 14,410,970 | In-kind | MOAFWM |
| | | | | | 239,400 | in-kind | PES |
| | 3.2 | 2,876,000 | 2,876,000 | Grants | | | |
| | 3.3 | 220,000 | 220,000 | Grants | | | |
| PMC | PMC | 4,185,983 | 1,142,183 | Grants | 3,043,800 | In-kind | MOAFWM |
| **Indicative total cost (USD)** | | 83,811,581 | 25,000,000 | | 58,811,581 | | |

## C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

| | |
|---|---|
| C.3.1 Does GCF funding finance capacity building activities? | Yes ☒ No ☐ |
| C.3.2. Does GCF funding finance technology development/transfer? | Yes ☒ No ☐ |

87. The project will invest an estimated amount in capacity development and technology transfer to Serbia respectively of USD 8,889,081 and USD 3,992,991. It will support the knowledge/technology transfer and capacity development of over 4250 persons (at least 30% women) from both public and private institutions. Furthermore, the project will ensure transfer to new generations of professionals by including introduced technologies and practices into national curricula ensuring scale up of knowledge transfer to over 10,000 students that will form the next generation of professionals in Serbia.

## D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

### D.1. Impact potential

88. This project will help contribute to GCF's portfolio-level mitigation and adaptation results objectives identified under the Updated Strategic Plan (USP) 2024-2027 to mitigate 1.5 to 2.4 gCO2eq and enhance the resilience of 570 to 900 million people over the 2024-2027 period. It will also contribute to the USP targeted results areas: 5 (Ecosystems), 7 (Clean Energy), and 10 ( Local private sector early-stage ventures and MSME innovation).

89. The **cross-cutting approach of the project**, focusing 67% of GCF proceeds on adaptation and 33% on mitigation, follows the country's NDC commitments to address the forestry sector through cross-cutting processes by enhancing ecosystem services provided by Serbian forests, including carbon removals. The selected approach also aligns with Serbia's Low Carbon Development Strategy, specifically its fourth objective to "Preserve the potential of mitigation measures, [...] by increasing the resilience to climate change of the priority sectors," including forestry. In line with the national strategic framework, the project aims to maximize investments by improving the resilience of vulnerable populations, particularly through natural hazard protection, provision of sustainable fuelwood and non-timber forest products, as well as carbon removals and sinks. This balanced approach demonstrates the project's alignment with national climate priorities and its comprehensive strategy to address both immediate adaptation needs and long-term mitigation goals in the forestry sector.

90. **Mitigation Impact:** The project will increase the capacity of public and private forests to remove CO2, it will reduce the negative impact of drivers of forest degradation, and it will support Serbian agribusiness companies to decarbonize their processes and increase their resilience. . With regards to mitigation impacts calculations, the project used FAO NEXT, a tool specifically designed for evaluating the carbon impact of projects in the AFOLU sector that uses IPCC methodology. Annex 22 reports the impacts of the different forest and decarbonization activities of the project in terms of CO2 removals and avoided emissions corresponding in total to -8.4 mln tCO2eq. Project investments will focus on: (i) enhancing investments in forest management and expansion via afforestation, forest restoration, development of high stand forests from not managed / degraded coppice stands, rehabilitation of degraded and unfarmed lands through agroforestry and development of a network of shelterbelts in agricultural areas with local administrations and land owners; and (ii) managing forests to stabilize and increase ecosystems services (e.g. carbon removals and sink) and upgrading the country's carbon finance mechanism. This also entails greening the biomass value chain and implementing rules and regulations for the efficient use of fuelwood; (iii) increasing private sector engagement in de-risking investments and decarbonization practices and; (iv) matching agrifood companies with available carbon financing to decrease emissions of production processes.

91. **Adaptation Impact:** Adaptation investments will help modernize the forestry sector to reduce its vulnerability to climate change and prepare forests to withstand forecasted changes, without compromising key ecosystem services, including those that are critical to poverty alleviation (i.e., energy, NWFP). Furthermore, the project will reduce the vulnerability of rural communities and the poor and their exposure to energy price shocks. Finally, the proposal will reduce the exposure and vulnerability of private forests smallholders as well as small and medium sized agrifood enterprises by reducing their overall climate risks and allowing them to use forests for additional income as well as to contribute to the decarbonization efforts. The transformative path proposed by the project will ensure investments around: (i) reducing the bottlenecks currently preventing climate adaptive forest management and sustainable provision of fuelwood; (ii) expanding nurseries' capacity to produce climate adaptive seedlings and increasing the number of available species (from 4 to 24+ Annex 2, appendix 7) for forestry investments, (iii) introducing CAS practices and technologies to: (a) climate proof public and private forestry investments, (b) enhance the forest governance, and (c) ensure the provision of ecosystem services that are paramount for rural communities and the most vulnerable and poor segments of the population (i.e. along the dimensions of energy, and livelihoods); and (iv) increasing the participation of communities and the private sector (table 6) in forest investments and management to ensure forest resilience, reduce over extraction of fuelwood and introduce forestry and agroforestry as climate resilience tools to restore degraded lands and reduce the risk of investments in the agrifood sector. Proposed investments are prerequisites to enable the forestry sub-sector to: (i) continue sustaining the country's energy security, particularly of the country's most poor and vulnerable; (ii) support the increase of renewable energy sources (RES) in the national energy mix (par.19); and (iii) maintain and expand the sector's contribution to reducing national net emissions by 2030 and neutralizing them by 2050. Finally, each adaptation investment will contribute directly to carbon removal. Refer to Annex 25 for further details of direct and indirect beneficiaries.

### D.2. Paradigm shift potential

92. **The paradigm shift** of the project is built on three pillars. **First,** the adaptation, renewable energy, poverty alleviation and decarbonization strategies of Serbia are strictly connected to the health of forests and their capacity to provide for key ecosystem services such as carbon removals, energy, livelihood, water and protection. Therefore, the

adaptation of the forestry sector, the expansion of forest areas and the reduction/removal of the negative impacts of drivers of forest's degradation are preconditions for the adaptation of the most vulnerable people and communities as well as decarbonization in Serbia. **Second**, the governance of forests as well as provision of ecosystem services (i.e. carbon removals, biodiversity, non-wooded forest products, protection and fuel biomass) can no longer be delegated to central institutions only (Component 1). It requires new paradigms where communities, local authorities and the private sector could effectively participate and invest. Therefore, the project will enhance stakeholders' participation to provide the forestry sector with new approaches for people to engage and additional financial resources making the sector less dependent on national budget or external donors. **Third,** transparency and evidence-based forest monitoring and reporting are paramount to secure the trust of national and international stakeholders and to assure that reported data about forests and emissions are verified (Component 1). Through the three pillars the project will support transforming the forest from "mines" to effective and transversal engines of sustainable and climate resilient development.

93. **Innovation:** The project offers an innovative approach to forest investments, forest management and mobilization of private actors in getting involved in decarbonization processes. Coherently the project will introduce a number of innovations: (I) introducing CAS practices, technologies and investments; (II) addressing forests with a crosscutting approach to ensure that investments in adaptation also yield mitigation and poverty alleviation results; (III) involving the diverse actors of the sector in the decision-making process; (IV) ensuring the instruments and technical assistance to private sector companies to engage in decarbonization also via forestry; (V) establishing digital tools to aggregate and engage private owners of land and forests; (VI) creating the linkages with national banks and investors with private sector companies interested in decarbonization and forestry; and (VII) guaranteeing with no additional costs quantifiable benefits such as: (a) increased biodiversity in forest and agriculture land; (b) expansion of bio-corridors; (c) socio-economic condition of rural households; (d) protection from disasters; and (e) soil erosion control. Each innovation is detailed with the FP and the Economic and Financial Analysis of the project. Finally, supporting the country in establishing a national offsetting and insetting mechanism will allow it to include soil carbon and, therefore, expand the mechanism to the entire agriculture sector.

94. **Potential for scaling up and replication:** The established mechanisms and introduced practices / technologies will allow the country to scale up introduced investments on over 1,717,360 ha representing the MOAFWM's target for 2022-2030[39] and expand it to all forests and the agriculture lands. Furthermore, the proposed theory of change and intervention strategy are replicable in the entire Balkan where the forestry sector is confronted with similar bottlenecks and drivers of degradation, the project will support the knowledge sharing at regional level to contribute to immediate scaling of activities through FAO offices. In terms of adaptation, the activities of the project will allow stakeholders to expand the adaptation benefits to all the rural communities in Serbia (2,966,839 individuals of which 1,513,088 are women). In addition, within the lifespan of the project - this proposal will create high potential for attracting relevant companies and contribute to the national offsetting opportunities. The project will work with public and private stakeholders including key associations such as the Chamber of Agriculture, the Serbia Grain Association, the PKS, Association of Financial Institutions, the Forest Owner Guild, SerBio and other associations as well as with renowned NGOs such as WWF-Serbia, BirdLife Serbia and NALED to ensure immediate scale up to the whole country by embedding in their processes and action the knowledge and technologies introduced by the project. This will increase the ownership and participation of the private sector into the process supporting the scale up and replication at the national level of the project's activities. Finally, the FAO is currently in the process of discussing the development of similar projects with the NDAs of the Republic of Moldova, North Macedonia and other countries applying a blueprint approach[40]. In addition, the project will enable companies to access already available green loans aimed at decarbonizing their processes. This will create a high potential for attracting other companies in the future and contribute to the national offsetting market.

95. **Potential for knowledge sharing and learning:** The project will invest in training and capacity development of both public and private stakeholders in each component. As reported in Section C3.2, the project will build and develop capacities of over 4250 persons, investing over USD 8 million. Additionally, to further ensure long-term sustainability beyond the execution of funded activities and the project life cycle, the project will fund and coordinate with the MESTD the upgrade of national curricula (universities and vocational schools) related to topics covered by the project such as SFM practices. Partnering with communities, CSOs and universities will prove critical in maximizing the project's impact in terms of disseminating knowledge and retaining implementation best practices related to decarbonization. Finally, the project will share lessons learned and best practices in the region to support and enhance knowledge sharing in the region.

96. **Contribution to the creation of an enabling environment:** The establishment of the National Forest Monitoring (NFM), the upgrade of the MRV system, the establishment of the offsetting/insetting scheme as well as the standards and strategy for fuelwood production/handling/management/use will strengthen the regulatory framework of the forest sector and will have a spill-over effect in other sectors of the Serbian economy. Combined with the decarbonization strategies (Component 3) of companies, the project will set up the enabling conditions for the private sector to keep being engaged in

the long-term decarbonization of the national economy, for the agriculture sector to contribute to the decarbonization process and to create new market opportunities in the forestry sector (e.g. green biomass) and for private forest owner to engage in both decarbonization and forest management.

97. **Contribution to the regulatory framework and policies:** the project will abide to and contribute to the execution of the climate change policy framework of Serbia (table 7 – Section D5) contributing to a few processes such as the forest management, the energy efficiency, the renewable energy, the NDC, and others. Furthermore, the project will ensure the provision of non-financial incentives (e.g. technical assistance and evidence based decarbonization governance) and will contribute to de-constrain key financial support (leveraged private finance) from national finance institutions to the private sector to promote investments in low-emissions and climate resilient development. Overall contribution to climate resilient development pathways consistent with a country's climate change adaptation strategies and plans: the project will induce a radical transformation of how forests contribute to adaptation, mitigation and poverty reduction in Serbia. By involving stakeholders according to their functions and potential, the project will ensure a considerable reduction of national expenditures to secure high carbon removals from forest. Finally, the project will radically change the way private sector owners of forests participate and engage in the management of forests and in the provision of ecosystem services (e.g. fuelwood production, carbon removals).

## D.3. Sustainable development

98. **Environmental Co-benefits:** In addition to the presented positive impacts in terms of CCA and CCM, the project will have positive impacts on biodiversity[41], on soil quality[42] and water availability, decrease of evapotranspiration and slow down soil erosion, increase agricultural yields, and protection of rural communities and infrastructures from flash floods, floods and landslides. To this end the project will collaborate with the Serbian Water Management Directorate to ensure, among the areas to be afforested, those identified by the Water Management Strategy (2017-2034) and paramount to prevent and mitigate soil erosion and floods. Furthermore, via afforestation activities and shelterbelts/windbreakers, the project will support the active protection and conservation of biodiversity. These will create corridors and shelter for wild animal species and flora. Forestry investments will be executed in accordance with the Law on Nature Conservation and the Decree on the Ecological Network (Official Gazette of RS, No. 102/2010). Finally, the project will partner during the entire execution with key NGOs such as WWF-Serbia that will support and advice on community participation as well as biodiversity and nature conservation.

99. **Economic Co-benefits**: Directly monitorable benefits will originate from: (**I**) the improved efficiency of wood biomass used for fuel that will reduce the energy expenditures of the poor and the most vulnerable[43]; (**II**) the increased value added from NWFP markets (example, fruits, beekeeping etc.) and eco-tourism thanks to the increased biodiversity and greater forest cover; (**III**) the potential benefits that will originate from the degraded private lands converted to bioenergy plantations and from the agricultural lands protected by shelterbelts. With regards to energy, the lower unit cost of energy will enhance affordability of energy for the poorest segments of the population. In Serbia, households currently spend more than 10% of their average expenditure on energy. Coupled with environmental / energy efficiency awareness campaigns to avoid increase of consumption of fuel wood, lower fuel costs will also imply a lower share of overall spending going toward energy, freeing up resources for other household needs.

100. **Social co-benefits:** The introduced practices and technologies under this project will contribute to creating new jobs and new markets (e.g. $CO_2$ management, green biomass, climate adaptive nurseries). Furthermore, relevant co-benefits of reducing the adverse impacts of fuelwood include distinctive social benefits as project's activities will help reduce poverty in Serbia in three ways: **(i)** improved quality of fuelwood with energy efficiency gains; **(ii)** greater transparency of the solid biofuel value chain, and **(iii)** enhanced economic opportunities through the sector's modernization. First, the availability of higher quality fuelwood (drier with higher calorific potential) will increase energy efficiency at the household level. Households are expected to face a lower unit cost for energy output produced by fuelwood. This lower unit cost of energy will enhance affordability of energy for the poorest segments of the population. In Serbia, households currently spend more than 10% of their average expenditure on energy. Coupled with environmental / energy efficiency awareness campaigns to avoid increase of consumption of fuel wood, lower fuel costs will also imply a lower share of overall spending going toward energy, freeing up resources for other household needs. Second, the project will enhance transparency around fuelwood availability and quality. Specifically, rural consumers will have greater knowledge on the correct user behavior and around fuelwood types and heating appliances to make more informed consumption decisions. Third, the modernization of the biomass sector will create economic opportunities in rural areas, directly benefiting the population of these areas.

101. **Gender empowerment co-benefits:** The project will benefit Serbia with some specific focus on sectorial stakeholders and private companies. In all training and investments, when possible, the project will give higher priority to

women owning degraded coppice stands or unfarmed/degraded lands cultivation of wooden species for bioenergy or other purposes and will ensure that at least 30% of beneficiaries are women**.**

102. **Sustainable Development Goals (SDGs)**: Climate change impacts and obsolete management practices threaten ecosystem services of forests in Serbia, including energy security, livelihood and carbon sinks. Some of the wider benefits of the project are expected to assist in mitigating some of these effects and help in reducing poverty (SDG 1), enhance food security **through rehabilitation and protection of agricultural soils** (SDG 2) **reduce health impacts of particulate matter through increasing efficiency of biomass combustion and the occurrence of disasters through enhanced ecosystems (SDG 3) promote high quality biomass fuels and** investments in **renewable energy and energy efficiency in the frame of industrial decarbonization practices** (SDG 7) undertake several key investments for combatting climate action (SDG 13) promote sustainable use of terrestrial ecosystems, combat desertification, and halt and reverse land degradation and biodiversity loss through **CAS practices** (SDG 15). For more details see Table 29 in Annex 2.

## D.4. Needs of recipient

103. Due to the barriers and adverse impact described in section B, the forestry sector is among the most vulnerable in Serbia, further increasing the vulnerability of rural communities and the poor that depend on forests for energy, NTFPs and livelihoods. Furthermore, the reported decrease in carbon removal (-19.4% compared to 2010) is limiting the capacity of Serbia to execute the LCDS of its economy. The project will address the vulnerability of the forestry sector by introducing CAS technologies, processes and practices and will support the country in expanding/enhancing/establishing the needed policy and legal reforms to remove the bottlenecks that are at the root of the identified climate change adaptation deficit of the sector. The project will address the three main priority areas identified by the country for the forestry sector: (i) Risk Reduction; (ii) Monitoring and Research; and (iii) Policies, capacity building and awareness raising. Furthermore, the project will support the country in addressing the reported forest's overexploitation risk due to fuelwood needs as well as the overall vulnerability of the population to natural hazards. Finally, the project represents an opportunity to further enhance the role of women in forest management and decarbonization investments. Thanks to a targeted capacity development focus (Annex 8) the project will ensure the participation and empowerment of women forest owners, service providers and entrepreneurs in the private sector.

104. In 2022, Serbia's public debt accounted about 60 % of its GDP [IMF, 2022]. The government launched several programs to address structural weaknesses, increase public sector efficiency, and eliminate bottlenecks to private sector growth, while maintaining macroeconomic stability. These include a "green growth" program to support post-COVID-19 economic recovery efforts while addressing a shrinking population, labor shortages, and climate change [WB, 2021]. Therefore, as reported in the NDC, the country will be able to expand its climate change mitigation efforts if it has financial resources. The project will only cover the additional costs of investments in climate action. Further, given the upcoming imposition of the Carbon Border Adjustment Mechanism (CBAM), Serbia faces an economic incentive to decarbonize. Starting with the implementation of CBAM, countries will have to pay a tax on the carbon content of exports to the EU. Given that the EU is Serbia's largest export market, decarbonization will improve the export competitiveness.

105. Despite growing demand and the economic necessity to decarbonize, Serbian firms report facing challenges in accessing finance for investment, including for decarbonization. Access to finance was one of the top five most frequently cited business environment obstacles, according to the 2019 World Bank Enterprise Survey in Serbia. Only 52.3% of domestic firms included in the survey reported using bank loans and 50.7% reported using supplier credit.[44] Credit from financial institutions is critical to meet Serbia's decarbonization goals in line with its NDC and other policy commitments. Yet, demand- and supply-side challenges create mismatches between firms and financial institutions. Overall, the gap in the market is due to limited information on both the demand and supply side, regulatory uncertainty and difficulties in identifying and matching financial institutions with firms.

## D.5. Country ownership

106. The climate change policy and strategy framework of Serbia is articulated and has at its core: (I) The Climate Law (03/2021), the LCDS (2023), the NDC (03/2021 – First update) and the National Adaptation Plan (NAP). Furthermore, the country has a number of laws and strategies that are relevant to climate change adaptation and mitigation. The project will contribute to several of Serbia's efforts to adapt to climate change and maximize mitigation benefits through measurable results. Each reported climate policy or legislation places forests at its core. They highlight the critical nature of forests in guaranteeing the country's energy security, increasing the use of renewable energy, enhancing livelihoods, removing carbon and decarbonizing the economy to make Serbia carbon neutral by 2050. Further, national strategies related to natural resource management and biodiversity point out the importance of healthy and adaptive forest ecosystems to reach national, regional and international environmental targets. The project will address Serbia's needs and priorities listed in its

NDCs, National Communications, National Adaptation Plans (NAP, 2015 and 2024) and national policy framework. As demonstrated by the letters of support (Annex 1.b) and by the partnerships that the project will establish once operative including with CSOs, non-governmental organizations, associations and communities (i.e. municipalities) (Annex 1.b). In addition, the project will contribute to the implementation of the GCF Country Programme of Serbia, by supporting the Cluster 1 (Energy efficiency and use of renewable energy sources) and Cluster 3 (NEXUS Water Resources – Agriculture – Forestry).

107. The NDA has been involved in the preparation of this proposal since the Concept Note phase. During each phase of the preparation of this proposal (Annex 2 and Annex 7), the NDA and FAO ensured the participation of public and private stakeholders in the process. The project was developed and prepared following a request to FAO by the GoS in 2019. During project elaboration, key stakeholders dealing with the forestry, energy and agriculture sectors in Serbia were consulted. Consultations occurred with the MOAFWM, the MEP, the METSD, the ME as well as local institutions such as municipalities and provinces. Other stakeholders include the PES and PEV, academia, CSOs (e.g. IUCN-Serbia, WWF-Serbia, BirdLife, NALED) related to environment and rural development as well as sectors associations, PKS and unions (Annex 7, Appendix 3). The consultations verified the technical feasibility of project component activities and allowed us to obtain feedback from stakeholders on all aspects of the project: climate rationale, relevant climate change adaptation, and mitigation targets, the project approach, including the expected paradigm shift, target areas, and site selection. Stakeholders agreed on needs to be addressed, targets, implementation arrangements and modalities, timeframe, and budget.

108. During project execution, consultations with stakeholders will take place yearly, at the time of the preparation of the Annual Work Plan and Budget (AWPB) – i.e., at the beginning of each of the six project Fiscal Years (FY). Before becoming a final AWPB, all activities will be discussed, reviewed, and validated. The AWPB will be presented by the PMU and reviewed by all stakeholders, including at the national, Governorate, and community levels. During these stakeholder engagement consultations, the Environmental and Social Management Framework (ESFM) – including relevant ESMPs prepared for sub-activities and the GRM - and the Gender Action Plan (GAP) - will be shared with stakeholders and explained. A complete description of the stakeholders' engagement plan during project execution is available in Annex 7–IV. Stakeholder consultation will take place in Belgrade and Novi-Sad. Participants will include relevant Ministries and, representatives of local institutions, CSOs as well as companies, associations and other concerned stakeholders. The PMU Monitoring and Evaluation (M&E) Unit will hold annual consultations in project areas to support planning and monitor the execution of activities. The ESMF, GRM and GAP will be presented and explained (FY1). The GRM and any ESMPs will be validated at each consultation during FY 2-7. Therefore, community consultations will feed into the review and preparation of the AWPBs. Details of the AWPB consultations are available in Annex 7 – Section IV. The PMU will be responsible for receiving all stakeholder feedback on any issues that may arise regarding the GRM. The PMU Environmental and Social Safeguards Specialist, together with the Gender Specialist and the PMU M&E specialist, will be responsible for ensuring that the ESMF, eventual ESMPs, and GAP are carried out and that the GRM is communicated to all stakeholders.

## D.6. Efficiency and effectiveness

109. Forestry is one of the priority areas identified by Serbia in its updated NDC (e.g. Section 3 (d)). Coherently, Serbia agreed to focus most of its forestry budget over next 7 years to the FP target areas and contribute to FP results. The co-financing committed by the MOAFWM reflects the amount that Serbia will invest in this project over the next 7 years. These contributions, earmarked to project activities, are essential to ensure the additionality of GCF investments, such as implementing climate-adaptive silviculture, promoting biodiversity in natural forests over monoculture, facilitating knowledge transfer, and climate-proofing assets.

110. The project will promote a more efficient and effective way to address forests, fuel wood and carbon removals. The private sector includes stimuli for private sector investments. Private sector investments will be incentivized and mobilized as an impact of the project via the offsets and insets, including leveraged private finance from private banks and forest owners' associations. The Project's economic and financial benefits have been estimated through four types of models, related to (1) investing in forestry through the project's activities, (2) investing in decarbonization, particularly in the agrifood sector, (3) enhancing the supply of scarce commodities (e.g., fuelwood) and (4) quantifying the value of incremental ecosystem benefits generated by the project's activities that are not captured in the individual models. A detailed analysis of the option analysis is available in Annex 2, 18-22.On the financial side, the individual models show that the project exhibits efficiency in the achievement of its mitigation targets. Overall, the activities stimulated by the project investment are financially profitable for the private sector and the beneficiary households. The models show positive financial parameters (including for forestry after applying the estimated required concessionality). They present 20-year IRRs higher than the financial discount rate of 7%[45] used as a relevant cost of capital for private investment decision.[46] The forestry investments demonstrate high net ecosystem services production with a low financial value of economic benefits deriving from

ecosystem services. The project's efforts on policy and regulatory framework will reduce the incentives for unsustainable practices (e.g., illegal logging) and strengthen the economic opportunities related to the forestry and energy sectors.

111. On the economic side at aggregate level and accounting for all relevant economic and ecosystem benefits (including the valuation of $CO_{2eq}$ and different sensitivity scenarios detailed in Annex 3, chapter D), the project shows very solid economic parameters, with a 20-year Economic IRR of 15.0 percent (higher than the social discount rate), with US$ 78.4million NPV, and a 40-year Economic IRR equivalent to 17.6 percent with US$ 227.5 million NPV.[47]

112. The project maintains a positive stream of aggregate economic returns under different sensitivity analyses, even when factoring in fluctuating carbon prices and climate change scenarios. Additionally, scenarios assuming 75%, 50%, and 0% loan disbursements for the decarbonization investments were evaluated. In all cases—except for the 10-year scenarios—the aggregated economic outcomes remain positive, regardless of variations in carbon prices.

**Table 4. Aggregated project benefits (with all project costs)**

**All aggregated economic benefits including all project costs (CO2e=40US$/t)**

| | 10 year results | | 20 year results | | 40 year results | |
|---|---|---|---|---|---|---|
| | IRR | NPV | IRR | NPV | IRR | NPV |
| (CO2e=40US$/t) | -6.4% | (28,156,115) | 15.0% | 78,441,519 | 17.6% | 227,522,945 |
| Increase of costs by 10% | -8.5% | (34,933,491) | 13.8% | 71,664,143 | 16.7% | 220,745,569 |
| Increase of costs by 20% | -10.4% | (41,710,867) | 12.7% | 64,886,767 | 15.8% | 213,968,193 |
| Increase of costs by 30% | -12.1% | (48,488,243) | 11.7% | 58,109,391 | 15.0% | 207,190,817 |
| Decrease of benefits by 10% | -8.7% | (32,117,879) | 13.7% | 63,819,991 | 16.5% | 197,993,274 |
| Decrease of benefits by 20% | -11.3% | (36,079,644) | 12.2% | 49,198,463 | 15.4% | 168,463,604 |
| Decrease of benefits by 30% | -14.2% | (40,041,408) | 10.6% | 34,576,935 | 14.2% | 138,933,933 |
| If 75% of loans are disbursed | -8.4% | (30,467,680) | 14.3% | 68,756,279 | 17.1% | 211,073,730 |
| If 50% of loans are disbursed | -10.7% | (32,779,245) | 13.4% | 59,071,039 | 16.5% | 194,624,515 |
| If 0% of loans are disbursed | -16.9% | (37,402,375) | 11.5% | 39,700,558 | 15.2% | 161,726,085 |

**In EFA document**

**All aggregated economic benefits including all project costs (CO2e=80US$/t)**

| | 10 year results | | 20 year results | | 40 year results | |
|---|---|---|---|---|---|---|
| | IRR | NPV | IRR | NPV | IRR | NPV |
| (CO2e=80US$/t) | -2.0% | (18,988,798) | 18.2% | 117,836,760 | 20.5% | 334,974,515 |
| Increase of costs by 10% | -4.2% | (25,766,174) | 16.9% | 111,059,384 | 19.4% | 328,197,139 |
| Increase of costs by 20% | -6.2% | (32,543,550) | 15.8% | 104,282,008 | 18.5% | 321,419,763 |
| Increase of costs by 30% | -8.1% | (39,320,926) | 14.7% | 97,504,632 | 17.6% | 314,642,387 |
| Decrease of benefits by 10% | -4.5% | (23,867,294) | 16.8% | 99,275,708 | 19.3% | 294,699,688 |
| Decrease of benefits by 20% | -7.2% | (28,745,791) | 15.2% | 80,714,656 | 18.0% | 254,424,860 |
| Decrease of benefits by 30% | -10.2% | (33,624,287) | 13.5% | 62,153,604 | 16.7% | 214,150,033 |
| If 75% of loans are disbursed | -4.2% | (22,679,974) | 17.4% | 105,647,922 | 19.9% | 315,086,045 |
| If 50% of loans are disbursed | -6.8% | (26,371,149) | 16.4% | 93,459,084 | 19.2% | 295,197,575 |
| If 0% of loans are disbursed | -13.4% | (33,753,499) | 14.3% | 69,081,409 | 17.7% | 255,420,634 |

**In EFA document**

**All aggregated economic benefits including all project costs (CO2e=10US$/t)**

| | 10 year results | | 20 year results | | 40 year results | |
|---|---|---|---|---|---|---|
| | IRR | NPV | IRR | NPV | IRR | NPV |
| (CO2e=10US$/t) | -10.0% | (35,031,602) | 12.2% | 48,895,088 | 15.0% | 146,934,267 |
| Increase of costs by 10% | -12.1% | (41,808,978) | 11.1% | 42,117,712 | 14.1% | 140,156,891 |
| Increase of costs by 20% | -13.9% | (48,586,354) | 10.0% | 35,340,336 | 13.3% | 133,379,515 |
| Increase of costs by 30% | -15.6% | (55,363,730) | 9.1% | 28,562,960 | 12.6% | 126,602,139 |
| Decrease of benefits by 10% | -12.3% | (38,305,818) | 11.0% | 37,228,203 | 14.0% | 125,463,464 |
| Decrease of benefits by 20% | -14.7% | (41,580,033) | 9.6% | 25,561,318 | 13.0% | 103,992,661 |
| Decrease of benefits by 30% | -17.5% | (44,854,249) | 8.1% | 13,894,434 | 11.8% | 82,521,859 |
| If 75% of loans are disbursed | -11.9% | (36,308,459) | 11.5% | 41,087,546 | 14.5% | 133,064,493 |
| If 50% of loans are disbursed | -14.1% | (37,585,317) | 10.7% | 33,280,004 | 13.9% | 119,194,720 |
| If 0% of loans are disbursed | -19.9% | (40,139,032) | 8.8% | 17,664,920 | 12.7% | 91,455,173 |

## E. LOGICAL FRAMEWORK

### E.1. Project/Programme Focus

☒ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

### E.2. GCF Impact level: Paradigm shift potential

| Assessment Dimension | Current state (baseline) | | Potential target scenario (Description) | How the project/programme will contribute (Description) |
|---|---|---|---|---|
| | Description | Rating | | |
| **Scale** | Forest management is highly fragmented with limited investment capacities. Instead, as increasing in size and health as foreseen by the different sectoral policies and strategies, forest is degrading due to climate change, fuelwood over extraction and lack of management instruments. | Low | Forest cover in Serbia is increased and sustainable and resilient forest management practices are included in forest management plans within an upgraded forest and carbon finance policy framework with direct and monitorable impacts on most vulnerable persons, ecosystems, energy and decarbonization of the Republic of Serbia. | Proposed actions will enable Serbia to reduce the exposure and vulnerability of its forestry sector and to enhance income and energy security of poor households while increasing total $CO_2$ removals and biodiversity. The project will aggregate and engage private owners and invest at scale at the national level in technologies, practices and investments ensuring knowledge transfer as well to the next generation of practitioners. |
| **Replicability** | Climate change actions are currently limited to project experience due to fragmentation of stakeholders, lack of mainstreaming across sectors and incomplete policy framework. | Low | Proposed actions will become part of the national policy and management framework of forests and form the carbon governance of the country. This will ensure replicability both at the national and regional scale. Proposed actions will enhance the role and contribution of the private sector and local administrations to both forest management and decarbonization. Administration, municipalities and private landowners as well as companies and agribusiness require time and incentives but specifically a clear governance to operate and shift from the BAU to a low carbon development path. Through the project the country will dispose of an advanced forest monitoring system that will also factor in non-wood related benefits (e.g. energy security, NTFP, protection and others) | The project will provide the public and private sector with the enabling framework to facilitate CAS and decarbonization across sectors and with a variety of tools, financing mechanisms and practical examples that can be replicated throughout the country and the region. To be highlighted in this regard the knowledge sharing platforms and the updating of the technical curricula ensuring that outcomes of the projects will be disseminated locally and in partner countries beyond the project implementation phase. |
| **Sustainability** | Forest and its ecosystem services are at the core of the sustainable development, decarbonization and poverty alleviation of Serbia.<br><br>Incomplete and exclusive forest and carbon finance management policy framework are hindering the expected contribution of forest to the overall | Medium | The paradigm shift would see cooperation between private and public actors by catalyzing a shift away from the current management framework toward a system were forests are sustainably managed under the same policy umbrella by all actors and that could effectively and efficiently contribute to both sustainable development and decarbonization of the country. Furthermore, with the creation of a carbon market, | The introduction of new forest governance and management methods and practices will increase the effectiveness of forest investments and will enhance the role and contribution of communities, the private sector and its associations to forest management, while at the same time reducing the adverse impacts of drivers of forest degradation and ensuring faster response of |

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| | sustainable development and decarbonization of Serbia.<br><br>Agrifood sector is not sufficiently supported in its ambitions to contribute to decarbonization and does not dispose of the capacities, tools or access to finance to contribute to sustainability and decarbonization. | | additional financial income streams are created to ensure investments in nature base climate action. | concerned stakeholders at no additional cost. Finally, the project will introduce the policy tools and incentives for private sector operators in the agrifood sector to contribute to forest development and to embed in their practices key adaptation and mitigation measures. |

**E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)**

| GCF Result Area | IRMF Indicator | Means of Verification (MoV) | Baseline | Target | | Assumptions / Note |
|---|---|---|---|---|---|---|
| | | | | Mid-term | Final | |
| Total Project Beneficiaries | Core 2: Direct and indirect beneficiaries reached | Project M&E system that will include ongoing data collection culminating in an annual update of the project beneficiaries. The systems integrate independent field surveys to inform the mid-term and final evaluations containing:<br>Centralized database supported by geospatial tools and developed to track beneficiaries<br>Disaggregation by gender, age, location, and type of benefit received<br>Unique identifiers to avoid double-counting<br>- External MoVs, such as national statistics and national household surveys | 0 | Direct:<br>312,456 total beneficiaries (159,353 women)<br>Indirect:<br>1,219,877<br>(622,137 women) | Direct:<br>729,064<br>total beneficiaries (371,823 women)<br>Indirect:<br>2,846,380<br>(1,451,654 women) | Economic, social and political context remains stable.<br>The total number of direct beneficiaries includes: (i) the number of persons in communities living in and in proximity of forest areas that the project will address (ii) the number of people currently using fuelwood as their primary source of energy for heating; and (iii) smallholders and forest owners that dispose of increased value of the forest due to increased climate resilience of their properties; (iv) the rural population that will increase its resilience due to enhanced forest ecosystem services (more details in Annex 25). |
| ARA1 Most vulnerable people and communities | Core 2: Direct and indirect beneficiaries reached | | 0 | Direct:<br>312,456 total beneficiaries (159,353 women)<br>Indirect:<br>1,219,877<br>(622,137 women) | Direct:<br>729,064<br>total beneficiaries (371,823 women)<br>Indirect:<br>2,846,380<br>(1,451,654 women) | |
| ARA4 Ecosystems and ecosystem services | Core 2: Direct and indirect beneficiaries reached | | 0 | Direct:<br>312,456 total beneficiaries (159,353 women)<br>Indirect:<br>1,219,877<br>(622,137 women) | Direct:<br>729,064<br>total beneficiaries (371,823 women)<br>Indirect:<br>2,846,380 | |

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| | | | | | (1,451,654 women) | |
| ARA4 Ecosystems and ecosystem services | Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice | Project M&E reports with details from the georeferenced database of project intervention areas and related monitoring via remote sensing analysis as well as ground visits.<br>Field survey carried out by independent consultants to inform the mid-term and final evaluations | 0 ha | Total hectares:<br>279,000<br>Of which<br>Forest:<br>250,00 ha under SFM practices<br>3,500 ha reforested<br>Agricultural land:<br>250 ha of shelterbelts established<br>250 ha of degraded and unfarmed agriculture land rehabilitated via agroforestry investments<br>Coppice stands converted to high forests:<br>25,500 ha | Total hectares:<br>559,000<br>Of which<br>Forests:<br>500,000 ha under SFM practices<br>7,000 ha of land reforested<br>Agricultural land:<br>500 ha of shelterbelts established<br>500 ha of degraded and unfarmed agriculture land rehabilitated via agroforestry investments<br>Coppice stands converted to high forests:<br>51,000 ha | Absence of major natural disasters including forest fires<br>No major changes in utilization of forest resources<br>State budget allocated to fulfill NDCs is guaranteed during and after the project.<br>Activities prepare the private Privat forest owners adequately to adopt the CAS practices<br>Economic, social and political context remains stable.<br>Ha accounted if: (i) included in forest management plans; (ii) proper execution is verified by M&E data (Remote sensing plus ground truthing)<br>Remote sensing analysis on each selected areas calculating the normalized difference vegetation index (NDVI)<br>Collect Earth applications can utilize freely accessible high-resolution space-borne imagery to allow cost-effective analysis of forest canopy cover and hence to verify impact of CAS practices on 500 kha of forest. |
| ARA3 Infrastructure and built environment | Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions | The indicator will be informed by two (midterm and final) independent surveys commissioned by the project to external consultants | Current value of forest roads:<br><br>USD 5.45 M | 182 Km of forest roads rehabilitated and valued at USD 6.9 M | 545 Km of forest roads rehabilitated and valued at USD 20,725 M | Absence of major natural disasters including forest fires<br>The GOS provides all clearances and approvals for street rehabilitation in a timely manner.<br>It has been estimated that about 545 km of forest roads in the project areas require upgrading and maintenance. The cost of the forest roads has been calculated based on a cost of USD 10,000 per km. The total initial value of the forest roads is, therefore, USD 5,45 M.<br>Added with the investments foreseen of USD , the total value of the forest roads at final evaluation will be USD 20,725 M. |

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| MRA1 Energy generation and access | Core 1: GHG emissions reduced, avoided or removed/sequestered<br>- | Project M&E systems including independent expert surveys at mid-term and termination | Reduced emission from decarbonization practice<br>0 | Reduced Emissions from decarbonization practices<br>-46,297 ($tCO_{2eq}$) | Reduced Emissions from decarbonization practices<br>-108,025 ($tCO_{2eq}$) | The economic, social and political context in the country and project areas remains stable.<br>Companies continue to be interested in decarbonizing processes.<br>It can be assumed that 50% of the decarbonization loan (=USD 25 M) will be dedicated to renewable energy interventions. Reductions (416,667 $ktCO_{2eq}$ over the whole lifetime of the project of 27 years) are calculated dividing the decarbonization loans (USD 25 M) with the estimated costs (USD 60) of mitigating 1 $tCO_{2eq}$ (Annex 22). Average reduction per year is 15,4 $ktCO_{2eq}$. |
| MRA3 Buildings, cities, industries and appliances | Core 1: GHG emissions reduced, avoided or removed/sequestered | Project M&E systems including independent expert surveys at mid-term and termination | Reduced emission from decarbonization practice<br>0 | Reduced Emissions from decarbonization practices<br>-46,297 ($tCO_{2eq}$): | Reduced Emissions from decarbonization practices<br>- 108,025 ($tCO_{2eq}$) | The economic, social and political context in the country and project areas remains stable.<br>Companies continue to be interested in decarbonizing processes.<br>It can be assumed that 50% of the decarbonization loan (=USD 25 M) will be dedicated to energy efficiency interventions. Reductions (416,667 $ktCO_{2eq}$ over the whole lifetime of the project of 27 years) are calculated dividing the decarbonization loans (USD 25 M) with the estimated costs (USD 60) of mitigating 1 $tCO_{2eq}$ (Annex 22). Per year, an average 15,4 $ktCO_{2eq}$ will be reduced. |
| MRA4 Forestry and land use | Core 1: GHG emissions reduced, avoided or removed/sequestered | NEXT tool informed by annual reports from the Forestry Department and Project M&E systems | Net national annual forest sector GHG emissions of -4,550,000 $tCO_{2eq}$ | -295,057 ▼ $tCO_{2eq}$ | -1,172,818 ▼ $tCO2_{eq}$ | Absence of major natural disasters and no major changes in utilization of forest resources and state budget for NDCs.<br>Private forest owners adequately prepared to adopt CAS practices.<br>Economic, social and political context remains stable.<br>The calculation (see Annex 22) includes the following activities: (i) 7 kha afforested (ii) 51 kha private and public coppicing stands converted into high forests (iii) 500 kha forest under CAS management (iv) 500 ha shelterbelts; (v) 500 ha land rehabilitation.<br>Total emission reduction over lifetime (27Y) of the project is -7,6 M $tCO2_{eq}$ and average per year is 0.28 M $tCO2_{eq}$ |

## E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

| Core Indicator | Baseline context (description) | Rating for current state (baseline) | Target scenario (description) | How the project will contribute | Coverage |
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| Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner | A series of important laws and policies have been approved/adopted in the last years. Nonetheless, the path towards sustainable and climate adaptive forest management, energy security from forests and decarbonization by 2050 is not completed and, due to lack of in country technical capacity and integrated coordination, Serbia is missing key operational tools and mechanism allowing the state to be capable of monitoring and accounting forest resources and the private sector to engage not just as a passive element, but as an actor of change in sustainable management. | medium | Integrated regulatory framework is in place allowing the monitoring and management of forest resources as well as the mitigation of forest degradation drivers | The establishment of the National Forest Monitoring System (NFM), the establishment of the offsetting/insetting scheme as well as the standards and strategy for fuelwood production/handling/management/use will strengthen the regulatory framework of the forest sector, and will have a spill-over effect in other sectors of the Serbian economy | National level (one country) |
| Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation | There is limited capacity for the development and financing of mid/long term decarbonization and climate resilient investments in the forestry. Forest resources are therefore managed still according to obsolete and inefficient practices leading to low economic and environmental benefits. | low | The forestry and private sector dispose of a wide variety of tools, instruments and services to contribute in an integrated and cross-sectoral manner to national climate commitments | A number of innovations will be introduced: (i) CAS practices, technologies and investments; (ii) a crosscutting approach to ensure that investments in adaptation also yield mitigation and poverty alleviation results; (iii) digital tools to aggregate and engage private land and forest owners; (iv) linkages of investors with private sector companies interested in decarbonization and forestry. | National level (one country) |

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| Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level | Almost entirely unregulated fuelwood market with associated weak value chains and weak linkages between private sector and forestry investments. | low | A growing and dynamic role of the private sector as an actor of change in the forestry sector, contributing to sustainable wood resource value chains for the benefit of the rural economy and for poverty alleviation. The private sector has at its disposal a diverse set of public and private investment opportunities to fully decarbonize activities and to increase long-term international market competitiveness. | The project will establish links between actors from the public, private and banking sector and identify business partnerships along the forestry resources value chain. Main activities are capacity development, provision of loans and instruments / mechanisms for smallholder aggregation. | National level (one country) |
| Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards | While there is growing interest in decarbonization activities there is currently low awareness among all actors on how to define and implement bankable and integrated strategies. | medium | All actors are aware of the possibilities to contribute to decarbonization pathways from a technical, financial, and planning point of view. Project outcomes are mainstreamed in sectoral programs, continuing in this way to enhance long term sustainable impacts of activities. | The project will train actors from government, private and banking sector, academia, and service providers on decarbonization actions benefiting the resilience of natural resources in the country. Knowledge sharing will occur throughout components, | National level (one country) |

## E.5. Project/programme specific indicators (project outcomes and outputs)

| Project/programme results (outcomes/ outputs) | Project/programme specific Indicator | Means of Verification (MoV) | Baseline | Target | | Assumptions / Note |
|---|---|---|---|---|---|---|
| | | | | Mid-term | Final | |
| **Outcome 1 – National-level upscaling of sustainable and climate adaptive** | Ha of forests included in the forest management plans with increased | Data records of MOAFWM collected on annual basis | 0 | 250,000 ha | 500,000 ha | The economic, social and political context remains stable. |

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| **silviculture and carbon finance framework achieved** | species composition and changes in structure | by the M&E staff of the project<br>Results from the Remote Sensing Analysis performed over identified forest management plans.<br>Independent expert surveys at mid-term and termination | | | | Political focus on CAS approaches maintained.<br>Timely approval of project documentation by counterparts.<br>Target originates from the estimation that every year 1/10 of the Forests Management Plans (established with the 2010 Law on Forests) will be developed with CAS practices and executed. Calculation is based on current capacities of forestry institutions.<br>GIS database of the MOAFWM Forestry Department will be a source for assessment.<br>The NFM system will continuously collect and analyze data related to forest structure changes following FAO good practices. |
| | # of national policies and regulatory frameworks integrated with CAS approaches | Technical reports by the project<br>Documentation of Policies and regulatory frameworks<br>Reports from the national engagement process of each policy document, guidelines and standard developed by the project | 0 | 1 | 3<br>(National carbon finance, short rotation Plantation strategy, Firewood quality rules and regulations) | The economic, social and political context remains stable.<br>In case of governmental change, political leadership maintains its current focus on CAS approaches.<br>Timely approval of project documentation by counterparts.<br>In addition to the processes that will be established by the project, the national engagement process will align with the protocols defined by the Law. |
| | % reduction of fuelwood consumption of HH in pilot areas | Technical reports by the project – baseline and final surveys in pilot area<br>Independent expert surveys at mid-term and termination of the project. | 0 | 6% | 6% | Regulatory Framework for Firewood quality rules and regulations is functioning.<br>Assessments evaluate the consumption of wood fuels based on FAO WISDOM methodology.<br>Using dry firewood instead of raw wood reduces energy consumption per $m^2$ of heated surface by 16% to 24%. Assuming at least 40% of households would increase fuelwood quality results in 6% average reduction. |
| | # of Academic and training institutions integrate introduced technologies, practices, standards and protocols in their curricula | Review of course curricula and promotional material publicizing the course content in publications and websites by the selected institutions. | 0 | 0 | 3 | Universities, vocational/training institutions have necessary resources to execute the courses.<br>Collaborations focus on main universities, vocational schools/training centres focussing on agriculture, forestry, economy and engineering (e.g. Universities of Belgrade and Kragujevac). |
| Output 1.1 - Forest management and monitoring policy framework for climate | NFM updated and operational | Final NFM<br>Governmental reports | 0 | 1 | 1 | Timely approval of project documentation and plan by governmental counterparts.<br>No major turnover of staff dedicated to the project activities. |

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| adaptive silviculture enhanced and disseminated | | Independent expert surveys at mid-term and termination | | | | |
| | # Guidelines for national decision makers on AFOLU to prevent soil degradation | Final guidelines/standards developed<br>Reports of the validation workshops and minutes of meetings<br>Governmental reports<br>Independent expert surveys at mid-term and termination | 0 | 1 | 1 | The economic, social and political context remains stable.<br>Timely approval of project documentation and plans by governmental counterparts.<br>In case of governmental change, political leadership maintains focus on CAS approaches. |
| | # Standards for fuelwood finalized and validated | | 0 | 1 | 1 | |
| | # Guidelines for SRP developed and validated | | 0 | 1 | 1 | |
| | # of citizens reached by gender | Reports/Documentation from the organization in charge of the energy days (Activity 1.1.3) (Disaggregated by origin, gender, age, regions)<br>Independent expert surveys at mid-term and termination | 0 | 3,000 | 10,000 | Municipalities in target areas are willing to host events and to contribute to organization.<br>Assessments will include key parameters such as number of people reached, and level of knowledge reached. |
| Output 1.2 - Enabling framework for national institutions to engage with carbon finance for AFOLU created | # of mechanisms elaborated and submitted for council of minister approval | Project documentation<br>Independent expert surveys at mid and end term | 0 | 2<br>(1 policy framework for domestic offsets<br>1 action plan on carbon finance opportunities) | 2<br>(1 policy framework for domestic offsets<br>1 action plan on carbon finance opportunities) | Timely approval of project documentation and plans by governmental counterparts.<br>Focus on CAS approaches maintained even in case of leadership change |
| | # of guidance documents elaborated and submitted for council of minister approval | | 0 | 2<br>(1 carbon finance opportunities,<br>1 insetting) | 2<br>(1 carbon finance opportunities,<br>1 insetting) | |
| | # of public and private stakeholders capacitated on carbon finance | Documentation of the organization responsible for providing the training<br>Disaggregated by origin, gender, age, regions and including capacity assessment | 0 | 30 | 108 | Stakeholders see sufficient value to make themselves available to participate in the training |

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| | | Pre and post training surveys and interviews with communities disaggregated by gender | | | | |
| **Outcome 2 - Improved energy security and livelihood from climate resilient forest ecosystem and GHG emission reductions from increased carbon sinks and decarbonization opportunities** | % inability to keep house warm | Independent expert surveys at end or mid term<br>National and international statistics | % of inability to keep house warm was 9.5% in 2022 in Serbia | 9.3% | 9% | Economic and political stability in the country |
| | Average increase of biomass growth per hectare in M3 | Data records of MOAFWM collected on annual basis by the M&E staff of the project<br>Results from the Remote Sensing Analysis performed over identified forest management plans.<br>Independent expert surveys at mid-term and at termination comparing sample target areas with reference ones | 4 | 5.2 | 5.2 | No major change in energy provision of the rural population<br>Supply/demand chains for biomass purchase are functioning<br>The interest of the population to buy thinning wood fuel |
| | Average survival rate per ha (%) of planted seedling | Independent expert surveys at mid-term and termination of the project comparing sample target areas with reference ones established at start-up. | 50% | 85% | 85% | Baseline Source: PES and Department of Forests of the MOAFWM |
| | # of Academic and training institutions integrating climate change in curricula | Review of course curricula and promotional material publicizing the course content in publications and websites by the selected institutions.<br><br>Pre and post training surveys and interviews with communities disaggregated by gender | 0 | 4 | 4 | Institutions have modified course content and are interested in offering courses and students are interested in participation<br>Involved will be faculty of forestry of the university Belgrade and vocational schools working on forestry, agriculture and accounting |

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| Output 2.1 - Production of climate-adaptive seedlings enhanced | # of public and private operators with increased capacities in the production of climate adaptive forestry seedling | Attendances sheets of training activities and records of the organization in charge of the training. Disaggregated by type, gender, regions | 0 | 70 (of which at least 28, or 30% women) | 140 (of which at least 42, or 30% women) | The operators see sufficient value to make themselves available for capacity training activities<br>Assessments will be conducted to identify capacitated operators to be counted (assumed 70% pass assessment) |
| | # Public Nurseries upgraded | Documentation of the M&E unit of the project<br>Photographic and remote sensing documentation<br>Independent surveys at mid-term and termination | 0 | 2 | 2 | Timely governmental approval of activities and purchase of equipment in a cost and time effective manner from the local markets |
| Output 2.2 – Knowledge on climate adaptive silviculture (CAS) of key stakeholders enhanced | # of public and private stakeholders with increased capacities in CAS | Documentation of the organization responsible for the capacity development<br>Data of participating persons from communities will be retrieved disaggregated by origin, gender, age, regions<br>Pre and post training surveys and interviews with communities disaggregated by gender | 0 | 350 | 1,750 | Stakeholders see sufficient value to make themselves available to participate in the training<br>Assessments will be conducted to identify capacitated stakeholders to be counted (assumed 70% pass assessment) |
| | # of Communities involved in climate adaptive forest investment and management trainings | | 0 | 34 | 174 | Local communities are interested in the project and in sharing information |
| | # of Guidelines to support CAS approaches published online | Final Guidelines<br>Project documentation | 0 | 4 | 4 | Timely approval of project implementation plan by counterparts |
| | # of upgraded national curricula of the faculty of forestry and vocational schools | Independent expert surveys<br>Documentation of project and schools | 0 | 0 | 6 | Schools maintain their interest to be involved in project |
| Output 2.3 - Public forest land restored and afforested in a climate adaptive and participatory manner | # of ha of degraded public coppice stands converted into high forest | Data records of MOAFWM collected on annual basis by the M&E staff of the project<br>Results from the Remote Sensing Analysis performed over identified forest management plans. | 0 | 11,000 | 33,000 | Absence of major natural disasters including forest fires in target areas.<br>Timely approval of project implementation plan by governmental institutions<br>The NFM system will continuously collect and analyze data of converted coppice and afforested areas following FAO good practices and assess forest structure changes |
| | # ha afforested | | 0 | 2,330 | 7,000 | |

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| | | Independent surveys at mid-term and termination | | | | |
| | # of trainers capacitated on reforestation and conversion techniques | Documentation of the organization responsible for the training<br>Disaggregated by type, gender, age, regions | 0 | 16 | 56 | Trainers see sufficient value to make themselves available to participate in training<br>Assessment will identify trainers capacitated to be counted (assumed 80% pass assessment) |
| | Km of forest roads rehabilitated and climate proofed | Data records of MOAFWM collected on an annual basis by the M&E staff of the project.<br>Results from the Remote Sensing Analysis<br>Independent expert surveys at mid-term and project termination | 0 | 125 km | 376 km | Necessary governmental approval process concluded without delay<br>Absence of major natural disasters in target areas |
| **Outcome 3 - Private sector is engaged in climate adaptive silviculture and decarbonization investments** | Increase of investments in USD of national finance institution in support to agrifood companies' adaptation and decarbonization strategies | Documentation provided by partner banks<br>Independent expert surveys at mid-term and termination of the project<br>Data disaggregated by type of investment (forestry, soil, energy, others) | USD 355 M<br>Current Loans in support of Agriculture and Agribusiness (Annex 2, Appendix 12, p. 18) | USD 15M | USD 50M | Source of data: National finance institutions, chamber of commers, associations, private companies<br>The project will ensure that at least 50 USD mln will be dedicated to decarbonization and resilience investments, aligned with project |
| | # ha with improved soil quality (Phosphorus, Reactive Carbon, Soil Electrical Conductivity, Soil Nitrate and Soil PH) | Project documentation<br>Independent expert surveys at mid-term and termination of the project comparing sample target areas with reference ones established at start-up | N/A | 286 | 1000 | Baseline data will be collected once land will be identified and agreed with its owner(s).<br>Data will be collected and processed based on existing methodology identified by the MOAFWM and FAO at start-up. |
| | Average increase of biomass growth per hectare in M3 of converted coppice stands | Data records of MOAFWM collected on annual basis by the M&E staff of the project<br>Results from the Remote Sensing Analysis performed over identified forest management plans. | 4 | 5.2 | 5.2 | No major change in energy provision of the rural population<br>Supply/demand chains for biomass purchase are functioning. |

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| | | Independent expert surveys at mid-term and at termination | | | | |
| | Total volumes in t of sustainable fuel biomass traded through the biomass trading platforms | Project documentation | 0 | 500 | 1000 | No major change in energy provision of the rural population<br>Supply/demand chains for biomass are functioning. |
| Output 3.1 - Private sector contribution to climate resilience of forests enhanced through climate adaptive and diversified investments and greening of the biomass value chain initiatives | # of ha of not managed and degraded private coppice stands converted into high forests | Data records of MOAFWM collected on annual basis by the M&E staff of the project<br>Results from the Remote Sensing Analysis<br>Independent expert surveys at mid-term and at termination | 0 | 6,000 | 18,000 | Coppice conversion will be assessed through continuous data collection and analysis with the NFM system following FAO good practices.<br>Changes in land use (i.e. shelterbelts, agroforestry, SRP) will be assessed following FAO guidelines integrating remote sensing and field verification. |
| | # of ha of shelterbelts established | | 0 | 200 | 500 | |
| | # of ha rehabilitated through agroforestry, SRP | | 0 | 200 | 500 | |
| | # of gender-inclusive biomass platforms established and /or strengthened | Records of municipalities<br>Official documentation by the biomass platforms<br>Independent expert surveys at mid-term and at termination of the project | 0 | 1 | 1 | Collaboration between municipalities and forestry actors remains stable<br>M&E unit will monitor performances and governance changes<br>The surveyor will collect data from the project, SERBIO and the municipalities in the areas of operations of the platforms |
| | # companies trained on utilization of biomass residues for energy purposes | Documentation of the organization in charge of training<br>Data of companies will be retrieved disaggregated by origin, gender, age, regions<br>Pre and post training surveys and interviews with communities disaggregated by gender | 0 | 30 | 80 | Companies dispose of sufficient processing residues for energy purposes and are interested in participating in trainings |
| | # of extensionists capacitated on CAS through training | Documentation of the organization responsible for providing the training. Disaggregated by origin, gender, age, regions and including capacity | 0 | 21 | 70 | Extensionist see sufficient value to make themselves available to participate in the training<br>After finishing training the extensionists will be subject to an exam to verify the level of capacity development. (assumed 70% pass assessment) |

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| | | assessment through project activities | | | | |
| | Km of forest roads rehabilitated and climate proofed | Data records of MOAFWM collected on an annual basis by the M&E expert of the project.<br>Results from the Remote Sensing Analysis<br>Independent surveys at mid-term and project termination | 0 | 56 km | 169 km | Necessary governmental approval process concluded without delay<br>Absence of major natural disasters in target areas |
| Output 3.2. - Mobilized private finance for agribusinesses involved in decarbonization | # of enterprises benefitting from services to develop decarbonization strategies | Project documentation<br>Documentation provided by the companies | 0 | 10 | 5 | The economic, social and political context in the country and project areas remains stable.<br>Companies maintain interested in decarbonization strategies |
| | USD loan disbursed to support decarbonization of companies | Independent expert surveys at mid-term and termination of the project.<br>Documentation provided by banks and companies | 0 | 15,000,000 USD | 50,000,000 USD | The economic, social and political context in the country and project areas remains stable.<br>Smooth approval process of project implementation plan by governmental counterparts.<br>National and international FIs maintain the availability of credit lines and other financial instruments to support companies.<br>Data obtained from , banks and companies involved in the process. |
| Output 3.3 - Financial institutions, consultancy service providers, and academia capacitated on climate-related challenges and opportunities | # of capacitated trainees on climate-related risks | Documentation from the PKS<br>Data of partners organizing the trainings disaggregated by origin, gender, age, regions, including capacity assessment | 0 | 10 | 10 | The economic, social and political context in the country and project areas remains stable.<br>The beneficiaries will be subject to a capacity assessment exam to verify their level of improvement. |
| **Project/programme results co-benefit indicators** | | | | | | |
| Co-benefit 1: Biodiversity conservation | Increase (%) in species diversity per hectare in project areas | Independent expert field-surveys at mid-term and termination of the project comparing sample target areas with reference ones established at start-up | According to the endemicity score there are on average 3.06 unique species | 5% | 10% | Absence of major natural disasters in target areas.<br>Surveyors collaborate with union of beekeepers to analyze pollen collected by bees, and map biodiversity changes of entomophile species.<br>Assessment will include remote sensing analysis and randomized ground sampling |

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| | | | present in Serbia per km2 | | | |
| Co-benefit 2<br>Poverty Reduction | % decrease in share of persons at-risk-of poverty in project intervention areas due to savings in fuelwood expenditures, increased availability of NTFP and employment opportunities in forestry related sectors. | Independent expert surveys at mid-term and termination of the project comparing sample target areas with reference ones established at start-up. | 21.7% | -2.5% | -5% | Data will be disaggregated by origin, gender, age, regions |
| Co-benefit 3<br>Gender Empowerment | % Increase participation of women in executing bodies deciding on investments in forestry and decarbonization | Independent expert surveys at mid-term and termination of the project comparing sample target areas with reference ones established at start-up | N/A | 30% | 30% | Assessment will verify how often women have been involved in executing bodies deciding on investments in forestry and decarbonization.<br>The executing bodies are those committees that give final decision about the adoption of an investment in forestry or decarbonization, e.g. Board of Directors of forest state owned companies and other companies, steering committees of Ministries, municipal assemblies. |

## E.6. Project/programme activities and deliverables

| Activities | Description | Sub-activities | Deliverables |
|---|---|---|---|
| **COMPONENT 1 – National level upscaling of sustainable and climate adaptive silviculture and carbon finance framework** | | | |
| **Output 1.1 - Forest management and monitoring policy framework for climate adaptive silviculture enhanced and disseminated** | | | |
| Activity 1.1.1: Establish the National Forest Monitoring System (NFMA) | Provide technical assistance and capacity development to design, start-up and implement a methodologically and statistically sound NFM to objectively assess the conditions and dynamics of forests at the country and regional (Oblast) levels, with the integration of private forest activities and non-forest-activities (e.g. windbreaks and insetting practices) | 1.1.1.1: Design and set-up of NFM system<br>1.1.1.2: Technical assistance for NFM System design<br>1.1.1.3: Roll out NFM system across AP Vojvodina and Central Serbia.<br>1.1.1.4: Technical assistance for NFM System implementation<br>1.1.1.5: Support to forest health/pest control and monitoring; field investments control | 1 NFM designed and started up<br>NFM staff trained to plan and implement Collect Earth (CE) and field survey<br>2 field survey teams for Vojvodina and 10 for Central Serbia trained and fully operational<br>Monitoring of 330,000 plots via Collect Earth and at least 8,820 plots via field survey<br>1 study tour for 4 experts to learn from European NFM best practices |
| Activity 1.1.2: Develop guidelines for decision makers on AFOLU to prevent soil degradation | Downscale International policies to the Serbian context and develop and disseminate guidelines for national and local decision makers regarding AFOLU and prevention of soil degradation | 1.1.2.1: Prepare and publish guidelines on the prevention of soil degradation<br>1.1.2.2: Disseminate knowledge on prevention of soil degradation through national and local Workshops | 1 guideline published online<br>120 persons from institutions and other stakeholders trained on AFOLU organized |

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| Activity 1.1.3: Create national standard for biomass production / handling for energy purposes | Establish appropriate rules, inspection mechanisms, and complaints procedures that sellers and distributors of firewood must abide to. | 1.1.3.1: Develop national standards for biomass production / handling and use<br>1.1.3.2: Support line agencies in the introduction of national standard for biomass production / handling and use<br>1.1.3.3: Evaluate impact of the standards introduced<br>1.1.3.4: Carry out Renewable energy days | 1 standard for biomass utilization for energy purposes<br>1 impact assessment of introduction of biomass standards<br>10,000 citizens reached through RES days |
| Activity 1.1.4: Develop the national strategy, action plan and execution guidelines for Short Rotation Plantations (SRP) | Develop a strategy and guidelines for SRPs. defining the required steps and policy framework to promote practices. Dissemination of the guidelines and strategies through local trainings and awareness raising events. | 1.1.4.1: Develop the National Strategy for SRP<br>1.1.4.2: Train stakeholders on the implementation SRP activities | 1 National Strategy for SRP<br>1 geospatial map indicating soils/lands with high potential for wood energy plantations<br>100 local experts trained on SRPs |
| **Output 1.2 - Enabling framework for national institutions to engage with carbon finance for AFOLU created** | | | |
| Activity 1.2.1: Support the design of an offset mechanism [for AFOLU] as part of the domestic carbon pricing framework and institutionalize knowledge in the national curricula. | Based on Serbia's decision on the carbon pricing framework (e.g. ETS / Carbon Tax), help design an offset mechanism for forestry, and strengthen capacity of private sector and academia to support implementation including MRV.6 | 1.2.1.1: Develop specific policy options for domestic offset mechanism for forest and soil carbon linked to Serbia's evolving ETS plans.<br>1.2.1.2: Strengthen capacity of private services providers and academia to support national and international carbon finance for AFOLU | 1 draft policy framework for domestic offsets<br>Existing MRV system upgraded with features specific offsetting and insetting<br>3 Serbian officials trained on best practices in 2 countries and exposed to global practices through the visit of 2 international experts<br>30 private experts trained to use MRV for carbon finance and AFOLU offset opportunities<br>1 curriculum developed to integrate into relevant graduate courses |
| Activity 1.2.2: Support the development of Serbia's Article 6 strategy related to "AFOLU" opportunities | Facilitate opportunities for Serbia's forest sector to access carbon finance through Article 6 under different scenarios of rules and regulations, including through capacity development for national negotiators and formulation of technical inputs for Serbia's next NDC update. | 1.2.2.1: Develop guidelines and action plan for Serbia's forest sector to access carbon finance through Article 6 under different scenarios of rules and regulations<br>1.2.2.2: Provide capacity development for national negotiators<br>1.2.2.3: Prepare technical inputs for Serbia's Article 6 strategy to be reflected in the next NDC update and national strategy documents | 1 policy development guidelines and action plan document on Serbia-specific carbon finance opportunities for AFOLU in emerging Article 6 frameworks<br>1 technical document on Serbia's Article 6 strategy for the next NDC update, based on stakeholder consultation<br>15 UNFCCC negotiators and relevant national experts trained on Article 6 options |
| Activity 1.2.3: Promote and support knowledge sharing at the regional level | Facilitate or make use of existing knowledge sharing networks (if available) for an exchange on offsetting and insetting that will further support the enhancement of the Serbian experience and is expected to create spill-over effects to other Balkan countries. | 1.2.3.1. Support regional knowledge-sharing through existing platform or channels<br>1.2.3.2.: Carry out regional communication campaigns to promote / disseminate knowledge | regional knowledge-sharing platform available<br>2 regional communication campaigns to promote platform |
| Activity 1.2.4: Support access to voluntary carbon finance for forestry in Serbia to enable sequestration beyond NDC targets | Facilitate access for Serbia's forest sector to carbon finance from evolving high-integrity Voluntary Carbon Markets (VCM). | 1.2.4.1: Define entry points for Serbia's forest sector to access carbon finance from VCM.<br>1.2.4.2: Based on national capacity gaps assessment for MRV systems, prepare costed systems and capacity development plan to address identified gaps | 1 set of guidance/database on accessing carbon finance linked to (evolving) high integrity standards<br>1 costed capacity development plan for MRV for VCM<br>40 national stakeholders trained on identified priority areas for VCM |

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| and to ensure long-term sustainability | | 1.2.4.3: Provide technical assistance to strengthen data systems and capacities in priority areas to enable better MRV for carbon finance for AFOLU | |
| Activity 1.2.5: Enable insetting as part of company decarbonization strategy support | Facilitate insetting for major value chains through removing information barriers and capacity constraints to policy options and opportunities. | 1.2.5.1: Document policy options to provide incentives for insetting at national level and develop appropriate mechanisms if conducive to enhancing climate results beyond NDC targets<br>1.2.5.2: Produce guidance on good practices for carbon accounting and MRV for insetting<br>1.2.5.3: Strengthen capacity of local service providers/consultants | 20 national experts/local service providers from trained to assist companies with establishing insetting approaches<br>1 set of guidance materials for companies on good practices in insetting / and related carbon accounting, in line with national requirements / international standards.<br>1 curriculum designed on insetting – building upon guidance (and complementing the decarbonization curriculum (3.3.2.)) for professional/advanced academic training |
| **COMPONENT 2- Improving energy security and livelihood from climate resilient forest ecosystem and GHG emissions reductions from increased carbon sinks and decarbonization opportunities** | | | |
| **Output 2.1 - Production of climate-adaptive seedlings enhanced** | | | |
| Activity 2.1.1: Upgrade public nurseries (Vojvodina/C. Serbia) | Support investments and increase the capacity of the Public Enterprises (PE) to ensure production of the necessary quantity and quality of seedlings for the project's forest restoration interventions and beyond. | 2.1.1.1: Install purchased equipment and upgrade related infrastructure.<br>2.1.1.2: Quality control of Seedling production | 2 nurseries upgraded (1 in Central Serbia and 1 in Vojvodina)<br>35 Mio. of climate adaptive seedlings produced |
| Activity 2.1.2: Train and support operators of public and private nurseries in the production of diverse and climate adaptive forestry seedlings | Organize four hands-on trainings on the production of high-quality plant material (seeds, seedlings and cuttings) to be delivered by the Chamber of Forestry Engineers to refresh the knowledge of staff of PES and ILFE in charge of the nursery works, and to train workforce from private nurseries to spread the knowledge on up-to-date seedling production methods and technologies | 2.1.2.1: Develop 2 training modules (high-quality plant material / high quality plant production)<br>2.1.2.2: Develop climate adaptive seed production protocols<br>2.1.2.3: Carry out 4 trainings for operators of nurseries for production of high-quality plant material | 200 operators trained on high-quality plant/seedling production<br>Production protocols for each selected species (at least 24 species) |
| **Output 2.2 - Knowledge on climate adaptive silviculture (CAS) of key stakeholders enhanced** | | | |
| Activity 2.2.1: Produce guidelines on climate adaptive silviculture | Develop four guidelines to support the smooth and swift transition to CAS approaches and disseminate the knowledge created. The four guidelines will cover the following 1) climate smart nursery production including seed selection; 2) soil preparation for planting on extreme sites; 3) effective and efficient planting methods; as well as 4) maintenance after planting and first thinning operations. | 2.2.1.1.: Develop 4 guidelines on CAS approaches<br>2.2.1.2: Carry out transfer of knowledge to over 500 people in 174 rural communities | 4 guidelines to support CAS approaches published online<br>On the job-training on CAS approaches to 40 professionals<br>500 people of 174 communities trained on climate adaptive forest investments, ecosystem management and management approaches. |

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| Activity 2.2.2: Train public and private stakeholders in climate adaptive silviculture (CAS) | Contribute to collaboration with line agencies and institutions to the capacity strengthening of stakeholders by establishing a consistent process for professional training and education on CC-related issues. | 2.2.2.1: Train key stakeholders for leveraging CAS<br>2.2.2.2: Expand training on CAS to forest managers | 2,500 public and private forest managers trained on climate adaptive forest management |
| Activity 2.2.3: Upgrade the National curricula of the faculty of forestry and vocational schools working on forestry, agriculture and accounting with introduced practices and technologies | Involve national institutions to ensure that capacity development needs are addressed, and up-to-date knowledge and skills are included in national technical curricula related to forestry and forest plant production | 2.2.3.1: Upgrade national curricula<br>2.2.3.2: Train teachers and professors in climate adaptive silviculture | 4 national curricula upgraded<br>40 teachers/professors trained in CAS |
| Activity 2.2.4: Facilitate regional knowledge-sharing dialogue on CAS | Establish a regional knowledge-sharing dialogue on CAS approaches for Serbia and neighboring countries. | 2.2.4.1: Design and startup of regional knowledge sharing on CAS | 1 position brief on CAS |
| **Output 2.3 - Public forest land restored and afforested in a climate adaptive and participatory manner** | | | |
| Activity 2.3.1: Carry out afforestation activities on public land | Support the efforts of the Serbian government to increase national forest cover with the afforestation of 7,000 ha with climate adaptive seedlings of tree and shrub species. | 2.3.1.1: Identify the specific afforestation sites with landowners, municipalities and civil society organizations concerned<br>2.3.1.2: Organize 3-days hands-on-training (training of trainers) on planting techniques in afforestation/forest restoration<br>2.3.1.3: Seedlings distribution (Vojvodina Shume)<br>2.3.1.4: Seedlings distribution (Serbia Shume)<br>2.3.1.5: Carry out site preparation and plantation activities as appropriate (includes fencing of afforestation sites in Vojvodina)<br>2.3.1.6: Container Seedlings distribution (MOAFWM) | 50 trainers capacitated on correct planting techniques<br>7,000 ha of currently set aside land afforested |
| Activity 2.3.2: Convert degraded coppice stands on public forest land accessible and converted into high forest | Work with the Ministry of Agriculture, Forests and Water Management ( MOAFWM ) and the respective PEs to restore at least 33,000 ha of forest stands, the majority of which will be degraded coppice stands, for conversion into high forest, but also forest stands damaged by abiotic or biotic factors with urgent need of restoring their ecological functioning. | 2.3.2.1: Identify the specific forest restoration site with the PEs concerned and local stakeholders as appropriate<br>2.3.2.2: Organize meetings in the neighboring communities of the selected restoration sites to present the planned restoration actions<br>2.3.2.3: Provide seedlings for forestry investment<br>2.3.2.4: Guarantee access for management and fire protection of forests under conversion<br>2.3.2.5: Implement hands-on training of the Trainers (training of trainers) on best silvicultural practices in conversion.<br>2.3.2.6: Seedlings distribution (Serbia Shume)<br>2.3.2.7: start up conversion of coppice forests through thinning and enrichment planting | At least 20 trainers capacitated on correct conversion techniques<br>33,000 ha of coppice stands concerted to high forests<br>545 km of forest roads rehabilitated and climate proofed |

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| **Component 3 – Engaging Private sector in climate adaptive silviculture and decarbonization investments** | | | |
| **Output 3.1 - Private sector contribution to climate resilience of forests enhanced through climate adaptive and diversified investments and greening of the biomass value chain initiatives** | | | |
| Activity 3.1.1: Convert degraded private coppice stands into high forest | Involve on a voluntary basis the private sector to promote the conversion of degraded private coppice stands to high forests. | 3.1.1.1: Establish the first Serbian aggregation platform based on the available electronic cadaster.<br>3.1.1.2: Carry out awareness raising campaigns to attract interest of private owners in the conversion of coppice forest<br>3.1.1.3: Guarantee access for afforestation, management and fire protection of forests under conversion<br>3.1.1.4: Seedlings distribution (Directorate of Forest)<br>3.1.1.5: Seedlings distribution (Serbia Shume)<br>3.1.1.6: Carry out conversion of coppice forests through thinning and enrichment planting with native species<br>3.1.1.7: Carry out monitoring activities of forest interventions | 1 communication campaign to attract private sector investment<br>1 Aggregation platform available<br>18,000 ha of conversion sites<br>100 extensionists trained on CAS |
| Activity 3.1.2: Rehabilitate unfarmed private lands through forestry investments such as short rotation plantations, agro-forestry or soil rehabilitation purposes (Vojvodina) | Identify at least 500 ha of unfarmed private lands to be cultivated with wooden species for short rotation forestry, agro-forestry, or soil protection or rehabilitation, or a combination of the mentioned purposes. | 3.1.2.1. Identification of degraded and unfarmed land sites for rehabilitation<br>3.1.2.2. Carry out site preparation activities including fencing<br>3.1.2.3 Distribute seedlings to rehabilitation sites (Directorate of Forests) | 1 geospatial map with potential intervention sites developed in collaboration with municipalities<br>500 ha of unfarmed private lands rehabilitated |
| Activity 3.1.3: Establish shelterbelts in agricultural landscapes | Identify at least 500 ha of agricultural lands for the establishment of shelterbelts, mainly in almost tree-less agricultural landscapes of Vojvodina region with the main aim of soil protection (aeolian erosion control) but also to increase biodiversity, provide pollination services as well as (re)establish more suitable habitats and migration routes for wildlife. | 3.1.2.1. Identification of sites for shelterbelts investments (PMU)<br>3.1.3.2: Carry out site preparation activities<br>3.1.3.3: Distribute seedlings to shelterbelt sites (MOAFWM)<br>3.1.3.4: Carry out monitoring activities of restoration interventions | 1 geospatial map with potential intervention sites developed in collaboration with municipalities<br>500 ha of shelterbelts established |
| Activity 3.1.4: Engage private actors in sustainable biomass value chains | Provide training for agro-industrial associations on possibilities for valorization of biomass residues for energy purposes among their members. | 3.1.4.1. Develop guidelines and training material for the energetic valorization of biomass residues<br>3.1.4.2. Train agrobusiness actors | 6 trainings for private sector actors provided involving 60 agrobusiness actors<br>Guidelines for the valorization of biomass residues for energy purposes |
| Activity 3.1.5: Support platforms involving stakeholders of the forestry and agricultural sector for a modern and transparent biomass energy value chain | Support the Biomass Producers Association in establishing the platform, involving stakeholders in the sector | 3.1.5.1 Support the identification of the territory for the establishment of the platform<br>3.1.5.2. Support the platform in the form of feasibility studies, business plan development and marketing | 1 platform established/enhanced<br>6 local workshops to train forest/landowners about economic and environmental benefits of sustainable biomass value chains |

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| **Output 3.2 - Mobilized private finance for agribusinesses involved in decarbonization processes.** | | | |
| Activity 3.2.1: Involve agribusiness and other companies in the decarbonization process of the private sector | Startup involvement of the private sector through capacity development of climate risk and feasibility of decarbonization investments assessment and dissemination of knowledge through conferences and workshops | 3.2.1.1: Update the existing "Digital Platform on Circular Economy" to include and disseminate information about decarbonization practices<br>3.2.1.2: Engage the private sector through the organization of communication campaigns, workshops and trainings | Digital Platform on Circular Economy, operated and managed by the PKS, updated to contain information about decarbonization practices and financing possibilities<br>150 companies involved through awareness raising and capacity development activities |
| Activity 3.2.2: De-constrain access to credit for agribusiness and other companies | Provide TA to agribusiness to detail strategies for decarbonization, with the elaboration/implementation of decarbonization strategies, budgets, and action plans. | 3.2.2.1: Support agribusiness companies' baseline for the decarbonization strategies.<br>3.2.2.2: Support firms in preparing their decarbonization strategies. | 50 decarbonization strategies developed<br>50 business plans to implement decarbonization strategies developed |
| **Output 3.3 - Financial institutions, consultancy service providers, and academia capacitated on climate-related challenges and opportunities** | | | |
| Activity 3.3.1: Support the chamber of commerce and industry and the association of financial institutions to assess climate-related risk of banks and ensure client engagement | Support financial institution to overcome lack of information and technical know-how concerning climate risks and decarbonization strategies, low awareness around compliance with new international standards and supporting clients in engaging in forestry-related activities. | 3.3.1.1: Capacity development to financial institutions on climate proofing their portfolio | 1 curriculum valid also for university courses on climate related risks in lending activities<br>5 financial institutions trained on climate related risks assessment |
| Activity 3.3.2: Support to the chamber of commerce and industry and the association of financial institutions and academia to train decarbonization service providers. | Ensure the creation of a local market of decarbonization technical service providers through the tailored capacity development of local technical experts/firms. | 3.3.2.1.: Develop and carry out a tailored training for service providers on decarbonization processes and strategies | 1 capacity needs assessment for decarbonization advisory service providers<br>Tailored capacity development of 5 decarbonization service providers<br>1 guideline on assessing / accompanying the implementation of decarbonization strategies |

## E.7. Monitoring, reporting and evaluation arrangements

113. **M&E Structure:** A monitoring and evaluation system will be established for the project in keeping with the guidelines of GCF to report on its Integrated Results Management Framework designed to measure the core indicators as well as all other indicators identified in Section E. The PMU established in Serbia will be responsible for monitoring of the project activities with the oversight of FAO- Serbia Office and technical backstopping by the regional office (REU) where required. An M&E system will be developed with an M&E Officer and a Monitoring Information System (MIS) to keep track of performance and core indicators at the national and province level. The M&E officer of the project will work in parallel with the M&E officer of the Forestry Directorate that will hold specular functions and support integrating the MIS with data related to cofinancing activities. All service contracts, Letters of Agreements and others with implementing partners will specify their responsibility with respect to sex-disaggregated data collection and reporting. The execution partners will submit reports to the PMU which will prepare a consolidated report on an annual basis. Regular meetings for monitoring and follow-up will be organized where problems will be discussed and, when needed, corrective measures will be recommended. FAO, as the main implementing agency, will be responsible for maintaining records of all project activities in standard reporting formats. All implementing partners will be required to provide information on the core indicators, impact, outcome and output level indicators specified in the IRMF. FAO-HQ will support the PMU in reviewing and analyzing progress reports and to assess performances against baseline and targets. FAO will manage and coordinate reporting to the GCF according to its standards procedures.

114. **Types of Reports**: The PMU at the FAO office in Serbia will formulate an annual work plan and budget based on the annual physical targets based on the implementation plan (Annex 5) which will be approved by FAO-Serbia/FAO-REU and the PSC. Formats will be developed for each of the reports, namely the quarterly statistical and narrative reports and Annual Performance Reports (APRs). These reports will be prepared by the technical staff at the PMU under the guidance of the M&E Specialist at the PMU. The key reports that will be submitted have been identified in the M&E Reporting Matrix given below together with their timelines and reporting responsibility (Table 14). More details are provided in Annex 11. The APRs will document the progress towards achieving the indicators in GCF's IRMF and any additional project level indicators that have been selected for the project. APRs will also contain a narrative with updates on the progress of each output and outcome envisaged at the project level. The contracts with the service providers will specify their reporting responsibilities, the frequency of the reports to be produced and provide them with the formats to be used for reporting. All partners will be required to review the GAP and report on its implementation. All data will be disaggregated by sex to enable an assessment of the progress in inclusion of women in the project.

115. **MIS System**: An MIS system will be developed for the project to record key information of all beneficiaries (e.g. MOAFWM , Municipalities, Company, Household). The M&E Unit in the PMU will coordinate and produce a consolidated MIS report for the project on an annual basis. Within the first quarter of the second year, when activities have been initiated and sufficient outreach has been achieved and the M&E data base begins to get populated, thematic maps will be generated by the project and will be monitored through consolidated remote sensing practices or geospatial analysis. This is expected to yield a better understanding of trends and patterns and make the analysis more meaningful in understanding the relationship between climate parameters and the pattern of adoption and participation in project activities. The MIS system will geo-reference all activities using FAO's Remote Sensing application- Earth Map. The MIS system will also record beneficiary phone numbers for feedback from participants. The MIS system will also be used for tracking beneficiaries over time and assessing impact.

116. **Survey Methods**: The project will construct a baseline using primary and secondary data against which subsequent changes and impact will be measured. To measure attributable changes, the evaluations will draw on mixed-methods, using qualitative methods (e.g. participatory rural appraisal, the use of control groups, focus group discussions, key informant interviews, etc.) and quantitative (e.g. site-scale survey). Information on some of the key GCF indicators like awareness about climate information, resilience to climate risks will be measured through specific questions on these elements. All surveys and assessments will be sex-disaggregated and key gender-sensitive indicators both quantitative and qualitative outlined in the GAP will be captured in the initial and subsequent surveys and findings. In addition, external parties will conduct all evaluations to ensure that there is no bias in the findings.

117. **Mid-term and final evaluations:** To provide an external viewpoint on the progress of the project and the achievement of its objectives, and in line with the AMA signed with the GCF, two independent project evaluations will be conducted by FAO – interim and final evaluations. In line with the FAO policy on evaluations. They will be carried out by a team of independent external consultants. Both the mid-term and final evaluations must be consistent with GCF requirements as outlined in the GCF Evaluation Policy, Evaluation Standards, and Evaluation Operational Procedures and Guidelines. The project's studies and baseline, mid-term and final surveys will constitute important

inputs for the interim and final evaluations. Mid-term and Terminal Evaluation will be arranged in compliance with OED Evaluation Policy requirement with support from the REU Regional Evaluation Coordinator.

118. **Beneficiary Feedback**: FAO will establish a mechanism for beneficiary feedback and demonstrate how they have incorporated the feedback in improving their implementation approach. The M&E staff of the PMU will undertake periodic visits to the project areas to discuss with communities their views regarding project activities and to confirm their direct involvement during the site selection phases. The beneficiary feedback will also entail discussions with partners and executing entities about their experience with the project and the way partners engage with them during the implementation of the various components. The beneficiary feedback will be organized so that the reports provide sex-disaggregated perspectives. The PMU will also establish a GRM which ensures confidentiality. FAO's Guidelines for Compliance Reviews will follow the procedures for Complaints Related to the Organization's ESS.

## F. RISK ASSESSMENT AND MANAGEMENT

### F.1. Risk factors and mitigations measures

**Selected Risk Factor 1: Change in political leadership could shift focus on CAS**

| Category | Probability | Impact |
|---|---|---|
| Technical and operational | Medium | Medium |

**Description**

Change in political leadership could shift focus to other priority investments and lead to rapid turnover of staff in public offices, potentially affecting the capacity of the project to respect the policy investments.

**Mitigation Measure(s)**

The project will not engage in the creation of additional policy frameworks, but in creating the conditions to execute the existing one. Additionally, the project will work prevalently with the technical offices of partner institutions (e.g. MOAFWM), with sectorial associations and with communities. Therefore, the project will only engage relevant high-level officials for the approval process, minimizing the risk of blockages during execution and ensuring that implementation can withstand structural changes that may take place in the government during implementation. Furthermore, the project is aligned with long term policy strategies, in particular in the frame of the EU-accession process. It is therefore likely that even in the event of changes in political leadership and further staff changes, the overall policy strategies will remain favorable.

**Selected Risk Factor 2: Rising energy prices exacerbate pressures on forest resources**

| Category | Probability | Impact |
|---|---|---|
| Technical and operational | Medium | Medium |

**Description**

Given the current conflict in Ukraine and the resulting tension in the region, as well as other factors impacting energy costs, Serbia could be exposed to economic and social shocks that may affect the country's ability to guarantee affordable energy prices. While the country has the tools and experience to cope with such risks, rising energy prices may increase pressure on forest resources for fuel, as they remain the cheapest energy source.

**Mitigation Measure(s)**

The strategy of the project stands on three main pillars that serve to mitigate these risks: *(i) Enhancing monitoring of resources use.* One of the main problems related to fuelwood is that its extraction occurs often in an uncontrolled way. This is evident by the drastic underestimation of consumption in national statistics. Component 1 establishes the basis for an efficient, dynamic and continuous monitoring and assessment of resources, enhancing control over misuse of resources that could be managed in a more efficient way. *(ii) Increasing fuelwood quality.* Fuelwood is often sold and consumed freshly cut, or air dried for few months. This leads to the low energy content of wood and consequently to an increase in consumption needs for the same amount of energy. With the enhancement of the regulatory framework the project will provide the rules and regulations that biomass sellers must abide by, giving the consumer more possibilities to take informed choices and hence possibilities to reduce consumption. *(iii) Promoting sustainable energy use*. The project will support the enhancement of the regulatory framework and the strategies of SRPs and create

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| capacities for the utilization of residues from agroindustry's, which can complement the utilization of solid biomass from forests and furthermore offer benefits to local economic development.<br><br>The project team, through the PMU, will collaborate with line ministries to regularly monitor energy prices and fuelwood extraction and discuss the data obtained during PSC meetings to seek sustainable and viable options to guarantee energy to the most vulnerable. Ad hoc PSC meetings will be called as necessary. The project team will be available to collaborate with the GoS and use the project as a platform to expand public communication campaigns addressing sustainable fuelwood use, energy-saving practices and sustainable renewable energy alternatives. | | |
| **Selected Risk Factor 3: Novelty of Carbon Governance process** | | |
| **Category** | **Probability** | **Impact** |
| Technical and operational | Medium | Low |
| **Description** | | |
| Serbia may become the first country in the Balkans to seize such an opportunity. However, the development of strategies and policies may face delays, as such processes often benefit from exchanges with comparable experiences, which may be lacking in this context. Additionally, the evolving policy frameworks within the EU, particularly concerning the implementation of the CBAM, could impose restrictions on the use of offsets in certain sectors. | | |
| **Mitigation Measure(s)** | | |
| FAO is cooperating with national and international finance institutions, private sector associations and CSOs on potential carbon market developments in Serbia. The project will work with stakeholders in understanding already existing models. Furthermore, the project will establish regional policy dialogues motivating other countries to apply similar approaches and exchange lessons with realities that have already advanced. | | |
| **Selected Risk Factor 4: Resistance to change** | | |
| **Category** | **Probability** | **Impact** |
| Technical and operational | Medium | Low |
| **Description** | | |
| Administration, municipalities and private landowners as well as companies and agribusiness will require time and incentives to shift from the BAU scenario to a low emission sustainable development. In the past decades forests have been considered a mining resource and the shift to a new governance will require transparency and clear engagement of all stakeholders including the private sector and civil society as a whole. | | |
| **Mitigation Measures** | | |
| ▪ The project will invest in awareness and education activities, always emphasizing investments and beneficiaries with an evidence-based approach and replicating local and international good practices. Additionally, the project will work within the framework of existing laws on forests, agricultural land and RES.<br>▪ As agreed in the national engagement process, concerned state actors will be involved in the whole project cycle process including monitoring and evaluation of executed activities.<br>▪ The project will be the vehicle for new forest and landscape restoration approaches and for involving the private sector at a larger scale in reaching national forestry and biomass targets. Women groups that are already actively participating in the national engagement process, will provide a further impetus to this end.<br>▪ The project will create the enabling conditions for the private sector to engage in the country's decarbonization process effectively and efficiently through adaptation and mitigation investments reach rural areas with.<br>▪ Besides working with the State authorities at the national and the local levels, the Project will develop strong and active cooperation with local elected bodies (i.e. Municipalities), communities and the civil society. | | |
| **Selected Risk Factor 5: Delays in governmental approval of project activities/outcomes** | | |
| Category | Probability | Impact |
| Technical and operational | Low | low |
| Description | | |

Directorate of Forests does not endorse in a timely manner the CAS guidelines and their integration in the national procedures for forest management.

**Mitigation Measure(s)**

FAO has involved all relevant stakeholders in the elaboration process of the guidelines and established a close relationship with the Directorate of Forests as well as the PES and PEV, the main responsible entities for managing state-owned and supervising private forests and forest lands at field level. The project will maintain this close relationship and ensure that all partners are informed in a transparent and timely manner to anticipate the approval of important milestones that are crucial for the further advancement of project activities.

**Selected Risk Factor 6: Delays in production of seedling material**

| Category | Probability | Impact |
|---|---|---|
| Technical and operational | Medium | high |

Description

A consistent and essential part of the activities is dependent on the production of high-quality seedlings. Any delays in the production of these would have a very high impact on all other project outcomes.

Mitigation Measure(s)

Although the project intends to solely use container seedlings for all other than vegetative propagated species (poplar species, shrubs), bare-root seedlings will complement the nursery production at all three nursery production sites to compensate for accidental loss or damage of the production in a greenhouse (e.g. due to pathogenic fungi). In addition, technical assistance will be maintained from year two through year four until staff of all three nurseries has acquired sufficient expertise to continue high-quality seedling production autonomously.

**Selected Risk Factor 7: Lack of Private sector engagement in decarbonization process**

| Category | Probability | Impact |
|---|---|---|
| Technical and operational | Low | Low |

**Description**

Currently there is substantial interest by the private sector – including financial institutions and industry associations -- to invest in decarbonization. However, due to limited technical knowledge, strategies and financial instruments available, the private sector may be disengaged or lose interest in investing in the process and prefer to invest in business as usual.

**Mitigation Measure(s)**

The project will build the capacities of all stakeholders on decarbonization and reduce the financial and technical barriers related to private investment in decarbonization. Through its awareness and capacity development activities, the project will emphasize and disseminate the economic advantages of investing. The project will work together with the PKS and its circular economy platform, financial institutions and other industry/services/commerce associations to ensure regular engagement of private sector operators. Therefore, the private sector will be engaged via its association and financing actors. Furthermore, as reported in section B.4, private sector organizations can serve as observers to the project steering committee. These actors are currently active in the country to respond to the companies' interests and needs to invest in reducing their overall carbon footprint as well as climate adaptation deficit. To this end, the project will work with identified institutional stakeholders as well as with companies providing, among other, technical assistance to finance institutions, associations and firms to assess climate risks and to monitor emissions or potential carbon sequestration from their activities, and the cost of Environmental, Social, Governance (ESG) compliance.

The project will also involve service providers active in the provision of technical and financial services to private sector operators in providing GHG accounting and ESG compliance support. Overall, compliance will follow the laws and standards reported in Annex 6. The project will supply the banks, companies and service providers with templates, based on recognized international standards like Greenhouse Gas Protocol Corporate Standard, ISO 14064 and European Union Standards, to report on decarbonization measures. Training will be provided on how to complete these templates as well as the importance of accurate and transparent reporting and compliance, including on reputational benefits and risks to noncompliance.  The reports will be verified by project experts and by Independent Experts at Mid and End Term to verify GHG emission mitigation occurred. Additionally, the project will work with partners in strengthening stakeholder networks between GoS, banks, and the PKS to promote additional accountability.

This will create sustained connections with private sector actors, including financial institutions. Key mitigation measures are embedded in component 3, focusing on raising awareness and training private sector operators on carbon reduction investments. Financial institutions have shown interest in sustainable investments, as indicated in letters of intent (Annex 1). Rising energy costs further incentivize investment in efficient technologies (Annex 3 – EFA), and the risk of waning private sector interest in decarbonization is considered minimal. The project's PMU and relevant team members will maintain regular communication with private sector partners and financial institutions, providing updated information, data, and technical assistance on decarbonization and climate risk mitigation. They will also gather feedback on concerns and needs to identify effective solutions and sustain engagement. Additionally, the project's M&E unit will monitor beneficiary engagement through semiannual questionnaires and interviews with companies and financial institutions.

**Selected Risk Factor 8: Ineffective and uncompleted Measuring, Reporting and Verification (MRV) system**

| Category | Probability | Impact |
|---|---|---|
| Technical and operational | Low | Low |

**Description**

The Country is still building its MRV system. Budget constraints and lack of adequate capacities in the line ministries could slow down the process impacting the activities of the project, related to the carbon governance and framework.

**Mitigation Measure(s)**

FAO has already established a strong collaboration and exchange on MRV with the line agencies to proactively anticipate any possible delays and provide support if needed. Through component 1, the project will also give a significant contribution to enhance the integration of carbon governance in forestry in the MRV.

**Selected Risk Factor 9: Carbon price volatility and decrease in carbon price**

| Category | Probability | Impact |
|---|---|---|
| Technical and operational | Medium | Medium |

**Description**

The volatility in carbon prices during the project implementation period could shape investment, by adding a dimension of uncertainty to returns on investments. Additionally, carbon prices have in general been rising over time. A potential drop in carbon prices would affects economic returns to a project, potentially reducing the attractiveness and viability.

**Mitigation Measure(s)**

The interventions to provide information provision and sharing and identifying potential investments with positive financial returns, even after excluding carbon related benefits, will reduce the effects of price volatility and decreases.

**Selected Risk Factor 10: Money laundering and countering the financing of terrorism (ML/TF)**

| Category | Probability | Impact |
|---|---|---|
| Governance | Low | Low |

**Description**

Risks of money laundering and countering the financing of terrorism.

**Mitigation Measure(s)**

FAO includes in the project agreement signed between FAO and the GoS clauses related to AML/CFT, as follows:

a) The Government shall comply, and shall require all persons and entities engaged in its activities under the Project to comply, with all internal anti-money laundering, counter-terrorism financing laws, rules, and regulations;

b) The Government confirms it has obtained sufficient undertakings from all persons and entities involved in its activities under the Project that they shall not engage in any prohibited practices; the Government undertakes and confirm that it shall comply with the substantive objectives of the GCF's Policy on Prohibited Practices;

c) Consistent with numerous United Nations Security Council resolutions adopted under Chapter VII of the UN Charter, the Government and FAO are firmly committed to the international fight against terrorism and, in particular, against the financing of terrorism. It is the policy of the Government and FAO to seek to ensure that none of their funds are used, directly or indirectly, to provide support to individuals or entities: i) associated with terrorism, as included in the list maintained by the Security Council Committee established pursuant to its Resolutions 1267 (1999) and 1989 (2011); or ii) that are the subject of sanctions or other enforcement measures promulgated by the United Nations Security Council. This provision must be included in all agreements concluded with third parties under the Project.

During project implementation FAO, will ensure close monitoring and supervision through its offices in the regional office and HQ in order to ensure that the activities are implemented in full compliance with the signed project agreement.

## G. GCF POLICIES AND STANDARDS

### G.1. Environmental and social risk assessment

119. The project carried out its environmental and social assessment following FAO's environmental and social safeguards (FAO, 2015), and prepared its ESMF. The ESMF identifies policy triggers for the project, screening criteria for activities, environmental and social impacts of the activities, and measures to mitigate identified risks. Mitigation actions will avoid, minimize and mitigate negative impacts during project implementation and operation. Mitigation actions will be in line with FAO and GCF ESS policy, and national legislation, and adhered to whichever is most stringent. The ESMF also sets out the modalities for stakeholder engagement, and the procedure and process for dealing with complaints, through the GRM. To ensure a smooth and effective ESMF process, an ESS specialist will be engaged by the project to be responsible for the environmental and social safeguards process (including GRM), interacting on a regular basis with key stakeholders and being available to respond to any grievances. The ESMF will be disclosed on relevant portals, and shared during stakeholder consultations, to increase awareness of potential consequences of project activities. During these stakeholder consultations, the GRM will also be presented and explained.

120. There will be **zero tolerance of sexual exploitation, abuse, and harassment (SEAH),** and the project's ESMF and consequent ESMPs will mainstream SEAH risk mitigation, in accordance with the FAO FESM. The project will support gender sensitization and trainings for project staff and beneficiaries on gender equality and social inclusion and SEAH and will elaborate a code of conduct for the implementation of the project. Specific procedures to minimize SEAH risk will be developed for the project GRM, to ensure the mechanism is survivor-centered and gender-responsive (including confidential reporting), and to facilitate linkages to related services and redress for anyone affected by SEAH.

121. **Grievance and Redress Mechanism (GRM):** The GRM is an integral project management element that intends to seek feedback from beneficiaries and resolve complaints on project activities and performance. The mechanism is based on FAO requirements and most importantly, it is based on existing, community-specific grievance redress mechanisms preferred by the local beneficiaries. FAO and EEs will inform communities about the GRM through culturally appropriate mechanisms, ensuring information on mechanisms at all three levels is communicated (i.e. project-based, and FAO-level redress mechanisms and GCF's Independent Redress Mechanism). The PMU and FAO Serbia will be responsible for managing the grievance and redress mechanism. Project-related SEAH and gender-based violence (GBV) grievances will be managed through the existing FAO GRM system, which will also be strengthened to include a procedure for handling SEAH that is inclusive, survivor-centered and gender-responsive, complemented by GBV referral pathways (see Annex 6: ESMF).

122. The project will identify and mitigate SEAH risks or potential adverse impacts on women, men, girls and boys as early as possible as a part of environmental and social risk screening and reflect such risks or impacts in relevant safeguards instruments (including ESMF and ESMPs, GAPs and others as appropriate), and propose mitigation measures, differentiated by gender and age where relevant as follows:

- Include measures (including pre-project implementation awareness raising for host communities and project workforces) to enhance gender equality, and to prevent, address and eliminate SEAH in the relevant projects or programs and safeguards instruments;
- Implement, monitor and continuously improve all measures to mitigate the identified SEAH risks and impacts;
- Ensure sufficient and adequate financial and human resources are allocated for SEAH-related compliance.
- Ensure that stakeholder consultations prior and during project implementation include awareness raising and stakeholder-differentiated understanding of SEAH related risks and mitigation measures.

123. Project components were identified through consultative processes and are prioritized in Serbia's national and climate policy. There will be no significant or irreversible negative environmental impacts associated with the project. Proposed project investments are designed to have positive social and environmental benefits; the project has however been classified as ***moderate risk (Category B)*** mainly related to on-ground activities in the forestry sector. Potential impacts are limited to the project footprint and could occur as a result of forest-related activities, but these are localized and are mitigated by selecting local species with wide ecological range and higher drought resistance, considering the bioclimatic type of each site and projected shifts in potential tree species range limits due to climate change. No downstream nor cumulative effects are envisaged. FAO ESS triggered are:

124. ESS 1. Biodiversity conservation, and sustainable management of natural resources. The project will support Serbia in enhancing the resilience of its forest ecosystems introducing CAS and SFM practices; it will not engage in poor natural resources management practices nor have negative impacts on natural resources. All project investments aim at restoring forests, increasing biodiversity as means to resilience. Forestry investments are designed to enhance biodiversity with specific priority to those areas that will act as corridors among existing forests. Therefore, activities will not impact protected areas of natural habitat or sensitive ecosystems. However, this safeguard is triggered because of afforestation/forest rehabilitation activities, and environmental and social assessments undertaken at the sub-project level, once identified, will consider biodiversity. While the Project will establish and/or manage planted forests, it will only plant with native or locally adapted species and involving local communities. Activities will be executed according to the responsible management of planted forests. Project activities will only include forestry investments in existing forest areas or in areas previously covered with forests. Therefore, livestock and aquatic genetic resources will not be impacted.

125. ESS 2. Resource efficiency and pollution prevention and management. The project will promote CAS. It will not lead to increased use of pesticides through intensification/expansion of production. In upgrading nurseries, no significant increase in water consumption is envisaged. No seeds will be procured and no new planting material (tree, shrub, crop varieties) will be introduced into the country. With regards to the establishment or management of planted forests/CAS – the Project will select local species with wide ecological range and higher drought resistance, considering the bioclimatic type of each site and projected shifts in potential tree species range limits due to climate change (e.g. avoiding planting seedlings from species in the lower limit of their ecological range; planting seedlings from species somewhat above the upper limit of their ecological range). No GMO or nor seeds with insecticidal seed coatings will be used in the project. No significant waste will be generated – with regards to road clearing, this will involve clearing of biological debris (vegetation) which will be composted or integrated into the environment. With regards to fencing, these will be of different forms including biological depending on specific context of the forestry investments. Each will be defined during the inception phase when precise sites will be formalized at which time appropriate screening for types of fencing and their handling and disposal will be conducted. This safeguard is triggered to account for potential waste disposal.

126. ESS 4. Decent work. This safeguard is triggered to ensure that the project will promote, respect and realize fundamental principles and rights at work. The employment of project workers will be based on the principle of equal opportunity and fair treatment, and there will be no discrimination with respect to any aspects of the employment relationship. Hiring of workers will be made following the laws and regulations of Serbia (Labour Law 24/05, 61/05 and 54/09) and workers will need to abide with the FAO code of conduct and FAO policies. All workers will be above 18 years old. Worksites must be accessible by road and transport from collection points in accessible areas to worksites will be guaranteed by the project through its partners and service providers.

127. ESS 5. Community health, safety and security. This safeguard is triggered to ensure that adverse impacts on health, safety and livelihoods of involved and affected communities are anticipated and avoided. Community exposure to health risks is not envisaged, however occupational health and safety (OHS) risks need to be considered with regards to afforestation/reforestation activities; these will be dealt with by providing training and protective measures and gear as needed. The project does not have planned any activities that will trigger fires in areas where it will work. Project activities will be in remote forested areas generally far from houses and communities. All workers in project areas will be selected among men and women from local communities, within a 25 km radius; the establishment of camps or other temporary accommodation structures will not be required. As works will occur in remote forested areas of the country, the project does not expect to have migrant workers.

128. The Project is designed to operate at the national level on public and private lands, with a focus in the AP of Vojvodina and Central Serbia, ensuring benefits to all people that will be impacted by Project activities. The presence of Indigenous Peoples is not foreseen, however before implementing field level activities, stakeholder consultation and second-level screening will be held once specific project sites are identified by Government. In the event that Indigenous Peoples are identified, all subsequent actions will be taken in accordance with both FAO Environmental and Social Management Framework and GCF Indigenous Peoples Policy[2]. Annex 6 provides the ESMP, and Annex 7 provides a summary of consultations and the Stakeholder Engagement Plan (SEP).

## G.2. Gender assessment and action plan

2 https://www.greenclimate.fund/sites/default/files/document/ip-policy.pdf

129. The project was designed in line with GCF's Gender Policy and considering that gender equality considerations should be mainstreamed into the entire project cycle to enhance the efficacy of climate change mitigation and adaptation interventions. A gender analysis was conducted to identify potential for promoting gender equity and mitigate potential risks. This was elaborated through a desk review of relevant national policies, legal and regulatory frameworks, and studies conducted in the sector, and considering the consultations with the project team with stakeholders in the Government and CSOs, including rural development, forestry and environmental sectors. The assessment was followed by GAP design addressing gender gaps and maximizing benefits of women's engagement (Annex 8).

130. Compared to EU Member States, the United Kingdom, and other countries in the region, Serbia ranks 21st. Compared to Croatia, Serbia has lower values on the index in the sub-domains of participation (77.0 versus 79.6), but higher values in the sub-domain of segregation and quality of work (62.5 versus 61.4). Compared to North Macedonia, Serbia has higher values in the sub-domain of participation (77.0 vs. 68.2), but significantly lower in the sub-domain of segregation and quality of work (62.5 vs. 70.7). Compared to first-ranked Sweden, Serbia has this domain of Index lower by 13.5 points, while compared to last-ranked Italy, it has an index higher by 6.1 points. According to findings, women have lower resilience and often lower adaptation capacities to respond to climate change and related emergencies. Due to their lower preparedness to act in emergency situations, a lack of information and relevant resource, and weaker economic and financial capacities, women are more vulnerable to the impacts of climate change (FAO 2021). Reportedly [Babović and Reljanović, 2019], over a sample of 2023 women in Serbia, 41.8% of women since the age of 15 (782 women in the sample) were exposed to sexual harassment, while in the 12-month period prior the survey, 18% of women (313 women in the sample) were exposed to this type of violence. According to UN-Women [UNW, 2023], lifetime Physical and/or Sexual Intimate Partner Violence concerned in 2019 17 % of ever-partnered women aged 18-74 years while 34% while 2% of the same age cohort experienced non-partner sexual s violence (lifetime).With regards to the risk of sexual exploitation, abuse and harassment (SEAH) risk, Serbia disposes of a recent legal framework to prevent, monitor and address the issue. These include the family law (2005), the law on Amendments to the Criminal Code (2016), the law on preventing domestic violence (2017) and the law on free legal aid (2019).

131. The project will assist with gender mainstreaming of climate change policies and mechanisms to respond to recommendations of the Committee on the Elimination of Discrimination against Women (CEDAW, 2019) and will provide incentives for the improvement of the situation of rural women in terms of advice for the access to land ownership, its sustainable use, employment in forestry sector and models for obtaining economic security Annex 8 page 48 and 49). The project will give priority to women owning degraded coppice stands or lands for conversion and will ensure that at least 30% of beneficiaries are women. Women percent was determined via consultation with national experts and consulting available databases related to women's participation in the different sectors/topics addressed by the project. Table 10 section D1 and related notes report the detailed data used to determine values. Women forest owners will be contacted and invited, and priority will be guaranteed for raising quality forests and forest production systems with establishing full value chains and partnerships and making them operative. The opportunity to restore their forests will increase value of the lands owned by women, reducing women's vulnerability.

132. More specifically, the GAP ensures that, in project Component 1: i) enhanced forest management and governance is mainstreamed across youth and includes women landlords. This will be done also ensuring a gender perspective – granted by ensuring consultation with women groups and associations - into addressed policies and standards; in Component 2: i) women owning forests/degraded lands are given priority; and ii) greening recommendations take into account the needs and the role of women. In Component 3, in the enhancement of private sector engagement in decarbonization: i) women-led enterprises are engaged and produce their carbonization strategies ii) consulting firms and their female consultants are given priority in trainings.

133. In terms of project management, gender training and expertise will be provided to all project team members and partners on a regular basis, moreover, stakeholder engagement strategy and communication strategy will be prepared and implemented in a gender-responsive manner. Gender activities will be under the responsibility of the project Gender Equality and Women's Empowerment Specialist in the PMU. The project will establish a grievance mechanism at field level to file complaints. Contact information and information on the process to file a complaint will be disclosed in all meetings and other related events throughout the life of the project. In addition, it is expected that awareness raising material be distributed to include the necessary information regarding the contacts and the process for filing grievances. The Project will include sexual exploitation and abuse awareness raising, and stakeholder-differentiated understanding, during stakeholder engagement. With regards to the prevention of sexual exploitation and abuse (PSEA), through its GRM the Project will ensure that all concerns and/or incidents will be reported to the PSEA focal point and the FAO Office of the Inspector General, as appropriate.

### G.3. Financial management and procurement

134. FAO will ensure that financial management and procurement of goods and services using GCF resources adhere to relevant FAO rules and regulations, as well as relevant provisions in AMA signed between FAO and GCF. These rules and regulations were reviewed and deemed satisfactory by the GCF Secretariat and Accreditation Panel as part of FAO's accreditation process. This includes financial management and procurement performed by FAO-REU/SERBIA Office as EE. FAO has deployed an Oracle-based Enterprise Resource Planning (ERP) system, the 'Global Resources Management System' (GRMS), which provides all FAO employees in all locations globally with travel, human resources, procurement, and finance functionalities. Using GRMS improves the flow of financial information, supports financial monitoring and reporting, increases transparency and visibility, and strengthens internal control. FAO maintains a Chart of Accounts which is used by the whole Organization and that allows for a separation of income and expenditure by donor and project, and it provides a standardized coding structure that enables data to be recorded, classified, and summarized to facilitate internal management and external reporting requirements.

135. Direct procurement by FAO is done in accordance with its Manual Section 502 on "Procurement of Goods, Works, and Services". To sub-contract the delivery of specific activities using Letters of Agreement, FAO operates in accordance with its Manual Section 507 on "Letters of Agreement". Such services are managed by the FAO Procurement Service, which provides policy and operational support to ensure that the Organization procures goods, works, and services based on "Best Value for Money" principles. Financial management and procurement executed by FAO-REU/SERBIA Office as Executing Entity will be overseen and supervised by the FAO-GCF project supervision team. The FAO-GCF project supervision team will undertake regular supervision missions and recruit an internationally recognized auditing firm to perform frequent spot checks and audits to ensure financial management and procurement by the PMU and executing entities are conducted in line with agreed standards and practices. This will be governed by the agreements signed between FAO and other executing entities before the project becomes operational.

### G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

## H. ANNEXES

| | | |
|---|---|---|
| **H.1. Mandatory annexes** | | |
| ☒ | Annex 1 | NDA no-objection letter(s) **(template provided)** |
| ☒ | Annex 2 | Feasibility study - and a market study, if applicable |
| ☒ | Annex 3 | Economic and/or financial analyses in spreadsheet format |
| ☒ | Annex 4 | Detailed budget plan **(template provided)** |
| ☒ | Annex 5 | Implementation timetable including key project/programme milestones **(template provided)** |
| ☒ | Annex 6 | E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):<br>**(ESS disclosure form provided)**<br>☐ Environmental and Social Impact Assessment (ESIA) or<br>☐ Environmental and Social Management Plan (ESMP) or<br>☐ Environmental and Social Management System (ESMS)<br>☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.) |
| ☒ | Annex 7 | Summary of consultations and stakeholder engagement plan |
| ☒ | Annex 8 | Gender assessment and project/programme-level action plan **(template provided)** |
| ☒ | Annex 9 | Legal due diligence (regulation, taxation and insurance) |
| ☒ | Annex 10 | Procurement plan **(template provided)** |
| ☒ | Annex 11 | Monitoring and evaluation plan **(template provided)** |
| ☒ | Annex 12 | AE fee request **(template provided)** |
| ☒ | Annex 13 | Co-financing commitment letter, if applicable **(template provided)** |
| ☒ | Annex 14 | Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule |
| **H.2. Other annexes as applicable** | | |
| ☒ | Annex 15 | Evidence of internal approval **(template provided)** |
| ☒ | Annex 16 | Map(s) indicating the location of proposed interventions |
| ☐ | Annex 17 | Multi-country project/programme information **(template provided)** |
| ☐ | Annex 18 | Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project |
| ☐ | Annex 19 | Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity |
| ☒ | Annex 20 | First level AML/CFT (KYC) assessment |
| ☐ | Annex 21 | Operations manual (Operations and maintenance) |
| ☒ | Annex 22 | Assessment of GHG emission reductions and their monitoring and reporting |
| ☒ | Annex 23 | Theory of Change |
| ☒ | Annex 24 | Climate Change Analysis |
| ☒ | Annex 25 | Estimation of beneficiaries |

ENDNOTES

1 Adaptation deficit in the context of the project is defined as follows: Lack of technical and knowledge capacity, economies of scale and access to finance to adapt in efficient manner to challenges of climate change.

2 FAO 2021 refers to Annex 24 – Climate Scenario "Past, present, and projected temperatures, precipitation and winds in the Republic of Serbia". FAO analysis is based on the Meteorological yearbook of the Republic Hydrometeorological Service of Serbia (2020).

3 The SNC, 2016, used data from meteorological stations operated by the Republic Hydrometeorological Service of Serbia.

4 The UNDP, 2018 and TNC, 2020 used 3 data sets to highlight climatic changes: 1) Average monthly values at the meteorological station of the Belgrade Observatory - through observations since 1888, when the Observatory was founded. Data from this station were used because they represent well changes in the entire territory of Serbia; 2) E-OBS grid climatology (horizontal resolution 10 km, time resolution 1 day) – series for the 1950-2017 period (Copernicus Programme of the European Union); 3) DanubeClim grid climatology (horizontal resolution 10 km, time resolution 1 day) – series for the 1961-2010 period.

5 https://www.hidmet.gov.rs/index_eng.php

6 A heat wave is a period of not less than 6 days during which the maximum daily temperature was higher than the expected maximum temperature for the time of year in which the heat wave was observed [UNDP, 2018].

7 Extracted from the TNC, 2020, on the methodology used: Climate projections were made for the periods: 2016-2035 (near future), 2046-2065 (mid-century) and 2081-2100 (end of the century), and the 1986-2005 period was taken as the reference period. The indices were calculated as the most probable value, a set (ensemble) of values obtained using nine regional models, which are among the EURO-CORDEX database models. The global *SPEI* database http://spei.csic.es/database.html was used for the analysis of drought. The values of the SPEI index for Belgrade were calculated from the observed monthly values of precipitation and temperature.

8 As in the TNC, the analyses of future climate change are aligned with the latest, Fifth Assessment Report of the Intergovernmental Panel on Climate Change. The results presented here represent the most likely value from the set (ensemble) of solutions obtained using daily values of temperatures and precipitation from nine regional climate models that can be downloaded from the EURO-CORDEX database. The reference period with respect to which the change in future climatic conditions is analysed is 1986-2005 and the analysed future periods are: 2016-2035 (near future), 2046-2065 (mid-century) and 2081-2100 (end of century).

9 According to Stjepanović et al. (2021), the Ellenberg climate quotient (EQ) and the Forest Aridity Index (FAI) can be used to represent climate characteristics impacting growth of trees. EQ is defined as: EQ=(Tmax. month / Pannual)∗1000, where Tmax. month is the average temperature of the warmest month (in °C) and Pannual is the annually accumulated precipitation (in mm).

10 FAI considers the average temperature of the critical months (July and August) and precipitation during the main growth cycle (May to July), as well as precipitation during the critical months (July and August).

11 The analysis of Forzieri et al. (2022) concluded that "tropical, temperate and arid forests underwent a decline in resilience probably related to the concomitant increase in water limitations and climate variability."

12 In particular, pest and diseases outbreaks have increased steadily with recurrent picks when the gipsy moth, favoured by drought, spread in Serbian forests [NAP, 2015; GoS, 2020].

13 About 60% of Serbia's endemic plant species remain endangered.

14 Detailed description of forest related policy frameworks is available in Annex 2, Appendix 7.

15 Specific objective 4 (out of 5) of the Low Carbon Development Strategy is to "preserve the potential of mitigation measures, determined for 2030 and 2050, by increasing the resilience to climate change of the priority sectors." Forestry is included next to agriculture and water resources as one of the three priority sectors.

16 Adaptation deficit in the context of the project is defined as follows: Lack of technical and knowledge capacity, economies of scale and access to finance to adapt in efficient manner to challenges of climate change.

17 From 2010 to 2020 total woodfuel consumption by HH has increase by 5.1%. Branko Glavonjić, 2023. Inventory of wood energy consumption and GHG emissions from wood fuels in Serbia

18 According to official statistics, the share of wood fuels in final energy consumption has increased from 11% in 2017 to 11.86% in 2020. Statistical Pocketbook of the Republic of Serbia, 2022 and 2018

19 Financial institutions that have expressed interest include Raifeissen. Beograd, Erste Bank Serbia, UniCredit Bank Serbia and Banca Intesa a.d. Beograd.

20 The 600 beneficiaries represent the total corpus of forestry engineers in force at the Ministry as well as forest corporations and local authorities. The intervention size is determined by the number of forest engineers currently working in forestry institutions/enterprises in Serbia. The project aims at training all, therefore no selection criteria other than this will be applicable. Priority will be given to women involved in the sector and workers of both public and private forestry companies.

21 One of the most important current examples is the The World Bank's First Serbia Green Transition Programmatic Development Policy Loan that is crucial for Serbia's sustainable energy development. It provides EUR 149.9 million to support Serbia's efforts in strengthening public sector institutions for resilient, greener, and more inclusive economic growth. The loan aims to accelerate the clean energy transition through energy market reforms, faster adoption of renewables, and protection of vulnerable consumers

22 The proposal will only rehabilitate existing roads and will not support the opening or construction of new ones. Likewise, the proposal will not support the enlargement or expansion of road networks in project areas. The existing forest road infrastructure in Serbia is below state-of-the-art standards and does, therefore, not allow sustainable forest investments and interventions. It has been estimated that about 515 km

of forest roads in the project areas in Central Serbia at cost of USD 35 per meter of road and about 30 km of forest roads in Vojvodina at cost of USD 90 per meter of road require "upgrading" (improving carrying capacity and climate proofing). The relatively high cost for "upgrading" the existing road infrastructure is caused by applying environmentally sound upgrading techniques, complexity of terrain and slope and – in the case of Vojvodina - the required long-distance transport of required material for forest roads. A forest road refers to any road located on forest lands or other areas that are used periodically for transporting forest goods, as well as for forest management and monitoring activities. This term does not include paved roads maintained for general public transportation or roads used exclusively as residential driveways.
23 The proposal will only rehabilitate existing roads and will not support the opening or construction of new ones. Likewise, the proposal will not support the enlargement or expansion of road networks in project areas. The existing forest road infrastructure in Serbia is below state-of-the-art standards and does therefore not allow sustainable forest investments and interventions. It has been estimated that about 515 km of forest roads in the project areas in Central Serbia at cost of USD 35 per meter of road and about 30 km of forest roads in Vojvodina at cost of USD 90 per meter of road require "upgrading" (improving carrying capacity and climate proofing). The relatively high cost for "upgrading" the existing road infrastructure is caused by applying environmentally sound upgrading techniques, complexity of terrain and slope and – in the case of Vojvodina - the required long-distance transport of required material for forest roads. A forest road refers to any road located on forest lands or other areas that are used periodically for transporting forest goods, as well as for forest management and monitoring activities. This term does not include paved roads maintained for general public transportation or roads used exclusively as residential driveways.
24 Digital Platform for Circular Economy (CE-HUB), Chamber of Commerce and Industry of Serbia. https://circulareconomy-serbia.com/
25 World Bank, World Bank Supports Serbia's Move Toward Greener, More Resilient, and Inclusive Growth
26 Serbia raises EUR 1 billion in its first green bond auction
27 Serbia 2021 Article IV Consultation, p. 19: https://www.imf.org/en/Publications/CR/Issues/2021/06/21/Republic-of-Serbia-2021-Article-IV-Consultation-and-Request-for-a-30-Month-Policy-461077
28 The IMF projects GDP growth of 4.5% in 2022 and 2023. Two leading ratings agencies – Standard and Poor's and Fitch Ratings – have assigned a positive or stable outlook (BB+) to Serbia's credit.
29 Data on imports from WITS, World Bank and data on GDP from World Development Indicators. Both reported in current USD.
30 World Bank, 2022. "Western Balkans Regular Economic Report: Steering through crises" Spring 2022.
31 World Bank, 2022. "Western Balkans Regular Economic Report: Steering through crises" Spring 2022.
32 Serbia 2021 Article IV Consultation, p. 19: https://www.imf.org/en/Publications/CR/Issues/2021/06/21/Republic-of-Serbia-2021-Article-IV-Consultation-and-Request-for-a-30-Month-Policy-461077
33 The IMF projects GDP growth of 4.5% in 2022 and 2023. Two leading ratings agencies – Standard and Poor's and Fitch Ratings – have assigned a positive or stable outlook (BB+) to Serbia's credit.
34 E.g. the Chamber of Agriculture, the Serbia Grain Association, the Chamber of Commerce, Forest Owner Guild and SerBio.
35 The platform involves stakeholders in the sector for the production and marketing of solid biofuels at standardized high quality for the local population. These platforms will have a production line for biomass and also dispose of storage and logistic facilities.
36 https://circulareconomy-serbia.com/. The Digital Platform for Circular Economy (CE HUB) recently created by the Chamber of Commerce and Industry of Serbia showcases circular-economy-related information for the business community, including on financial support, possibilities of improving knowledge and practice and possible savings by shifting to a CE business model.
37 The platform, in use since 2015, is a good example of how digital technologies could become instrumental in enabling small forest owners to aggregate their lands under the same management objective (conservation, production, recreation, maintenance and others) supporting at the same time the goals of countries
38 This innovative climate investment approach can be replicated and scaled up in the other countries in the region. The blueprint of this project approach is schematically described in Annex 4. The approach can be applied in a form of single-country project as well as programme (multi-country/ regional/ global). It is currently planned that the blueprint architecture will be applied for Moldova and possibly for North Macedonia.
39 In the period 2022-2030, Serbia has planned to invest (if resources will be available) in: 48,000 ha of afforestation, 4,000 ha of shelterbelts, 199,200 ha of natural regeneration, 6,000 ha of plantation (mainly poplar), 1,258,440 ha of coppice stands restored of which 922,400 converted into high stands, and protection of forests from weather hazards on 201,720,000 ha.
40 The current project blueprint approach as well as the main architecture of this concept note have been discussed and agreed with the NDA of the Republic of Moldova and it's currently being discussed with the NDA of the Republic of North Macedonia.
41 Activities will follow specific protocols that will guarantee the use of local and species that will be selected based on the characteristics of existing forests. The project will not negatively impact ecosystems
42 Converting unfarmed degraded agricultural lands that have been abandoned into biomass forests, will allow the lands to maintain value and produce income and for the soil to recover and gradually recover sufficient quality to sustain again agriculture. Furthermore, the activity will protect soils from erosions and will contribute in mitigation the adverse impacts of winds.
43 The project will have a positive impact on households that will be expected to face a lower unit cost for energy produced by fuelwood. This lower unit cost of energy will enhance affordability of energy for the poorest segments of the population. In Serbia, households currently spend more than 10% of their average expenditure on energy. Coupled with environmental / energy efficiency awareness campaigns to avoid increase of consumption of fuel wood, lower fuel costs will also imply a lower share of overall spending going toward energy, freeing up resources for other household needs.
44 World Bank Enterprise Survey 2019, Serbia.
https://www.enterprisesurveys.org/content/dam/enterprisesurveys/documents/country/Serbia-2019.pdf
45 As documented in Annex 3, the financial discount rate has been set at 7%, corresponding to the average interest rates for short-medium term loans (relevant to the private businesses and consumption patterns of the context) and their trends in the last years (Source: CBA compendium of interest rates).
46 See section D, key parameters.
47 The 40 years time horizon is used to capture the full deployment of benefits deriving from forestry investment.

**No-objection letter issued by the national designated authority(ies) or focal point(s)**

**Republic of Serbia**
**MINISTRY OF AGRICULTURE, FORESTRY AND WATER MANAGEMENT**
Date: 25th September 2023
Belgrade

**To: The Green Climate Fund (GCF)**

**Re: Funding proposal for the GCF by the Ministry of Agriculture, Forestry and Water Management of the Republic of Serbia regarding the project "Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration"**

Dear Madam, Sir,

We refer to the project titled the project "Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration" in Serbia as included in the funding proposal submitted by the Ministry of Agriculture, Forestry and Water Management of the Republic of Serbia.

The undersigned is the duly authorized representative of the Ministry of Agriculture, Forestry and Water Management of the Republic of Serbia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of the Republic of Serbia has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Republic of Serbia;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme. We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

MINISTER
Jelena Tanasković
GCF National Designate Authority / Focal Point

**Environmental and social safeguards report form pursuant to para. 17 of the IDP**

| **Basic project or programme information** | |
|---|---|
| **Project or programme title** | Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration (FOREST Invest) |
| **Existence of subproject(s) to be identified after GCF Board approval** | Yes |
| **Sector (public or private)** | Public |
| **Accredited entity** | Food and Agriculture Organization of the United Nations (FAO) |
| **Environmental and social safeguards (ESS) category** | Category B |
| **Location – specific location(s) of project or target country or location(s) of programme** | Districts of: Belgrade, Bor, Braničevo, Central Banat, Jablanica, Kolubara, Moravica, Nišava, North Bačka, North Banat, Pčinja, Pirot, Pomoravlje, Rasina, Raška, South Bačka, South Banat, Toplica, Zaječar, Zlatibor in the Republic of Serbia |
| **Environmental and Social Impact Assessment (ESIA) (if applicable)** | |
| Date of disclosure on accredited entity's website | Friday, January 17, 2025 |
| Language(s) of disclosure | English and Serbian |
| Explanation on language | Serbian is the official language of Serbia, with English widely spoken, and are the languages understandable to affected peoples/stakeholders. |
| Link to disclosure | English:<br>• https://openknowledge.fao.org/handle/20.500.14283/CD3951EN<br><br>Serbian:<br>• https://openknowledge.fao.org/handle/20.500.14283/CD3951SR |
| Other link(s) | FAO disclosure portal (English and Serbian):<br>• https://www.fao.org/environmental-social-safeguards/project-detail/enhancing-the-resilience-of-serbian-forests-to-ensure-energy-security-of-the-most-vulnerable-while-contributing-to-their-livelihoods-and-carbon-sequestration-(forest-invest)/en<br><br>Ministry of Agriculture, Forestry and Water Management of the Republic of Serbia website (English and Serbian):<br>• Оквир управљања заштитом животне средине и социјалним питањима - Управа за шуме Републике Србије<br><br>Vojvodinasume website (English and Serbian): |

| | |
|---|---|
| | • Serbian: Annex-6-ESMF_Serbian_Final.pdf<br>• English: Annex-6-ESMF_English_final.pdf<br><br>Srbijasume website:<br>• Serbian: https://srbijasume.rs<br>• English: https://srbijasume.rs/en/ |
| Remarks | An ESIA consistent with the requirements for a Category B project is contained in the "Annex 6: Environmental and Social Management Framework (ESMF)". |
| **Environmental and Social Management Plan (ESMP) (if applicable)** | |
| Date of disclosure on accredited entity's website | Friday, January 17, 2025 |
| Language(s) of disclosure | English and Serbian |
| Explanation on language | Serbian is the official language of Serbia, with English widely spoken, and are the languages understandable to affected peoples/stakeholders. |
| Link to disclosure | English:<br>• https://openknowledge.fao.org/handle/20.500.14283/CD3951EN<br><br>Serbian:<br>• https://openknowledge.fao.org/handle/20.500.14283/CD3951SR |
| Other link(s) | FAO disclosure portal (English and Serbian):<br>• https://www.fao.org/environmental-social-safeguards/project-detail/enhancing-the-resilience-of-serbian-forests-to-ensure-energy-security-of-the-most-vulnerable-while-contributing-to-their-livelihoods-and-carbon-sequestration-(forest-invest)/en<br><br>Ministry of Agriculture, Forestry and Water Management of the Republic of Serbia website (English and Serbian):<br>• Оквир управљања заштитом животне средине и социјалним питањима - Управа за шуме Републике Србије<br><br>Vojvodinasume website:<br>• Serbian: Annex-6-ESMF_Serbian_Final.pdf<br>• English: Annex-6-ESMF_English_final.pdf<br><br>Srbijasume website:<br>• Serbian: https://srbijasume.rs<br>• English: https://srbijasume.rs/en/ |
| Remarks | An ESMP consistent with the requirements for a Category B project is contained in the "Annex 6: Environmental and Social Management Framework (ESMF)". |
| **Environmental and Social Management System (ESMS) (if applicable)** | |
| Date of disclosure on accredited entity's website | N/A |
| Language(s) of disclosure | N/A |

| | |
|---|---|
| Explanation on language | N/A |
| Link to disclosure | N/A |
| Other link(s) | N/A |
| Remarks | N/A |
| **Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)** | |
| Description of report | N/A |
| Date of disclosure on accredited entity's website | N/A |
| Language(s) of disclosure | N/A |
| Explanation on language | N/A |
| Link to disclosure | N/A |
| Other link(s) | N/A |
| Remarks | N/A |
| **Disclosure in locations convenient to affected peoples (stakeholders)** | |
| Date | Friday, January 17, 2025 |
| Place | The hard copies will be made available in English and Serbian in the following places:<br><br>**Directorate of Forests,** Omladinskih brigade 1, Belgrade<br><br>**FAO Project Office,** Bul. Zorana Djidjica 64, Belgrade<br><br>**FAO National Correspondent Office in Serbia,** Ministry of Agriculture, Forestry and Water Management, 22-26 Nemanjina St. 11000 BELGRADE<br><br>**Agricultural Extension Offices** in 34 regional centers in Serbia: |

| Region | Address |
|---|---|
| Niš | Leskovačka 4 |
| Valjevo | Birčaninova 128 A |
| Vranje | Marička 1 |
| Kraljevo | Zelena Gora 29 |
| Jagodina | Kapetana Koče |
| Mladenovac | Stojana Novakovića 2 |
| Kragujevac | Cara Lazara 15 |
| Kruševac | Čolak Antina 41 |
| Leskovac | Jug Bogdanova 8a |
| Užice | Dimitrija Tucovića 125 |
| Negotin | Bukovski put bb |

| | | |
|---|---|---|
| | Novi Pazar | 7. jul bb |
| | Čačak | Hajduk Veljkova 43 |
| | Pirot | Srpskih vladara 98 |
| | Smederevo | Železnička bb |
| | Požarevac | Dunavska 91 |
| | Šabac | Vojvode Putnika 26 |
| | Prokuplje | Hajduk Veljkova 43 |
| | Zaječar | Nikole Pašića 37/IV |
| | Knjaževac | Knjaza Miloša 75 |
| | Loznica | Slobodana Penezića bb |
| | Padinska Skela | Industrijsko naselje bb |
| | Bačka Topola | Glavna 103 |
| | Kikinda | Kralja Petra I 49 |
| | Novi Sad | Temerinska 13 |
| | Pančevo | Novoseljanski put 33 |
| | Ruma | Željeznička 12 |
| | Senta | Poštanska 24 |
| | S.Mitrovica | Svetog Dimitrija 22 |
| | Sombor | Staparski put 35 |
| | Subotica | Trg cara Jovana Nenada 15/3 |
| | Vrbas | Kucarski put bb |
| | Vršac | Žarka Zrenjanina 27 |
| | Zrenjanin | Petra Drapšina 15 |
| **Date of Board meeting in which the FP is intended to be considered** | | |
| Date of accredited entity's Board meeting | N/A | |
| Date of GCF's Board meeting | Monday, February 17, 2025 | |

**Note: This form was prepared by the accredited entity stated above.**

# Secretariat's assessment of FP260

| | |
|---|---|
| Proposal name: | Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration (FOREST Invest) |
| Accredited entity: | Food and Agriculture Organization of the United Nations |
| Country: | Serbia |
| Project/programme size: | Medium |

## I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

| Strengths | Points of caution |
|---|---|
| The project targets two of Serbia's most vulnerable sectors, agriculture and forestry, which are critical to the country's economy, food and energy security, and the well-being of both rural and urban communities. These sectors play a pivotal role in reducing poverty and enhancing livelihoods, while also safeguarding environmental health and biodiversity. Moreover, the forestry sector is a key line of defence against floods and landslides, providing essential protection for the landscape and communities. | |
| To accelerate national climate adaptation and mitigation efforts, including through the private sector, the project will:<br>(i) Introduce innovative climate-adaptive silviculture and carbon finance solutions;<br>(ii) Apply a cross-cutting approach that maximizes adaptation, mitigation and poverty reduction benefits;<br>(iii) Strengthen partnerships between national banks, investors and private sector companies focused on decarbonization and forestry; and<br>(iv) Develop digital tools to engage and aggregate private land and forest owners. | |
| The project benefits from significant co-financing from the Government of Serbia in in-kind contributions, underscoring the | While the significant co-financing from the government and targeted leveraged financing from the financial sector and private forest |

| | |
|---|---|
| government's commitment to its national adaptation plan and nationally determined contribution priorities, particularly for the agriculture and forestry sectors. This co-financing will support critical activities, including the rehabilitation of access roads, production and distribution of climate-adaptive, biodiverse seedlings, and monitoring of forest health for pests and diseases. These actions are essential for ensuring the additionality of GCF investments, such as implementing climate-adaptive silviculture, promoting biodiversity in natural forests over monocultures, facilitating knowledge transfer and climate proofing assets.<br><br>Furthermore, the project aims to attract USD 50 million in leveraged private finance from financial institutions in Serbia, as demonstrated by letters of interest. This funding will be directed towards loans for decarbonization investments across various sectors. An additional USD 0.8 million is expected to be leveraged from private forest landowners to support forestry investments. | landowners are key assets of the project, the accredited entity must ensure the timely realization of these funds to meet project objectives. This will require close monitoring and proactive mitigation actions. |

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity (AE) and, if considered appropriate, subject to the conditions set out in Annex II to document GCF/B.41/02.

# II. Summary of the Secretariat's assessment

## 2.1 Project background

3. Serbia is a country with a significant forest cover that plays a critical role in its energy security, ecosystem health and economy. With over 40 per cent of the population, especially rural communities, relying on fuelwood for heating, forests are essential to Serbia's energy supply. However, climate change has intensified the degradation of these forest resources through higher temperatures, irregular rainfall and prolonged droughts, threatening biodiversity and increasing vulnerability to extreme weather events. Over-extraction of fuelwood further endangers forest sustainability, leading to soil erosion, habitat loss and increased carbon emissions.

4. The project titled "Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration (FOREST Invest)", developed in partnership with the Food and Agriculture Organization of the United Nations (FAO), aims to address these challenges through a nexus approach that integrates climate adaptation and mitigation. Key interventions include upscaling climate-

adaptive silviculture, strengthening energy security through climate-resilient forestry and engaging the private sector in decarbonization investments. The project introduces innovative solutions, such as carbon finance frameworks, digital tools to engage private forest owners, and partnerships between national banks and private sector actors. The project also aims to improve the energy supply chain through sustainable biomass products, reducing dependence on unsustainable fuelwood. Targeted support for private sector investments in biomass facilities will enable the creation of a sustainable market for biomass energy, enhancing Serbia's carbon sink capacity while meeting energy needs.

5. Through the proposed interventions, the project aims to benefit a total of 3.57 million people (54 per cent of the total population), with direct benefits reaching over 729,000 individuals, including 371,000 women. Over 27 years, the project is expected to sequester approximately 8.4 million tonnes of carbon dioxide equivalent (t $CO_2$ eq). Co-benefits include biodiversity conservation, gender empowerment and poverty alleviation, aligning with Serbia's climate goals and nationally determined contributions.

6. The total funding required for this project is USD 83.8 million, with a request for USD 25 million from GCF. The Serbian Government will contribute in-kind co-financing amounting to USD 53 million, alongside in-kind support of USD 4.8 million from Serbia Shume and Vojvodina Shume, and a USD 0.4 million grant from FAO. Additionally, USD 50 million is expected to be leveraged from private sector investments. FAO will serve as the AE, while the Serbian Government, through the Ministry of Agriculture, Forestry and Water Management, along with Serbia Shume and Vojvodina Shume, will be executing entities (EEs).

7. The project is classified under environmental and social safeguards (ESS) category B, indicating moderate impacts that are site-specific and manageable through mitigation measures. By enhancing sustainable forestry management and improving energy security, this project will contribute to Serbia's climate resilience, benefiting vulnerable local communities and fostering long-term environmental and economic sustainability.

## 2.2 Component-by-component analysis

*Outcome 1: National-level upscaling of sustainable and climate-adaptive silviculture and carbon finance framework achieved (total cost: USD 9.2 million; GCF cost: USD 1.7 million)*

8. To achieve this outcome, the project will enhance forest management practices and establish a carbon finance framework, enabling Serbia to align with international standards and meet its climate commitments. The outputs and activities set out below are designed to support this goal.

9. **Output 1.1: Forest management and monitoring policy framework for climate-adaptive silviculture enhanced and disseminated**. This output focuses on establishing a continuous National Forest Monitoring system to provide reliable, up-to-date data on forest health, biodiversity and carbon storage. Leveraging FAO Collect Earth software, the system will utilize permanent sample plots to assess changes in forest area, carbon storage and ecosystem health. Activities include training field teams for consistent data quality, with the Directorate of Forests ensuring ongoing monitoring post-project. The project will also develop guidelines to promote sustainable land and forest use, reduce soil degradation and align with Intergovernmental Panel on Climate Change standards, ensuring that these practices are shared widely through national workshops and outreach efforts.

10. **Output 1.2: Enabling framework for national institutions to engage with carbon finance for agriculture, forestry and other land use created**. This output will develop Serbia's capacity to leverage carbon finance by establishing frameworks for carbon offsetting and domestic carbon pricing, specifically within the forestry and the agriculture, forestry and other land use sectors. Key activities include designing a policy framework for carbon offset

mechanisms linked to Serbia's evolving emissions trading system, and providing training to service providers and academic institutions to support long-term carbon finance initiatives. The project will also facilitate Serbia's engagement in international carbon markets through Article 6 of the Paris Agreement, offering technical support for participation in regional carbon finance mechanisms and knowledge-sharing across the Western Balkans.

11. **Output 1.3: National standard for biomass production and handling for energy purposes established**. To improve energy efficiency and reduce emissions, this output will introduce standards for biomass production that mandate a maximum moisture content for fuelwood, enhancing calorific value. Technical regulations, including inspection mechanisms, will be established for biomass suppliers, along with consumer awareness campaigns to promote efficient biomass usage. Pilot programmes in selected municipalities will evaluate the impact of these standards, with renewable energy days held to raise awareness of best practices in biomass usage alongside other renewable energy options.

12. **Output 1.4. National strategy and action plan for short rotation plantations developed**. This output focuses on diversifying Serbia's biomass resources by developing sustainable short rotation plantations (SRPs) for energy, soil rehabilitation and agroforestry. Activities include crafting a national SRP strategy, creating guidelines for sustainable wood energy plantations and training experts in SRP management practices. This strategy is expected to support Serbia's energy security and reduce pressure on natural forests, contributing to the country's carbon reduction and energy diversification goals.

*Outcome 2: Improved energy security and livelihood from climate-resilient forest ecosystem and greenhouse gas emission reductions from increased carbon sinks and decarbonization opportunities (total cost: USD 48 million; GCF cost: USD 14.3 million)*

13. Under this outcome, the project aims to improve energy security and livelihoods by fostering a climate-resilient forestry sector that enhances carbon sinks and decarbonization opportunities. To achieve this, the project will deliver the outputs and activities set out below.

14. **Output 2.1: Production of climate-adaptive seedlings enhanced**. This output focuses on increasing Serbia's capacity for climate-resilient forestation by upgrading public nurseries and seed centres to produce diverse, climate-adaptive seedlings. These facilities will produce over 33 million container seedlings and bare-root plants, targeting 7,000 hectares for afforestation and 51,000 hectares for forest restoration. Activities include selecting native species suited for Serbia's changing climate, upgrading nursery infrastructure and training staff in modern seedling production techniques. This output is designed to enhance biodiversity and improve forest resilience, directly contributing to Serbia's energy security and adaptation goals.

15. **Output 2.2: Knowledge on climate-adaptive silviculture enhanced**. Under this output, the project will equip forest stakeholders with the knowledge and skills to implement climate-adaptive silvicultural practices. Key activities include developing guidelines for climate-smart nursery production, soil preparation, efficient planting methods and maintenance of forest ecosystems. Additionally, capacity development training programmes will be conducted for over 600 individuals in rural communities, prioritizing women and vulnerable groups. These programmes will cover sustainable forest management and include study tours to countries with successful forest and landscape restoration practices. This output aims to support a smooth transition to climate-adaptive forestry, reducing risks to forest health and productivity, and to facilitate regional knowledge-sharing dialogue on climate-adaptive silviculture.

16. **Output 2.3: Public forest land restored and afforested in a climate-adaptive and participatory manner**. This output involves large-scale afforestation and restoration efforts on pre-identified public lands, totalling 7,000 hectares. The project will develop detailed planting plans, conduct consultations with local communities and use climate-adaptive seedlings to ensure ecosystem resilience. A significant focus will be on converting degraded coppice forests into high forests, promoting sustainable management practices and actively involving local

communities in restoration activities. This effort will be supported by substantial co-financing from the Ministry of Agriculture, Forestry and Water Management, which will enable climate proofing and maintenance of existing forest roads. These improvements are critical to ensuring access to restoration sites and facilitating sustainable forestry investments.This transformation is expected to create long-term decarbonization opportunities by shifting from fuelwood to industrial timber production, while maintaining fuelwood access in the short term.

*Outcome 3: Private sector is engaged in climate-adaptive silviculture and decarbonization investments (total cost: USD 22.4 million; GCF cost: USD 7.8 million)*

17. This outcome aims to strengthen the resilience of Serbia's forestry sector by involving private sector actors in sustainable forestry and decarbonization activities. The project will work with private landowners, agribusinesses financial institutions and service providers to foster investments in forestry, promote sustainable biomass value chains and increase decarbonization finance. The outputs and activities set out below are designed to support this goal.

18. **Output 3.1: Private sector contribution to climate resilience of forests enhanced through climate-adaptive and diversified investments and greening of the biomass value chain initiatives**. Under this output, the project will encourage private landowners and businesses to engage in sustainable forestry through activities like converting degraded coppice stands into high forests and rehabilitating unfarmed lands with short rotation plantations. The project will implement a digital cadastre platform to identify interested landowners and facilitate investment clustering for efficient forest management. Key activities include converting degraded private coppice stands into high forest, establishing shelterbelts in agricultural areas and engaging private businesses in sustainable biomass value chains, thereby contributing to Serbia's climate resilience targets.

19. **Output 3.2: Mobilized private finance for agribusinesses involved in decarbonization processes**. This output focuses on building agribusinesses' capacity to invest in decarbonization through technical assistance, strategy development and financial linkage support. Activities include outreach through the existing Digital Platform on Circular Economy technical assistance for decarbonization plans, and collaboration with financial institutions to facilitate access to finance for green investments. The project will prioritize agribusinesses and forest-related firms, supporting their efforts to adopt energy-efficient technologies, reduce greenhouse gas (GHG) emissions and align with international climate compliance standards.

20. **Output 3.3: Financial institutions, consultancy service providers, and academia capacitated on climate-related challenges and opportunities**. This output will enhance the capacity of Serbian financial institutions to assess climate-related risks and invest in decarbonization projects. Activities include training for financial institutions on climate risk assessment, engaging with the Chamber of Commerce and Industry of Serbia, and supporting consultancy providers in environmental, social and governance compliance and GHG accounting. By strengthening the expertise of banks, consultancy services and academia, the project aims to ensure a sustained knowledge base for future private sector decarbonization and resilience investments.

# III. Assessment against investment criteria

## 3.1 Impact potential *Scale: High*

21. The project aims to deliver transformative contributions to Serbia's climate mitigation and adaptation, aligning with targeted results of the updated Strategic Plan for the GCF 2024–2027 (5: Ecosystems; 6: Infrastructure; 7: Clean energy; and 10: Micro, small and medium-sized

enterprises). By addressing critical vulnerabilities in the forestry and energy sectors, the initiative supports Serbia's transition towards a resilient, low-carbon future.

22. On the mitigation front, the project prioritizes afforestation, forest restoration and degraded coppice management, with a targeted net carbon impact of 8.4 Mt $CO_2$ eq. Over 27 years, these efforts will increase carbon removals from the forestry sector by 7.6 Mt $CO_2$ eq and reduce net emissions from the private sector by 0.8 Mt $CO_2$ eq. Key strategies include greening the biomass value chain, decarbonizing agribusiness and introducing carbon finance mechanisms to enable Serbian companies to mitigate emissions in production. Several commercial banks in Serbia have expressed written commitments to support these efforts, demonstrating strong private sector engagement. By mobilizing both public and private financing, the project aligns with Serbia's net zero target by 2050.

23. In terms of adaptation, the project will modernize forestry practices to enhance resilience for rural communities that rely heavily on fuelwood. Climate-adaptive silviculture, coupled with diverse climate-adaptive seedlings, will strengthen forest governance and ecosystem services. Private and community-led sustainable forestry and agroforestry practices will improve energy security, diversify sustainable energy sources and enhance resilience for the most vulnerable populations in rural areas.

24. The project is expected to reach 729,064 direct beneficiaries and 2.8 million indirect beneficiaries, contributing to enhanced resilience, expanded reforested areas and significant $CO_2$ removals. Through its integrated approach to carbon finance, sustainable forest management and adaptive capacity-building, the initiative positions Serbia for a paradigm shift in climate adaptation and mitigation. This aligns local economic resilience with national and international climate targets, driving a sustainable and inclusive transition.

## 3.2 Paradigm shift potential *Scale: Medium to high*

25. The project demonstrates a high potential for paradigm shift by introducing climate-adaptive silviculture practices and continuous forest monitoring, aligning Serbia with international forest management and carbon finance standards. These innovations will enhance ecosystem services, including carbon removals, energy security and livelihoods, while directly supporting Serbia's climate adaptation and decarbonization strategies.

26. By engaging private sector actors and strengthening biomass value chains, the project creates a replicable and scalable model for forest adaptation across Serbia and the wider Balkans. The establishment of national biomass standards and the integration of private sector carbon finance mechanisms lay the groundwork for sustainable forest management, encouraging regional adoption and replication.

27. Key indicators such as climate-adaptive silviculture adoption, implementation of biomass standards and increased private sector participation underscore the project's transformative potential. By embedding sustainable forestry practices at the national level, the project establishes a strong foundation for enhanced climate resilience, adaptation and long-term environmental and economic sustainability.

## 3.3 Sustainable development potential *Scale: High*

28. The project demonstrates strong sustainable development potential, rated as high owing to its socioeconomic and environmental contributions. By supporting energy security for rural and low-income households through high-quality biomass sources, the project aims to reduce energy costs and support rural incomes, contributing to poverty alleviation. The improved efficiency of biomass also reduces household energy expenditure, freeing up

resources for other needs. This socioeconomic impact aligns with the project's goal to improve livelihoods, particularly in rural areas.

29. Environmental co-benefits are substantial, as the project restores degraded lands, establishes shelterbelts and promotes biodiversity. These measures not only enhance local ecosystems but also offer protective services against climate hazards like soil erosion, floods and landslides. By collaborating with the Serbian Water Directorate, the project ensures that afforestation and shelterbelt areas align with national flood and erosion prevention priorities, directly benefiting communities vulnerable to climate impacts.

30. Social and gender empowerment co-benefits are core components of the project. By creating jobs in climate-adaptive nurseries, green biomass and modernizing the biofuel sector, the project will boost rural employment and support poverty reduction through more efficient fuelwood use and enhanced transparency in the biomass sector. Gender inclusion is prioritized, with at least 30 per cent of beneficiaries expected to be women, particularly in activities like land restoration for bioenergy. These efforts align with multiple Sustainable Development Goals (SDGs): supporting sustainable energy (SDG 7), ecosystem restoration (SDG 15) and climate action (SDG 13), while also promoting poverty alleviation and food security (SDGs 1 and 2).

## 3.4 Needs of the recipient *Scale: Medium to high*

31. Serbia is highly reliant on its forestry sector, which faces significant vulnerability due to climate change impacts, especially in rural areas where communities depend on fuelwood, non-timber forest products and other forest-based livelihoods. This project addresses these needs by introducing sustainable forest management practices, developing alternative energy solutions and enhancing climate-adaptive silviculture to build resilience within the forestry sector. By focusing on critical areas – risk reduction, monitoring and policy reform – the project strengthens Serbia's capacity to protect forest resources, ensuring long-term energy security and ecosystem health.

32. The project also addresses key barriers to resilience by establishing a national forest monitoring system, facilitating access to carbon finance and building private sector capacity in sustainable forestry and decarbonization. By reducing technical and financial obstacles, it enables broader adoption of climate-resilient forestry practices. Given the high vulnerability of rural communities and the urgent need for climate adaptation, this project's impact potential is rated medium to high. Key indicators include improved energy access for rural households, funding leveraged for resilience and increased private sector participation in sustainable forestry.

## 3.5 Country ownership *Scale: High*

33. The project aligns closely with Serbia's updated climate goals, including its Low Carbon Development Strategy, Climate Law and national adaptation plan. These frameworks emphasize sustainable forest management, decarbonization and the expansion of renewable energy sources, making forestry a core element in Serbia's ambition to become carbon neutral by 2050. This project, designed through extensive stakeholder consultations, will support Serbia's climate resilience goals by advancing climate-adaptive silviculture practices and introducing innovative carbon finance mechanisms. By addressing forest degradation, enhancing ecosystem services and expanding carbon sinks, the project directly supports Serbia's nationally determined contributions and the broader goal of adapting to climate change while achieving energy security.

34. The Government of Serbia recognizes the importance of sustainable forest ecosystems in ensuring energy security and reducing vulnerability to climate risks. Supported by key

ministries, including the Ministry of Agriculture, Forestry and Water Management, and civil society organizations, the project will implement climate-adaptive silviculture and establish a national forest monitoring system. With national institutions, such as the Directorate of Forests, closely involved, the project integrates seamlessly into Serbia's regulatory framework and monitoring processes. By aligning with the GCF country programme and Serbia's broader climate commitments, the project ensures that forestry investments contribute to long-term resilience, economic development and environmental protection.

## 3.6 Efficiency and effectiveness *Scale: High*

35. The project demonstrates strong economic and financial efficiency, leveraging co-financing from the Serbian Government and private sector investments to achieve its objectives. Through mechanisms such as offsets, insets and concessional finance, it mobilizes private sector participation and encourages forest-owner associations to adopt sustainable practices.

36. Although it has a high level of in-kind contribution as co-financing from the Ministry of Agriculture, Forestry and Water Management, financial models reveal profitable returns, with 20-year internal rates of return (IRRs) exceeding the 7 per cent financial discount rate, making investments in forestry, decarbonization and ecosystem services attractive to private stakeholders and households. This aligns with GCF standards, ensuring cost-effectiveness and efficient resource use in achieving sustainable outcomes.

37. Regarding profitability and viability, the AE provided a detailed analysis of the financial viability of decarbonization and forestry investments, with and without GCF concessionality. For decarbonization loans, the profitability indicators for agribusiness investing in decarbonization technologies, such as energy-efficient boilers, show IRRs of above 30 per cent, with payback periods of approximately five years. These results indicate strong financial viability, even in scenarios with moderate cost increases. As for the forestry investment, financial returns for forestry investments are constrained in the absence of GCF concessionality, with positive IRRs achieved only in the 40-year horizon under 95 per cent concessionality. This justifies the need for financial support from GCF to de-risk these long-term investments.

38. Economically, the project shows robust long-term viability, with 20-year and 40-year economic IRRs of 15.0 and 17.6 per cent respectively, significantly above the 6 per cent social discount rate used in the analysis, and net present values of USD 78.4 million and USD 227.5 million. These figures underscore the significant ecosystem and carbon benefits generated by the project. Sensitivity analyses demonstrate resilience to fluctuating carbon prices, seedling survival rates and changes in labour costs. The results indicate that the project remains robust under most scenarios, although lower carbon prices (USD 10/t $CO_2$ eq) slightly reduce its economic returns.

# IV. Assessment of consistency with GCF safeguards and policies

## 4.1 Environmental and social safeguards

39. **Project overview.** The project aims to support Serbia in enabling the forestry sector to contribute to the country's low-carbon strategy by stabilizing and increasing carbon removals, upgrading management capacities of key institutions and communities and incentivizing private sector companies to engage in the decarbonization process. The main environmental and social co-benefits include biodiversity conservation through restoration of local forests and afforestation in a climate-adaptive way; soil quality improvement; increased water availability; decrease in evapotranspiration; protection of communities from floods and landslides; poverty

reduction through increased livelihood and employment opportunities; and gender empowerment through the prioritization of women's participation in project activities.

40. **Environmental and social risk category and safeguard instrument**. On the basis of the AE assessment, the project is categorized as B, having moderate impacts largely due to works associated with the forest establishment and management of smallholder forest owners and municipalities in Central Serbia and Vojvodina and agroforestry in agricultural areas. This will include upgrading the capacity of existing public nurseries in support of forest expansion in the target areas , afforestation of at least 7,000 hectares of land, and restoration of at least 51,000 hectares of degraded forests, establishment of planted forest on at least 500 hectares of unfarmed lands and establishment of 500 hectares of shelterbelts in agricultural lands. At the national level, the project will also involve regulatory framework improvements, capacity development of forest operators, policy upgrades and institutional strengthening. Potential environmental and social impacts are limited, localized and can be readily mitigated. No significant adverse environmental and social impacts are foreseen. The proposed categorization is aligned with the accreditation level of the AE and the GCF Environmental and Social Policy. An environmental and social management framework (ESMF) has been prepared and budgeted; specific sites and relevant activities will be identified during implementation.

41. **Compliance with GCF's ESS standards**. The following paragraphs describe how the project complies with the standards.

42. **ESS 1: Assessment and management of environmental and social risks and impacts**. The project's ESMF provides a general description of the project areas; potential environmental and social risks and impacts; environmental and social standards triggered; mitigating measures; implementation arrangements with regard to safeguards implementation; procedures for screening and assessment once specific sites and activities have been identified; capacity-building needs for safeguards implementation; grievance redress mechanism; information disclosure requirements; and the stakeholder engagement process, which is further detailed in annex 7 to the funding proposal package. The project used the FAO Environmental and Social Screening Form and will continue to use it during implementation, supplemented with a non-eligibility list. The AE has initially identified potential environmental and social impacts related to on-the-ground activities in Vojvodina and Central Serbia. These will be re-assessed and documented through the subproject-specific environmental and social impact assessment and environmental and social management plan, to be prepared prior to the commencement of subproject activities. The AE will ensure that the mitigation measures and requirements will be in line with FAO and GCF ESS policies and standards and national legislation, whichever is the more stringent.

43. **ESS 2: Labour and working conditions**. The project will operate in areas where there is gender inequality and in situations where youth work as unpaid family workers, lack access to decent jobs and are increasingly abandoning agriculture and rural areas. Labour and working conditions will be addressed as per ESS 4 (Decent work) of the AE. The project will promote, respect and realize fundamental principles and rights at work. The employment of project workers will be based on the principle of equal opportunity and fair treatment, and there will be no discrimination with respect to any aspects of the employment relationship such as recruitment and hiring, compensation, working conditions and terms of employment, access to training, job assignment, promotion, or termination of employment or retirement. Workers will be engaged following Serbian rules and regulations under the Labour Law (Official Gazette of the Republic of Serbia 24/05, 61/05 and 54/09) and workers will need to abide with the FAO code of conduct and FAO policies. In hiring workers, priority will be given to locals, specifically from within a 25-kilometre radius for each activity, as described in the eligibility criteria. The project will ensure that no workers under the age of 18 will be employed. The non-eligibility list in the ESMF includes all harmful or exploitative forms of child labour or forced labour.

44. With regard to worker's health and safety, FAO has identified the following mitigating measures: (i) comply with general rules and regulations on Occupational l Health and Safety Risks (OHSR); (ii) ensure workers are equipped with protective gear, as applicable; (iii) ensure the availability of first aid kits at work sites and necessary information on rescue during emergencies; (iv) ensure workers are trained on Occupational Health and Safety Risks (OHSR) prevention and management on site; and (v) follow World Health Organization and national guidance on prevention of the spread of communicable diseases such as coronavirus disease 2019, as applicable.

45. **ESS 3: Resource efficiency and pollution prevention**. The project will promote climate-adaptive silviculture, which enhances resources efficiency. The project will not lead to significant impacts on water resources or increased use of pesticides through intensification or expansion of production. In the establishment and management of forests, only local species with wide ecological ranges and higher drought resistance will be selected. The project will not procure or use introduced species or genetically modified organisms. Waste generation is expected from activities such as upgrading of nurseries, thinning operations in regular intervals so that young forests can grow in optimal conditions and maximize biomass production, and from road rehabilitation and maintenance works (involving clearing of biological debris) and fencing (either biological or structural). These activities are not expected to generate significant amounts of waste; any waste will be disposed of in designated areas. Available good international industry practices and guidelines will be adopted to mitigate impacts, including those of the World Bank Group. Specific mitigating measures will be detailed in the subproject environmental and social management plan to be prepared.

46. **ESS 4: Community health, safety and security**. The forest activities are generally located in remote areas, far from residential areas and communities. Planting techniques will take into consideration fire prevention through tree spacing and fire breaks. As reflected in the eligibility criteria for afforestation of public lands, a buffer zone without forest cover will be maintained in afforestation areas to prevent potential forest fire expansion. There is no significant influx of workers requiring the establishment of camps or temporary accommodation, and engagement of migrant workers is not expected. Potential risks related to community health and safety will be addressed through the provision of training and personal protective equipment , as needed. Road clearing activities are seen to be beneficial in ensuring accessibility in the case of fire occurrence.

47. **ESS 5: Land acquisition and involuntary resettlement**. The ESMF reflects that there will be no land acquisition or involuntary resettlement or displacement resulting from the project activities. Landowners in Serbia are clearly identified via the digital cadastre and no investment will be made without the consent of landowners, communities and municipalities to ensure that lands are free from any encumbrances. The project activities will be executed only on land owned by the State or by farmers with clear ownership free from any dispute. The project will not result in expropriation of lands or planting on land of dubious ownership. The project will work on land that is no longer suitable for agriculture and not in use from a productive perspective. Exercise of eminent domain or any other permanent or temporary, economic or physical displacement due to involuntary resettlement will not be supported under the project. With regard to road rehabilitation, works will be small in scale, limited to existing access roads, and will only involve cleaning of drainage systems, using equipment such as shovels, hoes and rakes, hand tampers and portable pumps. Roads will be cleared from debris to provide access for afforestation, management and fire protection of newly established forests. Road users from the communities will be consulted and informed of schedule of works ahead of time.

48. **ESS 6: Biodiversity conservation and sustainable management of living natural resources**. The ESMF describes Serbia as a global centre for biodiversity in the region characterized by great genetic, species and ecosystem diversity. However, climate change and

the rate of forest degradation are causing biodiversity issues, such as less diverse species composition and less structured forests, that impact the habitats of wildlife and wildlife migration, and occurrence of pest and disease outbreaks. The ESMF indicates that the project will not significantly impact biodiversity and natural habitats but will support enhancing the resilience of forest ecosystems through sustainable forest management practices and adoption of climate-adaptive silviculture. The project will restore forests which will act as corridors within existing forests. The project will plant only with native or locally adapted species, involving local communities in the responsible management of planted forests. The non-eligibility list covers the destruction of protected areas or other high biodiversity and high conservation value areas. Afforestation activities and shelterbelts are intended to create corridors for wild flora and fauna. Potential negative impacts associated with the on-the-ground forest activities related to afforestation, restoration of damaged forests, shifting degraded coppice stands to high forest and establishment of shelterbelts will be mitigated following good international industry practices, and guidelines will be adopted, including those of the World Bank Group. Any potential negative environmental and social impacts on a specific subproject site and its area of influence will be discussed in detail during preparation of the environmental and social impact assessment and environmental and social management plan. Biodiversity assessments will also be conducted to ensure that the project areas are not considered as critical habitats and that appropriate mitigating measures are adopted. An international expert will be engaged to undertake biodiversity assessments, ensuring that the mitigating measures will be appropriate and in line with national and GCF ESS requirements.

49. **Indigenous Peoples Policy and ESS 7: Indigenous Peoples**. The AE will meet the requirements of the GCF Indigenous Peoples Policy by applying its relevant policies in a manner that at all times meets GCF requirements. The AE has legally committed to ensuring that the funded activities are at all times in compliance with the applicable requirements of the GCF Indigenous Peoples Policy. This means that the mechanisms in place can be considered sufficient to safeguard Indigenous Peoples. The AE engaged with authorities and civil society organizations to determine the presence of Indigenous Peoples. Responses from consulted groups confirm the data extracted through desk research that no Indigenous Group is known in the target areas. Stakeholder consultation and second-level screening will be held once specific project sites are identified. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the AE, who is also encouraged to share emerging good practices and success stories with the group.

50. **ESS 8: Cultural heritage**. The project does not anticipate impacts on cultural heritage sites. However, in the case of any discovery of artifacts, chance find procedures will be followed. The project will exclude any activity that is considered as illegal under host country laws, regulations or ratified international conventions and agreements with regard to cultural heritage.

51. **Implementation arrangements**. A project management unit (PMU) will be established consisting of an ESS Specialist to implement the ESMF for the entire project duration. The ESS Specialist will be responsible for ensuring overall compliance with the ESMF, provide capacity-building with regard to the execution of environmental and social management plans, disseminating and managing the grievance redress mechanism, and monitoring of safeguards performance, among other tasks described in the ESMF. The ESS Specialist will work closely with the Gender Specialist and Monitoring and Evaluation Specialist on matters regarding stakeholder engagement. Safeguards implementation at the PMU-level will be overseen by the FAO Serbia Representative Office, the Project Steering Committee representing the line ministries and other stakeholders, including representatives of the private sector, and the FAO Regional Office. A monitoring and evaluation system will be established as part of the project, which includes collection and consolidation of safeguards performance. The implementing partners will be submitting reports to the PMU as a basis for the annual performance reports as required by GCF. Such monitoring reports will serve as a basis for any corrective actions needed

during implementation. Capacity-building activities on safeguards implementation (including training and awareness-raising) will be held at different management levels and at the national level.

52. **Stakeholder engagement**. A series of engagements with direct and indirect stakeholders, including representatives from government offices, non-governmental organizations, community-based organizations, universities and research institutes, conservation groups and private sector operators, as well as associations, unions and service providers, etc., were undertaken at the project design and formulation stage. Key concerns raised were on climate rationale, relevant climate change adaptation and mitigation targets, the project approach, including the expected paradigm shift, as well as target area and site selection. Stakeholders agreed on needs to be addressed, targets, implementation arrangements and modalities, time frame and budget. Such key concerns were addressed and incorporated into the project design. The inclusion of civil society and private sector organizations in the Project Steering Committee will contribute to ensuring that consultations remain free, open, inclusive and well documented. Stakeholder consultations will continue during implementation and will be guided by a stakeholder engagement plan.

53. **Grievance redress mechanism**. A project-level grievance redress mechanism will be implemented at the field level to address project-related grievances during implementation, as reflected in the ESMF and stakeholder engagement plan. The PMU, through its Safeguards Specialist, who will be acting as the grievance redress mechanism focal point, together with the Project Coordinator, will be jointly managing grievances related to environmental and social standards. Any aggrieved party may also resort to the FAO Serbia Representative Office, the FAO Regional Office for Europe and Central Asia and the FAO Office of the Inspector General, if needed. The GCF Independent Redress Mechanism will also be made available to the stakeholders. The grievance redress mechanism will not impede access to the country's judicial or administrative remedies. Information on the project's grievance redress mechanism will be communicated during implementation, as described in the stakeholder engagement plan.

54. **Sexual exploitation, abuse and harassment**. The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from sexual exploitation, abuse and harassment (SEAH) in GCF-financed activities. The AE provided SEAH safeguarding in its submission for this funding proposal. The ESMF of the project assessed the potential SEAH risks of this specific project, identified measures for preventing SEAH and provided measures on how the project will adequately respond to SEAH cases. Among other SEAH safeguarding measures throughout this project, the AE developed a comprehensive checklist outlining all the existing SEAH frameworks and gaps in the country from a SEAH risk management perspective. These are outlined at different levels, including stakeholders engaged from a SEAH risk assessment perspective, country-level baseline data and project-specific SEAH risks. The AE has put in place various SEAH-related mitigation measures (included in the gender action plan). The project will also leverage on other like-minded institutions in the country, and capacity-building activities such as SEAH training will be done in coordination with the United Nations Population Fund, the United Nations Entity for Gender Equality and the Empowerment of Women, the United Nations Office for the Coordination of Humanitarian Affairs and other community stakeholders. The SEAH grievance redress mechanism will be survivor centred and gender responsive in nature – with confidentially at the core of the Investigation Guidelines of the FAO Office of the Inspector General as a guiding framework. The AE has SEAH protection and gender-based violence capacity to facilitate the SEAH measures put in place for the project.

## 4.2 Gender policy

55. The AE has provided a gender analysis and action plan and therefore complies with the GCF Gender Policy. The gender assessment highlights significant barriers to gender equality in

Serbia's rural areas and forestry sector. Key challenges include entrenched patriarchal norms and stereotypes limiting women's roles in decision-making, economic participation and resource ownership. Rural women face restricted access to education, technology and training, leading to lower employment rates and fewer leadership opportunities. Women are also overrepresented in unpaid or informal work and underrepresented in forestry education and technical roles. Additionally, institutional weaknesses, such as inconsistent gender mainstreaming and inadequate coordination among public institutions, further exacerbate gender gaps in rural and forestry-focused interventions.

56. The gender action plan outlines measures to address these challenges by integrating gender equality into forestry governance and rural development. It defines activities, indicators and targets for women's participation and benefit with corresponding timelines, budget and gender expertise. Key actions include targeted capacity-building initiatives for women, promoting their involvement in decision-making processes, and fostering access to education, training and green job opportunities. The plan emphasizes gender-responsive budgeting, the use of sex-disaggregated data for better policy implementation and strengthening institutional mechanisms to promote inclusion. Furthermore, it supports increasing women's leadership and entrepreneurship opportunities while challenging gender stereotypes through community engagement and awareness-raising campaigns.

## 4.3 Risks

### 4.3.1. Overall programme assessment (medium risk)

57. The project aims to enhance climate resilience and sustainable livelihoods for vulnerable populations by promoting climate-adaptive forestry practices, improving energy security and mobilizing private sector investments for decarbonization in Serbia. The project is designed to have three components: (i) national-level upscaling of sustainable and climate-adaptive silviculture and carbon finance framework; (ii) improving energy security and livelihoods from a climate-resilient forest ecosystem and GHG emission reductions through increased carbon sinks and decarbonization opportunities; and (iii) engaging the private sector in climate-adaptive silviculture and decarbonization investments. Total project costs are estimated at USD 83.8 million. The total comprises (i) a GCF grant of USD 25.0 million; (ii) FAO contribution of USD 0.5 million; and (iii) in-kind contribution from the Government of Serbia of USD 53.5 million and two State-owned companies of USD 4.9 million for a co-financing ratio of 1:2.35. The proposed co-financing amounts from other sources are considerable and will be provided in-kind. However, the vast majority will come via government contributions, which will be sourced from the approved budget programme deemed essential to achieving the climate impact, including the rehabilitation of rural roads, investment in forest fire monitoring and monitoring of forest health. Regular monitoring of financial contributions, close collaboration with co-financiers and maintaining contingency funds to cover cost overruns will be conducted.

### 4.3.2. Accredited entity/executing entity capability to execute the current programme (low risk)

58. FAO serves as the AE for the project in Serbia, overseeing its execution and collaborating with the Ministry of Agriculture, Forestry and Water Management and two public entities, Serbia Shume and Vojvodina Shume (acting collectively as EEs). Serbia Shume and Vojvodina Shume have been playing a key role in the supervision of forests in Serbia (at both the national and the regional level). With a dedicated PMU and technical experts, FAO ensures effective implementation, drawing from its extensive track record in similar contexts, including past collaboration with GCF on 20 projects. This experience underscores the capacity of FAO to

implement the project and navigate challenges, ensuring compliance with GCF policies, regulations and the accreditation master agreement, including the funded activity agreement, fostering confidence on the part of the Secretariat in the AE and the capacity of the EEs to execute the project activities.

#### 4.3.3. Programme-specific execution risks (medium risk)

59. **Co-financing**: the Secretariat acknowledges the sizable co-financing totalling USD 58.4 million from the government and public entities, resulting in a co-financing ratio of 1:2.35. If co-financing does not materialize, this may impede project implementation. As such, the incorporation of covenants and conditions regarding the reporting of the co-financing status in the term sheet/funded activity agreement are considered adequate mitigation measures. The funds are also already earmarked by the government and its entities. Additionally, it should be noted that Serbia has been upgraded from BB+ to BBB– by Standard & Poor's (Ba2 and BB+ by Moody's and Fitch respectively), thus gaining an investment grade rating on the back of strong gross domestic product growth and foreign direct investment. The government debt dropped to a moderate 48.5 per cent of gross domestic product, providing additional room for fiscal spending in the mid-term.

#### 4.3.4. Compliance risk (medium risk)

60. The inherent risk for the proposed project primarily arises from its diverse stakeholder landscape, which includes both government and selected private sector involvement in forest and biomass investments. While the project operates in a relatively low-risk jurisdiction, according to the FAO risk assessment, the inclusion of private sector partners introduces potential vulnerabilities in anti-money-laundering and countering the financing of terrorism (AML/CFT) compliance, as these entities may have varying degrees of experience and adherence with AML/CFT standards. This diversity in compliance maturity could lead to vulnerabilities in monitoring financial flows, increasing the possibility of unintended money-laundering and financing terrorism (ML/FT) exposure. While FAO internal controls and oversight mechanisms provide a strong foundation for risk management, the project's reliance on varied partners necessitates continuous vigilance to effectively address potential AML/CFT vulnerabilities associated with both public and private sector involvement.

61. To address these risks, FAO has instituted measures to mitigate ML/FT risks, which are embedded within its existing internal controls and reinforced by project-specific strategies. Key controls include binding AML/CFT clauses within all agreements with the Government of Serbia and relevant third parties, mandating adherence to FAO policies on prohibited practices and AML/CFT compliance. FAO internal protocols, such as an annual fraud prevention plan and a semi-annual review of the risk log, establish proactive mechanisms to detect and respond to potential financial crime indicators. The FAO Enterprise Risk Management Unit further monitors compliance with these controls, ensuring alignment with international AML/CFT standards. Additionally, the integration by FAO of United Nations Security Council sanctions compliance requirements into project agreements strengthens the project's capacity to prevent any indirect support to entities associated with terrorism. This multi-tiered control framework is complemented by rigorous first-level due diligence assessments and oversight from FAO regional offices, ensuring compliance with AML/CFT policies during the entire project lifecycle.

62. Considering the extensive controls and mitigation strategies in place, the residual risk level for this project is assessed as medium. This level reflects the project's reliance on multiple stakeholders to uphold stringent AML/CFT policies across both public and private sectors. However, the consistent application of FAO internal policies, including a zero tolerance approach to fraud, regular audits and inclusion of AML/CFT clauses in all agreements with third parties, significantly reduces the probability of non-compliance. With the FAO established

multilayered approach to AML/CFT risk management, the project aligns with the GCF high standards for fiduciary and compliance integrity, positioning it effectively to meet the GCF AML/CFT and governance requirements.

#### 4.3.5. GCF portfolio concentration risk (low risk)

63. The impact of this proposal on the GCF portfolio concentration in terms of results area and single proposal is estimated as immaterial.

#### 4.3.6. Recommendation

64. It is recommended that the Board consider the above factors in its decision.

| Summary risk assessment | | Rationale |
|---|---|---|
| Overall programme | Medium | The funding co-financing amounts from other sources is considerable. The proposal has an overall risk assessment of medium based on the issues highlighted. |
| Accredited entity/executing entity capability | Low | |
| Project-specific execution | Medium | |
| GCF portfolio concentration | Low | |
| Compliance | Medium | |

## 4.4 Fiduciary

65. FAO will serve as the AE for the project and will be responsible for its overall oversight. It will also be the EE of the GCF funds. The Government of Serbia, acting through the Ministry of Agriculture, Forestry and Water Management, together with Serbia Shume and Vojvodina Shume, will be EEs implementing activities funded by their own grants and in-kind co-financing.

66. As AE, FAO will carry out supervision of the project, and will ensure that financial management and the procurement of goods and services using GCF resources adhere to relevant FAO rules and regulations, as well as relevant provisions in the accreditation master agreement between FAO and GCF.

67. Financial management and procurement executed by FAO as the EE will be overseen and supervised by the FAO-GCF project supervision team. The project supervision team will undertake regular supervision missions and recruit a qualified, internationally recognized auditing firm to perform frequent spot checks and audits to ensure that financial management and procurement by the PMU and EEs are conducted in line with agreed standards and practices.

68. The Secretariat notes that the entire share of co-financing amounts to USD 58,811,581, which is 70 per cent of total funding proposed for the project, is contributed in the form of in-kind contribution, mainly by the Ministry of Agriculture, Forestry and Water Management.

## 4.5 Results monitoring and reporting

69. As a cross-cutting initiative, the project aims to achieve both mitigation and adaptation results. It is expected to reduce or avoid 7.6 Mt $CO_2$ eq over its 27-year lifespan. Additionally, an estimated total of 3,475,444 direct and indirect beneficiaries are anticipated to receive adaptation benefits, representing 11 per cent (direct beneficiaries) and 43 per cent (indirect

beneficiaries) of the target country's population. The AE has provided definitions and methodologies to estimate these adaptation benefits and the corresponding beneficiaries. Several rounds of discussions were conducted before the final numbers were determined, and the AE was very responsive to the comments and guidance from the Secretariat.

70. The theory of change diagram effectively captures different levels of expected changes and the logical linkages between them, from the goal statement to the proposed activities. It is commendable that the diagram identifies co-benefits achievable alongside climate results and illustrates the causal relationships between outcomes and co-benefits. The logical framework included relevant GCF core indicators to monitor the key mitigation and adaptation results within the chosen GCF results areas. Furthermore, it is laudable that the AE has established project-specific outcome indicators to measure various aspects of change that cannot be captured by GCF core indicators. Additionally, relevant indicators have been identified for project co-benefits.

71. It is recognized that the project will mobilize private finance for agribusinesses involved in decarbonization processes as part of its activities. This result is also captured in the logical framework as one of the outcome indicators. Since the AE has already included the avoided emissions through decarbonization of the private sector in the GHG accounting as a component of the overall private investment in annex 3, the Secretariat would like to emphasize that the process of monitoring GHG emissions from private investment should be carefully designed and agreed upon with relevant private sector partners.

## 4.6 Legal assessment

72. The accreditation master agreement was signed with the AE on 8 June 2018 and it became effective on 4 October 2018. The five-year accreditation term of the AE lapsed on 3 October 2023 but was extended through decision B.37/18, paragraph (q), until the earlier of three years from the date of lapse of the five-year accreditation term or the date on which a revised accreditation framework is adopted by the Board. The Board approved the re-accreditation of the AE pursuant to decision B.37/18, and negotiation of the amended and restated accreditation master agreement is ongoing.

73. The AE has provided a certificate confirming that it has obtained all internal approvals, and it has the capacity and authority to implement the project.

74. The proposed project will be implemented in Serbia, a country in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in such countries may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

75. Therefore, it is recommended that the Board consider whether disbursements of GCF proceeds should only be made after GCF has obtained satisfactory protection against litigation and expropriation in the country or has been provided with appropriate privileges and immunities for GCF and its personnel.

76. To address the matters raised in this section, it is recommended that any approval by the Board be made subject to the following conditions:

(a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

# Independent Technical Advisory Panel's assessment of FP260

| | |
|---|---|
| Proposal name: | Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration (FOREST Invest) |
| Countries | Serbia |
| Accredited entity: | Food and Agriculture Organization of the United Nations |
| Project/programme size: | Medium |

## I. Assessment of the independent Technical Advisory Panel

1. Forests play a crucial role in Serbia's economy and energy security. Biomass accounts for nearly 12 per cent of the country's energy consumption. Firewood is used for heating by over 40 per cent of the households, most of them in rural areas. However, both climate change and excessive wood extraction contribute to forest degradation, increasing vulnerability and negatively affecting the contribution of forests to reduced net greenhouse gas (GHG) emissions.

2. By adopting climate-adaptive forest management practices and promoting renewable biomass as a sustainable source of energy, the proposed project – FOREST Invest – seeks to improve climate resilience, livelihoods and energy security in Serbia.

3. The project has three components whose combined outputs contribute to three broader outcomes, respectively:

(a) Outcome 1 involves national-level upscaling of the sustainable and climate-adaptive silviculture and carbon finance framework, including output 1.1 (Forest management and monitoring policy framework for climate-adaptive silviculture enhanced and disseminated) and output 1.2 (Enabling framework for national institutions to engage with carbon finance for the agriculture, forestry and other land use (AFOLU) sector created);

(b) Outcome 2 is "Improved energy security and livelihoods from climate-resilient forest ecosystems and GHG emission reductions from increased carbon sinks and decarbonization opportunities", including output 2.1 (Production of climate-adaptive seedlings enhanced), output 2.2 (Knowledge on climate-adaptive silviculture of key stakeholders enhanced), and output 2.3 (Public forest land restored and afforested in a climate-adaptive and participatory manner); and

(c) Outcome 3 involves private sector engagement in climate-adaptive silviculture and decarbonization investments. It includes output 3.1 (Private sector contribution to the climate resilience of forests enhanced through climate-adaptive and diversified investments and greening of the biomass value chain initiatives), output 3.2 (Mobilized private finance for agribusinesses involved in decarbonization processes), and output 3.3 (Financial institutions, consultancy service providers, and academia capacitated on climate-related challenges and opportunities).

4. The programme will be implemented through a set of executing entities: the Food and Agriculture Organization of the United Nations (FAO), the Ministry of Agriculture, Forestry and Water Management (MOAFWM), and two public forest management enterprises (Serbia Shume

(PES) and Vojvodina Shume (PEV)). FAO will manage the GCF funds, and the other executing entities will manage interventions to be implemented through their own co-financing.

5. The project has an implementation period of 7 years (84 months) and a total lifespan of 27 years. The total budget amounts to USD 83.8 million, counting on a GCF grant contribution of USD 25 million.

6. The classification under environmental and social safeguards as category B suggests moderate, site-specific and manageable risks for adverse effects on people and the environment.

7. In its assessment, the independent Technical Advisory Panel (iTAP) considered the document package as submitted on 25 November 2024, along with some complementary analysis and minor edits to some of the documents on 20 December. Written responses to questions posed by iTAP were received on 13 December, and additional information was provided during a zoom call between the iTAP, the accredited entity and the GCF Secretariat on 17 December.

## 1.1 Impact potential *Scale: High*

8. This cross-cutting project contributes to both mitigation and adaptation results, with most of the funding (two thirds of the GCF contribution and nearly three quarters of the co-financing) aimed at producing adaptation results. The most important result areas in terms of budget are adaptation result area 3 (ARA3) (Infrastructure and built environment), using 52 per cent of the co-financiers' contribution, and ARA4 (Ecosystem and ecosystem services), using 50 per cent of the GCF allocation.

9. According to the Notre Dame Global Adaptation Initiative (ND-GAIN), Serbia has relatively low vulnerability to climate change and relatively high readiness, suggesting that adaptation challenges exist, but that Serbia is well positioned to adapt.[1] Yet, as reportedly pointed out in Serbia's national communications to the United Nations Framework Convention on Climate Change, agriculture and forestry are the most climate-vulnerable subsectors in Serbia. These sectors are also vital for the country's economy. Hence, whereas vulnerability may not be high in relative terms, the project specifically targets the most climate-vulnerable sectors in Serbia.

10. Over a million people are expected to benefit from the stakeholder involvement and dissemination of knowledge about climate-adaptive silviculture (output 2.2). Almost 800,000 people are expected to benefit from the enhanced production of seedlings (output 2.1), and the land restoration and afforestation activities (output 2.3) are expected to benefit some 660,000 people. Another 650,000 people are expected to benefit from the private sector's contribution to green and climate-adaptive investment (output 3.1), and nearly 560,000 people stand to benefit from the upscaling of the climate-adaptive silviculture and carbon finance framework, and especially the forest management and policy framework (output 1.1).

11. Altogether, the project is expected to benefit nearly 3.6 million people (0.73 million direct beneficiaries and 2.85 million indirect beneficiaries), which is close to half of the population of the country. The beneficiaries include forest owners, wood consumers, smallholders affected by soil degradation and those smallholders and municipalities interested in establishing windbreaks. Assumptions include directly benefiting the poor and those at risk. Indirect beneficiaries include 174 municipalities and 882,482 individual forest owners. Annex 25 specifies the assumptions and the calculation of the direct and indirect beneficiaries at the output level.

[1] Among the 187 countries ranked in the ND-GAIN index, Serbia is the 97th most vulnerable country to climate change and ranked 80th in readiness (https://gain.nd.edu/our-work/country-index/rankings/).

12. The FOREST Invest project also seeks to sequester around 8.4 million tonnes of carbon dioxide ($CO_2$) over its 27-year lifespan. This is to be achieved by greening the biomass value chain, decarbonizing agribusiness, enabling carbon finance mechanisms for Serbian companies to mitigate emissions in their production processes, as well as leveraging private sector financing. By engaging private commercial actors in sustainable forestry, the project seeks to mobilize both public and private sectors in working towards Serbia's climate goals, particularly its net-zero target by 2050.

13. The mitigation impact potential was prepared using the Nationally Determined Contribution Expert Tool (NEXT), which was recently launched and fits well with the specific context in Serbia. Annexes 22a and 22b to the funding proposal clearly document the methodology for determining the baseline, the data used for carbon accounting, and the calculation of the mitigation impact potential. Furthermore, the funding proposal clarifies how the monitoring information and data will be gathered and analysed in a way that facilitates the understanding of progress at the field level as well as links to the national measurement, reporting and verification system of Serbia.

14. The project stands to effectively work with the most climate-vulnerable parts of the Serbian economy. The impact potential is assessed to be high.

## 1.2 Paradigm shift potential *Scale: Medium to high*

15. The theory of change statement reads:

> "IF knowledge and technology, coupled with climate-sensitive forestry investments, are made available in Serbia, and transparent, inclusive governance mechanisms for forests and carbon are established, THEN the country will have an enabling environment to foster cooperation between private and public actors to increase and sustainably and adaptively manage forests, reduce the exposure of the most vulnerable, and achieve its strategic adaptation and mitigation targets, BECAUSE these investments will strengthen forest ecosystems, enhance energy security, and support vulnerable communities—particularly women and girls—while increasing carbon storage and advancing decarbonization within the agrifood sector" (revised funding proposal, section B.2 (a), para. 12).

16. The statement is rather complex and multilayered, but emphasizes knowledge and governance mechanisms, areas which stand to enhance the potential for paradigm shift in terms of changing behaviours and generating change beyond a one-off investment. In this sense, a paradigm shift stands to be the main outcome, to the extent that the ambition to decarbonize the wood-fuel industry, greening its value chain and reducing the climate vulnerability of the forestry sector, be realized.

17. The project aligns a set of activities to support forest management, monitoring and methods for climate-smart silviculture, and it engages with the enabling framework for AFOLU carbon finance. Several commercial banks in Serbia have expressed their interest through written commitments. This demonstrates high potential for the leveraging and mobilization of additional resources.

18. The project also engages with private/individual land and forest owners, for example through digital tools, to support the scaling up of technologies, practices and investments to the national scale and enable the private sector to transform into a change agent towards decarbonization and climate resilience.

19. Extensive training of foresters and working closely with and through the two major (public) enterprises in the country, PES and PEV, enhances the potential for sustaining the improved and climate-resilient forest management over time.

20. As discussed in section 1.3 below, given the male dominance in the forestry sector, there is an opportunity for a paradigm shift to the extent that the project successfully challenges entrenched stereotypes and produces gender-equal outcomes.

21. Altogether, the paradigm shift potential – making the forestry sector and wood-fuel production more sustainable and climate-resilient using proven experience – is assessed as medium to high.

## 1.3 Sustainable development *Scale: Medium to high*

22. The cross-cutting approach of the project involves ensuring that investments in adaptation also yield mitigation as well as poverty alleviation results. FOREST Invest seeks to contribute to several of the Sustainable Development Goals (SDGs). The funding proposal states that wider benefits of the project includes contributions towards poverty reduction (SDG 1), food security through management of agricultural soils (SDG 2), health gains from reduced air pollution resulting from improved biomass combustion efficiency (SDG 3), decarbonization through the promotion of renewable energy and energy efficiency (SDG 7), climate action (SDG 13), as well as the sustainable use of terrestrial ecosystems through climate-adaptive silviculture (SDG 15).

23. Environmental co-benefits are derived from positive impacts on biodiversity, soil quality and water resources. Climate-adaptive agriculture and forestry management stand to reduce soil erosion and the incidence of floods and landslides. Afforestation and shelterbelts/windbreakers will create corridors and shelter for wild flora and fauna.

24. Economic co-benefits are to be produced by improving the wood fuel quality (drier wood with higher calorific potential) and unit price of wood biomass used for fuel. Households in Serbia reportedly spend on average more than 10 per cent of their income on energy. Awareness campaigns to disseminate methods for improving combustion efficiency should also help reduce – or alleviate the increase in – the consumption of fuelwood. The project also seeks to support income diversification, for example by using forest land for beekeeping or developing ecotourism. The degraded private agricultural lands to be converted to bioenergy plantations also stand to generate income.

25. Social co-benefits: The methods, practices and technologies introduced by the project should contribute to the creation of new jobs and markets, for example involving climate-adaptive nurseries as well as green biomass and $CO_2$ management. The improved quality of firewood may alleviate the plights of poor households, with energy efficiency gains that stand to increase energy efficiency at the household level. Modernization and the greater transparency of the biofuel business should increase economic opportunities for more people. Consumers should also acquire greater knowledge about fuelwood types and heating appliances to support better-informed consumption decisions and firewood utilization.

26. The project will ensure gender-sensitive development impact. According to the written questions and answers between iTAP and FAO, the "specialized gender expert will ensure gender considerations are effectively integrated in all project activities" (response to question 3).

27. According to the Gender Assessment and Action Plan (see annex 8 to the funding proposal), there is a significant gender gap in the country, with rural women having an unfavourable economic situation. Also, gender inequalities in real estate ownership are very

pronounced, as "women are much less likely than men to own or co-own property" (see page 30 of the funding proposal).

28. There is no data on male/female ownership of forests in Serbia since forest land is not included in the farm registry. Related to farming, however, the Farm Structure Survey from 2018 suggests that 81 per cent of farm holders and 85 per cent of farm managers are men (see page 28 of annex 8 to the funding proposal).

29. Trained female forestry professionals tend to be more present in research and nature conservation than in timber production. Annex 8 to the funding proposal explains that while there is a lack of statistics, "available data indicates that women also tend to be more present in forestry societies, play greater roles in environmental non-governmental organizations and civil society groups, and may lead entrepreneurial associations" (see page 30 of the funding proposal). A recent FEM4FOREST study found that women account for a third of all forest administration employees: "Almost all of them (14 out of 16) are forest engineering graduates and 2 have a Master's degree." (see page 28).

30. Annex 8 further reports that:

> "One of the key gender issues in environmental protection is that benefits from natural resources exploitation are reserved for men and negative consequences and risks are shared with the whole community. This is coming from the fact that a very limited number of ecosystem services are utilized and reduced to quantifiable amounts of products" (see page 37).
>
> In the forestry sector, this is directly applicable to timber.

31. The Gender and Social Inclusion Action Plan states:

> "The project aims to upgrade Serbia's male-dominated forestry sector and associated decarbonization business by fostering inclusive and gender-responsive approaches that ensure no one is left behind. By promoting women's participation in developing innovative business models, the project will help create sustainable and equitable opportunities for all, contributing to both climate resilience and gender equality." (see page 46 of annex 8 to the funding proposal).
>
> It will be important that the full project management unit be committed and vigilant to take every opportunity for enhancing gender equality in project implementation and ensuring its translation into gender-equal outcomes.

32. The project, working within the given structure of the economy, should be able to contribute positively to income generation opportunities for forest owners and improve the quality of firewood for rural communities. The sustainable development potential is assessed as medium to high.

## 1.4 Needs of the recipient *Scale: Medium*

33. Serbia is an upper-middle-income country with a growing economy: annual growth in gross domestic product (GDP) per capita was 8.3 per cent in 2021 (see page 5 of annex 7 to the funding proposal). It can be argued that Serbia does not need grant funding to support climate action. The main justification for the GCF funding request is that in 2022, Serbia's public debt accounted for more than 60 per cent of its GDP (see para. 74 of the funding proposal). In the written question and answer exchange between iTAP and FAO, it was also suggested that the GCF grant will address "a complex blend of market failures by delivering public goods, such as by mobilizing forestry investment in remote areas, introducing modern and climate-adaptive silviculture techniques not adopted previously due to informational barriers, and enhancing the wood value chain as a source of renewable energy for the country."

34. The funding proposal further suggests that concessional financing will help enhance the energy security of the rural poor and set Serbia on a more sustainable forestry-energy-decarbonization path. In the longer term, however, the idea is that concessional finance should unlock additional funding.

35. Serbia's revised nationally determined contribution (NDC), submitted in August 2022, commits to "*Increasing* its ambition to the GHG emission reduction by 13.2 per cent compared to the 2010 level (i.e. 33.3 per cent compared to 1990) by 2030." [2] It also highlights the issue of energy, including the importance of hydroelectricity generation to the Serbian economy, the household sector's reliance on electricity for heating (suggesting that heating is an issue regardless of which fuel is used), and the low level of energy efficiency.

36. The updated NDC refers to the mitigation potential in the sectors of agriculture, forestry and water. Relating to 'forestry-bioenergy,' Serbia, according to the funding proposal (see para. 75), "plans to invest over USD 9 million annually into the sector, including a majority of its forestry budget over the next seven years into the target areas of the FOREST Invest proposal." Nonetheless, the country has not included the costs of adaptation and decarbonization related to forests, which, according to the funding proposal, "puts the overall government investment in forestry at risk."

37. External funding for Serbia's forests is hence argued to be critical to help Serbia achieve its current NDC targets and decarbonize its economy. By injecting initial concessional financing into the forestry sector, the project seeks to catalyse the leveraging of additional funds from the government and other external financiers, including the private sector, to improve carbon storage in forests and national capacities to manage them, and to reduce inefficiencies in the wood value chain.

38. As mentioned above (section 1.1, para. 9), Serbia, according to ND-GAIN, has relatively low vulnerability to climate change and is well positioned to address existing climate risk. In this context, the project aptly opts to reduce the exposure and vulnerability of Serbia's forestry sector and enhance resilience and energy security of households, while increasing total $CO_2$ removals from forestry and biodiversity.

39. Altogether, the needs of the recipient are assessed as medium.

## 1.5 Country ownership *Scale: Medium to high*

40. The proposed project was developed and prepared following a request to FAO by the Government of Serbia in 2019. During project elaboration, key stakeholders dealing with the forestry, energy and agriculture sectors in Serbia were consulted. The Summary of Consultations and Stakeholder Engagement Plan (see appendix 3 to annex 7 of the funding proposal) accounts for numerous consultation meetings held with national and international stakeholders. The stakeholder engagement plan foresees continuous national as well as community-level stakeholder consultations. The project implementation will also coordinate with other externally funded initiatives, including the Global Environment Facility (GEF). The anticipated activities of FOREST Invest build on the Contribution of Sustainable Forest Management to a Low Emission and Resilient Development in Serbia initiative, financed by the GEF Trust Fund and implemented by FAO. Forest areas across central Serbia and Vojvodina that are in need of urgent climate-proofing have been identified through an analysis performed by the Forestry Institute of Serbia.

41. The main institutional stakeholders include ministries, municipalities, industry associations and the chamber of commerce. MOAFWM is responsible for forestry, agriculture and water management policies in the country. The project will work mostly with the Forestry

[2] https://unfccc.int/sites/default/files/NDC/2022-08/NDC%20Final_Serbia%20english.pdf.

Directorate, with the support and involvement of the Agricultural Land Directorate, the Rural Development Directorate, and others as necessary. Other institutional stakeholders include the Ministry of Environmental Protection, which will be the main partner for the establishment of the carbon offsetting/insetting mechanism, and the Ministry of Mining and Energy, which is responsible for energy efficiency and security. The project also engages with the Ministry of Education, Science and Technology Development, which will be the main partner for the upgrade of university and vocational schools' curricula, and the Ministry of Economy, which will be engaged in decarbonization activities involving the private sector. The Ministry of Finance will be involved in the carbon finance options and the development of related guidelines.

42. Municipalities, in charge of municipal lands within their territory, will – to the extent that it is forested – ensure adherence to climate-adaptive silviculture and sustainable forest management. The project engages with the Serbian Chamber of Commerce and Industry as well as producers' or owners' associations, which will assist in the identification of private sector actors and relevant activities. Also, SerBio, an association of non-governmental organizations, companies and individuals in the field of biomass utilization, is engaged to help facilitate interactions with users of biomass. The iTAP positively notes the engagement with both the producer and consumer side among the institutional stakeholders, even though the project's emphasis lies on the side of production.

43. The project contributes to several of Serbia's efforts to adapt to climate change and maximize mitigation benefits. The country's climate policy places forests at its core, highlighting the critical nature of forests in guaranteeing the country's energy security, increasing the use of renewable energy, enhancing livelihoods, removing carbon and decarbonizing the economy to make Serbia carbon-neutral by 2050. National strategies related to natural resource management and biodiversity point to the importance of healthy and adaptive forest ecosystems, not least to reach national, regional and international environmental targets.

44. FAO has a solid presence and track record in Serbia, and has engaged with stakeholders to develop a project that is suitable to the needs of the country. The country ownership is assessed as medium to high.

## 1.6 Efficiency and effectiveness *Scale: High*

45. The coherent structure of the funding proposal and the logical way that activities feed into outputs, which in turn produce outcomes and co-benefits, speaks for a high likelihood of the project achieving its objectives. This contributes to project efficiency and effectiveness.

46. Moreover, the project sets out to promote a more efficient and effective way to address forests, fuelwood and carbon removals, including stimuli for private sector investments, which will be incentivized and mobilized via the offsets and insets, including leveraged private finance from private banks and forest owners' associations.

47. Forestry is one of the priority areas identified in Serbia's updated NDC, and the country has agreed to focus most of its forestry budget over next seven years on the target areas of the proposed project. The co-financing committed by the MOAFWM (in-kind) is important and amounts to nearly two thirds of the total project budget.

48. The economic and financial benefits have been estimated through four types of models, as explained in annex 2 to the funding proposal. Forestry investments demonstrate high net ecosystem services production with a low financial value of economic benefits deriving from ecosystem services. Also, the project's efforts on policy and regulatory frameworks is to disincentivize unsustainable practices, such as illegal logging. All in all, the analysis suggests financial efficiency in the achievement of its mitigation targets.

49. The funding proposal reports that the project shows very solid economic parameters, with a 20-year economic internal rate of return (EIRR) of 15.0 per cent (higher than the social discount rate), with a USD 78.4 million net present value (NPV) and a 40-year EIRR equivalent to 17.6 per cent with a USD 227.5 million NPV (see para. 111 of the funding proposal). The 40-year time horizon is used to capture the full deployment of benefits deriving from forestry investment.

50. The economic returns and leverage are presented as robust. The target for mobilization of leveraged loans is USD 50 million, but it is suggested that even in a scenario where no loans are disbursed, the project's aggregated economic results remain positive over a 20-year horizon, with an EIRR of 11.5 per cent and an NPV of USD 39.7 million (see para. vi of the funding proposal).

51. FAO has developed a solid project that effectively addresses the major climate sensitivities of Serbia through what seems to be an efficient institutional setup. The efficiency and effectiveness criterion is assessed as high.

# II. Overall remarks from the independent Technical Advisory Panel

52. Overall, the funding proposal is well-crafted and efficiently addresses problems of forest degradation and fuelwood consumption in the context of climate change in Serbia, which are leading to increasing climate vulnerability and GHG emissions. The cross-cutting endeavour, to be implemented by FAO, MOAFWM and the two public forestry management enterprises PES and PEV, is likely to achieve its objectives and generate significant adaptation and mitigation outcomes.

53. The iTAP recommends that the Board approve this funding proposal.

# Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP260)

| | |
|---|---|
| Proposal name: | Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration (FOREST Invest) |
| Accredited entity: | Food and Agriculture Organization of the United Nations |
| Country/(ies): | Serbia |
| Project/programme size: | Medium |

| Impact potential |
|---|
| The AE acknowledges that the overall impact potential is assessed as **high** by iTAP. |
| **Paradigm shift potential** |
| The AE acknowledges that the overall paradigm shift potential is assessed as **medium to high** by iTAP. |
| **Sustainable development potential** |
| The AE acknowledges that the overall sustainable development potential is assessed as **medium to high** by iTAP.<br>The AE will ensure that the project management unit be committed and vigilant to take every opportunity for enhancing gender equality in project implementation and ensure its translation into gender-equal outcomes. |
| **Needs of the recipient** |
| The AE acknowledges that the overall needs of the recipient are assessed as **medium** by iTAP. |
| **Country ownership** |
| The AE acknowledges that the overall country ownership is assessed as **medium to high** by iTAP. |

| **Efficiency and effectiveness** |
| --- |
| The AE acknowledges that the overall efficiency and effectiveness is assessed as **high** by iTAP. |
| **Overall remarks from the independent Technical Advisory Panel:**<br><br>The AE acknowledges iTAP's overall assessment and recommendation for the Board approval. The AE will ensure that the expected mitigation and adaptation outcomes are achieved. |

# Annex 8
# Gender Assessment and Action Plan

*For the GCF-FAO Project "Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration (FOREST Invest)"*

Table of Contents

## List of abbreviations

| | |
|---|---|
| UNFCCC | United Nations Framework Convention on Climate Change |
| NFI | National Forest Inventory |
| BAU | Business as Usual (scenario) |
| FAO | Food and Agriculture Organisation of the United Nations |
| NDC | Nationally Determined Contributions |
| GCF | Green Climate Fund |
| LULUCF | Land-use change and forestry |
| CEDAW | Convention for the Elimination of All Forms of Discrimination against Women |
| UN | United Nations |
| SDG | Sustainable Development Goals |
| GAP | Gender Action Plan |
| CSO | Civil Society Organisations |
| CBD | Convention on Biological Diversity |
| EU | European Union |
| COVID 19 | coronavirus disease caused by a virus named SARS-CoV-2 (WHO) |
| GEF | Global Environmental Facility |
| DRR | Disaster Risk Reduction |
| NARS | National Assembly of the Republic of Serbia |
| CBGE | Coordination Body for Gender Equality |
| AP | Autonomous Province |
| SCTM | Standing Conference of Towns and Municipalities |
| NWFP | Non-Wood Forest products |
| SME | Small and Medium Enterprise |
| NP | National Park |
| PFOA | Private Forest Organisations |
| MOAFWM | Ministry of Agriculture, Forestry and Water Management |
| MEP | Ministry of Environmental Protection |
| MME | Ministry of Mining and Energy |
| MESTD | Ministry of Education, Science and Technological Development |
| MoE | Ministry of Economy |
| GBV | Gender Based Violence |

## Introduction

Compared to the reference period used in the two available national communications to the United Nations Framework Convention on Climate Change (UNFCCC) 1961-1990, climate variables in Serbia have changed. Temperatures – both minimum and maximum – are increasing steadily while rainfall patterns appear more erratic and generally concentrated in autumn and spring. Projections to 2050 indicate that changes will worsen and that both temperatures and precipitation will further become an issue to address especially in those sectors, such agriculture, that are still largely dependent on climate.[1] This fact is raising the importance of mitigating measures country should take into consideration, especially those related to implementation of low carbon development strategies.

Serbia's low carbon development strategy promotes a transition towards a climate-neutral and resilient economy where decarbonization relies largely on increasing CO2 absorption by forests (+17% - 2030). Nonetheless, climate change is affecting Serbian forests, and the adaptation deficit is high across stakeholders. Forests are facing unprecedented threats due to climate change, and CO2 removals are decreasing due to forest degradation (-19.4% in 2015 compared to 2010) and, in a business-as-usual scenario (BAU) might reach -26.1% in 2050 compared to 2010.[2]

According to National Forest Inventory (NFI) 2008 results, forests in Serbia cover around 30% of the country's total area (Central Serbia 37 % forest cover; Vojvodina 6% of forest cover)[3]. Even though the forest cover was increased within the last 20 years in Serbia, forest degradation, along with resulting habitat loss and fragmentation, is one of the key environmental problems faced by Serbia at present. Forest degradation on a large scale has resulted in the loss of forest carbon, biodiversity and other key ecosystem goods and services, and has substantially reduced the potential of Serbian forests to act as carbon sinks.[4] Between the years 2001 and 2018, 63,280.94 ha of forest were lost in the national territory of Serbia. The tree cover loss in percentage would represent a 1.83% of the whole area[5].

Following the government request to Food and agriculture organisation (FAO) and the Nationally Determined Contributions (*NDC)*[6] *targets of Serbia*, as well as the *GCF Country Programme*, the project is under preparation, intending to address national forest's priorities and encourage private sector engagement in decarbonization, as envisaged by the Republic of Serbia. It is titled: ***Enhancing the resilience of Serbian forests to ensure energy security of the most vulnerable while contributing to their livelihoods and carbon sequestration.***

The objective of the project is to support the Republic of Serbia in **enabling the forest sector to contribute to the country's climate change adaptation and low carbon strategy** by:

- Reducing the vulnerability of vulnerable communities by increasing their energy security,
- Stabilizing and increasing carbon removals,
- Upgrading capacities of key institutions and private sector operators to manage the decarbonization process,
- Incentivizing private sector companies to engage in climate de-risking and decarbonization process.

[1] Monzini, Jacopo (2021): Background information for the FAO TCP/SRB/ 3801/C1
[2] Ibid
[3] Unpublished FAO 2019. Socio-economic perspectives of sustainable forest management & local development in Serbia.
[4] Project Document, GEF 6 Project: Contribution to Sustainable Forest Management to a Low Emission and Resilient Development
[5] https://earthenginepartners.appspot.com/science-2013-global-forest/download_v1.6.html
[6] FAO (2019) Regional analysis of the Nationally Determined Contributions of the countries in Southern Europe, Eastern Europe and Central Asia – Gaps and opportunities in the agriculture sectors. 132 pp

The project will address bottlenecks to **climate change adaptation and mitigation** through**:**

> **Component 1 -** Enhancing forest management and forest governance while reducing drivers of degradation;
> **Component 2 -** Enhancing LULUCF contribution to climate change mitigation and DRR;
> **Component 3 -** Supporting private sector engagement in climate derisking and in decarbonization.

The co-benefits of the project will include:

1. the increased participation of the private sector in sustainable and climate adaptive forest management
2. the greening of the wood-based biomass value chains.

The combined impact of these actions will **enable Serbia to reduce the vulnerability of its forestry sector and communities while increasing total CO2 removals**.

Given the distribution of forests in Serbia, **the project will work at the national level** in order to benefit the entire population of the Republic of Serbia. The project will ensure the participation of key sectoral actors from national and local institutions as well as from civil society organizations and academia.

The success of the project will, to a large extent, depend on the **practical integration of gender equality mainstreaming into forest governance and climate finance**. The value of its contribution will be increased, as the overall implementation plan and actions foreseen are sensitive to the diverse needs of the communities with regard to gender, ethnicity and economic status, so the project will strive to **promote gender balance** from the basics - stakeholder participation in meetings and other platforms relevant to decision-making through institutional structures improving for **achieving gender-responsive forest governance and in-depth technical knowledge extension, budget planning and actions** conducted.

To enable project to have a gender-sensitive approach and provide benefit to the wider community of the focused area, as the standard requirement of the GCF, this gender analysis provides recommendations and action plan for the for the funding application.

## Aims and targets of the proposed project

By enhancing the resilience of Serbian forests and rural communities while enhancing the carbon storage potential of the country to support and boost the decarbonization process through adaptation and mitigation investments, the project belongs to the result areas of **mitigation and adaptation**.

Transition towards a climate-neutral and resilient economy in Serbia is seen through a low carbon **development strategy which promotes primarily decarbonization through increasing the CO2 absorption by forests, but that requires an adaptive approach to ensure ecosystem services to rural and vulnerable communities**.

Therefore, the project aims at

- **A.** Increasing the resilience of 729,064 individuals (371,823 women),
- **B.** reducing national net emissions by increasing carbon removals from the forestry sector (7.6 MtCO2 (27Y).

Unfortunately, **Serbian forests are already affected by climate change** themselves and being degraded by fuelwood extraction. **Forests are degrading, making Serbian capacity to deal with required adaptation deficient**. Drought in Serbia has become more frequent, decreasing forest vitality and exposing them more often to forest fires. **Rural livelihoods have become increasingly uncertain**, pushing local communities to adopt **overexploitation patterns to provide food and energy.** For that reason, project targets are seen as possible to reach only by supporting the forestry sector by:

(a) increasing forest cover by 7,000 ha;
(b) enhancing about 52,000, ha of public and private degraded coppice stands into high forest;
c) ensuring sustainable forest management and climate adaptive silviculture on 500,000 ha of forests;
(d) creating an offsetting and insetting mechanisms that will allow the private sector to dispose of CO2 credits to complete their decarbonization.
e) engaging with private sector companies in climate de-risking and decarbonization processes.

These actions fall under the 3 main categories of adaptation measures of the Serbian Low Carbon Development Strategy among other:

1) Afforestation using location mapping and tree species adapted to climate change;
2) Introducing “Climate-Smart Forestry” approaches; and
3) Changing forest management practices following a Close-to-Nature-Forest-Management/Close-to-Nature-Forestry approach

Those will require full engagement of various local rural development stakeholders and governmental institutions, civil society, and academia. Also, complete knowledge/human and natural capital of local people/indigenous knowledge should be wisely employed in combination with the cutting-edge technologies, to conducts long term action for the purpose of supporting development of sustainable communities and healthy ecosystems capable to interact for the benefit of each other.

## Gender-related international frames of the proposed project

The Republic of Serbia is investing a lot of energy to come along with the contemporary trends of sustainable development, which also include maximizing utilization of the available human resources for the benefit of its communities. Serbia's gender equality policies are guided by international commitments that the state has undertaken for the ratification of key international conventions, such as the *Convention for the Elimination of All Forms of Discrimination against Women* (CEDAW) and the *Council of Europe Convention on preventing and combating violence against women and domestic violence* (the Istanbul Convention). Also, Serbia ensures active participation in international platforms, such as the *Beijing Declaration and Platform for Action*, and UN resolutions, including the *United Nations Security Council Resolution 1325 on Women, Peace and Security*. Serbia regularly submits its reports under these conventions.

Serbia's development policies have also been aligned with the Agenda for Sustainable Development until 2030[7] and the Sustainable Development Goals (SDGs). Therefore, Agenda 2030 represents a framework for its sustainable development throughout different sectors, including gender and forest protection. Gender equity is addressed under the SDG 5, but it is in practice a cross-cutting issue for all of the 17 SDGs. Regarding the sustainable forest conservation and management part of these goals, important in particular are:

- **Sustainable production and consumption** (SDG 12), which targets also empowerment of the local populations, including women, to provide rational natural resources exploitation and their longevity
- **Climate action** (SDG 13): which considers differences in women and local communities, and other groups affected by climate change, as much as the knowledge and experiences they possess, that can help find adapted solutions to climate change.
- **Conserving forests and biodiversity** (SDG 15): which is talking about the limiting the industrialization of agriculture and forestry in order to protect forests, biodiversity and preserve local communities' ways of subsistence.

Recommendations of international mechanisms indicate the direction that reforms in Serbia should take to improve gender equality in different areas. The Concluding observations and recommendations of the CEDAW Committee in 2019, "recognize specific forms of discrimination of women in rural areas and recommend priority actions for the improvement of the situation of rural women in various aspects, from access to ownership, employment, economic and social security, protection from violence, to increased representation in decision-making bodies, and similar"[8].

CEDAW Committee considers that gender equality has not been achieved sufficiently in the case of women in rural areas of Serbia since:

(a) Serbia missed to adopt measures, including temporary special measures, to ensure that rural women, including women employed in the informal sectors of the economy, have access to education, healthcare, housing, formal employment, social security and retirement schemes, life-long training opportunities, ownership and use of land on an equal basis with men, and that their specific needs are met;

(b) Serbia didn't ensure the equal participation of rural women in decision-making, including in relation to agricultural holdings, and involve them in the design, development,

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[7] Official Gazette of the Republic of Serbia, No. 104/2009. 10 Official Gazette of the Republic of Serbia, No. 22/2009. 11 Official Gazette of the Republic of Serbia, No. 4/2016.
[8] Committee on the Elimination of Discrimination against Women, 2019

implementation, monitoring and evaluation of all relevant plans and strategies, such as those relating to health, education, employment, retirement and social security;

(c) Serbia missed to strengthen data collection on rural women, disaggregated by age, gender and geographical area, to assess their situation and the progress made overtime.[9]

The NGO "Environmental Ambassadors for Sustainable Development" in their 2019 "shadow report"[10] emphasised that Serbia is committed to all SDGs but has not adopted specific national Sustainable Development Goals. The first Voluntary National Review was submitted to the High Political Forum in 2019 (Government of the Republic of Serbia). The report has emphasised that SDGs are important for reaching gender equality, human rights and environmental protection which is obviously not fully understood and applied in Serbia.

Another important international framework followed is provided through the *Convention on Biological Diversity* (CBD), which Serbia ratified in 2002. The four strategic objectives of the CBD Gender Plan of Action are[11]:

i. Integration of gender perspectives
ii. Promoting gender equality
iii. Demonstrate the benefits of integrating gender
iv. Increase the effectiveness of implementation efforts

Actions based on these objectives are related to policy, organizational, delivery, and constituency. Some *ad-hoc* activities in respect to biodiversity conservation related to gender is noticed in almost every project, yet usually with no relevant follow-ups. Project activities of the kind are mostly related to trainings or events, while recording of participating women is the only evidence of providing equal opportunities for benefiting through knowledge transfers.

GCF guidelines for the promotion of gender equality

At the Climate Change Conference in Marrakesh (COP22), Parties to the UN Framework Convention on Climate Change (UNFCCC) reiterated their commitment to mainstreaming gender in climate action and the UNFCCC process, providing substantial instructions in a stand-alone decision on gender. Parties gave specific guidance, including to the GCF as an operating entity of the Convention's Financial Mechanism, to enhance reporting on how gender considerations are integrated in all aspects of activities.[12]

Gender mainstreaming is central to the GCF's objectives and guiding principles, including through engaging women and men of all ages as stakeholders in the design, development and implementation of strategies and activities to be financed. The GCF Governing Instrument states that: "The Fund will strive to maximize the impact of its funding for adaptation and mitigation… promoting environmental, social, economic and development co-benefits and taking a gender-sensitive approach."[13]

Thus, gender equality considerations should be mainstreamed into the entire project cycle to enhance the efficacy of climate change mitigation and adaptation interventions and ensure that gender co-benefits are obtained. This applies to all projects, not just those intended from the outset to centre

[9] https://womenngo.org.rs/images/CEDAW/CEDAW_Concluding_observations.pdf
[10] Environmental Ambassadors for Sustainable Development: Serbia-shadow-report, https://www.women2030.org/wp-content/uploads/2019/07/Serbia-shadow-report.pdf
[11] Convention on Biological Diversity: 2015-2020 Gender Plan of Action
[12] GCF and UN Women (2017): Mainstreaming Gender in Green Climate Fund Projects - A practical manual to support the integration of gender equality in climate change interventions and climate finance, Korea
[13] Ibid

on women or to have a gender focus. Gender mainstreaming is fundamental to any project intervention and does not necessarily signify additional costs; in fact, mainstreaming gender makes climate interventions more effective and efficient.[14]

According to the GCF's Gender Policy, projects submitted to the Fund are required to be aligned with national policies and priorities on gender and fit with the Fund's gender policy". This includes a gender assessment and gender action plan (GAP), which should provide information on project responses to the needs of women and men in view of the addressed problematics.[15]

### Methodology

Gender analysis was conducted during the project preparation phase to identify potentials for promoting gender equity, advancing women's participation in project activities and to assess and mitigate potential risks.

In order to ensure relevance, quality and proper targeting, the gender analysis has been elaborated using a combination of a desk review of relevant national policies, legal and regulatory frameworks and studies conducted in the sector (secondary data) and direct consultations with relevant stakeholders, including government staff in charge of gender issues, representatives of NGOs and women from rural communities involved in forestry-related activities.

The gender assessment was followed by the development of a concrete gender action plan (GAP), intended to address gender gaps and maximize opportunities and benefits of women's engagement with boosting the decarbonization process, through enhancing the resilience of Serbian forests and the carbon storage potential of the country. To ensure ownership as well as proper identification of issues and targeted actions, development and finalization of the gender analysis and development of GAP was conducted through regular consultations between the project team and various stakeholders The process was further strengthened through direct consultations with NGOs in pilot regions communities in Banat (Kikinda) and Eastern Serbia (Dimitrovgrad) as typical regions for Voivodina plains and mountain regions of Central Serbia which were used to inform the preparation of the GAP. Questionnaires were used to assess position and needs of the private forest women and men holders. Further in-depth analysis is planned during the inception phase using the same approach, with the intention to further enrich the GAP per territory specific gender analysis data and define adequate measures. The Gender expert, with support from the FAO PMU will oversee this engagement.

**Stakeholders consulted during the project identification and formulation:**

During the preparation of this assessment and the subsequent gender action plan (GAP), the team engaged with a diverse set of stakeholders and their focal points for issues related to the social and gender dimension of the project. These included: (I) Institutions, (II) Local administrations, (III) Civil Society Organizations and (IV) Academia.
The feedback received was integrated into the actions and indicators included in the GAP, such as, for example, ensuring a gendered approach to updating syllabi, as well as the provision of data, contacts and documents.

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[14] GCF and UN Women (2017): Mainstreaming Gender in Green Climate Fund Projects - A practical manual to support the integration of gender equality in climate change interventions and climate finance, Korea
[15] Ibid

| Organization | Type of Organization | |
|---|---|---|
| Regional Rural Development Standing Working Group in Southeastern Europe (SWG RRD) | Organization | International |
| Faculty of Forestry | Academia | National |
| IUCN Serbia/NBS forest project | NGO | National |
| Ministry of Agriculture Forestry and Water - Rural Development/IPARD measure 7 | Institutions | National |
| Natura Balkanika/GIZ project team on gender | Association | National |
| Singidunum University/Env.and Sust.Development Studies | Academia | National |
| Pokret gorana, local branches | Association | National |
| Municipality Dimitrovgrad | Institutions | National |
| Municipality Kikinda | Institutions | National |
| Municipality Zrenjanin | Institutions | National |
| Hunting association Zrenjanin | Association | International |
| Forest institute Belgrade | Institutions | National |
| National Alliance for Local Economic Development | NGO | National |

## Part I - Gender Analysis

The following gender analysis should help understand relationships between men and women, their access to resources, their activities, and the constraints they face in Republic of Serbia. It is intended to promote gender – relevant entry points, policies and identify opportunities for enhancing gender equality in respect to enhancing the resilience of forests and the carbon storage potential of Serbia. Gender analysis of Serbian environment should help identify the multiple causes of vulnerability, including gender inequality, as much as to help identify and build on the diverse knowledge and capacities within local rural communities/households that can be used to make them more resilient to climate related shocks and risk while boosting the decarbonization process.

### 1. GENDER EQUALITY IN SERBIA

Serbia is a landlocked country in Southeast Europe (West Balkans), with a socialist heritage, which is relevant for the forms of gender inequalities and position of women. There is formal equality in rights (equality and prohibition of discrimination are integrated in the Constitution of the Republic of Serbia, and there are also Gender Equality Law and the Law on the Prohibition of Discrimination).

At first glance in the public sphere, men and women are formally equal. With more than 30% of women in the national parliament and woman in the position of Prime Minister, Serbia has recently scored a very good results in the EU Gender Equality Index[16] (in the domain of power). Education is one of the key emancipation strategies for women and in 2021 (i.e. more female than male graduated students). However, inequalities are visible in many aspects:

- The private sphere, family, parenthood, or unpaid work, which spills over into the field of paid work.
- Segregation in the education system and the labour market, where dominantly female jobs are paid less and discrimination of young women in the labour market, mostly due to expected motherhood. Women are in general less active in the labour market and less employed, regardless of education.
- Prevalence of gender stereotypes visible in the private and public sphere, at the discursive and practical level, and
- Violence against women.

The value of the Gender Equality Index for Serbia in the sphere of time is 48,7 out of 100 (European Union average 65,7). The Country Gender Assessment conducted by FAO during 2020-21, indicate significant gender gaps in rural areas of Serbia across diverse dimensions, including access to assets, economic participation, roles in and gains from agricultural production, the exercise of a range of welfare rights, political participation, access to social services, lifestyles and resilience to climate change and emergencies. It was also noted that the COVID-19 pandemic, has had profound impact on the rural population and the position of women in rural areas, also creating opportunities for innovative approaches and new practices that can improve the economic activity of rural women in the future, and consequently their overall wellbeing.[17]

Serbia is confronting several demographic challenges, such as emigration, rural depopulation, and demographic ageing. The current demographic trend in rural areas is within this most recent research estimated as unfavourable, characterized by continuing population decline, a rise in the average age

[16] According to the latest Gender Equality Index 2018, Serbia was still a country of pronounced gender inequalities in all domains. These inequalities were significantly more pronounced than the EU average (55.8 vs. 66.2), and progress that has been made (compared to 2016), was very small (3.4)[16]. One of the main axes of inequality noted by the Index refers to gender segregation, which is established during education and continues later in the labour market.
[17] Ibid

of the population, falling fertility and birth rates contributing to negative population growth, and high levels of migration from rural to urban areas and beyond to other countries. [18]Outward migration from rural areas is more prevalent among women than men, and the reasons for this can be found in women's lower ownership of assets, their weaker ties to land and estates, and their unequal participation in the rural economy. Rural women's living conditions are less adequate in comparison with urban women, especially in terms of access to employment in the non-agricultural sector, and access to education, social services and amenities which are important for quality of life, such as cultural and recreational amenities, all of which then act as pull factors towards urban areas[19]

Life expectancy at birth is 77.1 years for women and 72.0 years for men. The average age of the population in 2018 was 41.4 years, and it is worth noting that the average age of women was higher than the average age of men (42.7 and 40.0 years, respectively). In the same year, the share of the working age population in the total population was 65.5 percent. The ageing index (the ratio of the population aged 60 years or more to the young population aged.[20] Women in rural areas are on average older than women in the city, which is the consequence of migrations from rural to urban areas. In 2019 there was a slightly higher risk of poverty for women compared to men for almost all age groups. Only in the age group 55-64 are women less endangered than men. The risk of poverty rate for women aged 65 and over is 23.2%, and for men in the same group 18.3%.[21] The population in rural areas is at a higher risk of poverty and social exclusion.[22]

### *1.1. Common beliefs, perceptions, and stereotypes related to gender*

According to the Strategy for Gender Equality, patriarchal cultural patterns and norms, widespread gender stereotypes and prejudices are present in Serbia. Anti-gender discourse and misogyny are insufficiently effectively suppressed and represent some of the key challenges in Serbian society. Violence against women, sexism, gender stereotypes and prejudices are widespread in all spheres of public and private life in Serbia. The re-patriarchization of society is caused by power structures based on the logic of capital and the neoliberal state.[23]

Gender inequality rooted in patriarchal norms is still highly prevalent among Serbia's population in the following spheres: labour and employment, education, political and social participation; financial status, and income. It is also ingrained in the sphere of private life, manifested in the unequal distribution of unpaid family care and domestic work between women and men. In its worst form, it is a driver of violence against women, both in the public and private spheres, with dire consequences for the safety and wellbeing of women. [24]

Due to strong gender stereotypes, parenthood is still considered as the role and responsibility of the woman, which is the basic reason leading towards various forms of discrimination in Serbia.[25] Those are mostly women who leave jobs, take half-time jobs or are unemployed in order to be able to take care of kids or other family members. Indicatively, 80% of those who leave jobs due to "family reasons" are women, whereas 98% of those who work in households only are women. Also, 79% of single-parent families are of "mother and kids" type. Women own 29,7% of real estate in Serbia[26].

[18] Ibid
[19] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en
[20]Official Gazette of the Republic of Serbia, No. 54/2009, 73/2010, 101/2011, 93/2012, 62/2013, 63/2013, 108/2013, 142/2014, 68/2015, 103/2015, 99/2016, 113/2017, 95/2018, 31/2019, 72/2019 and 149/2020
[21] Strategy for Gender Equality for the period 2021-2030 year
[22] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en
[23] Dokmanović M. (2018): Gender equality in Serbia: achivements, obstacles and perspectives, Institute of social sciences, Belgrade
[24] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en
[25] Women2030, Serbia-shadow-report.pdf (women2030.org)
[26] Ibid

Rural women, single female-headed households, women with disabilities are particularly vulnerable and disadvantageous groups due to gender stereotypes and other factors of vulnerability and social exclusion ruling in Serbia. However, in both urban and rural areas, men demonstrate a significantly higher level of patriarchal attitudes than women[27]. It should be noted that, in the private sphere, both men and women in rural Serbia are predominantly oriented toward patriarchal values. When it comes to values associated with the public sphere, rural women are oriented toward non-patriarchal values (49 %), while men are predominantly indecisive, i.e., somewhere in between patriarchal, and non-patriarchal values. Older inhabitants are more likely to adopt patriarchal values in terms of both public and private patriarchy.[28] The acceptance of gender roles within the private sphere is much more widespread, compared to the public one.[29] Women express a more critical attitude towards gender roles in the public sphere (education, political, and labour) than in the private one.[30]

### *1.2. Status of women in Serbia - commitments, legislation, and policies*

The Constitution of the Republic of Serbia in its basic regulations (Article 15), among other things, guarantees equality of women and men and obliges the state to pursue a policy of equal opportunities, prohibits direct and indirect gender discrimination, guarantees equality before the law, guarantees equality of spouses, equalizes the status of extramarital union and marriage, guarantees the freedom to decide on childbirth as well as special protection (legal, social and health) of mothers, pregnant women, children and single parents.[31]

Laws and bylaws that regulate areas important for gender equality in the Republic of Serbia are numerous in Serbia and can be roughly divided into two groups. In one, there are those who are directly related to gender equality and regulate the competence of institutions in this area at all levels of government. In the second, those that regulate various areas of social life are relevant to gender equality and women's empowerment. The achievement of gender equality is governed by an umbrella law – the Gender Equality Law, while the prohibition of discrimination on any grounds, including sex, gender identity, or any other personal characteristic (which can also include the type of housing) is defined in the Law Prohibiting Discrimination. Gender responsive budgeting is obliged as part of the budget in accordance with the Law on the Budget System (2015).

The legal framework and public policies in the field of gender equality have been adopted, and there is a gender mechanism at all levels, including local self-governments. The new Law on Gender Equality, adopted in 2021 in Article 42, regulates the implementation of equal opportunities policies and measures and the obligation to integrate a gender perspective in the field of environment and environmental protection.

There is also progress in capacity development and gender mainstreaming procedures in public policies, mostly thank to UN Agencies and other international donors. Yet, despite evident progress in harmonizing with the international and European standards of gender equality, and establishing a legal, political, and institutional framework of gender equality, Serbia lags behind reaching *de facto* gender equality. Many sectoral policies are still gender-blind, perpetuating inequalities between women and men. [32]

According to Strategy for Gender Equality, the gender perspective is not embedded in the laws and regulations governing support for innovation. Women's innovative entrepreneurship support programs have been sporadic and with smaller individual allocations. Men’s majority-owned firms

[27] Ibid
[28] Ibid
[29] Petrović I., Radoman M. (2020): Patriarchy, authoritarianism and nationalism in Serbia - changes in value orientations, Institute for Sociological Research, Belgrade, https://www.researchgate.net/publication/339460079)
[30] Petrović I., Radoman M. (2020): Patriarchy, authoritarianism and nationalism in Serbia - changes in value orientations, Institute for Sociological Research, Belgrade, https://www.researchgate.net/publication/339460079)
[31] Constitution of the Republic of Serbia, Official Gazette of the Republic of Serbia No. 98/2006
[32] Dokmanović M. (2018): Gender equality in Serbia: achivements, obstacles and perspectives, Institute of social sciences, Belgrade

received support in one year as much as women's majority-owned firms received over five years. Recently, the Fund, in response to the results of the gender analysis, began to consider the coverage of companies owned by women in its information campaigns. However, this is still far from removing all barriers to greater participation of women, which will represent the challenge for this project.

There is still a lack of sex disaggregated data and gender statistics in majority of sectors relevant for this project, including the sector of environmental protection as much as forestry, green economy, and entrepreneurship in general. In very few cases, some evidence exists about achievements in implementing advanced gender policies. For instance, gender analysis of the Fund for Innovations of the Republic of Serbia shows, that male-owned companies received 9 times bigger amounts of grants than female owned businesses. Yet, in some other sectors, progress is obvious; although monitoring is still lagging behind; we know from the structure of available financial support and rules in place, that gender sensitivity exists. For instance, law on Agriculture and Rural Development regulates the objectives of agricultural and rural development policies and rules related to the special procedure for the implementation and control of the Instrument for Pre-Accession Assistance for Rural Development (IPARD) programme[33]. Within measures supporting agriculture and rural development, especially in the Axis 3 devoted to rural economy diversification, special attention is provided to women need to upgrade primary based rural economy through introduction of tourism and processing etc. Through additional points granted to the evaluation score/ranking of applications, women and youth are provided with the better chance to obtain financial support for their initiatives. The share of women recipients in total funding for rural development is 23.2 percent, which is above their average share among farm holders and managers (19 percent and 15 percent, respectively).[34] Although the specific window supporting forestry and adjacent activities from the rural development portfolio (both national and IPARD) is still missing[35], the existence of gender-sensitive approach to rural development incentives is making timing for the project very good.

Serbian strategic plans related to gender are finally completed during 2021. The last in a series of state documents dealing with the status of women,[36] - the Serbian government adopted on 14th of October 2021, the Gender Equality Strategy for the period from 2021 to 2030. The one aims overcoming of the gender gap and achieving gender equality, as a precondition for development of Serbian society and the improvement of daily lives of women and men, girls and boys in which women and men, girls and boys, as well as persons of different gender identities have equal rights and opportunities for personal development, provide equal contribution to the sustainable development of society and take equal responsibility for the future.[37]

The general objectives of the National Strategy on Gender Equality in Serbia are:

1. Changed gender patterns and promoted culture of gender equality
2. Increased equality of men and women in different areas through the implementation of equal opportunity measures.
3. Gender mainstreaming applied in the development, implementation, and monitoring of public policies.

Competent authorities and other entities involved are obliged to consistently consider the protection of human rights, gender equality and especially the rights of the poor, the elderly, children, persons with disabilities, refugees, displaced persons and other vulnerable groups.

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[33] Official Gazette of the Republic of Serbia, No. 98/2006 and Official Gazette of the Republic of Serbia, No. 41/09, 10/13 and 101/2016

[34] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en

[35] Dordevic-Milosevic (2019): Socio-economic perspectives of Sustainable forest management & local development in Serbia, Study for GCP/SRB/002/GEF: Contribution of sustainable forest management to a low emission and resilient development in Serbia (FSP), FAO Belgrade

[36] Strategy for Gender Equality for the period 2021-2030 year, Official Gazette of the Republic of Serbia No. 103/2021

[37] [2021-2030 Gender Equality Strategy adopted (srbija.gov.rs)]

Despite the efforts invested in promoting gender responsive budgeting, the national budget is not yet fully gender mainstreamed. The advances are even smaller at the local level, where the process of transitioning to the new budgeting system is slow. The Action Plan for the period 2021- 2023 is in the drafting process. On 20 May 2021, the country's parliament adopted two key laws paving the way for real gender equality in the country: The Law on Gender Equality and the changes to Law on elimination of discrimination.

Multi-year policy and programming documents take into consideration gender components and the need for the economic empowerment of women in rural areas. The development of women's and youth entrepreneurship is set as a specific operational objective under the priority action). An important mechanism for gender mainstreaming is provided in addition in the National Rural Development Programme, as well as IPARD, which also recaps the need for the economic empowerment of women in rural areas (especially through designing support measures). Decree on using funds to support the development of non-agricultural activities in rural areas for 2010, provided that associations could get incentives for projects focusing simultaneously on the purchase of raw materials and employment of women from rural areas doing handicraft, in which way incentives were given to seven agricultural holdings led by women and 12 women's associations.[38]

Furthermore, in 2010, the Decree on the conditions and manner of the use of incentives to support rural development through investment in agricultural holdings for the production of vegetables, industrial plants, and investments for procuring livestock unit for production, women holders of the agricultural holding would get 10 points more each, and 117 applications with women holders of agricultural holdings 15. Women were awarded 10 points more each, and the grant amount on this call was 100% of the value of the project.[39] A total of 223 applications were submitted, out of which in 80 cases the holders of agricultural holdings were women; i.e. women's associations. A total of 70 requests were approved, out of which 19 holders of agricultural holdings were women, or women's associations (7+12).

The Decree on the conditions and manner of the use of incentives to support rural development through investment in agricultural holdings for the production of milk and meat, stipulated that women would get additional five points, so three agricultural holdings with women holders were awarded funds.[40] Finally, the reply also stated that the government invested efforts in employing hard-to-employ individuals, including rural women, in accordance with the National Employment Strategy for the period 2011-2020; that the 2012 Work Plan of the Development Fund of the Republic of Serbia allocated resources to credit women's entrepreneurships and that the Gender Equality Directorate awarded funds to women working in tourism in rural areas. Unfortunately, the measures selected so far never included the forestry sector itself,[41] but have provided opportunity for women owners to diversify their forest economy (processing of forestry products/handicrafts, agroforestry, silvopastoral activities, tourism).

### *1.3. Institutional setup and gender equality mechanisms*

[38] A total of 223 applications were submitted, out of which in 80 cases the holders of agricultural holdings were women, i.e. women's associations. A total of 70 requests were approved, out of which 19 holders of agricultural holdings were women or women's associations (7+12)

[39] Ibid

[40] A total of 74 applications were submitted and 36 approved. Out of the total of 14 applications made by women holders of agricultural holdings, three were paid out

[41] Dordevic-Milosevic (2019): Socio-economic perspectives of Sustainable forest management & local development in Serbia, Study for GCP/SRB/002/GEF: Contribution of sustainable forest management to a low emission and resilent development in Serbia (FSP), FAO Belgrade

Gender equality mechanisms have been established in the legislative and executive branches of power, at national, provincial, and local levels, and within the independent bodies, such as the Ombudsperson and the Commissioner for the Protection of Equality.[42]

Within the Government of Serbia, the mandate for agriculture and rural development is assigned to the Ministry of Agriculture, Forestry and Water Management and following the establishment of a new Government in autumn 2020, the Ministry for Rural Welfare was formed. The Ministry of Agriculture, Forestry and Water Management is responsible for public administration affairs including, among others, strategies, and policies for the development of the agricultural sector and food industry and forestry. The Ministry for Rural Welfare has a mandate to monitor welfare in rural areas, and to propose measures and policies for improving the life and work situation of the rural population and preserving rural traditions and cultural life in rural areas.[43]

The highest level of the gender mechanism relevant for the project exists within the National Assembly of the Republic of Serbia (NARS). Two of its committees are of importance for gender issues in rural environment - Committee for Agriculture, Forestry and Water Management and the Committee for Human and Minority Rights and Gender Equality, which are both functioning as regular NARS working bodies. These committees check expenditures and monitor the implementation of Governmental policies, the enforcement of laws, and other general acts (under the responsibility of ministries, state agencies and bodies).

The Coordination Body for Gender Equality (CBGE) provides horizontal and vertical coordination of the mechanisms for gender equality. Horizontal coordination includes a network of gender focal points that are assigned duties to advance gender equality and mainstream gender in the work of each ministry. Vertical coordination includes local gender equality mechanisms in towns and municipalities.

The Government recognized in 2020, has also established the Ministry for Human and Minority Rights and Social Dialogue. The mandate of this Ministry is to implement public administration around human and minority rights, anti-discrimination, and gender equality.

At the level of the Autonomous Province (AP) of Vojvodina, a Committee for Agriculture and a Committee for Gender Equality are active in the AP Vojvodina Assembly, while in the provincial executive branch, the mandate for agriculture and rural development is assigned to the Secretariat for Agriculture, Forestry and Water Management, and the mandate for gender equality to the Secretariat for Social Policy, Demography and Gender Equality. The Ombudsman of AP Vojvodina has a Deputy for gender equality. The independent institutions of Protector of Citizens, or Ombudsperson, and of the Commissioner for the Protection of Equality, are also vital for gender equality and protection against discrimination and an essential condition for an inclusive society and inclusive development.

According to the Gender Equality Law, local governments are required to establish local gender equality mechanisms. There is the Gender Equality Network, which consists of representatives of local gender equality mechanisms, whose role is to provide support to local governments to improve gender equality (SCTM, 2020)[44]. A key factor recognized for the success of the gender mechanism[45] is effective coordination among institutions. In Serbia, however, institutional coordination is weak, and not all relevant actors are fully involved in these processes. Public institutions lack a systematic approach to addressing gender equality and inclusion in forestry and climate change. Additionally, it is essential to secure consistent budget funding for institutions and organizations, rather than relying

[42]FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[43] Ibid
[44] Statistical Office of the Republic of Serbia (2017): Women and men in the Republic of Serbia, ISBN 978-86-6161-166-7
[45] within the EU funded project "Support to Priority Actions for Gender Equality in Serbia, phase I" (GEF II project)

on ad-hoc project-based financing. This project will create an opportunity to expand discussions on gender equality in forestry beyond the existing network to include the business sector, focusing on reducing gender gaps in the field and making it more appealing to young women and men in rural areas.

The project will work to provide and interactive and long-lasting link between The Ministry of Agriculture, forestry and water management, PE “Serbia Shume” and “Vojvodina Shume” and gender focal points, as well as the focal points in local administrations (officers in charge of human rights and gender) for upgrading capacities of these key institutions and private sector operators, but also for continuous further provision of gender data to support further evidence based inclusive policies upgrade on the national and local levels.

March 2021 marked a beginning of the second phase of a three-year EU funded project “Support to Priority Actions for Gender Equality in Serbia, phase II” (GEF II project) that supports the Government of the Republic of Serbia in effective implementation of the EU Gender Equality Acquis, adoption and implementation of the new legal and strategic framework for Gender Equality, Gender Equality mechanisms on national and local levels, and gender mainstreaming in policies and EU funds programming, implementation and monitoring. Furthermore, the project will advance the position of women and will support local communities in fulfilling their commitments on gender equality. The timing is perfect for creating synergy with the project working in practice with local communities as well as introducing novelties into the national frames for supporting gender mainstreaming environmental and forestry sectors. The UN Women Office in Serbia implements this project, in close cooperation with the Coordination Body for Gender Equality, the Ministry of European Integration, the EU Delegation in Serbia, and other partner institutions as well as with women’s civil society organizations.

Following the new Action Plan on Gender Equality and Women’s Empowerment in External Action 2021–2025 (GAP III) adopted by the EU at the end of 2020 and to provide a systemic basis for gender-responsive EU support to Serbia, Gender Country Profile has been developed. The document is providing a comprehensive and structured gender analysis in the six intervention areas identified in the GAP III, including promoting economic and social rights and empowering girls and women; promoting equal participation and leadership, and Climate change and environment and digitalisation.[46]

UN Women supported the Statistical Office of the Republic of Serbia to finalize, design and print the publication “Women and Man in Serbia” offering data for the following areas: Population, Health, Social protection, Education and science, Employment, Wages and pensions, living standard, Time-use, Judiciary, Decision-making, and international indices. In addition, draft Gender Equality Index for Serbia has been developed jointly with the Statistical Office, the Social Inclusion and Poverty Reduction Unit, and the Coordination Body for Gender Equality.

Consultations during preparation of the GAP showed that partner institutions on the local level e.g. local administration (municipalities) have limited capacities to deal with gender issues. Typically, these have experience with one or more small scale ad hoc projects which offered information on inclusion of women in socioeconomic life or participated and supported by their presence during activities CSO sector implemented. Most of these activities were considering economic empowerment of women in rural areas, rarely some documents were produced and published through public campaigns to raise awareness of the public about the local champions of small businesses such as processing, tourism and marketing.

[46] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en

Academic institutions through their individual researchers were involved with few projects doing research about women and their engagement in forestry, tourism, agriculture, handicrafts, rural development etc. As institutions they respond to legal obligations to possess documents (plans) related to gender issues, but no specific activities are recorded as a follow-up.

Institutions on the national level regularly fulfil legal requirements to create gender related policy, but no activities are recorded either within these institutions or extension in the sector they are responsible for. The Ministry of Agriculture, Forestry and Water Management have rural development measures planning teams which are taking care to include in every measure an advantage for women which apply for financial support, yet no further actions are taken to promote these opportunities in the field. This was the task of the Rural development network, which is not operative in the sense of providing regular support to rural development on the field anymore.

Majority of CSO working in related areas such as forestry, rural development, environment, hunting, nature protection has no experience with gender issues if not engaged with projects financed from international organizations, where coverage of gender issues was obligatory at least through provision of women participation in trainings or measuring of impact of conducted activities on women.

### *1.4. Gender equality among the target group*

The project will bring benefits to the whole of Serbia, although its activities besides work with the administration is targeting primarily rural spaces[47] in which the majority of its interventions will occur. Stakeholders to be involved, no matter if they are public or private, mainly live and operate in rural areas dealing with agriculture and forestry which represent the fundament of local economies, while agriculture and forestry lands are the main natural resource local communities base their livelihoods on. Therefore, rural women and men are, as a beneficiaries of local ecosystems services and their managers, the main target group of this project. Rural areas of Serbia are not the space that can boast of gender equality. Stakeholders differ in their inclusion and rights to make decisions; positions they hold in the hierarchy for the businesses, or resources management - leadership or simple representation differ between women and men. Gender inequalities are rather common for the sector in Serbia, but also the feature of the overall community from which they are spilled over or reflected in the economy sector. Role of women and men in the rural communities of Serbia differ, creating a need for further analysis of the gender equality which follows.

#### Education

Out of 127,463 illiterate over the age of 10 in Serbia (1.96% of the total population after the 2011 Census), 104,632 were women (3.12% of the total number of women).[48] There are distinct gender inequalities in education, evidenced in: higher illiteracy rates among rural women compared with both rural men and urban women; the greater share of rural women with lower education levels relative to their male counterparts; and gender segregation by field of study, reflected in the higher concentration of girls in the social services, social sciences and humanities subjects, and determined by traditional perceptions of typical “female” and “male” occupations.[49] Women are concentrated in the social sciences and humanities, as well as in health and social care services (where employment opportunities and wages are lower), and few women are enrolled in the fastest-growing fields of study, such as engineering, technology, a and information and communication technology.[50] Boys are

[47] According to the Law on Territorial Organization the entire territory of the Republic of Serbia can be considered as rural territory, excluding the territories of administrative centers of 24 cities. The total area of the Republic of Serbia is 88,499 km$^2$, and the rural areas are about 75,000 km$^2$ or 85% of the territory.
[48] Statistical Yearbook 2021, Statistical Office of the Republic of Serbia
[49] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[50] FAO (2021): Ibid

more numerous in the fields of education: electrical engineering, mechanical engineering and metalworking, geodetics and construction, transport, forestry and woodworking, and geology and mining[51].

In the general population, 97.7% of boys and girls attended high school. 88.9% of the total available girls were in secondary education. Child labour in rural areas jeopardizes children's educational attainment and also their safety and development. Boys are more exposed to this risk than girls, due to traditional attitudes that men should manage farms and boys' early engagement in work is considered to be an introduction to managerial, decision-making roles in adult life.[52] Among students who complete general secondary four-year education (high school), there are more girls (58%) than boys (42%). The proportion of boys to girls graduating from three-year secondary vocational schools is three to one.

Girls enrol and graduate colleges and universities more often than boys (57% of students and 59% of graduates). Among those who had a doctorate there were 57% of girls compared to 43% of boys. Among graduate students, women dominate in fields of education, while men prevail in the fields of: Informatics and Communication Technologies (74%) and engineering, manufacturing, and civil engineering (63%). In the field of services, the shares of women and men are equal (50%). In 2016, more women (57%) than men (43%) became holders of PhD degrees. Women make majority in many academic fields; however, men are more represented in engineering, manufacturing and construction (57%), and in services (69%).[53]

Men are dominant among the members of the Serbian Academy of Science and Art (SANU). In 2016, over 90% of all members were men. There is not a single woman in the Department of Social Sciences. Among those employed in the field of R&D, in 2016, men are dominant with a 52% share. Almost the same proportion is among scientific researchers. The largest share of women researchers is in medical sciences, almost 60%, and their share is smallest in engineering and technology, about 37%. In the majority of age-groups, women are more computer-literate than men. In older age groups, there are more men among computer users. The majority of internet users are also women.[54]

The educational structure of the rural population is less favourable than the urban population (a higher share of persons with low qualifications and a lower share of persons with tertiary education), and the share of the population with no education or lower education levels is higher among rural women than among men. The highest rate of illiteracy is found among rural women (compared with both rural men and urban women), which is the result of the characteristics of education of the older generations. There are significant gender differences with respect to secondary school and higher education attendance,[55] yet the share of the female population participating in postgraduate education – formal and non-formal is extremely low, except in cases when women is employed in a state institution, education, or administration when additional education and on-job trainings are obligatory.  This is usually justified with obligations within households and lower mobility as stated above. The Project should be able to provide trainings in locations accessible to majority of stakeholders and/or a transport for all interested women in accordance with their needs.

***Education and training in the field of forestry*** There are 4 career opportunities for forestry education in Serbia: (a) vocational training as forest technician; (b) higher education for forest engineering at BSc level; (c) higher education for forest engineering at MSc level; (d) higher education for a doctor of biotechnical sciences - Ph.D. level. The student can start his higher education at the University of

[51] Ibid
[52] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[53] Statistical Office of the Republic of Serbia (2017): Women and men in the Republic of Serbia, ISBN 978-86-6161-166-7
[54] Ibid
[55] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en

Belgrade - Faculty of Forestry with an undergraduate study program at the Department of Forestry, and to continue its education through master's degree studies and doctoral studies at the same Faculty. In the last 10 years, 850 students graduated from secondary schools (training for forestry technicians), 101 of whom were women. The highest number (92) of secondary school students was recorded in the school year 2011/2012 and the lowest in 2015/2016. On average, the share of female students in these schools is 12%. The highest share (16%) is recorded in the current school year (2020/21) and the lowest (9%) in the previous school year (2019/20).[56]

Future women's access to information, education, technical knowledge, and/or skill upgradation

Serbian Gender Equality Strategy for the period from 2021 to 2030 anticipate besides the overall need for inclusion of women (especially women from vulnerable groups including rural women) also the specific need for their inclusion in green jobs, circular economy, and new technologies as the important sectors for increasing their chances in employment and self-employment. That is related also to the implementation of the Western Balkan Green deal and other strategies and programmes developed by the Government in the field of environmental protection and climate change in which the proposed project fits perfectly. Rural women in Serbia are, however, less likely to access and use information about opportunities for sustainable development in their environments. They also lag behind in using communication technologies. This makes rural communities more vulnerable and capable of accessing information and using opportunities which exist, neither to express their interests nor to participate in their creation. Computer literacy will be one of the crucial points in building skills of rural women which are expected to enter new green businesses.

Strategy for the development of education and upbringing in the Republic of Serbia until 2030 in its general and specific goals does not deal with gender issues related to the number of boys and girls, and later men and women, covered by all levels of education from preschool to university level as well as those in the adult education program. The situation is the same with the Action Plan for the period 2021-2023 for the implementation of the Strategy for the development of education and upbringing in the Republic of Serbia until 2030.

Gender imbalance is obviously highly expressed in the information and computer technology sector, which represents the best prospects for progress both economically and in terms of participation in creating the future. Data on gender structure indicate the dominant participation of men in the ICT sector, which makes up two thirds of employees in the entire sector. In the narrower field of information technologies, men occupy as much as 70% of jobs; i.e. the participation of women in ICT is 30%. It is estimated that less than 20% of women are in managerial positions, while less than 10% of women are in the leadership position of a company director. [57]

Economic participation, employment

Per FAO's National gender profile of agriculture and rural livelihoods: Serbia (2021) aging index ((the ratio of the population aged 60 years or more to the young population aged 0 to 19 years old) is 142.9, with significant differences between women and men (165.0 and 122.1, respectively). The unemployment and inactivity rates are considerably higher among "non-urban" women" (13%.3. and 41.6%, respectively). Overall, the employment rate of women is 38.1%, which is by 14.7 % less than the employment rate of men. The greatest gender gap on the labour market in regard to employment is noted in the age category 55–64 years, in which the employment rate of women is 32.5% and the employment rate of men is 52.8%. Unemployment is quite prominent among the subgroups per gender and age among younger women aged 15 to 24; the unemployment rate is 39.5%. /While men

[56] FEM4FOREST

[57] Strategy for Gender Equality for the period 2021-2030 year

of the same age group have a slightly better indicators, they are in no better position, for men between 15 to 24 is 32.2%. There are twice as many self-employed among men than among women (in the age group 15 to 64, 28% of men and 13% of women are self-employed).[58]

The share of informally employed women and men is larger among self-employed persons than among those employed in all age groups. The largest number of informally employed are aged 15 to 24 (67.4% women and 54.5% men). The inactivity rate for women is higher by 16.3 p.p. than for men (54.6% vs. 38.2%). The largest gender gap considering inactivity occurs in the age group 55 to 64, where the inactivity rate for women is 64.8% and for men it equals 40.4%.[59] In the period 2016-2019, the number of registered employees among both sexes increased, with the growth of registered employment being more pronounced among women - about 93 thousand or 10.3% compared to about 70 thousand or 6.4% among men.[60]

The gender gap in employment is more pronounced in rural areas than in urban areas. There are limited employment opportunities outside agriculture for women living in rural areas and agricultural work accounts for a significant share of their employment (one-third of rural women). This type of work usually takes place in family holdings, where women are seldom registered as holder or manager, and more often in the role of unpaid family worker. Rural women are less likely than urban women to find employment in highly skilled jobs, and are more likely to be employed as farmers, in manual and elementary occupations. In comparison with men living in rural areas, women are less likely to be employed as farmers and in manual jobs, and more likely to be employed in unskilled and elementary occupations.[61]

The activity rate of women in the labour market was 46.5% and by 15.5 percentage points was lower than the male activity rate (62%) in 2020 in the working age population older than 15 years. The female employment rate was 42.2% and by 14.4 percentage points was lower than the male employment rate (56.6%). The female unemployment rate was 9.4% and by 0.7 percentage points. was higher than the male unemployment rate (8.7%).[62]

The gender pay gap for 2014 was 8.7%, which means that women were paid 8.7% less than men. Serbia is among countries with the lowest gender pay gap in Europe. However, if earnings are observed according to educational attainment or occupations, the difference in earnings between women and men is significantly higher than the average gender pay gap, most often in favour of men. „ The proportion of women with low wages in the total number of employed women (24.4%) is higher than the proportion of men with low wages in the total number of employed men (21.6%). „ Despite the fact that women, on average, live longer than men, it is noticeable that in all categories of old-age pension beneficiaries, women live, on average, shorter than men.[63]

Women work daily 5 hours more than men unpaid and spend on average 42 minutes less on paid work than men. Women work 40% more on unpaid jobs and are engaged 2 hours and 18 minutes more time on unpaid household work than men. Their total work hours are longer. 95% of women and 77% of men participate daily in unpaid jobs[64]. There are differences between urban and rural areas, with

[58] Statistical Office of the Republic of Serbia (2017): Women and men in the Republic of Serbia, ISBN 978-86-6161-166-7
[59] Ibid
[60] Employment Strategy of the Republic of Serbia for the period 2021-2026 year, Official Gazette of RS No. 18/21 and 36/21 - correction
[61] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en
[62] Report on the implementation of the National Action Plan for Employment from 2020, https://www.minrzs.gov.rs/sr/dokumenti/izvestaji/sektor-za-rad-i-zaposljavanje-0, Ministry of Labor, Employment, Veterans and Social Affairs of the Republic of Serbia
[63] Statistical Office of the Republic of Serbia (2017): Women and men in the Republic of Serbia, ISBN 978-86-6161-166-7
[64] Gender analysis of economic value of unpaid work in Serbia, UN Women, 2020. Available at: http://socijalnoukljucivanje.gov.rs/wp-content/uploads/2020/08/Analiza_monetarne_vrednosti_neplacenog_rada_UN_Women_SRB.pdf

rural women spending more time on unpaid household work than any other category[65]. Taking into consideration the average time spent on unpaid care activities, UN Women in Serbia 2020 have estimated the value of unpaid care activities to15,1% of GDP where 10,4% of GDP is unpaid work of women and 4,7% of men.[66]

There are significant regional disparities in the indicators on women farm holders/managers and women employed in farming (higher share of women was recorded in medium and large-sized holdings in the Vojvodina region, where women are younger, better educated and, as a result, more likely to use modern management practices. The South-East Serbia region is distinct from other regions due to its higher share of women across all indicators, which is likely to be related to negative demographic trends and women's longer lifespans. (Life expectancy in Serbia for the year 2020 was 71.4 years for men and 77.2 years for women.[67]). Conversely, women in the Šumadija and Western Serbia region are underrepresented among farm holders and managers, probably due to the higher share of younger members in the household and the higher rural population density in this region[68].

The COVID-19 pandemic has had a significant impact on rural women. The most severely affected sectors were wholesale and retail trade, accommodation, transport, food and beverages, service activities, forestry and logging, and crop and animal production. In these sectors, over 700 000 workers are estimated to be at immediate risk because of the characteristics of their jobs, which include informal employment, short-term contracts, and working in micro enterprises which are particularly vulnerable in this crisis.[69] Both women engaged in agricultural activities and those working in non-agricultural sectors have been significantly affected by the crisis. Women living in rural areas and working in non-agricultural sectors have been at the highest risk of losing a job compared with women from urban areas and men from both types of settlements. In addition, more rural women than rural men have had to leave their jobs and stay at home to take care of children.

During the state of emergency, rural women have also faced many challenges considering maintaining value chains for their products (starting farming activities on time, finding the necessary workforce, and selling their products). Some innovative solutions were used, such as marketing online, but these were not available to the majority of rural women. Rural tourism got the new chance,[70] but investments required to get more rural households on board could not be obtained, while direct financial support and loans offered were insufficient - rural women need additional support to fully recover from this crisis.

Division of labour among women and men

Rural women's roles are limited to households and farms. They also often sell agricultural products on local markets. Sometimes within agriculture products one can also find wild fruits or eatable greens, while mushrooms and medicinal plants in the form of herbal mixtures are more often offered by men. Women move into the forest to collect forest products just in company with a man from the family (rarely other women; it is more typical for women returnees from urban areas). Although many rural women drive vehicles, they rarely use their family cars to go alone to the forest, or elsewhere. Of course, differences exist and reflect the culture, religion, and ethnicity of the community. Villages which are side by side, however, thanks to the history of their inhabitants, can differ in habits. For

[65] Time use survey, Republic Office for Statistics, 2015.
[66] Gender analysis of economic value of unpaid work in Serbia, UN Women, 2020. Available at: http://socijalnoukljucivanje.gov.rs/wp-content/uploads/2020/08/Analiza_monetarne_vrednosti_neplacenog_rada_UN_Women_SRB.pdf
[67] Statistical Yearbook 2021, Statistical Office of the Republic of Serbia
[68] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en
[69] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[70] Đorđević-Milošević et al. (2020). Economic diversification policies and rural tourism in South East Europe, https://seerural.org/wp-content/uploads/2020/11/Economic_diversification_policies_and_rural_tourism_in_SEE_2020.pdf

instance, villages from which families worked abroad, have more emancipated women population which use vehicles and agriculture machinery regularly, while a neighbouring traditional village will have even women going to locally organised training for women producer groups on milk quality accompanied by men. When it comes to employment in enterprises dealing with forestry, women deal usually with administration and accountancy.[71]

In the period 2015-2019 the share of informal employment of women of working age was 15.6% (the share of informal employment of men was 15.8%), which is a result of higher growth of formal employment of women (19.4%) in that period than the growth of formal employment men (11.5%). The structure of informal employment differs significantly in terms of professional status: mostly informally employed women work as assisting household members, and informally employed men are predominantly self-employed. The position of these women is particularly vulnerable, as they not only do not exercise their formal employment rights (as do other informally employed) but are not paid for their work.[72]

Employment status, including unpaid employment, of men and women in urban and rural areas in Serbia in 2018, is given in Table 1.

Table 1. Employment status, by sex and area of living in Serbia in 2018

| Employment status, by sex and area of living in 2018 in [%] | | | | |
|---|---|---|---|---|
| Status | Men in urban areas | Men in rural areas | Women in urban areas | Women in rural areas |
| Salaried workers | 79.4 | 54.6 | 89.3 | 55.6 |
| Self-employed with employees | 6.1 | 2.8 | 3.1 | 1.4 |
| Self-employed with no employees | 13.9 | 36.8 | 6.4 | 21.9 |
| Contributing household members | 0.7 | 5.8 | 1.2 | 21.1 |

Source: FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en

Contributing household members/unpaid family workers are a specific category in the workforce. They are engaged in some form of family business, and work without a labour contract, salary or social benefits based on labour status. Rural women working on family farms form a significant proportion of this category. When family farms are registered, they have a legal obligation to pay social contributions themselves (for example, pension, and disability insurance). However, if the farm is not registered, their economic engagement is fully unformal, and they are not legally required to pay social contributions. Even on registered farms, informal workers often do not pay social contributions and therefore do not exercise their welfare rights based on employment.[73]

Longitudinal data from the Labour Force Survey indicate a steady fall in the number of unpaid family workers. In 2007, there were 132,553 women and 47,528 men in this employment status, while in 2019 there were 99,200 women and 37,900 men. The majority of unpaid family workers are located in rural areas, working on family farms - 91,400 women and 32,600 men in 2019. Their share in the total number of employed persons is 7 percent for women and 2 percent for men.[74]

[71] Ibid
[72] Ibid.
[73] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[74] Ibid.

During the first half of the 1990s, the economic position of farmers improved relatively, compared to the lower strata of manual workers (unskilled and semi-skilled), as a result of less elastic demand for agricultural products. The fact that farmers are no longer at the very bottom of the material position hierarchy has persisted to this day (with a further significant reduction in their numbers and an increase in inequality within the group, due to the formation of a subgroup of large landowners). [75]

Women made up the majority of employees in administrative services (60%), among professionals and artists (59%), in service and trade occupations (57%), among engineers, professional associates and technicians (53%), and in simple occupations they made up half of the employees in 2019. Men were in the majority among artisans (83%), machine operators (78%), executives (directors), officials and legislators (67%) and farmers (58%).[76]

There are differences between rural and urban women in terms of the type of occupation they perform. Employed women in rural areas are less likely than employed women in urban areas to work as professionals, artists, engineers, technicians, associate professionals, and clerical support workers, and much more likely to work as farmers, crafts and related trade workers, machine operators and unskilled workers in elementary occupations. Relative to rural men, rural women are more likely to be employed as professionals, engineers, clerical support workers, service and sales workers, and in elementary occupations, and are less likely to be employed as trade workers and machine operators.[77]

In the process of implementation and monitoring of measures, activities and interventions in employment policy, a gender-sensitive approach will be applied and strive to achieve gender-balanced results based on gender-disaggregated data. The principle of gender responsive budgeting, which implies equal inclusion of unemployed women and men in measures, including gender-balanced opportunities in the employment process, will continue to be applied.[78]

<u>Political participation: decision-making and leadership</u>

Serbia has made the greatest progress in the subdomain of political power, in comparison with its progress in other domains and when compared with the results of many other countries in Europe in 2018. The reason for this is the enforcement of legal provisions on the 30 percent minimum quota (and since 2021, 40 percent) for women in Serbian National Assembly and AP Vojvodina Assembly. Women members of parliament account for a 37.6 percent share in the legislative branch, Serbia's National Assembly. The current National Assembly of the Republic of Serbia consists of 98 women (39.2%) and 152 men (60.2%). The President of the National Assembly is a man. The National Assembly has 20 committees headed by 5 women (25%) and 15 men (75%). Among the deputy chairmen of the board are ten women and nine men, and among the secretaries of the board are 16 (80%) women. On the board, women make up the majority of the six boards. The Prime Minister of the Republic of Serbia is a woman. Of the five Deputy Prime Ministers, two are women (40%). Out of a total of 23 ministries, 10 were led by women (43.4%).[79] The participation of women in political life begins with their engagement with political parties and their associated bodies. Membership in political parties is, however, not monitored systematically by gender. Data on membership in political parties and their bodies are not disaggregated by gender; they are often out of date or unavailable on the parties' websites.[80] According to the social stratification survey conducted by the Institute for Sociological

[75] Lazić M., Cvejić S. (2019): Stratification changes in the period of consolidation capitalism in Serbia, Institute for Sociological Research, Belgrade
[76] Strategy for Gender Equality for the period 2021-2030 year
[77] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[78] Employment Strategy of the Republic of Serbia for the period 2021-2026 year, Official Gazette of RS No. 18/21 and 36/21 - correction
[79] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[80] Strategy for Gender Equality for the period 2021-2030 year

Research (ISR) in 2018 ,the share of rural women's party affiliation is lower than that of women living in urban areas (6.8 percent and 10.6 percent, respectively).

Both rural and urban men are more politically engaged – 11.6 percent of men in urban areas and 13.7 percent of men in rural areas are affiliated to political parties, out of which 6 percent are active members. Looking at specific forms of political participation over the last three years (for instance, petitioning politicians or government officials, signing appeals/petitions and joining a strike or blockade), we can see that the urban population, both women and men, are more likely to engage in these forms of political activism. The survey reveals that around 9.4 percent of women and 9.8 percent of men have signed some form of appeal or petition which is significantly lower than in urban Serbia. Only 1.5 percent of rural women engaged in diverse protests in public (double less than urban), while rural men are also less active in this way (0.8 percent), compared with 3.6 percent of urban.

Looking at the share of women elected as presidents of municipalities or mayors, there is a substantial gender gap; only 6.6 percent of women hold these positions. The share of women as municipal presidents/mayors equals 6.6% only, and 31.2% of councillors in local government bodies are women.

Environmental authorities provide equal opportunities for women and men to participate in the natural resource management system and in the right to be informed about the state of the environment. The Ministry in charge of environmental protection implements gender mainstreaming during the planning, management and implementation of plans, projects, and policies for environmental protection.

Leadership of women in local governance/political systems and formal/informal institutions

Gender Equality at the institutional level was shown by the Research of the Commissioner for the Protection of Equality from 2017. In 150 local self-government units in Serbia, there are 12 women (only 6.6 percent) in the positions of mayors (mandate from 2016-2020) which is evidence of a substantial gender gap.[81] 13.2 percent of women are in the position of the president of the Municipal / City Assembly. The data from this research shows that there are only 12.7 percent of women on local community councils, which is about 2,000 women compared to 19,000 men. Also, when it comes to managerial positions, research data show that men make up 83.7 percent of directors of public companies. Another survey conducted by the same institution shows that one fifth of Serbian citizens still think that managerial positions are intended for men and that the best role for a woman is to be a housewife.[82] At the level of towns and municipalities, 33 percent of towns and municipalities have not enforced the statutory quota.[83]

Considering the small share of women in local community bodies, rural women experience far fewer opportunities to improve their lives in accordance with the needs of the communities in which they live. According to the findings of a qualitative study of women living in the Zlatibor district, for instance, the patriarchal values typical of rural communities represent a significant barrier to their more active involvement in political life. Moreover, the barriers to political engagement identified by the women included: the absence of family and community support; the ingrained opinion in rural areas that a woman's place is at home; a disregard for women's opinions and proposals; and the heavy combined burden of work on the farm, household work and caring for children or other relatives.[84]

[81] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[82] Đajić M. (2020): Women's leadership in the context of social cohesion – public policy proposal, UNPBF, https://cmv.org.rs/wp-content/uploads/2020/12/Z%CC%8Censko-liderstvo-i-drus%CC%8Ctvena-kohezija-Milos%CC%8C-%C4%90ajic%CC%81.pdf
[83] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[84] Ibid.

There is a considerable opportunity to promote women leadership, following the results of various projects supporting entrepreneurial activities of rural women, especially of the existing cooperatives and associations women initiatives in the last two decades. During 2020, GIZ regional campaign "*Empowered women, empowered economy*"[85] it was noticed that rural women united in women entrepreneurship organisations had much more follow-ups after the projects supporting their work were finished then average. However, these initiatives failed to expand on local and regional communities (men) afterwards, even in cases when local communities did recognise their achievements meaning that leaders of these initiatives no matter of achieved results or increased capacities are considered local community leaders even when no other local leadership exist.

An example of good practice in promoting the leadership of women in politics is the establishment of the Women's Parliamentary Network in the National Assembly of the Republic of Serbia. It was founded in 2013 by the decision of all women deputies, and still exists today. The Women's Parliamentary Network was the first multi-party group to submit joint amendments to the budget and managed to negotiate their adoption with the Government of Serbia. A good social example is also the Women's Platform for the Development of Serbia. It is an informal network that seeks to connect women's voices through dialogue, within the women's movement and with decision-makers. It seeks to include and reinforce the voices of women from vulnerable groups primarily through cooperation and dialogue with women's organizations and women experts. It is based on the "Palić process" which gathered women from Serbia and the region for the first time in 2000 to talk about what kind of future they want - what public policies, what institutions, what services, what everyday life.

<u>Women's and men's participation in household and community level decisions</u>

Rural women in Serbia are much more oriented to the local community because their mobility is lower. They are mostly unemployed/housewives and work informally in agriculture and their primary role is taking care for children and household, food processing. They are often engaged with work in stables or orchards in the yard. They also have low levels of education (mostly elementary school) because they leave high school and marry young. If they manage to educate further, they often leave their villages and move to towns, except if these are suburban or close to some industrial centre when they combine employment on the farm and in households with formal employment. Being preoccupied they are basing their primary farm production on lands in vicinity of their households, contributing to their degradation due to intensity of their use and mismanagement of its fertility and pollution (particularly in case of milk and pig production in stables which is often combined with off-farm employment). In such a household organisation, abandonment of remote plots of agricultural land (arable and grasslands) is regular, which is leading to their economic degradation (transformation in coppices). Forest exploitation is also oriented towards closest stands and those easily accessible due to existing roads.

The main use of private forests in rural areas (and as a consequence degradation factor) is the use of wood in the industry and for the heating. This activity is based on mechanization and physical strength and that reduces women's ability to make decisions about use of the forest or land, even if they are formal owners of the land. Because of the role in the household and gender roles, they have lower levels of the autonomy and influence in decision-making at the household and/or community level.

In some projects in Serbia is already noticed that women with knowledge (technologies), information (available funds) and skills (running machinery for instance, or introducing processing, handcrafts, tourism, or other forms of adding value to primary products) have much more favourable position in

[85] GIZ (2020): Support to Economic Diversification of Rural Areas in Southeast Europe" (SEDRA), implemented jointly by the Regional Rural Development Standing Working Group in Southeast Europe (SWG) and GIZ Open Regional Fund for South-East Europe – Legal Reform in the framework of a regional awareness campaign: "Empowered women, empowered economy"

making decisions. Also, men informed about the importance of the role of women in value chains (for instance, quality of food products, food safety, promotion, and marketing skills) start reducing their resistance to include women members of their families in decision-making[86]. In the public sector women with additional skills and higher education do not necessarily benefit in getting more attention and opportunity to participate or make decisions, except in cases when they are politically engaged. Therefore, the introduced rule of having more than 30% of women taking decision-making positions in the Government does not necessarily respond to women skills, similarly to men.[87]

Women's civic participation is characterized by predominantly traditional patterns, reflected in their participation in church-affiliated and other religious organizations and associations, in events related to handicrafts, and in humanitarian actions and similar forms of civic activity. Women do express a readiness to join women's cooperatives, but these types of cooperatives are very rare. Types of decisions usually made by rural women are linked to the daily routine in households. If large investments are to be made, women and men mainly share responsibilities. Investments on the farm are more in the hands of man; rarely women are asked for the opinion. Community decisions are under men's jurisdiction, except in cases where single women run household. Constraints restricting women's active participation in household and community level decision are rather cultural then economic or political (patriarchal culture as described above).

In local self-governments, during the planning, management and implementation of plans, projects and environmental policies, gender mainstreaming is applied. This is the result of much stronger environmental awareness and involvement in environmental organisation of women than men. Recent struggles for water in Serbia (protests against derivation hydroelectric power plants) have increased synergy in communities under threat, mobilising both men and women. Yet men were those who were traditionally expected to defend physically their properties and interests; so, activism is in vulnerable communities still more in the hand of men, which is a logical consequence of the violent character of recent events. The project should go further in empowering women and all other members of local communities in decision-making considering environmental protection, natural resources management, forestry, and rural development policies. For this purpose, ecological movements of various kinds could be used as a driving force for the project activities and a vehicle for getting on-board rural communities.

## 2. THE CONTEXT OF THE PROJECT

### *2.1. Situation of women and men in rural areas and the forestry sector*

According to the Gender Equality Strategy for the period 2021-2030. year, the **economic position of women in the Serbian countryside (rural areas) is unfavourable** due to less chances for sustainable employment and less personal ownership of resources (land, arable land, and real estate). In addition, there is inadequate access to: community services that would increase women's chances in the labour market (care and nursing services for children, the sick and the elderly), public transport, information on rights and available support, information technology, knowledge and lifelong learning programs would create opportunities for sustainable employment, self-employment, cooperatives, and social entrepreneurship. Current support for women in agriculture and rural development is insufficient and needs to be replaced by systemic support that contributes to sustainable change and improves the quality of everyday life in rural areas. All measures proposed to reduce the gender gap in the economy should especially include women from all age groups in the countryside.

[86] SIDA project "River of Milk" in South-East Serbia
[87] Commissioner for the Protection of Equality, 2017

The share of women farm holders in Serbia in general is comparable to that found in countries with similar cultural, economic, and social heritage. According to the Farm Structure Survey (FSS) in Serbia, in 2018, women farm holders accounted for 19 percent of total farm holders and women farm managers for 15 percent of the overall total. There has been a slight increase in the share of women farm holders since 2012, but the average age of women farm holders is 65 years, suggesting that this increase is primarily related to the depopulation of villages and traditional inheritance patterns in which the farm holding is passed to the eldest family member.[88] **There is no data on female ownership over forests in Serbia**.

According to the study conducted by the project FEM4FOREST in 2021, in the forestry sector (generally defined as covering forestry, sawmilling, wood processing and furniture manufacture, as well as pulp and paper production), data are available on positions and gender distributions in forest administrations and state-owned forest companies. There are no exact data of share of female workers in SMEs in forestry and NWFPs-based private enterprises in Serbia. Jobs in the public forest sector seem to be more attractive to women. Data presented in this study give an indication of the number of jobs in the public forest sector and the shares of women and men, without an insight into the type of jobs and the qualifications of employees. This analysis will also inform the integration of relevant modules in the syllabi prepared within the project frames.

<u>Women in forest administration</u>
The share of women in forest administrations in Serbia is 36% (3 out of 4 female employees in the forest administration are trained foresters, and it seems common that women hold leadership positions in forest directorates at provincial and state levels). Around ¾ (76%) of women employed in the Forest Directorate have professional forestry education. Almost all of them (14 out of 16) are forest engineering graduates and 2 have master's degree. Around ¼ (24%) of employees in the Section of forestry and hunting inspection are women. All of them (8) are forest engineering graduates.

<u>Women in forest decision-making</u>
Women are heads of the Department for forestry policy and implementation of measures for forestry improvement and the Department of the Section of forestry and hunting inspection in Belgrade. Apart from these positions, there were no other management positions where women had been in the previous 10 years. In addition to management positions, women (with professional forestry education) work on the following tasks within the Forest Directorate: forest protection, monitoring and analytical data in the field of forestry; monitoring, the implementation of forestry policy measures; protection and improvement of forestry; planning activities in the field of forest reproductive material, republic forestry, and hunting inspector.

<u>Women in the public forest user sector</u>
In State Forest companies only 20% of women (trained foresters) work, while in both public and private companies. The share of women in the total number of employees in Public Enterprise (PE) Serbia Shume and PE Vojvodina Shume" is 3,191 and 1,443, of which women are 699 and 239 respectively. Total number of women with professional forestry education is 188 and 42, respectively, of which no one holds PhD degree, while MSc degree has 5 and 10 employees and BSc/graduate forest engineer degree 127 and 22. Forest technicians were 56 and 10 in PE Serbia Shume, 22% of the total number of employees are women, and 27% of them have a professional education in forestry (67% are graduate forest engineers).

Currently, women (with professional forestry education) hold following management positions: head of gene fund and nursery production department, head of development department, head of foreign trade and marketing department, head of forest management and forest management planning

[88] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en

section, head of forest utilisation section, head of commercial affairs section, head of forest administration. Women were in the same management position in the last 10 years, as well as assistant director for forestry and head of department for forest protection and protected areas.

In PE Vojvodina Shume, 17% of the total number of employees are women, and 18% of them have a professional education in forestry (52% are graduate forest engineers). In this enterprise, 3 women (with professional forestry education) currently hold management positions: head of forest management and forest management planning section; head of section for hunting and fishing; head of commercial and marketing section. As well, women were in the following management position in the last 10 years: director-general (2005-2019), assistant director for forestry (2015-2019), head of forest estates and forest administrations, head of forest management and forest management planning section, head of section for hunting and fishing, head of commercial and marketing section.

Employment in the public forestry and logging sub-sector accounted for 0.3% in Serbia in 2019 (SO, 2020)[89]. Employment of women in forestry decreased in Serbia, from 28 % to 27 %[90] in the period 1990-2010. Between 1990 and 2015, the share of women employed in the forestry and logging was ranging between 14 and 16%[91]. In the sub-category "silviculture and other forestry activities", the share of female workforce was 16% in 1990 (the highest), and decreased 2000 (9% in 2000 and 2010 and 10% in 2015). This sub-category has the highest share of the female workforce, compared to other sub-categories[92]. According to available data, no female workers were recorded in the period 1990-2015 in sub-category "logging"[93]. The share of female workforce in the sub-category "gathering of NWFPs" was the highest (6%), while in other years the share dropped (3% in 2000, and 4% in 2010 and 2015)[94]. In the sub-category "support services to forestry", the share of female workers was 10% in 1990 (the highest), 5% in 2000 and 6% in 2010 and 2015.[95]

<u>Women in the private forest user's sector</u>

The data for female employment in the private forest sector is limited and can't be properly even estimated, since there is no active register of forest owners,[96] while forest land is not included in the Farm registry. Plus, due to that fact, female forest ownership shares do not correspond with equivalent shares of women in the forest workforce,[97] so any cross-calculation might not be precise either.

Women are generally not owners of the land; they are more often in the role of the unpaid family worker on the family farm than in the role of farm head or manager[98]. Therefore, it is difficult to assess how the lower number of women employed in forestry and logging. The share of women doing lumber jack work or being engaged as technicians in forestry operations is rather low. Women are much more represented in informal forest work related to non-timber forest products (collection, processing, gastronomy, and tourism), which is much less visible and hardly possible to quantify.

Women have lower potential to carry out the processes of rural household economy diversification, even though diversification is an important driver of rural development in the family farming sector.

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[89] SO (2020): Report on registered employment - annual average 2019, Statistical Office of the Republic of Serbia, Belgrade

[90] FAO and UNECE (2020): Forest sector workforce in the UNECE region Overview of the social and economic trends with impact on the forest sector, UN Forestry and Timber Section, Geneva, Switzerland

[91] FRA (2020): Global Forest Resource Assessment 2020 Report Serbia, FAO, Rome.

[92] Ibid

[93] Ibid

[94] Ibid

[95] FEM4FOREST (2021): Country report for Serbia, Forest in Women's Hands Report on Current Situation and Position Of Women in Forestry in Danube Region, Belgrade

[96] UNECE (2020): Who owns the forest? Forest ownership and tenure in the UNECE region. UNECE/FAO/FACESMAP study, United Nations Economic Commission for Europe, Geneva

[97] Country report for Serbia in Fem4Forest Status report (2021)

[98] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en

Nevertheless, the share of women managers is higher than average in relation to holdings specialized in certain types of primary production, such as poultry, sheep, goats, cereals, oleaginous and protein crops, root crops, arable crops, flowers and decorative plants, mushrooms, grapes, nuts, and fruits; and this is in line with traditional forms of gender segregation in performing economic activities on family farms. The types of production with an above-average share of women holding managers vary from region to region.[99]

<u>Women in nature conservation and forests related services, research, and education</u>
Trained female forestry professionals are more prevalent in research, knowledge transfer, pedagogy, and nature conservation. This fact might imply that women coped with the lack of job opportunities in traditional forestry, by shifting to other subfields. The women participation in forest pedagogy (employees) is relatively high (forestry engineering (B.Sc.) level from 26% at the Forestry Faculty of University of Belgrade; share of women with doctorates in forest sciences is also considerable). Given the high shares of female graduates in forestry programs, the future of the sector may become more female. The forest sector in the Balkan countries is seen as significantly potent in terms of self-employment in wood production and collecting and processing non-wood forest products, especially in rural areas. Research in Serbia showed that 12,300 self-employed workers in rural areas were engaged in firewood production in private forests. This is important, as these workers are not reflected in national employment statistics (Vasiljević, 2015 through UNECE, 2020).[100]

For female working in nature conservation is more attractive than in timber production. The share of women in a total number of employees in National parks (NP) PE “NP Fruška gora” / 42 of 140, PE “NP Đerdap” 24/85, PE “NP Kopaonik” /71 and PE “NP Tara” 37/185. In PE “NP Fruška gora”, 30% of the total number of employees are women, and 14% of them have a professional education in forestry (50% are graduate forest engineers). Currently, 1 woman (with professional forestry education) holds a management position, working as nursery manager. In the last 10 years, 2 women were holding management positions (head of protection and improvement unit). In PE “NP Đerdap”, 28% of the total number of employees are women. However, none of them has a professional forestry education. In the last 10 years, women with professional forestry education did not hold any management position.

In Public Enterprise “National Park Kopaonik”, 13% of the total number of employees are women. However, none of them have a professional forestry education. In the last 10 years, 1 woman (with professional forestry education) has held a management position, working as head of the section for improvement of natural values. In PE “NP Tara”, 20% of the total number of employees are women, and 27% of them have a professional education in forestry (70% are graduate forest engineers). Currently, 2 women (with professional forestry education) hold a management position, working as assistant director for planning, protection and development and head of private forests section. In the last 10 years, 4 women (with professional forestry education) were holding management positions (assistant director for planning, protection, and development; head of private forests section; a nursery manager, and head of landscape planning section).[101]
Available data indicates that women also tend to be more present in forestry societies, play greater roles in Environmental NGOs and civil society groups and may lead entrepreneurial associations. However, there are no official statistics on private forest organisations (PFOAs) in Serbia, which are also registered as NGOs. Thus, the number of members, the share of women, forest area, etc. remain unknown.

[99] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[100] UNECE (2020): Forest sector workforce in the UNECE region - Overview of the social and economic trends with impact on the forest sector, Geneva timber and forest discussion paper 76, United Nations, Geneva
[101] FEM4FOREST (2021): Country report for Serbia, Forest in Women’s Hands Report on Current Situation and Position of Women in Forestry in Danube Region, Belgrade

### *2.2. Gender and climate change in Serbia*

Men's and women's vulnerability and capacity to adapt to climate change

Women have lower resilience and often lower adaptation capacities to respond to climate change and related emergencies, such as floods, droughts, or other extreme weather events. Due to their lower preparedness to act in emergency situations, a lack of information, a lack of relevant resource, and weaker economic and financial capacities, women are more vulnerable to the impacts of climate change and disaster events. At the same time, measures to decrease greenhouse gas emissions and reduce environmental pollution can have a more significant impact on rural women than men because they more frequently belong to groups that are particularly vulnerable to such processes. For example, women are more often the holders of smaller farms or businesses; on average, as farm holders, they are older; women tend to be at greater risk of poverty; and they are also more exposed to weather-related risks due to their work in open fields. Recent policy initiatives at a national level have started to integrate gender equality into climate change policies and mechanisms.[102]

One of the facts supporting the conclusion that women will express more vulnerability to any change including climate change-related events, is the fact that rural depopulation in Serbia is not solely related to low birth rates but also to migration trends which women are more susceptible to. Namely, the share of women in the total population of migrants leaving rural areas is higher than the share of men, and the same trend is observable in the younger population (20 to 35 years) compared with the population aged 35 years and older. These trends have significantly contributed to the loss of the working age population in their reproductive years in rural Serbia and have further aggravated depopulation trends (Commissariat for Refugees and Migrations, 2018).[103]

Gender inequalities that may be exacerbated by climate change impacts

Because of their higher exposure to the risk of poverty, and unavailability of climate change adaptation resources, women in general, and rural women in particular, are more vulnerable to the negative impacts of climate change, especially when their livelihoods depend on the use of natural resources, as is the case with agriculture. In addition, women are mostly excluded from the decision-making process, which means that they have less influence on policies designed to combat climate change and increase the population's resilience to climate change.[104]

In the last twenty years, Serbia has been hit by five major floods and five major droughts that have occurred alternately. Both floods and droughts occurred during the main agricultural season, significantly reducing the yields of fruits, vegetables, cereals, and other agricultural crops, which caused great economic losses to many farms and agriculture as a whole. In addition, the floods caused great damage, destroying fertile land, farmed and wild animals; rural and agricultural infrastructure, and also human casualties were recorded. The biggest direct negative impact of these natural disasters was on plant production. Livestock production was most often indirectly negatively affected - malnutrition, poor water quality, and unfavourable breeding conditions. Adverse impacts on plant and livestock production had a negative impact on food production, resulting in irregularities in the raw materials supply.

All of the above affected the socio-economic status of the population, whose livelihood depends on natural resources and agriculture. Although production and farm holders are prevalently men, women own about a quarter of small farms with a standard output of up to € 2,000, who are particularly vulnerable to climate change. Very often, they are completely dependent on agricultural production and natural resources. Seasonal workers in the agriculture sector are also among the most vulnerable

[102] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en
[103] Ibid
[104] Ibid.

socio-economic group. The women farm holders and family workforce members engaged in agricultural production in these holdings are often elderly women. In rural areas, there are numerous one-person households with elderly women. Such households are at higher risk of poverty, which limits their climate change adaptation capacities.

Gender inequalities in real estate ownership are very pronounced – women are much less likely than men to own or co-own property. No data disaggregated by rural/urban area are available but given the prevalence of traditional norms in rural areas, it is possible that the share of rural women owners of real estate is relatively small compared with the aggregate data for Serbia.[105]

Inequalities and the way they affect people's capacity to adapt to climate change

Various analyses and studies done in recent years in Serbia confirm the vulnerability of the population and national economy to climate change. The vulnerabilities of groups and individuals are different and depend on a series of internal and external factors. The level of vulnerability depends on the population/social group and activity, their income, and expenditures, i.e., the inequalities that exist between different social groups, as well as on a series of individual parameters for each individual.[106]

The global process of transformation into a carbon-neutral and climate-adapted society, as well as all other overall processes, can additionally endanger already vulnerable population groups - poor, very old, very young, with serious medical problems, poorly educated, without the possibility of retraining and without access or with limited access to modern education system for new practices, technologies, and sectors in which more employment is expected in the coming period.[107]

The impact of climate change on the population in rural areas is primarily reflected in lower income due to a drop in agricultural production. Also increased health risks due to a decrease in the availability of water resources and more difficult access to health care are feasible. Among them, farmers with minimal capital are certainly the most vulnerable because, most often, they are completely dependent on agricultural production and natural resources. Seasonal workers in the agriculture sector are also among the most vulnerable.[108] Job losses are expected primarily in sectors linked to fossil fuels, and in the agriculture sector, as well as in large enterprises. On the other hand, employment in forestry and small and medium-sized enterprises can be expected to increase.[109]

Floods and heat waves can have particularly negative effects on so-called socially isolated population groups, which by definition include the elderly, patients with chronic ailments and the disabled; those who depend on social assistance, live alone, members of ethnic minorities, homeless people, but also residents of remote and infrastructure-wise poorly connected areas.[110] Extremes coming with climate change as much as social risks of diverse kinds are threatening the struggle to improve gender equality/equity and the achieved results. For instance, the floods in 2014 in Serbia and other disasters in the last 5 – 10 years (including COVID-19 pandemic) showed significant gender-based differences in impact on men and women, as well as in coping mechanisms and participation in prevention and the response, between men and women in affected communities. Gender issues have been identified and acknowledged in several gender analyses and gender mainstreaming in disaster risk reduction has become a significant issue. Since the floods in 2014, DRR policy and strategic framework has been changed and included measures for gender responsive disaster risk management.[111]

[105] Ibid
[106] Božanić D., Mitrović Dj. (2019): Study on the Socio-economic Aspects of Climate Change in the Republic of Serbia, Belgrade
[107] Božanić D., Mitrović Dj. (2019): Study on the Socio-economic Aspects of Climate Change in the Republic of Serbia, Belgrade
[108] Ibid
[109] Ibid
[110] Ibid
[111] FAO (2021): National gender profile of agriculture and rural livelihoods – Serbia, Budapest https://doi.org/10.4060/cb7068en

*2.3. Role women and men are anticipated to play in the context of the project*

The project is designed to follow relevant Serbian national legal and policy frames, as well as international good practices in achieving gender equality in the context of all its planned activities. However, the project has an intention to go beyond following the Committee on the Elimination of Discrimination against Women (CEDAW)[112] recommendations,[113] which have expressed their concern that rural women lack access to education, formal employment opportunities, access to land ownership on an equal basis with men, while their participation in decision-making is limited.

Gender inequality in the rural economy was also concluded earlier in the Strategy for Agriculture and Rural Development of the Republic of Serbia, 2014-2024. Among women there is less participation of active persons, less employees, and fewer ones who work outside agriculture than among men. From the perspective of regional differences, it is noticed that in Vojvodina the participation of unemployed persons among women is slightly lower than among men, which does not mean that their economic position is better, since less than men are employed in the non-agricultural sector, less engaged in agriculture and inactive in a significantly larger number comparing with men. The situation of women is much more unfavourable according to all indicators in South and East Serbia, where gender differences are particularly strong in all segments of the labour market.[114]

Engagement of women in forestry in Serbia remains limited by the needs of households, particularly in rural areas where basic social services and infrastructures are poorly developed. The responsibility carried by rural women in various forest value chains is limited to processing part of NWFP, while their role in firewood and game meat value chains is minimal. In the primary part of the NWFP it is considered that women should avoid going alone to forests due to risks of attack by humans, while animal attacks are not identified as relevant. Hunting and cutting of wood are also almost completely reserved for men, while on the rare occasion, women participating are considered as special attractions, so subject of sensational writing in written and electronic media. Very few women could be found to lead management of hunting grounds, do game wardens jobs or lead hunting societies (more in Vojvodina, then in the rest of Serbia), but they are regularly highly appreciated and valued for their work. Processing of meat as much as processing of timber is also a man's job in Serbia, and not just in rural households, but generally through the sector. It is considered that these jobs are too hard for the physical constitution of women, so they come to the value chains, eventually in the marketing segment, or in case of tourism and catering in the gastronomy part. The processing of the majority of non-wood forest products is mainly in women's hands, although men are also often involved in processing.[115]

In the context of Serbia's aging population and gender disaggregated labor division, rural women, particularly those living alone or with dependent children, are more exposed to energy insecurity due to their poor access to equipment to self-procure fuelwood from forests and therefore are more vulnerable. Furthermore, as forests in the proximity of rural settlements are being overexploited, fuelwood becomes more expensive, and women are more and more dependent on men to acquire such vital resources.

The project assessed that within rural communities, families, especially those headed by single female-headed households, assuring access to high quality fuelwood is an important way to contribute to

[112] EU WOMEN body

[113] CEDAW 2019 Recommendations provided for the upgrade of the Serbian National Programme of Rural Development

[114] Beker K. at all (2017): Situation of rural women in Serbia - report, The United Nations Entity for Women

[115] Dordevic-Milosevic (2019): Socio-economic perspectives of Sustainable forest management & local development in Serbia, Study for GCP/SRB/002/GEF: Contribution of sustainable forest management to a low emission and resilient development in Serbia (FSP), FAO Belgrade

their increased resilience and overall decrease in vulnerability. With the enhancement of the regulatory framework, the project will provide the rules and regulations that biomass sellers must abide by, giving the consumer more possibilities to make informed choices and, hence, possibilities to reduce consumption.[116] This can contribute to cost savings, time and labor efficiency, and an overall reduction of vulnerability. The measures will be accompanied by awareness-raising campaigns on the sustainable utilization of wood fuel. A more efficient biomass burning approach will, in addition to reduced costs, also lead to less indoor air pollution, potentially improving health outcomes for women and children who normally spend more time in their home.[117]

Furthermore, the support to the country in developing short rotation plantations for energy purposes will also provide income opportunities, particularly for women-headed households, to obtain incomes from previously unfarmed land. The activity will also rehabilitate soil quality, making the land more climate resilient and increasing its value.

### *2.4. Resources that women and men have access to*

Women and men have equal legal access to all resources: economic, financial, physical, natural, and other assets. How much of this access is in practice, as well as who manages or controls access to these resources is a question in itself. According to Gender Barometer data, differences in ownership between men and women in Serbia are linked to differences in inheritance and differences in registration of property.[118] According to available data, 25% of real estate is exclusively owned by women; 65% is exclusively owned by men, and 10% is jointly owned in 2019.[119] Men are more often owners of immovable property and automobiles, while women are more often co-owners.[120] Men own around 3/5 of the entire property, and women around 2/5, while the percentage of men owners of agricultural land is twice as high as the percentage of women.[121] UN Women's Shadow report from 2017 provides some indicative figures reflecting properties in rural areas of Serbia:

- 26.7% holdings registered to women
- 15.9% of women manage holdings
- men own around 3/5 of total property
- 88% of houses in rural areas are owned by men
- 84% of women do not own agricultural land[122] (no data for forests)

The gender gap in the economy in the Republic of Serbia is more visible through the unequal position of women and men in the labour market, differences in earnings, pensions and income in general, differences in entrepreneurial activity, access to resources to support employment and self-employment, participation in the informal economy, differences in ownership over real estate and land as well as participation in decision-making in the economy. In each of these observed elements,

116 *According to research in Serbia [E4tech, 2015] using high quality ("dry") firewood compared to low quality ("raw") wood reduced energy consumption per $m^2$ of the heated surface by 16% to 24%.*

117 The modern "top-down" approach for burning fuelwood produces 75% less Particulate Matter pollution then the traditional "bottom-fire" ignition method. https://uneuropecentralasia.org/sites/default/files/2022-11/Together_for_warm_homes_and_clean_air_Outreach%20brief.pdf

118 Marina Blagojević Hewson, Rodni barometar u Srbiji – razvoj i svakodnevni život

119 Strategy for Gender Equality for the period 2021-2030 year

120 Ibid

121 Gender Disaggregated Data – Western Balkans, Statistical Reports 2005-2013, FAO, available at: http://www.fao.org/fileadmin/user_upload/nr/land_tenure/Genders_Report_WB.pdf

122 UN Women (2017): Shadow Report to the Committee for the Elimination of All Forms of Discrimination against Women regarding the fourth reporting cycle of Serbia, ISBN 978-86-900100-0-4

women are still in a far more unfavourable position, while in the group of women there are particularly vulnerable groups of women whose economic situation is even more unfavourable.[123]

Declaratively, prevailing gender gap in education and in the labour market, the knowledge and empowerment of women in the field of green technologies and innovations is priority for gender equality are *ad-hoc* addressed in Serbia. Most activities in the field of encouraging women's entrepreneurship (consistent support measures envisaged for support in SME Development Strategy, Industrial Production Development Strategy and Regional Development Strategy) are still small scale and project-based, rather than systemic. Concrete support exists within rural development support measures (national and IPARD). Women's organizations and women's informal groups do fund raising addressing, traditionally primarily with the violence against women. Work on economic empowerment of women, especially rural women in agriculture, straightening the women’s activism in rural areas is limited in time and budgets, although often appear in various forms of calls for grants in diverse economy and social sectors, institutions, and organisations. This offer is not consistent, neither long-term, so that beneficiaries can plan their actions in long-term.

Yet, the legislative framework is formulated in such a way as to enable equitable access for all beneficiaries, with low eligibility thresholds and requirements, especially when it comes to accessing local and national financial incentives. Support for rural development in Serbia is designed to prioritize holdings registered to women specifically through the following measures: boosting economic activities in rural areas through support for non-agricultural activities; supporting young people in rural areas; and providing loan support schemes and incentives for the preparation and implementation of local rural development strategies. This type of support for women farm holders is reflected in the higher number of points awarded to women applicants when ranking potential recipients of subsidies, as well as in a lower interest rate on loans. Nevertheless, the available data provide an insufficient basis for drawing conclusions about whether these support measures are effective in facilitating the economic empowerment of women, particularly given that the share of women among farm holders and managers remains low. The share of women recipients in total funding for rural development is 23.2 %, which is above their average share among farm holders and managers (19 % and 15 % respectively)[124].

This indicates that women are actively engaged in achieving compliance with the eligibility criteria for financial incentives. The data indicate that the average amount of incentives per holding is higher for holdings registered to women than for holdings registered to men. The advantages (extra points) given to women in the application process for some of the grants might be a factor that motivates women to register their holdings. However, there is no evidence to support this assumption. Measures aimed at boosting non-agricultural activities are another special form of support for rural areas. These measures are intended to boost the development of the tourism sector in rural areas, as well as to revive one of its subsectors – arts and crafts, i.e., cottage industries. [125] Although these activities can also influence the forestry sector and its SFM, there are no data to illustrate this effect, while direct support for supporting forestry does not exist.

Annual reports on the state of small and medium enterprises, entrepreneurship, and competitiveness for the period from 2015 to 2020 do not show data on women's entrepreneurship, so there is no data on how companies operate in relation to the gender of owners, and there is no publicly available progress report on implementation of the Strategy for Small and Medium Enterprises and its pillar number 6, which refers to women's entrepreneurship. There are still no systematic gender statistics on women's entrepreneurship.[126]

[123] Ibid
[124] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en
[125] Ibid
[126] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en

Some progress has been made as a result of the gender responsive budgeting process in the Republic Statistical Office, with the introduction of new gender indicators in business statistics. Progress has also been made with the Public Procurement Office, which since 2020 has started monitoring the participation of women-owned enterprises in the total number of concluded contracts or framework agreements through public procurement procedures on an annual basis. The Public Procurement Office's Performance Report for 2020 states that this percentage was 23%, which is 8% more than the estimate.[127]

Although most of the difficulties faced by entrepreneurs are common to both sexes, there is a particular impact on women entrepreneurs due to sector choice, gender discrimination and stereotypes, underdeveloped and inflexible childcare, difficulties in reconciling family, private and work responsibilities, or differences in attitudes of women and men towards entrepreneurship. Women also have less property and real estate registered in their names, which makes it difficult for them to access finances.[128] Rural women have poor access to finance (primarily the use of credit cards and opportunities for securing a loan; for example, for housing or consumer purposes[129].

### *2.5. Specific needs of women and response strategies to their vulnerabilities*

Women have lower resilience and often lower adaptation capacities to respond to climate change and related emergencies, such as floods, droughts, or other extreme weather events. Due to their lower preparedness to act in emergency situations, a lack of information, a lack of relevant resources, and weaker economic and financial capacities, women are more vulnerable to the impacts of climate change and disaster events. At the same time, measures to decrease greenhouse gas (GHG) emissions and reduce environmental pollution can have a more significant impact on rural women than men because they more frequently belong to the groups that are particularly vulnerable to such processes: for example, if they run farms in Serbia, rural women are more often the holders of smaller farms. The same applies to businesses. On average, as farm holders, they are also older and often single. Women tend to be at greater risk of poverty; and they are also more exposed to weather-related risks due to their work in primary production.

Recent policy initiatives at the national level have started to integrate gender equality into climate change policies and mechanisms. The project will try to assist with gender mainstreaming of related policies to respond to recommendations of the CEDAW Committee in 2019 and will provide incentives for the improvement of the situation of rural women in terms of advice for the access to land ownership, its sustainable use, employment in forestry sector and models for obtaining economic security[130]. Recognizing specific forms of discrimination of women in rural areas this Project is taking into consideration needs of women for provision of knowledge and investments for starting entrepreneurial initiatives on the available land resources by starting primary production, adding value and marketing advice as a basic or additional employment.

In the course of the preparation of this assessment, people, both women and men, were involved in expressing their needs. Interested stakeholders, private and civil society stakeholders which were engaged primarily insisted on pairing financial with the technical support, which is fitting with the experience of the expert involved. The planning and implementation of activities in various development projects were usually decoupled – either capacity building or investment support was provided, generating limited results with no follow-up after the projects were finished.

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[127] Ibid.

[128] Strategy for support the development of small and medium enterprises, entrepreneurship and competitiveness for the period from 2015 to 2020

[129] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en

[130] Committee on the Elimination of Discrimination against Women, 2019, no page number

In all training and investments, when possible, the project will give higher priority to women[131] owning degraded coppice stands or lands for conversion to forest or bioenergy plantation and will ensure that at least 30% of beneficiaries are women, yet the project will not make specific window for women, since experience have shown that actions with an environmental impact demand strong synergetic action and support of whole local community. If just concentrated on the economic benefits of women, wider benefits might be overlooked. This is happening because there is a stereotype that novelties in resources management should not be brought by women, or to be more precise innovative, energetic women who are accepting the challenge to change are considered freaky and a threat by patriarchal society. This is even emphasized with the fact that large percent of women responding positively to initiatives such as those which the project will promote, are those which are either single or considered too dominant in their communities, being heir of land and properties in families which had no male offspring.

Engagement with women and women focused organizations in the geographic area and focal sectors of the Project is crucial to understand the challenges and barriers that they face. At project inception, women and women-focused organizations will be included in consultations to ensure that perspectives, needs and challenges are considered by the Project and in adaptive management. The project will also provide support to local partnerships and take care to properly inform both women organizations as those which have no gender earmark, considering that in situation in which women organizations are often excluded from the regular flows of information from the economic and environmental sector (even stigmatized for being established to make a mess in social sphere).

### *2.6. Opportunities to challenge gender stereotypes and increase positive gender relations*

Recent discussion over the implementation of Nature Based Solutions, by the ADAPT project of the IUCN, come out with the notion that one of the key gender issues in environmental protection is that benefits from natural resources exploitation are reserved to men and negative consequences and risks are shared to the whole community. This is coming from the fact that a very limited number of ecosystem services are utilised and reduced to quantifiable amounts of products.

In the forestry sector this is driving directly to timber and further to hard physical work only man a capable of doing. Stigmatic approach, however, lead us to the obvious replacement of theses, in which the majority of even this work I mechanised, so man does actually use machinery for this work and not their mussels. On the other hand, women are actually those who work mostly manually and men with mechanisation and heavy machines which with modernisation do not anymore require physical strength, but skills and knowledge women and men can obtain equally. Anyway, the perception is that male work is harder and therefore more important. If provided with the training how to use machines and financial support to purchase them, this could not just improve women's economic activities and increase income, but improve their position in the society and make them capable of using their resources without being exposed to extortion of money, which will decrease benefits (it's common that women forest owners, especially those which are single, due to lack of machinery and skills, but also safety issues engage man outside of their families to manage their forests, which is often resulting in frauds; if they are married in majority of cases their husbands make all decisions and manage forests alone).

The project should provide inputs for final including forestry measures in Rural development programme of Serbia, while also providing set of rules to help women get advantage in obtaining

[131] A preliminary list of beneficiaries disaggregated by gender will originate from the digital cadastre of the Republic of Serbia to ensure that gender accounting is well reflected in both the baselines and targets. Depending on the results of the analysis of the cadastre targets will be increased at design.

financial support for improving their economic position and participation in sectors of green and circular economy. Namely, the National Agriculture and Rural Development Strategy of Serbia for the period 2014-2024 defined two strategic goals very important for forestry:
1. Sustainable resources management and
2. Environmental protection and efficient public policy management and institutional framework improvement for agricultural and rural areas development.

The present institutional capacity for supporting rural development is so modest that specific forestry measures remain so far not in place. However, rural development measures which are active, supporting improvement of competitiveness also have space for forestry if properly upgraded, as much as those of environmental kind, rural economy diversification, and upgrading of life quality in rural areas. These could be related directly or indirectly to the forestry sector and also bring on board women and youth using National Rural Development Program, as well as the IPARD programme budgets.

Measures for improvement of protection and quality of agricultural land shall mean the type of subsidies which improve physical, chemical, and biological features of the land, meaning that they should also be in relation to the forestry, although forest land is not mentioned (for instance prevention of erosion and improvement of agriculture land include establishment of forest belts etc.). Under the conditions specified in the Law, structural subsidies may be directed to certain group of beneficiaries, such as farmers that predominantly produce for the market, cooperatives, local self-government units, associations and other beneficiaries engaged in forestry production, processing, and sale of forestry products. The upgrades should include opening Farm Registry to private forest owners and implementation of special conditions for providing advantage to women initiatives as already provided in some measures for women farmers dealing with agriculture (see page 40).

Rural development support to forestry might have two kinds of windows:

Direct support to forestry through forest measures:

- investments in afforestation
- investments in machinery and equipment for primary forest production
- investments in forest infrastructure
- investments in biomass-based energy material production (processing of wooden materials into pellets, briquettes, charcoal etc.)
- investments in strengthening other wood processing
- investments in wild game farming (at the moment it’s financed form the Fund for Hunting Economy)
- etc.

Indirect support to forestry supporting local community benefits through investments in strengthening food production competitiveness through investments in storing and processing of NWFP:

- diversification of rural economy through introduction of tourism related also to forest ecosystems and supporting non-consumptive activities (tourist accommodation, catering, infrastructure, heating equipment etc.)
- diversification of rural economy through introduction or strengthening food supply chains to tourism (investments in processing equipment for NWFP or gastronomy)

- support to HNVF and agroforestry systems for sharing benefits between agro-diversity conservation, nature conservation in nature protected areas dependable of pastoral activities, maintaining forests, and providing quality inputs for tourist activities & food production.
- etc.

Women forest owners will be invited, and incentives will be provided for raising quality forests and forest production systems with establishing full value chains and partnerships – horizontal and vertical, and making them operative. Opportunity to restore their forests including conversion of degraded coppicing stands into high forests and afforestation, including conversion of degraded agricultural lands into bioenergy plantations, will increase value of the lands owned by women, making them less vulnerable socially and economically. Consequently, their confidence and safety will be increased, while participation in decision-making and quality of life will be increased. With the additional knowledge and skills provided, they can establish or strengthen value chains for products they produce or diversify production and get new opportunities for employment. If additional incentives are provided for adding value to these products, women will be able to overcome the problem of lower mobility which is creating unemployment and get more options for income generation in their own environment, without even moving from their farms.

Opportunity for planning and investing in decarbonization of private sector companies, in particular through *insetting* (by **investing in the forest ecosystems)** women may use to provide **assured supply of raw materials for their businesses, while** providing significant, measurable benefits to communities surrounding their value chain which on the end will provide double benefit - **building a strong reputation of a responsible entrepreneur and (environmental) leader** of their communities.

To mitigate risk of not benefitting fully from these anticipated outcomes of the project for rural women, the project will pay special attention to providing good communication and dissemination and preventing lack of information. The project will build on experience gained from the implementation of various agriculture and rural development projects. For instance, when it comes to financing agricultural production and access to favourable credit lines (subsidized loans), women and persons under the age of 40 years are in Serbia granted preferential treatment for obtaining loans and lower interest rates, but access to such information become complicated for “digital illiterate” people after the printed version of Ministry of Agriculture, Forestry and Water Management (MOAFWM) publication for registered farmers (introduced with the farm registry) ceased to exist along with the Rural Development Offices Network (used to have continuous campaigns in all municipalities of Serbia).

An opportunity for rural communities to improve their access to information are represented by diverse associations such as women clubs and societies, entrepreneurs and craftsmen, private forest owner’s organisations and other branch organisations such as beekeeping, organic producers, etc. which are perfect hubs if regularly fed with relevant information. The situation in which Rural Development Network of Serbia is not anymore handy, support to the new networking of the kind within the project is necessary to provide benefit to rural population and decrease their vulnerability deriving from isolation. The synergy with interested of the MOAFWM in agriculture and rural development with the forestry sector, could be expanded to ministry in charge of environment, tourism, and rural affairs, especially because agriculture extension works poor and other ministries do not even have such an extension.

Lack of access to transportation also disempowers rural women: they are less likely to have a driving licence or own a car, and are consequently more reliant on the use of public transportation, which is unsatisfactory due to an insufficiently developed network of bus routes (limited frequency of buses;

inconsistent schedules), poor connectivity to the nearest cities, and long distances to bus stations.[132] This is the shortage which prevents women employment off-farm as much as access to information, financial and legal advice, etc. Creating more jobs through diversification of on-farm and off-farm local jobs through afforestation and adjacent services establishment in local communities is one of the top contributions of the project to vulnerable communities and marginalised groups, in particular rural women and vulnerable communities from remote areas (mountain and other marginal) or with the limited mobility (due to bad road infrastructure and transport, personal disabilities or exposed to various threats including those deriving from climate change such as natural disasters etc.).

Policy and programming documents pertaining to the development of Serbia's agricultural sector and rural areas take into consideration the gender components and the need for the economic empowerment of women in rural areas. Serbia's Agriculture and Rural Development Strategy, in its 12th priority action area pertaining to the Improvement of the Social Structure and Strengthening Social Capital, recognizes women as a category that needs to be empowered. Despite the efforts invested in promoting gender responsive budgeting, the national budget is not yet fully gender mainstreamed. The advances are even smaller at the local level, where the process of transitioning to the new budgeting system is slow.[133] In this respect, the project might contribute to a large extent to the gender mainstreaming of the existing budget intended for cooperatives from the portfolio of Ministry of Rural Affairs. In the end, an important mechanism for gender mainstreaming in national and local policies is gender-responsive budgeting is stipulated by the Law on Budget System.

## 3. SERVICES AND TECHNOLOGIES PROVIDED BY THE PROJECT

It is expected that the transferred technologies and interventions conducted will have impact on mitigation of risks of extreme climate-related disasters and climate change in the whole Serbia, which will have profound effect on vulnerable rural communities and support women and the energy security of their household to decrease the risks they face with due to climate change and mismanagement of key ecosystems services such as energy (e.g. fuelwood). Services and technologies provided by the project will be available and accessible to both women and men. The project will benefit the entire population of Serbia with a specific focus on sectorial stakeholders including vulnerable communities depending on fuelwood for energy, individual forest and agricultural landowners (farmers) and private companies interested in derisking their business and in decarbonizing their processes. In all training and investments, when possible, the project will give higher priority to women[134] owning degraded coppice stands or lands for conversion to forest or bioenergy plantations and will ensure that at least 30% of beneficiaries are women, where the percent originates from the average participation of women in the various sectors addressed by the project[135].

The project will provide knowledge through introduction of the up-to-date technologies related to afforestation, sustainable forest management and forest's ecosystem services (e.g. biomass for energy) use, and support investments in climate adaptive silviculture[136] practices and increasing the

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[132] FAO. 2021. National gender profile of agriculture and rural livelihoods – Serbia. Budapest. https://doi.org/10.4060/cb7068en
[133] Ibid
[134] A preliminary list of beneficiaries disaggregated by gender will originate from the digital cadaster of the Republic of Serbia to ensure that gender accounting is well reflected in both the baselines and targets. Depending on the results of the analysis of the cadaster targets will be increased at design.
[135] Percentage of women involved in forestry is between 14% and 16% [FRA, 2020], percentage of women leading municipalities (2021) is 13% [UNDP, 2021.], percentage of women employed 48.5% [WB, 2022].
percentage of women graduating from agriculture, forestry and accounting universities and vocational schools in 2021 is (aggregated average) 35% [FEM4Forest, 2021], percentage of women involved in forestry in public institutions and public/private forest companies is (aggregated average) 28%[FRA, 2020].
[136] Tailored to the Serbian context from the experience of the US Forest Department, of the Spanish, Italian and French forestry sector as well as from concreate experiences in Lebanon and Armenia where adaptation of forests is considered a series of practices and actions (from seedling to maintenance) needed to enhance the ability of forest ecosystems to adapt and survive in the projected climate scenario. These include: the preparation of drought resistant seedlings; the use of clear handling and planting procedures and specific maintenance protocols.

forest cover, including by establishing shelterbelts/windbreaks[137] in agricultural lands and nature-based solutions for disaster risks reduction and soil quality improvements (agroforestry and agroecology technologies).

Specifically, the following services will be provided on the national level to the MOAFWM and other responsible ministries without specific concern about the gender, since the administration have reached satisfactory level in respect to involvement of women in all capacity building activities. Women will be involved in all stages of the process including public meetings and consultations. Their participation will be visible, recognised and empowered through permanent communication with the team preparing these meetings and consultations.

All documents as much as materials following capacity building actions conducted in administration to be developed will be available on the internet, but since access to digital means of communication is not equally territorially provided, while vulnerable rural communities and particularly their women members often do not have opportunity to use them, project will provide regular bulletin and info sheets which will be distributed not just on-line, but also in printed version through local civil society partners, chambers, agriculture extension services, public and private enterprises and municipal authorities. This action will be preceded by encouraging MOAFWM to re-establish support to the Serbian rural development network as the one recently performed efficiently distribution of information about the support measures available for development in rural areas.

Stakeholders will be regularly informed about the services project has provided within the strengthening of the *Forest Monitoring and Assessment System* and the establishment of the *Monitoring, Reporting and Verification System*. *National strategy and action plans* will be developed with the extensive participation of representation of all relevant stakeholder groups through a national campaign, with the special attention to the representation of all groups, women, and men, while documents will be published and shared in wider population. Addresses for posting will be selected in a way to reflect at least an existing share of women landholders in the farm registry (agricultural land) and cadastre (forest land), namely, bulletin will be mailed to women landowners as much as to men. Certain effort will be needed to expand the address book during the span of the project to include more women.

The programming of the rural development in MOAFWM should finally include forestry which will require establishment of the working group capable of transferring interest of the sector into suitable measures to be included in both National and IPARD plans, while specific interests of the adding value and strengthening forest products value chains should be introduced with all suitable measures of all axes. The consultation with the stakeholders from the women entrepreneur organisations will be of immense importance for shaping new conditions which will provide a positive discrimination for the future project applications deriving from women, youth, and disabled. Revisions will be needed in axes two to accommodate interest of the project activities related to silviculture, establishment of agroforestry and silvopastoral systems, maintenance and revitalisation of traditional farming and introduction of innovative technologies based on forest ecosystems in general (interest of women dealing with the processing of non-wood forest products to grow mushroom, medicinal plants, wild fruits and greens, game etc.).

The positive impact of the project for energy security and biodiversity will include consultations with women about the selection of species for afforestation and reestablishment of forest ecosystems in general (climate adaptive forestry seedlings which also produce fruits which could be local processed

[137] Establishing shelterbelts/windbreaks will not only raise the share of forest cover where most needed (e.g: Vojvodina and Southern and Eastern Serbia [GFA, 2019; NAP report, 2020]), but also reduce the negative impact of wind erosion on agricultural production and prevents burying of drainage and irrigation canals.

or used through women entrepreneurship). Consultation with women should be organised for identifying main problems with land degradation and land use in villages and their surroundings and options within the project to address them by selecting economically most suitable solutions and modelling actions which will mostly tackle land which is regularly used and maintained by women for production or as an environment for their side activities which have to benefit of the well maintained countryside (fruits and vegetable production and tourism).

The national standard for biomass production / handling and use will be based on available international good practices, developed with international experts but adopted to the Serbian context using expertise of women and men working in the business sector and civil society in Serbia.

Design of *Execution guidelines for wood energy plantations* and *Guidelines for decision makers on LULCF to prevent soil degradation* will reflect needs and specific problems in different rural regions of Serbia following the set of natural constraints and specific problems identified during the National Forest inventory and from the work of Soil department in the MOAFWM. *Execution guidelines for wood energy plantations* will be sent, besides the previously mentioned group also to unemployed persons under 40 using local registries. New national curriculum will be promoted in a Bulletin, and schools through a campaign for scholars which are entering high schools and faculties through media and lectures during the life span of the project. Information on the establishment of the *Regional knowledge- for CAS* will be shared through the same system with the addition of the network of Chambers of commerce and business.

*The improved efficiency of wood biomass used for fuel* besides planned activities should occur not just with the business sector, but to include women in households in despite do they possess or not forests or agricultural land. Getting them on board should be done through a wide national campaign in both rural and urban areas for all users of wood biomass for fuel, but promotional activities have to be set in a way to promote territory specific and relevant options instead of general solutions which are not applicable in the certain territory (replacement if wood biomass with the by-products from agriculture in plains of Vojvodina, use of biogas produced from domestic animals excrements in zones with developed livestock husbandry, food waste in urban and suburban zones etc.) Also, solutions for different scales of use should be presented (households, industries).

Services provided by the project should reach diaspora and land and forest owners in urban areas, which are presently not informed about possibilities to properly manage their resources. Since the phenomenon of owners who are distant from their lands is very present in Serbia, mechanisms are needed to be created to activate their potentials as much as to protect further degradation of forests they own (legal frames and incentives).

The project will transfer to Serbia the knowledge, technology and CAS practices needed to reduce the climate change and adaptation deficit of public and private forest stakeholders, as well as to enable the forest sector to become more resilient to climate change and contribute effectively and efficiently to the national decarbonization process. Besides public natural forests, 18.3 thousand hectares of private forests will be under CSA/SFM. Project will support all women initiatives, since priority for support are women and single women households. Selected civil society organizations will be engaged to provide better access to private landowners willing to do afforestation or other interventions planned within the project.

Enabling and strengthening Rural development network is identified as a shared mechanism for mobilizing private sector in rural areas. Institutional capacity development of such a network will in future represent a challenge and innovative approach to the mobilization of private actors in getting involved in decarbonization processes, which might provide more certainty for synergetic activities in agriculture and forestry sectors (with the open possibility for getting on board also environmental one as much as tourism and services etc.). For the sake of enabling the forestry sector to fully benefit from rural development support provided nationally and through IPARD, Rural development network should be targeted also with at least all public awareness activities in this project.

Work with public and private stakeholders including key associations such as the Chamber of Agriculture, the Serbia Grain Association, the Chamber of Commerce, Forest Owner Guild and SerBio, etc. (the list from the Project concept might be expanded during the project implementation to accommodate also interest expressed by various women business associations) which are planned to provide smooth overtaking the ownership on and ensure participation of the private sector in their scaling up and replication at the national level of the project's activities and achievements.

Training and capacity development of both public and private stakeholders will be delivered regardless of gender. Specific attention will be paid to engaging as much as possible women landlords (landowners) using Cadastre and Farm Registry for transferring adaptive silviculture practices and technologies intended to increase the effectiveness and efficiency of investments in forests.

Investments will also be supported for the introduction of the combination of sustainable and climate adaptive silviculture practices (tailored forestry investments (afforestation / forest restoration / natural regeneration). Applications creating new job opportunities for women and young people (women and men) and new markets (e.g., $CO_2$ management, green biomass, climate adaptive nurseries, greening the fuel biomass value chain) will be particularly appreciated, advantage will be given through an addition of the points in scoring marks for the quality of proposal. The scoring system exercise will help create conditions for future use of similar support through national schemes for employment, use of rural development measures, etc.

Future investments should be expanded with the encouragement to municipal authorities to launch decarbonisation strategic planning. Design of a special set of measures could be encouraged for projects targeting efficient biofuels use in households for women and poor (such as, for instance, replacement of old with the more efficient heaters and stoves or introduction of alternative to wood fuels, small biogas production facilities, etc.).

Afforestation, reforestation, natural regeneration, sustainable forest management, and conversion of degraded coppice stands into high forests will be supported through transfer of technologies and financial support, followed by the technical assistance with the climate adaptive silviculture which will also include production of written guidelines on 1. nursery production, 2. soil preparation, 3. planting operations 4. management of the production of diverse and climate adaptive forestry seedlings. Women will be particularly encouraged to embrace these as a part-time jobs, while for young people (both women and men) incentives will be provided for start-ups in businesses of the kind and for generating full time employment.

## 4. PROTECTION FROM SEXUAL EXPLOITATION AND ABUSE (PSEA)

### *4.1. Assessment of risks related to SEAH*

The assessment of risks within this gender assessment did not find any risks related to sexual exploitation, abuse and harassments (SEAH). No indication of risks was found using presented methodology neither from the secondary resources, nor in discussions with implementing partners, nor beneficiaries in pilot regions communities in Banat (Kikinda) and Eastern Serbia (Dimitrovgrad) as typical regions for Vojvodina plains and mountain regions of Central Serbia which were used for a case study during preparation of the GAP. Civil society and public enterprises representatives also did not raise this SEAH risk as a question during preparation of the gender assessment report. The SEAH risk was never a topic of academic research, so most probably dealing with the PSEA may be considered as a non-specific problem for the forestry sector.

FAO and National Designated Authority consider SEAH risks to be minor, as both institutions have strong mechanisms for preventing SEAH under which project will also abide. In the communities surrounding the project, SEAH is also not anticipated, based on FAO experience on other projects in the country.

### *4.2. Mitigating risks and preventing SEAH*

The possibility for SEAH to appear is not specific to the project, but might appear in general, *a*s in any other human activity, so rising awareness considering its threat might be useful for all stakeholders and local communities involved in project to prevent SEAH. Therefore, the project will work to strengthen all involved parties to be able to deal with Protection from Sexual Exploitation and Abuse (PSEA).

The stakeholders engaged on SEAH safeguards within the project will be the Ministry of Agriculture, Forestry and Water Management, PE “Serbia Shume” and “Vojvodina Shume” gender focal points and the focal points in local administrations (officers in charge of human rights and gender) together with the project PMU, which will have a gender and social experts. Both will work with relevant institutions (e.g. UNFPA, UNWOMEN, UNIVEF, OCHA) to develop specific PSEA and SEAH trainings for extending necessary knowledge to project stakeholders and local communities. The project will have an ESS specialist point in PMU in contact with focal points in local administrations.

The constant coordination will be kept between the project gender and social experts, the National Gender Coordinator, and the Regional Gender Coordinator in FAO. The social expert will be in charge of keeping work with relevant gender/social welfare Government ministries and departments, other anti-gender-based violence organizations or networks while gender expert will work with focal points in local administration and project partners MOAFWM liaising with institutional stakeholders that are to project stakeholders and communities.

Preventing SEAH is envisaged through continuous work with women through the duration of the project – more specifically, stakeholder consultations prior and during project implementation will include awareness raising and stakeholder-differentiated understanding of SEAH related risks and mitigation measures. The project will ensure regular visits to communities and local institutions. The Gender and Social expert will work with local government or authorities and to sensitize community members on SEAH safeguards. The Gender and Social expert of the project will support local officials in campaigns on prevention of SEAH. Champions will be identified to, where applicable, act as allies on SEAH safeguarding. SEAH training on SEAH risks, how to report them and the services available including SEAH GRM established by the project will be provided to project stakeholders and communities.

Stakeholder consultations prior and during project implementation will include awareness raising and stakeholder-differentiated understanding of SEAH related risks and mitigation measures. The Grievance Redress Mechanism provides an accessible and inclusive survivor-centred and gender-responsive grievance redress mechanisms with specific procedures for SEAH including confidential reporting with safe and ethical documenting of such cases, that indicate when and where to report incidents, and what follow-up actions will be undertaken.
The Project will ensure that all concerns and/or incidents will be reported to the ESS specialist and the FAO Office of the Inspector General, as appropriate. The Project PMU will also have a gender specialist with PSEA expertise.

The Project Coordinator of the PMU will be responsible for documenting, monitoring and reporting on any grievances received and how they were addressed. AE is going to continuously monitoring and report on SEAH cases to the responsible Governmental body in the MOAFWM and if needed provide required documentation to Ministry of internal affairs, Ministry of justice, Ombudsman and other while keeping the confidentiality as required by the relevant policies and regulations in Republic of Serbia.

## Part II: Gender and Social Inclusion Action Plan

A project-specific GAP provided below should ensure gender mainstreaming in this GCF project. It mirrors the logical framework of the project and is an integral part of project/program design. It provides guidelines on priority actions within the logical framework, as well as those to be taken into account during implementation, monitoring and evaluation of the project.

| The project aims to upgrade Serbia's male-dominated forestry sector and associated decarbonization business by fostering inclusive and gender-responsive approaches that ensure no one is left behind. By promoting women's participation in developing innovative business models, the project will help create sustainable and equitable opportunities for all, contributing to both climate resilience and gender equality. | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Component / Outcome** | **Gender Sensitive Project Activities** | **Gender sensitive activities** | **Target** | **Indicators** | **Timeline** | **Budget** | **Indicative cost of the Component/ Output** | **Responsible partner** |
| **Inception phase** | Consultations with local parties (including women and women focused organizations, private forest land holders and CSOs) to upgrade GAP | Data on ownership of land for afforestation collected and gender disaggregated. Conducting consultations for identifying specific measures related to territory specific features of the Serbian territory out of reassessed pilot areas, to identify employability of women in forest sector, women's preferences related to use of forests to upgrade gender action plan with measures and to provide relevant support within the planned project activities. | At least 90% of interested parties on the local level, primarily forest land holders, are interested in afforestation and strengthening biomass for energy and other forest products processing value chains. | Number of participatory sessions with local CSO and community representatives<br><br>Number of participatory sessions with women focused organizations<br><br>Number of questionnaires filled and percentage of women included | Inception phase | N/A[138] | 8,501,506 | PMU |
| | Networking institutions and upgrading capacities of key institutions and private sector operators to manage the decarbonization process from a gender skills/capacity perspective and build and interactive national network providing inputs for the evidence-based policies through | Conducting regular consultations and monitoring through the network of core national gender responsible institutions, private businesses and those participating in the project | All institutions participating in project - The Ministry of Agriculture, Forestry and Water Management, PE "Serbia Shume" and "Vojvodina shume" and gender focal points and the focal points in local administrations have gender capacity enhanced | Number of reports provided to the network (percentage of participants actively participating in the network sessions and reports presented). Number of training participants | Entire project | 50,000 | N/A | PMU |

138 Included as part of inception phase

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | continuous contribution of disaggregated data from forestry business sector | | | Pre and post training survey (perception of trainees after trainings) Sex and age disaggregated. | | | | |
| | Survey of the national education curricula | Exploring gaps in addressing gender equality and inclusion specifically in the context of Serbia in all planned syllabi or introducing new topic as part of a separate module. Included as part of baseline survey | Whole education system in forestry assessed and targeted for introduction of gender related guidelines – high school legal aspects, academia access to financial and technical assistance, technique and technology employment in green business for women, young people and rural families | Number identified chapters for gender mainstreaming and social inclusion in Syllabi for schools and academic institutions involved | Y1 and Y2 | $ 50,000 | $ 8,501,506 | PMU |
| Component 1 - National-level upscaling of sustainable and climate adaptive silviculture and carbon finance framework | Policy Development Sub-activity 1.1.1.4, 1.1.2.2.,1.1.3.2., 1.2.4.3, .1.2.5.3 | Enhanced forest management and governance and carbon finance is mainstreamed across citizens and ensuring that it reaches all, including women | At least 60% of the women involved in the various sectors of interest are trained. | Number of training session, participating individuals and institutions ensuring that Pre and post training survey (sex- and age disaggregated perception of trainees after trainings) | Data collected annually by project M&E and Gender staff Y1-Y7<br><br>Gender sensitive survey carried out by external parties at mid-term and terminal evaluation Y3-Y7 | $ 331,007 | $ 9,196,453 | PMU |
| | Awareness and Behavioral Change Sub-Activity 1.1.3.3 | | At least 50% of women are reached by the awareness and behavioral change activities | # of women reached by the awareness, behavioral change and communication campaigns # of women applying practices and approaches introduced by the project | | | | |
| | Updates of the national education curricula Sub-activity 1.2.1.2 | | 100% of the syllabi updated by the project will incorporate the gender-related matters and the concept of no one left behind” | Number of upgraded syllabi directly and clearly addressing gender related issues | | | | |
| Component 2 - Forest structure | Innovation & Technology / knowledge transfer | Enhanced climate adaptive silviculture procedures and technologies are | At least 60% of the women involved in the various | Number of training session, participating | Data collected annually by | $ 20,115 | $ 48,002,928 | PMU |

<table>
<tr><td rowspan="2"></td><td>Sub-Activity 2.1.2.1, 2.1.2.2, 2.1.2.3, 2.2.4.1.</td><td rowspan="2">mainstreamed across youth and ensuring that it reaches all, including women</td><td>sectors of interest are trained.</td><td>individuals and institutions</td><td rowspan="2">project M&E and Gender staff Y1-Y7<br><br>Gender sensitive survey carried out by external parties at mid-term and terminal evaluation Y3-Y7</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td></tr>
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<tr><td rowspan="5">Component 3 - Private sector is engaged in sustainable and climate adaptive silviculture management of forest and decarbonization via AFOLU</td><td rowspan="3">Greening the wood biomass fuel value chain. Sub-activity 3.1. 4.1, 3.1. 4.2. 3.1. 5.1,3.1. 5.2.</td><td rowspan="3">Assessing VCs to identify stakeholders and their roles, gaps and needs of both men and women along value chains and compiling recommendations for addressing their specific needs</td><td rowspan="3">At least 40 enterprises involved in biomass production are assessed for identifying VCs gaps and needs, to ensure that greening recommendations will take into account needs and role of women</td><td>Number of value chains/products assessed</td><td rowspan="5">Data collected annually by project M&E and Gender staff Y1-Y7<br><br>Gender sensitive survey carried out by external parties at baseline, midterm and terminal evaluation Y3-Y7</td><td rowspan="3">$ 48,625</td><td rowspan="5">$ 22,426,217</td><td rowspan="5">PMU</td></tr>
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<tr><td>Number of recommendations provided to actors by gender</td></tr>
<tr><td rowspan="2">Forestry Investments. Sub-Activity 3.1.1.1, 3.1.1.2, 3.1.1.6, 3.1.2.1,3.1.2.2,3.1.2.3, 3.1.3.1, 3.1.3.2, 3.1.3.3, 3.1.3.4.</td><td>Rising awareness of women landowners (<30%) on the possibilities for benefiting of interventions<br><br>Raising awareness and changing mind-sets of women owning land to make decisions about its use in the context of the Project</td><td>90% of the women owning forests/lands are informed about the objectives of the project.</td><td rowspan="2">Number of applications for assistance received by women/men</td><td rowspan="2">$ 1,231,947</td></tr>
<tr><td>Implementing informational campaigns targeting women landlords (<30%) sharing information on:<br>• financial support<br>• knowledge and skills</td><td>30% of the women owning forests/lands engage in project investments.</td></tr>
</table>

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | needed to run new type of business | | | | | | |
| | | Providing investment support for forest restoration/ bioenergy plantations establishment (as a part time job or a start-up) for women entrepreneurs and landlords | | Land surface per activity included by women/men | | | | |
| | | | | Number of companies/farms involved per activity owned by women/men | | | | |
| | Capacity development and technical assistance for companies, service providers, national finance institutions, associations, and service providers. Sub-activity 3.2.1.1, 3.2.1.2,3.2.1.1, 3.2.2.1, 3.2.2.2, 3.2.2. 3.3.1.1. | Enhancing knowledge of private sector and institutional actors about climate risk assessment and decarbonization | 100% women led enterprises identified are engaged and have produced their decarbonization strategies | Number of awareness raising campaigns organized on decarbonization | | $ 774,000 | | |
| | Innovation & Technology / knowledge transfer. Sub-activity 3.2.2.2, 3.3.2.1. | Strengthening consulting services for supporting private sector engagement in climate risk assessment and decarbonization. | At least 10 consulting firms trained for providing services. All of their woman consultants are trained | Number of consulting firms involved | | | | |
| | | | 30% of women employed in concerned sectors and institutions are engaged in project's activities | Number of women engaged with designing, implementation and/or monitoring of climate risk assessment and decarbonization strategies of companies | | | | |
| **Project Management** | N/A | Improvement of technical and institutional capacities of the team members and partner institutions participating in project -in The Ministry of Agriculture, Forestry and Water Management, PE "Serbia Shume" and "Vojvodina shume " and gender focal points and the focal points in local administrations on the gender dimension | All team members and project partners have completed gender mainstreaming training in the forestry sector. The training will also include PSEA and SEAH module of trainers and have access to gender expert's support when needed. | Number of private stakeholders included in the database | Y1 – Y7 (support anticipated during the whole project duration) | $ 2,170,684 | $ 4,185,983 | PMU |
| | | Project team (e.g. gender and social expert) and | | Number of women and men entrepreneurs, | | | | |

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| | | stakeholders (government authorities) technical and institutional capacities increase on the gender concept and the gender dimension in development (including facilitation skills to engage women) | | farmers and their groups sensitized<br>Number of the project staff and staff of focal points trained on the gender mainstreaming demonstrating increased awareness of gender and inclusions issue via pre and post training survey (perception of trainees after trainings)<br>Sex and age disaggregated | | | | |
| | | Database of land properties and needs for their remediation or improvements gender enriching with disaggregated data | Gender expertise is permanently available to all project team members and partners (gender and social specialists engaged to organize trainings for project partners and provide regular expertise local communities)<br>Gender-responsive Stakeholder engagement strategy and Communication strategy are prepared and implemented | Gender specialist involved in the project | | | | |
| **Indicative total cost (USD)** | | | | | | $ 4,676,378 | | |

**ANNEX 1.** PSEA RISK SCREENING CHECKLIST

| Ensuring basic risk mitigation measures are in place ahead of stakeholder engagement | Responsibility | Comments | Link | Source |
|---|---|---|---|---|
| Does the AE have a SEAH Policy (or SEAH provisions in another policy)? | AE | Yes, FAO disposes of a SEAH policy.<br>Harassment in all its forms is contrary to the United Nations Charter, the FAO Staff Regulations and<br>Rules and the Standards of Conduct for the International Civil Service. In line with Article 1 of the FAO Staff Regulations, the Director-General will endeavour to always ensure the highest standards of conduct by staff members.<br>2. This Policy on Harassment, Sexual Harassment and Abuse of Authority is consistent with the principles and values of the UN system concerning the prevention of harassment and abuse of authority. | The relevant FAO policies that address SEAH are<br>Policy on Sexual Harassment<br><br>Policy on the Prevention of Harassment, Sexual Harassment and Abuse of Authority<br><br>Protection from Sexual Exploitation and Sexual Abuse (PSEA)<br><br>Whistleblower Protection Policy | FAO |
| If the AE has contracted out stakeholder consultations, does that entity have a SEAH Policy (or are they contractually bound to apply the AE's)? | AE/Consultant | As per contracts with external entities, PSEA measures also apply, in accordance with relevant contractual clauses in agreements. For this project, *stakeholder consultations were not outsourced.*<br><br>The FAO PSEA relevant policies are also binding to person of any contractual status with FAO. | | FAO |
| Does the AE have an employee Code of Conduct? | AE | Yes, FAO disposes of a personnel code of ethical conduct (2021) that provides clear indication about PSEA and Prevention of Sexual Harassment, Abuse of Authority and Harassment. | https://www.fao.org/3/cb4863en/cb4863en.pdf<br><br>FAO Code of Ethical Conduct | FAO |
| If the AE has contracted out stakeholder consultations, does | AE/Consultant | For this project, stakeholder consultations were not outsourced.<br><br>All FAO personnel (including, but not limited to, staff members, consultants, national project personnel (NPP), personal service providers, | | |

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| that entity have an employee Code of Conduct (or are they contractually bound to apply the AE's)? | | volunteers, and interns) are expected to behave in accordance with the ethical standards in the FAO Code of Ethical Conduct. | | |
| Have AE employees and consultants conducting stakeholder consultations been trained on preventing SEAH and the Code of Conduct? | AE/Consultant | PSEA training is among the mandatory trainings for all FAO personnel of all categories. In the next column is the list of mandatory trainings on SEAH and Ethical Code that all FAO employees must complete at the start of their employment. | https://www.fao.org/3/nd482en/nd482en.pdf<br><br>Prevention of Sexual Exploitation and Abuse (PSEA) (Mandatory)<br><br>Prevention of Harassment, Sexual Harassment and Abuse of Authority (Mandatory)<br><br>United Nations Course on Working Together Harmoniously (Mandatory)<br><br>Ethics and Integrity at the United Nations (Mandatory)<br><br>FAO Whistleblower Protection Policy (Mandatory) | FAO |
| Does the AE have a grievance mechanism in place in case of early SEAH complaints from stakeholder engagement? | AE | Yes, FAO has a GM in place for early SEAH complaints. FAO has a specific channel for SEA, which goes directly to the Office of the Inspector General. There is a 24h/ 7 days hotline for this. | https://www.fao.org/environmental-social-standards/en/<br><br>SEA complaints can be lodged through FAO's Office of the Inspector General by email, phone or online using EthicsPoint | FAO |
| Does the AE have a specialist on staff who can undertake the more advanced | AE | FAO confirms that sufficient technical resources and capacities to ensure compliance with GCF requirements regarding SEAH are available (see also the FAO Annual Report on Corporate Policy, Processes and Measures | https://www.fao.org/3/nk304en/nk304en.pdf | FAO |

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| assessment in Stage 4 as well as deal with early SEAH complaints if they arise; and if not, does the AE require budget and /or assistance with this? | | on the Prevention of Harassment, Sexual Harassment and Sexual Exploitation and Abuse)<br><br>FAO has PSEA specialists at global level that can support country-level PSEA Focal Points to undertake risk assessments. | | |
| Contextual Level (and Baseline Conditions) | Reference | Comments | | |
| Does the country have laws prohibiting sexual harassment / stalking generally? | National /State law (Gender Assessment) | Yes, Serbia disposes of number of laws, policies and strategies to contrast SEAH. These include - among others: the family law (2005), the law on Amendments to the Criminal Code (2016), the law on preventing domestic violence (2017) and the law on free legal aid (2019) | https://evaw-global-database.unwomen.org/es/countries/europe/serbia?pageNumber=2 | UNW |
| Do labour laws prohibit sexual harassment in the workplace? | National/ State law (Gender Assessment) | Yes, Serbia disposes of a specific law. The law which prohibits harassment in the workplace (including sexual harassment) is the Law on the prohibition of harassment at work (Official gazette of Serbia no. br. 36/10). This Law has been in force since 2010. | https://www.ilo.org/dyn/travail/docs/2403/Labour%20Law%20Republic%20of%20Serbia.pdf | ILO |
| Does the country have laws prohibiting intimate partner violence (IPV)? | National/ State law (Gender Assessment) | Yes, IPV is addressed by the law on Amendments to the Criminal Code (2016), the law on preventing domestic violence (2017) and the law on free legal aid (2019) | https://evaw-global-database.unwomen.org/es/countries/europe/serbia?pageNumber=2 | UNW |
| What is the prevalence of GBV in the country? | National statistics (Gender Assessment) | According to UN-Women [UNW, 2023], in Serbia, at the national scale, lifetime Physical and/or Sexual Intimate Partner Violence concerned in 2019 17 % of ever-partnered women aged 18-74 years while 34% while 2% of the same age cohort experienced non-partner sexual s violence (lifetime). | https://evaw-global-database.unwomen.org/es/countries/europe/serbia?pageNumber=2 | UNW |
| What is the legal age a person can marry? | National law | Serbia deposited the Convention on the Rights of the Child in 2001, which sets a minimum age of marriage of 18, and the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) in | https://evaw-global-database.unwomen.org/es/countries/europe/serbia/2005/family-law-official-gazette-of-the-rs-no-18-2005 | UNW |

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| | | 2001, which obliges states to ensure free and full consent to marriage. Finally, article 23 of the family law (2005) sets the minimum aga at 18. | | |
| Despite any laws, what is the prevalence of child marriage in the country? | National statistics | 1% of girls in Serbia are married before their 15th birthday and 8% before their 18th birthday. Statistics show a higher percentage in rural areas in both cases. Urban 0.7% and 4.6%, Rural 2.5% and 14%. | https://mics-surveys-prod.s3.amazonaws.com/MICS6/Europe%20and%20Central%20Asia/Serbia/2019/Survey%20findings/Serbia%20%28National%20and%20Roma%20Settlements%29%202019%20MICS%20SFR_English.pdf | Serbia Statistic Office, UNFPA, EU |
| What is the income level of the country? | World Bank ranking (H, HM, M, LM, L) | Serbia is an upper middle-income country | https://data.worldbank.org/country/RS | WB |
| Where does the country rank on global gender indices? | World Bank Reports / Other | Compared to EU Member States, the United Kingdom, and other countries in the region, Serbia ranks 21st, between Croatia and North Macedonia. Compared to Croatia, Serbia has lower values on the index in the sub-domains of participation (77.0 versus 79.6), but higher values in the sub-domain of segregation and quality of work (62.5 versus 61.4). Compared to North Macedonia, Serbia has higher values in the sub-domain of participation (77.0 vs. 68.2), but significantly lower in the sub-domain of segregation and quality of work (62.5 vs. 70.7). Compared to fi-rst-ranked Sweden, Serbia has this domain of Index lower by 13.5 points, while compared to last-ranked Italy, it has an index higher by 6.1 points. | https://eurogender.eige.europa.eu/system/files/events-files/gender_equality_index_for_serbia_2021.pdf | EU |
| Is there a national action plan on GBV and/or sexual harassment? | National government | Yes, Serbia is currently executing the National Strategy for the Prevention and Combating of Gender-Based Violence against Women and Domestic Violence 2021-2025. | https://evaw-global-database.unwomen.org/es/countries/europe/serbia/2021/national-strategy-for-the-prevention-and-combating-of-gender-based | UNW |
| Does the country have specialized services for survivors of GBV (at both the national and local level) including women's | Local gov / NGOs | Yes, there are at least 10 organization financed by both the state and international donors that are active and specialized in providing services for survivors of GBV (at both the national and local level) including women's shelters, adequate medical facilities and facilities which provide psycho-social support. | https://eca.unwomen.org/en/stories/feature-story/2023/07/prioritizing-womens-needs-through-local-gender-responsive-budgeting-in-serbia#:~:text=UN%20Women%2C%20under%20a%20project,to%20introduce%20GRB%20since%202015. | UNW |

| | | | | |
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| shelters, adequate medical facilities and facilities which provide psycho-social support? | | | | |
| Is the country currently experiencing war, internal conflict or humanitarian disaster? | National / Media | No | | |
| Project Level Risks | Responsibility | Comments | | |
| Are women concentrated in lower paid roles and mostly line-managed and supervised by men? | AE | Yes. In Serbia, women's wages are lower than men's even in professions where women make up the majority of the workforce, such as in the health care and welfare system. The wage gap between women and men is 8.8% | https://www.serbianmonitor.com/en/wage-gap-between-women-and-men-in-serbia-is-8-8-in-favour-of-men/ | Academia |
| Are piece-rate systems or other performance-related pay structures used where individuals are in control of how much other workers get paid? | AE | No | | |
| Will project workers have control over life-changing resources such as the allocation of compensation for displacement or access to basic or | AE | Project workers will not be displaced. All workers will be selected within a radius of 25 miles from the worksite. | | |

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| highly sought-after resources? | | | | |
| Will security personnel be used? Will they be armed? | AE | No, the project will not employ armed security personnel. | | |
| Will there be an influx of male workers into the project area (as opposed to only using local labour)? | AE | All workers in project areas (see selection criteria of activities) will be selected among men and women within a 25 km radius. Therefore, there will be no influx of male workers in project areas. Furthermore, project activities will be in remote forested areas generally far from houses and communities. | | |
| Are local communities poor and lacking basic resources? | AE | Although poverty rate is higher in rural areas, local communities, by large, do not lack basic resources. | | |
| Will migrant workers be employed by the project, especially those who may not speak the local language? Will they be employed on a temporary or daily basis? | AE | Hiring of workers will be made following the laws and regulations of the Republic of Serbia and workers will need to abide with the FAO code of conduct and FAO policies. As works will occur in remote forested areas of the country, the project does not expect to have migrant workers. | | |
| Will project workers all have formal contracts? | | Yes, hiring of workers will be made following the laws and regulations of the Republic of Serbia (Labor Law 24/05, 61/05 and 54/09). These regulate contracts, wages all the other aspects related to labour. Workers will need to abide with the FAO code of conduct and FAO policies. | | |
| Will goods frequently be transported over long distances, especially through poor and/or remote communities? | AE | No, the project will not require transport of good, people or materials over long distances. | | |

| | | | | |
|---|---|---|---|---|
| Are worksites or project activities based in remote locations? Will worksites be spread out, with isolated spaces? | AE | Worksites will be in remoted forest and rural areas of the country. Nonetheless, as workers will be selected from communities within a radius of 25 km from the worksite, these will not require the establishment of camps or other temporary accommodation structures. | | |
| Will project workers live in the community or in worker housing? If in worker housing, is it mixed sex? | AE | As workers will be selected from communities within a radius of 25 km from the worksite, these will not require the establishment of camps or other temporary accommodation structures. | | |
| Will workers be required to travel long and potentially unsafe distances, and at times of day when transport options may be limited? | AE | Workers will be selected from communities within a radius of 25 km from the worksite. Based on the criteria identified in the FFP, worksite must be accessible by road. Furthermore, transport from collection points in accessible areas to worksites will be guaranteed by the project through its partners and service providers. | | |
| Will the project operate in highly pressurized work environments, with tight seasonal deadlines? | AE | The project will work with tight seasonal deadlines, but it will not be in highly pressurized work environments. | | |
| Is the project located within a male-dominated sector where female workers will be employed? | AE | Forestry is a sector where women employment does not go above 14%. Nonetheless, employment will be open and accessible to all without any gender restriction. | | |
| Have communities, especially low income/ vulnerable | AE | Participation in the project is voluntary and involves both state owned forests and forest owned by municipalities and private operators. | | |

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| communities, voluntarily raised concerns in relation to SEAH/GBV during consultations? | | Therefore, the project will assess SEAH/GBV during final identification of forestry investment sites. | | |
| Have any changes been made to project design or adaptive management undertaken due to concerns of stakeholders and communities? (If yes, work through this checklist again) | AE | No, stakeholders have not raised concerns. | | |

## **ANNEX 2.** PSEA RISK MITIGATION MATRIX

| Description of [Potential] Risks | Likelihood (LMH) | Potential Impact (LMH) | Risk Mitigation Measures |
|---|---|---|---|
| Contextual Risks | | | |
| **National Level Risks**<br>• Lack of strong legal system to enforce laws<br>• Low levels of prosecution of SEAH incidents | L | L | • Ensure presence in the PMU of a gender and social expert with extensive experience of local context.<br>• Ensure constant coordination between the project gender and social expert, the National Gender Coordinator, and the Regional Gender Coordinator in FAO.<br>• Work with relevant gender/social welfare Government ministries and departments, other anti-gender-based violence organizations or networks.<br>• Strong enforcement of the AEs SEAH (and/or its equivalent) policy.<br>• Enforcement of SEAH related laws as it pertains to the project/program.<br>• Liaise institutional stakeholders with providers of SEAH training (e.g. UNFPA, UNWOMEN, UNIVEF, OCHA among others) to project stakeholders and communities. |
| **Societal Risks**<br>• Sociocultural norms that do not challenge SEAH<br>• Low levels of awareness on rights, SEAH etc.<br>• Limited services for SEAH survivors | M | M | • Ensure regular visits to communities and local institutions of the gender and social expert to work with local government or authorities and to sensitize community members on SEAH safeguarding.<br>• Identify champions where applicable to act as allies on SEAH safeguarding.<br>• Provide SEAH training to project stakeholders and communities. |
| **Project Risks** | | | |
| • Limited SEAH protection services in project/program area<br>• High rates of femicide or sexual violence (e.g., used as a tactic of war) in project/program areas<br>• Women fear that participation or employment in the project/program may exacerbate ongoing forms of SEAH. | L | L | As above societal risks and:<br>• Through the work of the Gender and Social expert of the project, support local officials in campaigns on prevention of SEAH;<br>• Leverage existing relationships with government stakeholders; identify champions / supporters / changemakers within the government (specifically on SEAH).<br>• Conduct SEAH awareness-raising and sensitization campaigns within the community.<br>• Inform the community the community on SEAH risks, explain how to report them and the services available including SEAH GRM established by the project. |

The project aims to upgrade Serbia's male-dominated forestry sector and associated decarbonization business by fostering inclusive and gender-responsive approaches that ensure no one is left behind. By promoting women's participation in developing innovative business models, the project will help create sustainable and equitable opportunities for all, contributing to both climate resilience and gender equality.

| Component / Outcome | Gender Sensitive Project Activities | Gender sensitive activities | Target | Indicators | Timeline | Budget | Indicative cost of the Component/ Output | Responsible partner |
|---|---|---|---|---|---|---|---|---|
| Inception Phase | Consultations with local parties (including women and women focused organizations, private forest land holders and CSOs) to upgrade GAP | Data on ownership of land for afforestation collected and gender disaggregated. Conducting consultations for identifying specific measures related to territory specific features of the Serbian territory out of reassessed pilot areas, to identify employability of women in forest sector, women's preferences related to use of forests to upgrade gender action plan with measures and to provide relevant support within the planned project activities. | At least 90% of interested parties on the local level, primarily forest land holders interested in afforestation and strengthening biomass for energy and other forest products processing value chains. | Number of participatory sessions with local CSO and community representatives<br><br>Number of participatory sessions with women focused organizations<br><br>Number of questionnaires filled and percentage of women included | Inception phase | N/A (included as part of inception workshop) | $ 8,501,506.00 | |
| | Networking institutions and upgrading capacities of key institutions and private sector operators to manage the decarbonization process from a gender skills/capacity perspective and build and interactive national network providing inputs for the evidence based policies through continuous contribution of disaggregated data from forestry business sector | Conducting regular consultations and monitoring through the network of core national gender responsible institutions, private businesses and those participating in the project | All institutions participating in project - The Ministry of Agriculture, Forestry and Water Management, PE "Serbia Shume" and "Vojvodina shume" and gender focal points and the focal points in local administrations have gender capacity enhanced | Number of reports provided to the network (percentage of participants actively participating in the network sessions and reports presented).<br><br>Number of training participants<br><br>Pre and post training survey (perception of trainees after trainings)<br><br>Sex and age disaggregated. | Entire project | $ 50,000.00 | N/A | PMU |
| | Survey of the national education curricula | Exploring possibility gaps in addressing to include gender equality and inclusion specifically in the context of Serbia inchapters in all relevant planned syllabuses or introducing new topic as part of a separate module. Included as part of baseline survey | Whole education system in forestry assessed and targeted for introduction of gender related guidelines – high school legal aspects, academia access to financial and technical assistance, technique and technology employment in green business for women, young people and rural families | Number of subjects identified to get new chapters for gender mainstreaming and social inclusion and in Syllabi for schools and academic institutions involved | Y1 and Y2 | $ 50,000.00 | $ 8,501,506.00 | |
| Component 1 - National level upscaling of sustainable and climate adaptive silviculture and carbon finance framework | Policy Development<br>Sub-activity 1.1.1.4, 1.1.2.2.,1.1.3.2., 1.2.4.3, 1.2.5.3 | | At least 60% of the women involved in the various sectors of interest are trained. | Number of training session, participating individuals and institutions ensuring that<br><br>Pre and post training survey(sex- and age disaggregated perception of trainees after trainings) | | | | |
| | Awareness and Behavioral Change<br>Sub-Activity 1.1.3.3 | Enhanced forest management and governance and carbon finance is mainstreamed across citizens and ensuring that it reaches all, including women | At least 50% of women are reached by the awareness and behavioral change activities | # of women reached by the awareness, behavioral change and communication campaigns # of women reached by the awareness, behavioral change and communication campaigns<br><br># of women applying practices and approaches introduced by the project | Data collected annually by project M&E and Gender staff<br>Y1-Y7<br><br>Gender sensitive survey carried out by external parties at mid term and terminal evaluation<br>Y3-Y7 | $ 331,007.00 | $ 9,196,453.00 | PMU |
| | Updates of the national education curricula<br>Sub-activity 1.2.1.2 | | 100% of the syllabi updated by the project will incorporate the gender-related matters and the concept of no one left behind." | Number of upgraded syllabi directly and clearly addressing gender related issues | | | | |
| Component 2 - Improving energy security and livelihood from climate resilient forest ecosystem and GHG emissions reductions from increased carbon sinks and decarbonization opportunities | Innovation & Technology / knowledge transfer<br>Sub-Activity 2.1.2.1, 2.1.2.2, 2.1.2.3, 2.2.4.1. | Enhanced climate adaptive silviculture procedures and technologies are mainstreamed across youth and ensuring that it reaches all, including women | At least 60% of the women involved in the various sectors of interest are trained. | Number of training session, participating individuals and institutions | Data collected annually by project M&E and Gender staff<br>Y1-Y7<br><br>Gender sensitive survey carried out by external parties at mid term and terminal evaluation<br>Y3-Y7 | $ 20,115.00 | $ 48,002,928.00 | PMU |
| | Updates of the national education curricula and regional knowledge. Sub-Activity 2.2.3.1, 2.2.3.2 | | 4 Syllabi are upgraded | Number of upgraded syllabi | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Component 3 - Engaging Private sector in climate adaptive silviculture and decarbonization investments | Greening the wood biomass fuel value chain. Sub-activity 3.1. 4.1, 3.1. 4.2. 3.1. 5.1,3.1. 5.2. | Assessing VCs to identify stakeholders and their roles, gaps and needs of both men and women along value chains and compiling recommendations for addressing their specific needs | At least 40 enterprises involved in biomass production are assessed for identifying VCs gaps and needs, to ensure that greening recommendations will take into account needs and role of women | Number of value chains/products assessed | Data collected annually by project M&E and Gender staff Y1-Y7 Gender sensitive survey carried out by external parties at mid term and terminal evaluation Y3-Y7 | $ 48,625.00 | $ 22,426,217.00 | PMU |
| | | | | Stakeholders by gender, roles and position in the value chain | | | | |
| | | | | Number of recommendations provided to actors by gender | | | | |
| | Forestry Investments. Sub-Activities 3.1.1.1, 3.1.1.2, 3.1.1.6, 3.1.2.1,3.1.2.2,3.1.2.3, 3.1.3.1, 3.1.3.2, 3.1.3.3, 3.1.3.4. | Rising awareness of women landowners (<30%) on the possibilities for benefiting of interventions<br><br>Raising awareness and changing mind-sets of women owning land to make decisions about its use in the context of the Project | 90% of the women owning forests/lands are informed about the objectives of the project. | Number of applications for assistance received by women/men | | $ 1,231,947.00 | | |
| | | Implementing informational campaigns targeting women landlords (<30%) | | | | | | |
| | | Providing investment support for forest restoration/ bioenergy plantations establishment (as a part time job or a start-up) for women entrepreneurs and landlords | 30% of the women owning forests/lands engage in project investments. | Land surface per activity included by women/men | | | | |
| | | | | Number of companies/farms involved per activity owned by women/men | | | | |
| | Capacity development and technical assistance for companies, service providers, national finance institutions, associations and service providers. Sub-activities 3.2.1.1, 3.2.1.2,3.2.1.1, 3.2.2.1, 3.2.2.2, 3.2.2. 3.3.1.1. | Enhancing knowledge of private sector and institutional actors about climate risk assessment and decarbonization | 100% women led enterprises identified are engaged and have produced their decarbonization strategies | Number of awareness raising campaigns organized on decarbonization | | $ 774,000.00 | | |
| | Innovation & Technology / knowledge transfer. Sub-activity 3.2.2.2, 3.3.2.1. | Strengthening consulting services for supporting private sector engagement in climate risk assessment and decarbonization. | At least 10 consulting firms trained for providing services. All of their woman consultants are trained | Number of consulting firms involved | | | | |
| | | | 30% of women employed in concerned sectors and institutions are engaged in project's activities | Number of women engaged with designing, implementation and/or monitoring of climate risk assessment and decarbonization strategies of companies | | | | |
| Project Management | N/A | Improvement of technical and institutional capacities of the partner institutions participating in project -in The Ministry of Agriculture, Forestry and Water Management, PE "Serbia Shume" and "Vojvodina Shume" and gender focal points and the focal points in local | All team members and project partners have completed gender trainingmainstreaming training in forestry sector. The training will also include (including PSEA and SEAH training module of trainers) and have access to gender advisoryexpert's support when needed. | Number of private stakeholders included in the database | Y1 – Y7 (support anticipated during the whole project duration) | $ 2,170,684.00 | $ 4,185,983.00 | PMU |
| | | Project team and stakeholders (government authorities) technical and institutional capacities increase on the gender concept and the gender dimension in development (including facilitation skills to engage women) | | Number of women and men entrepreneurs, farmers and their groups sensitized | | | | |
| | | | | Number of the project staff and staff of focal points trained on the gender mainstreaming demonstrating increased awaress of gender and inclusions issue via pre and post training survey (perception of trainees after trainings)<br><br>Sex and age disaggregated | | | | |
| | | Database of land properties and needs for their remediation or improvements gender enriching with disaggregated data | Gender expertise is permanently available to all project team members and partners (gender specialist engaged to organize trainings and provide regular expertise) | Gender specialist involved in the project | | | | |
| | | | Gender-responsive Stakeholder engagement strategy and Communication strategy | | | | | |
| **Indicative total cost (USD)** | | | | | | **$ 4,676,378.00** | | |